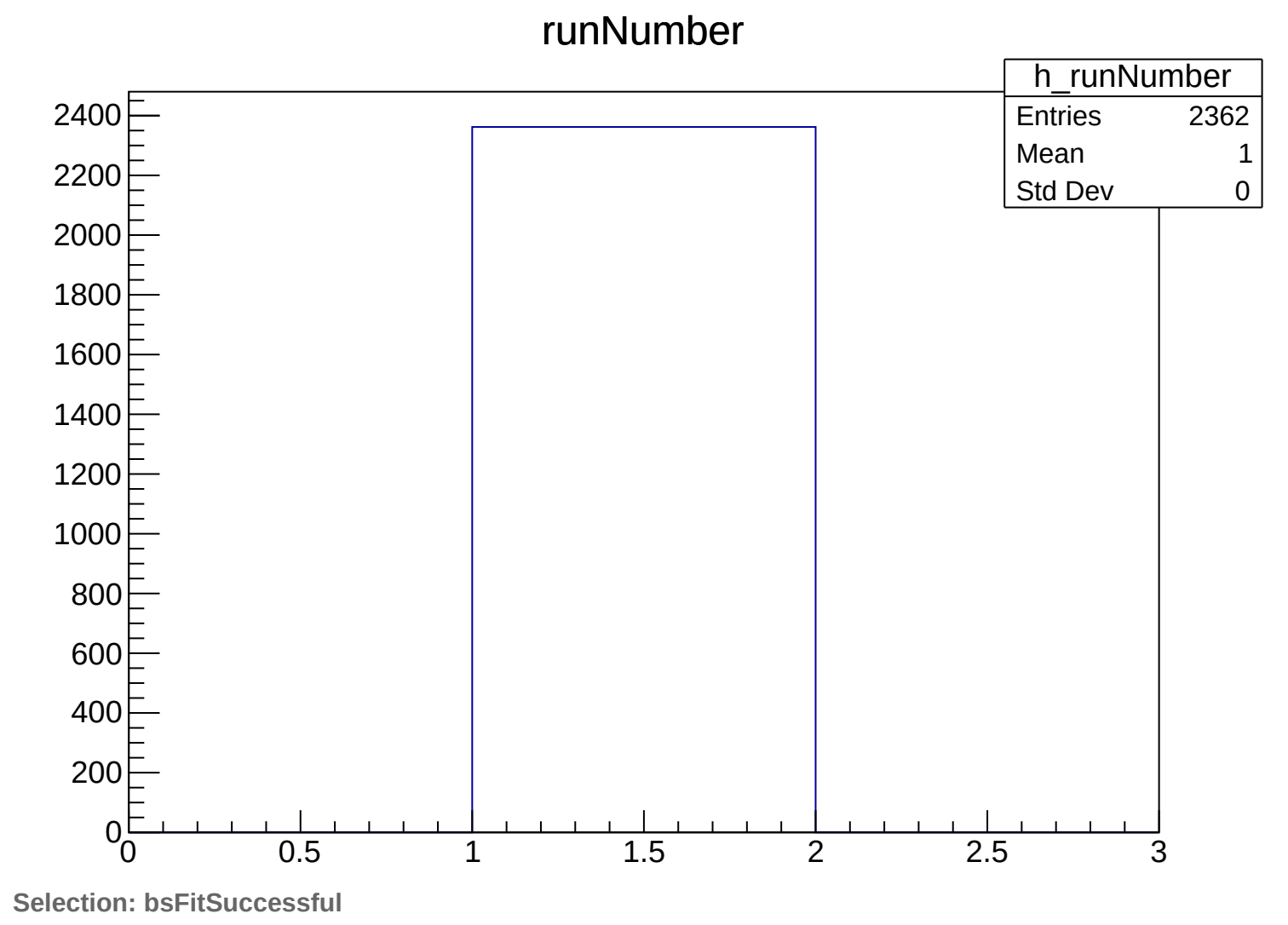
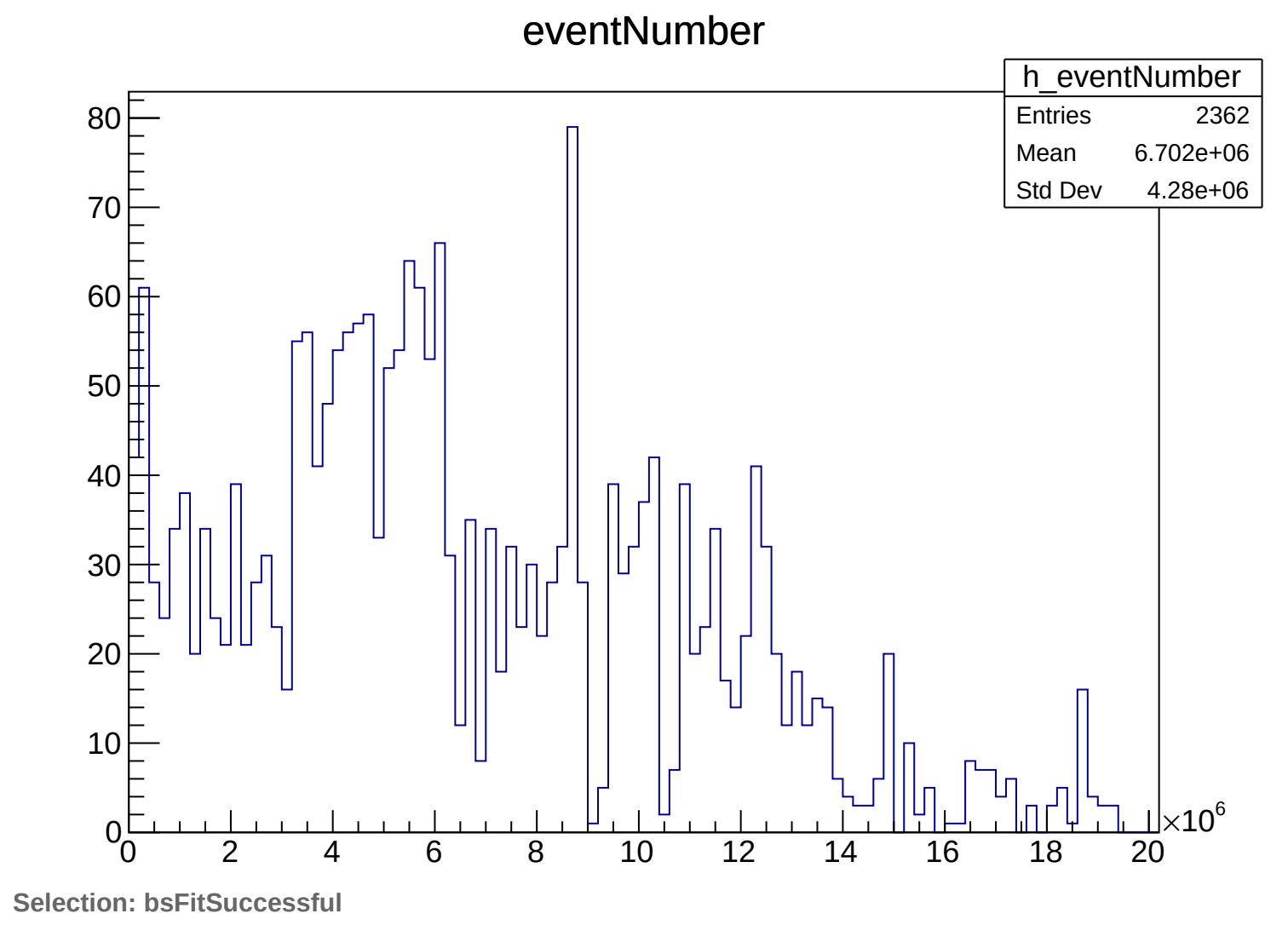
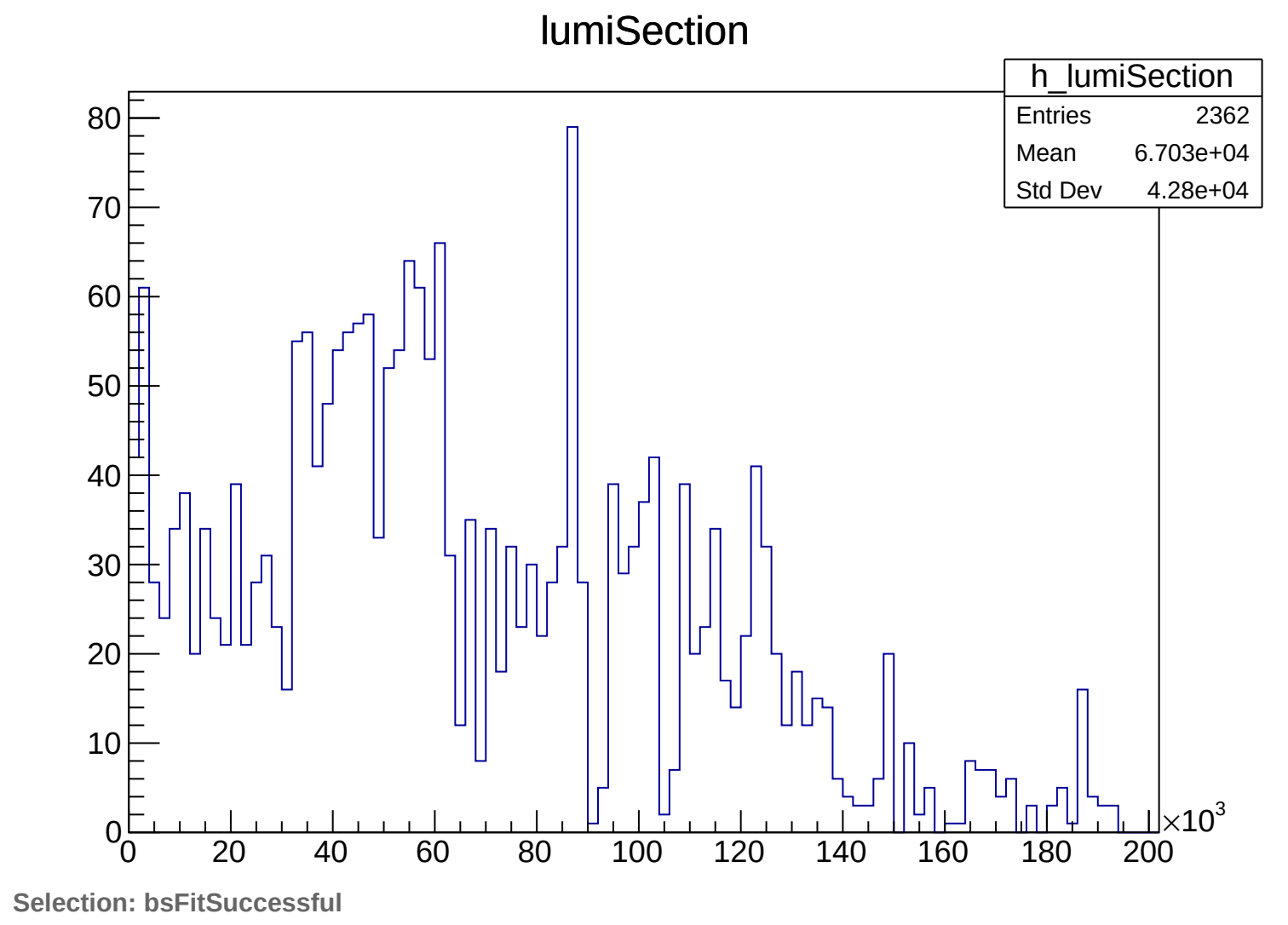
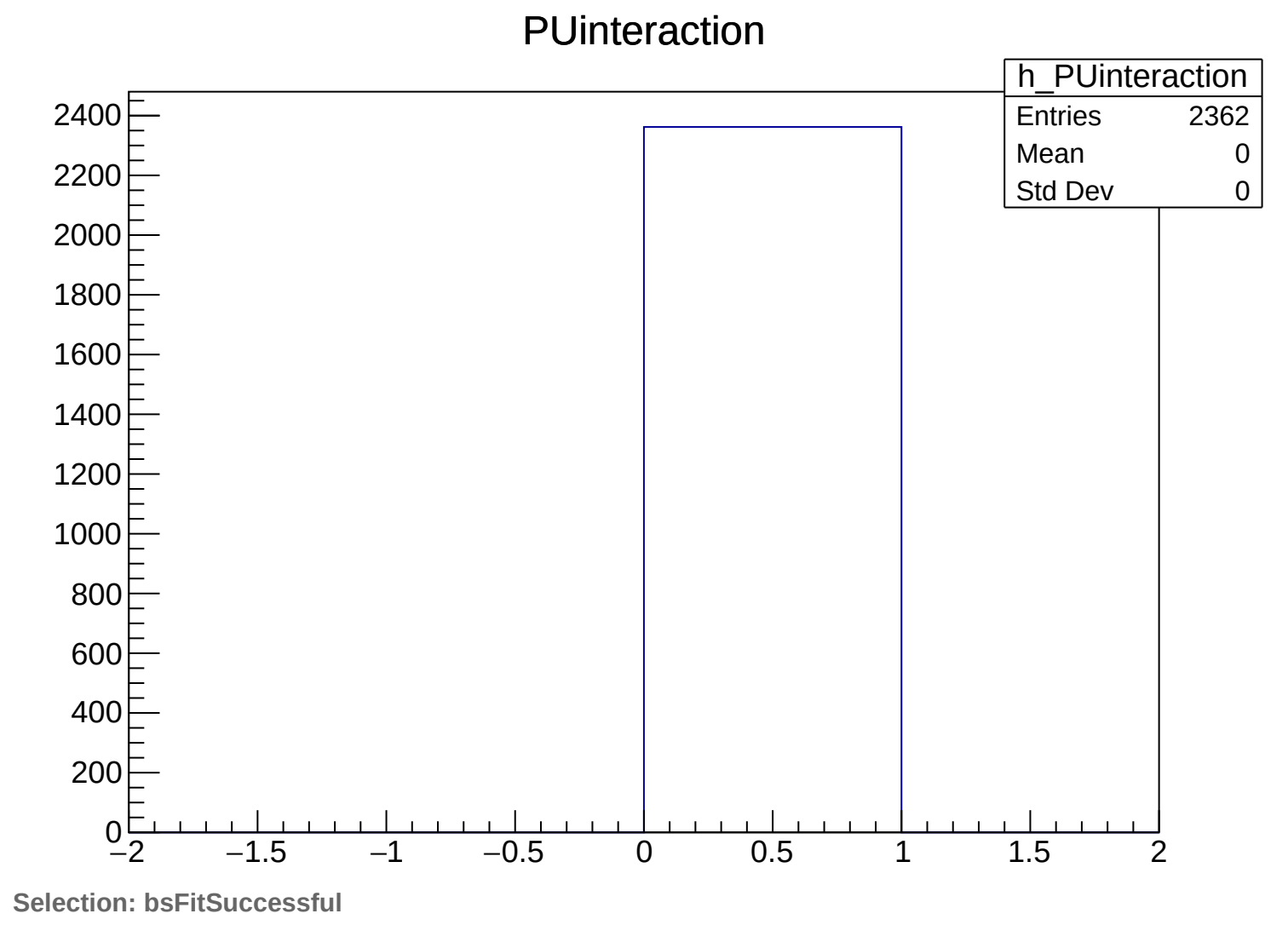
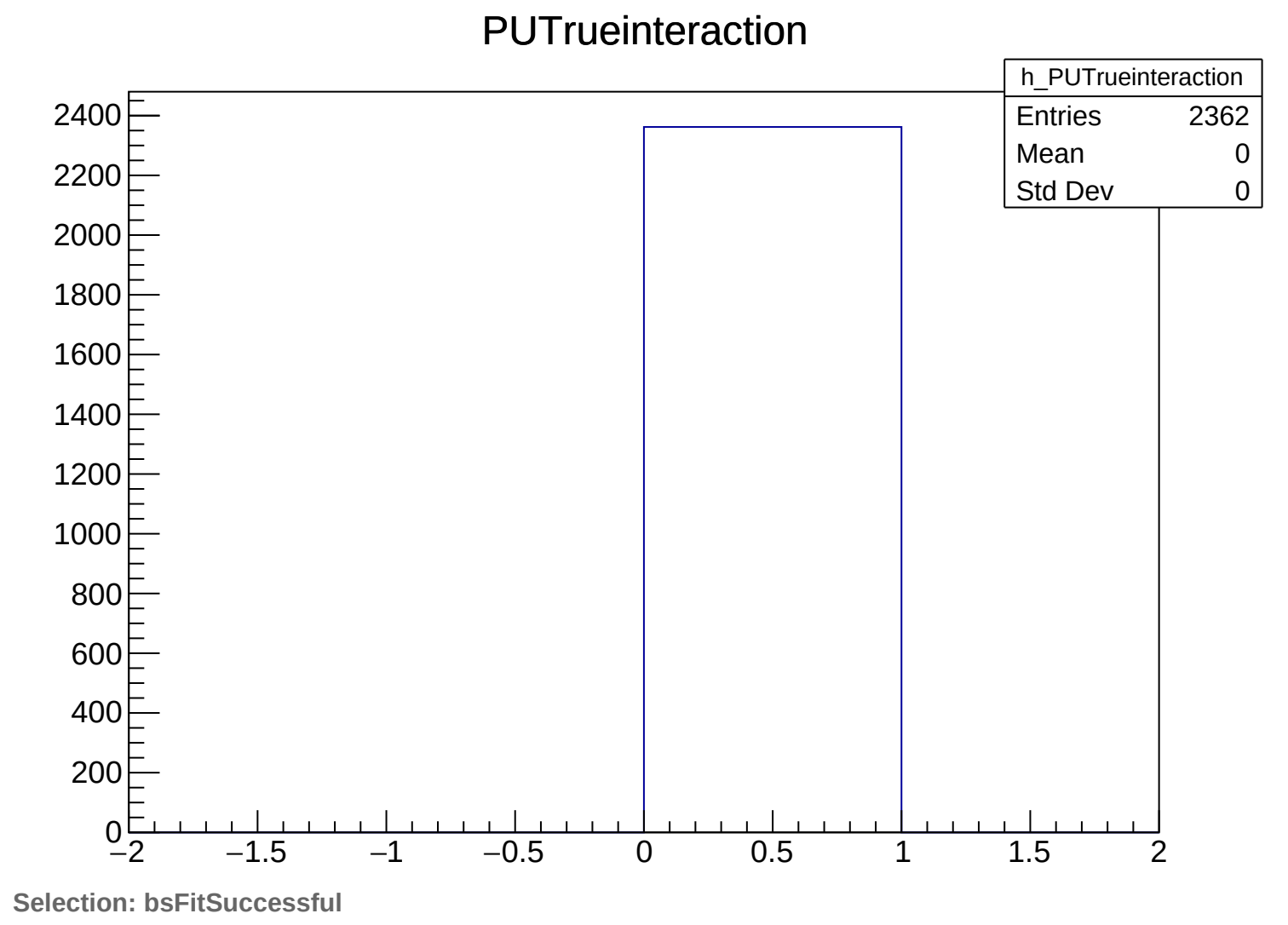
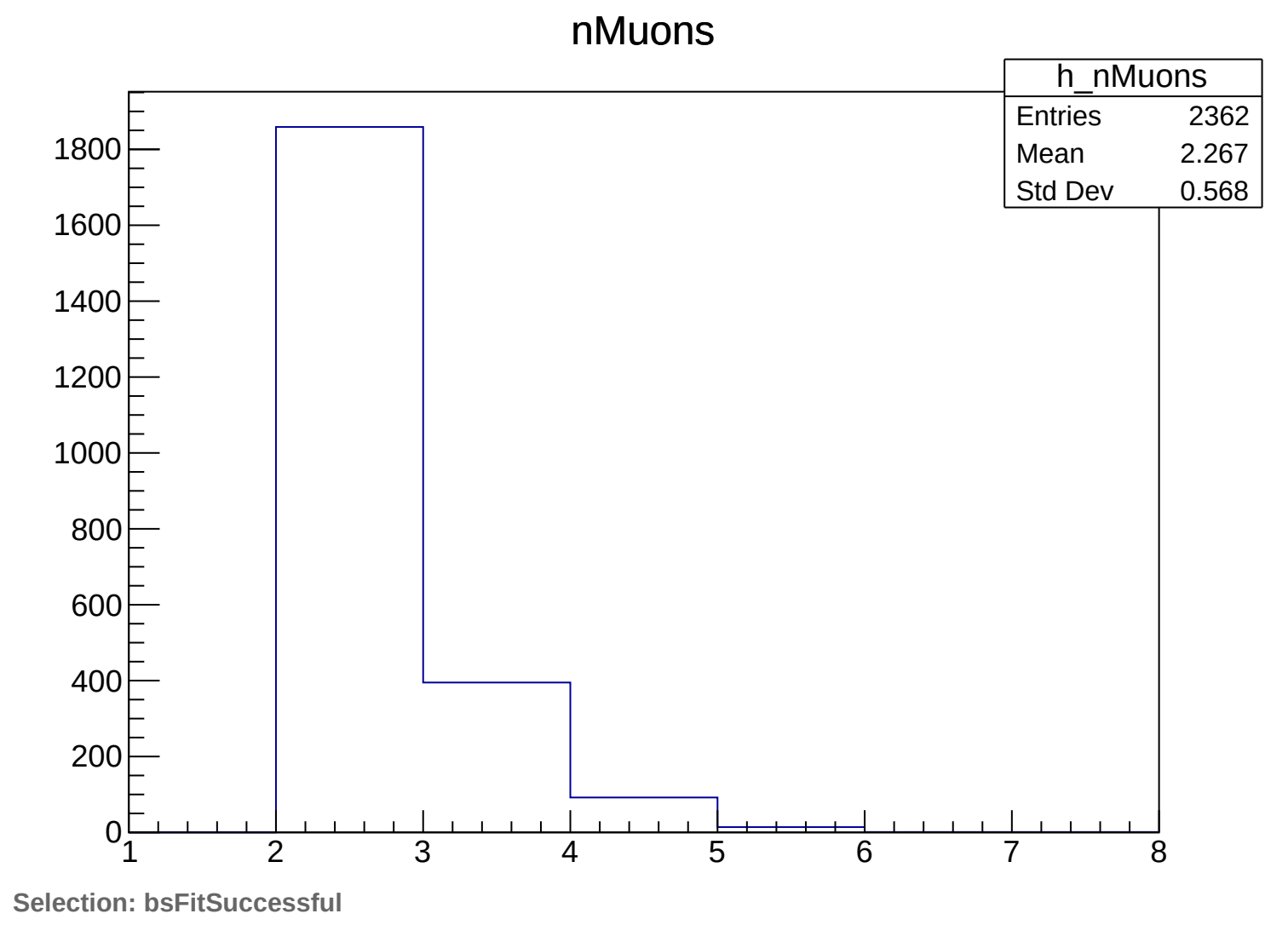

runNumber
h_runNumber
Entries 2362
Mean 1
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.5
1
1.5
2
2.5
3
Selection: bsFitSuccessful

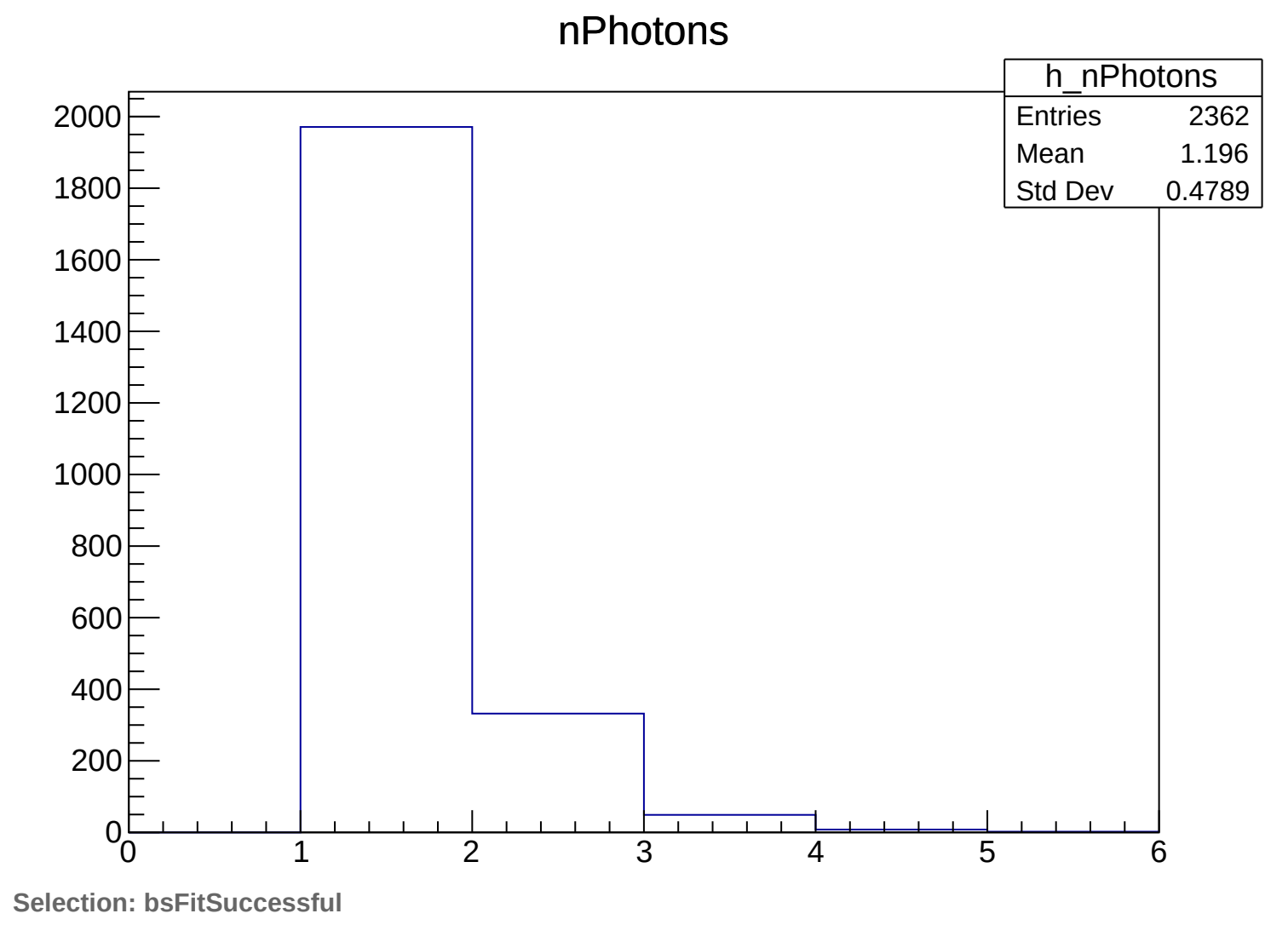

eventNumber
h_eventNumber
Entries 2362
Mean 6.702e+06
Std Dev 4.28e+06
80
70
60
50
40
30
20
10
0
0
2
4
6
8
10
12
14
16
18
20
×10^6
Selection: bsFitSuccessful

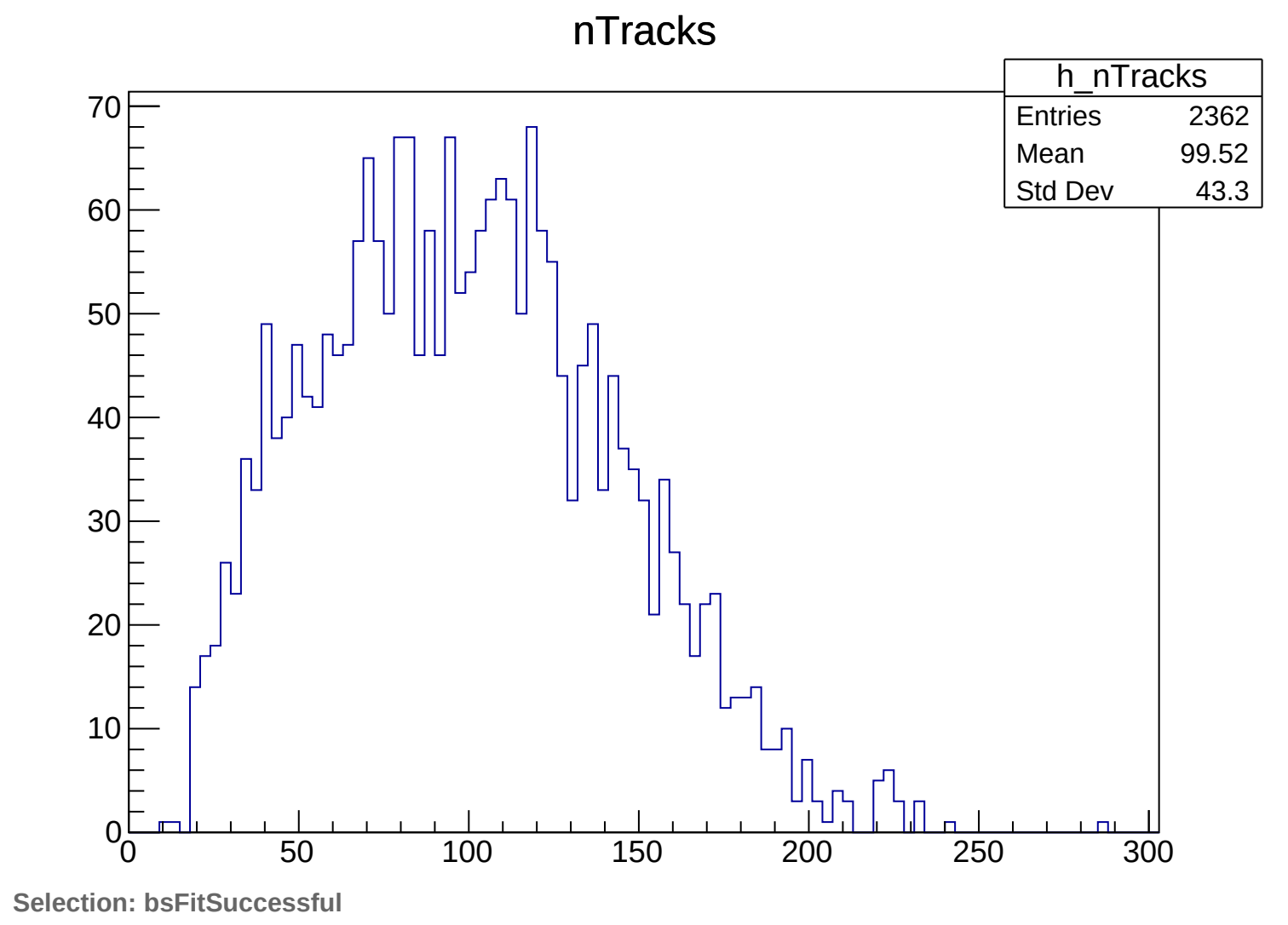

lumiSection
h_lumiSection
Entries 2362
Mean 6.703e+04
Std Dev 4.28e+04
80
70
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
180
200
×10^3
Selection: bsFitSuccessful

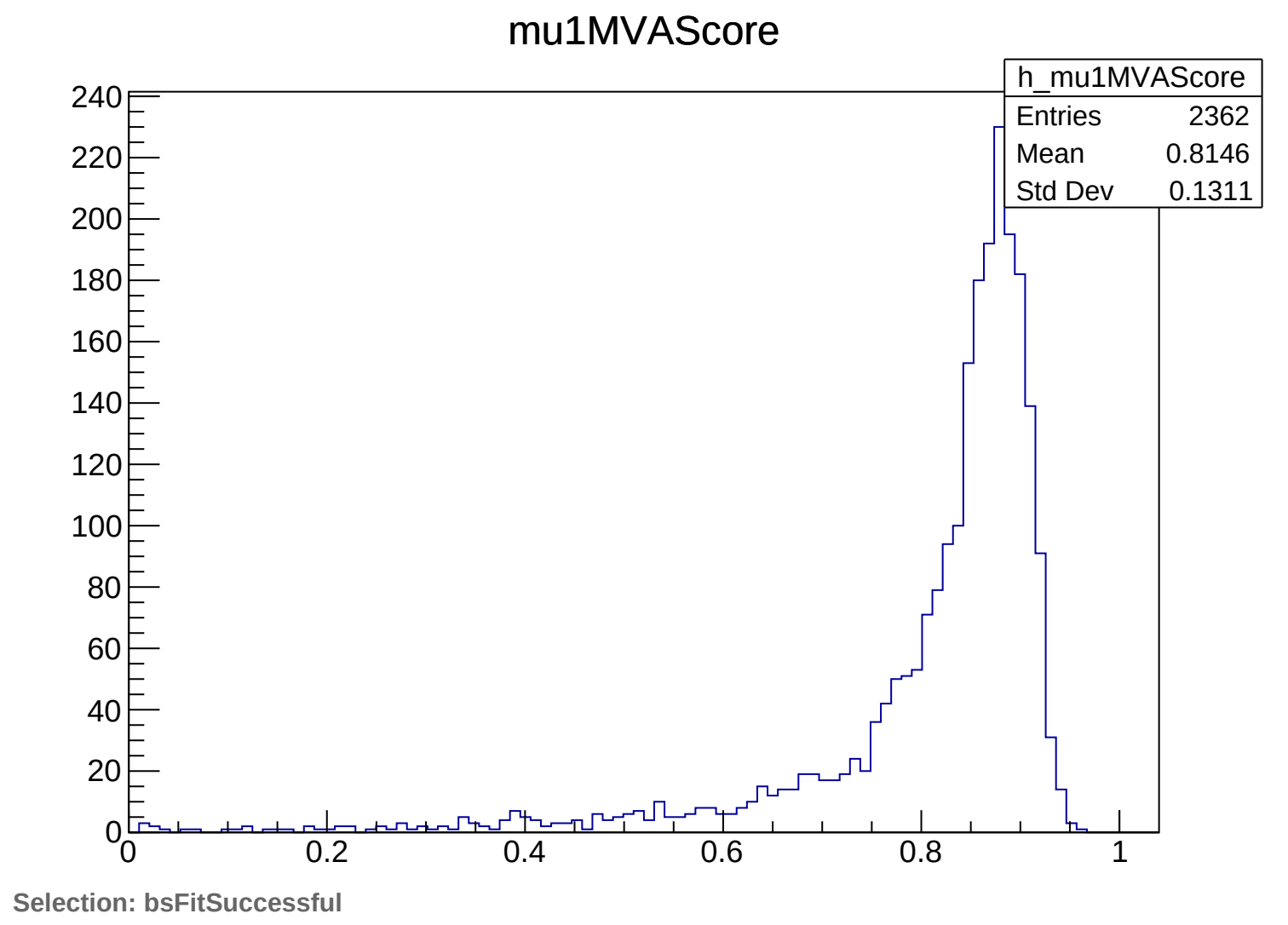

PUinteraction
h_PUinteraction
Entries 2362
Mean 0
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

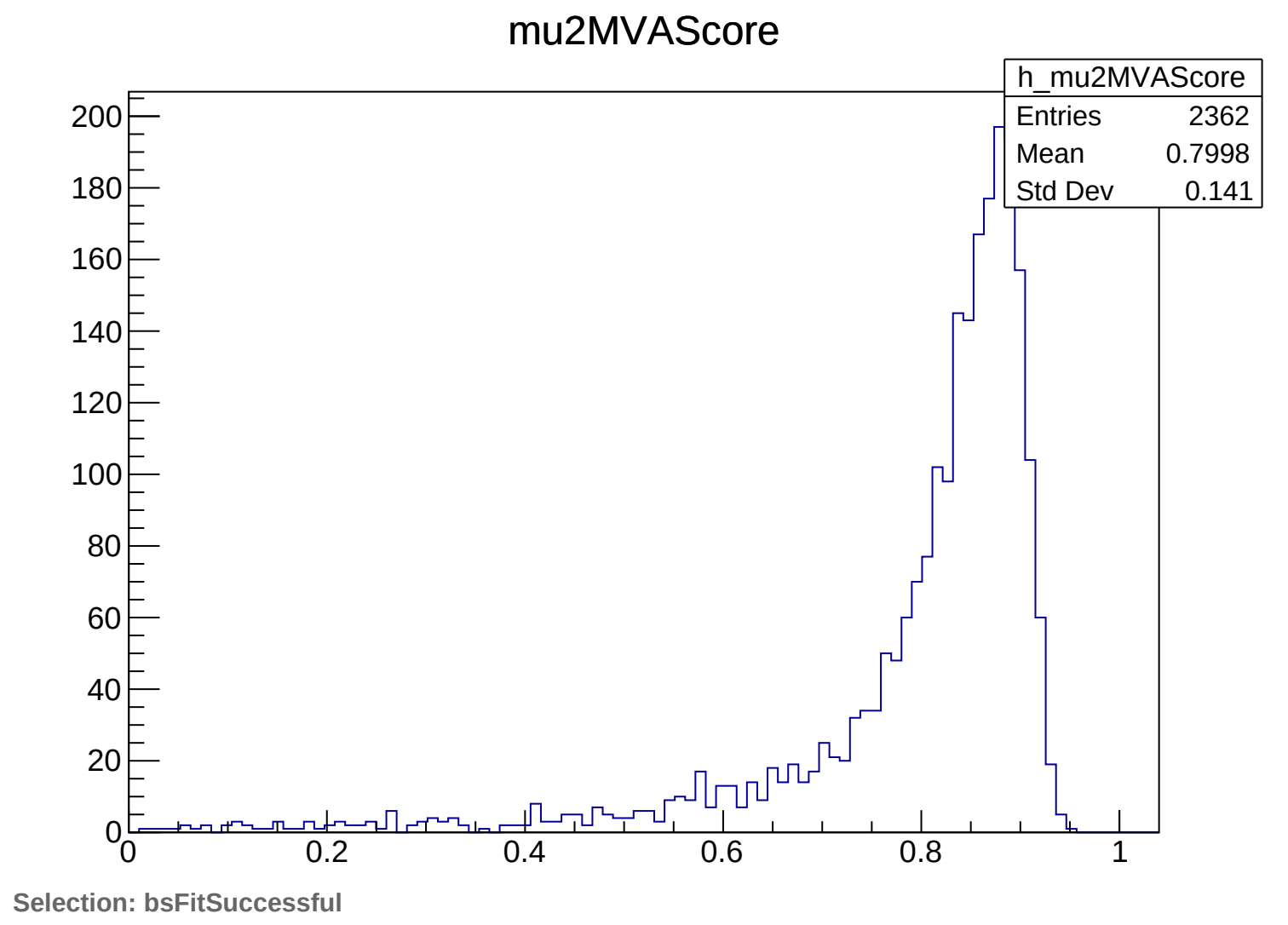

PUTrueinteraction
h_PUTrueinteraction
Entries 2362
Mean 0
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

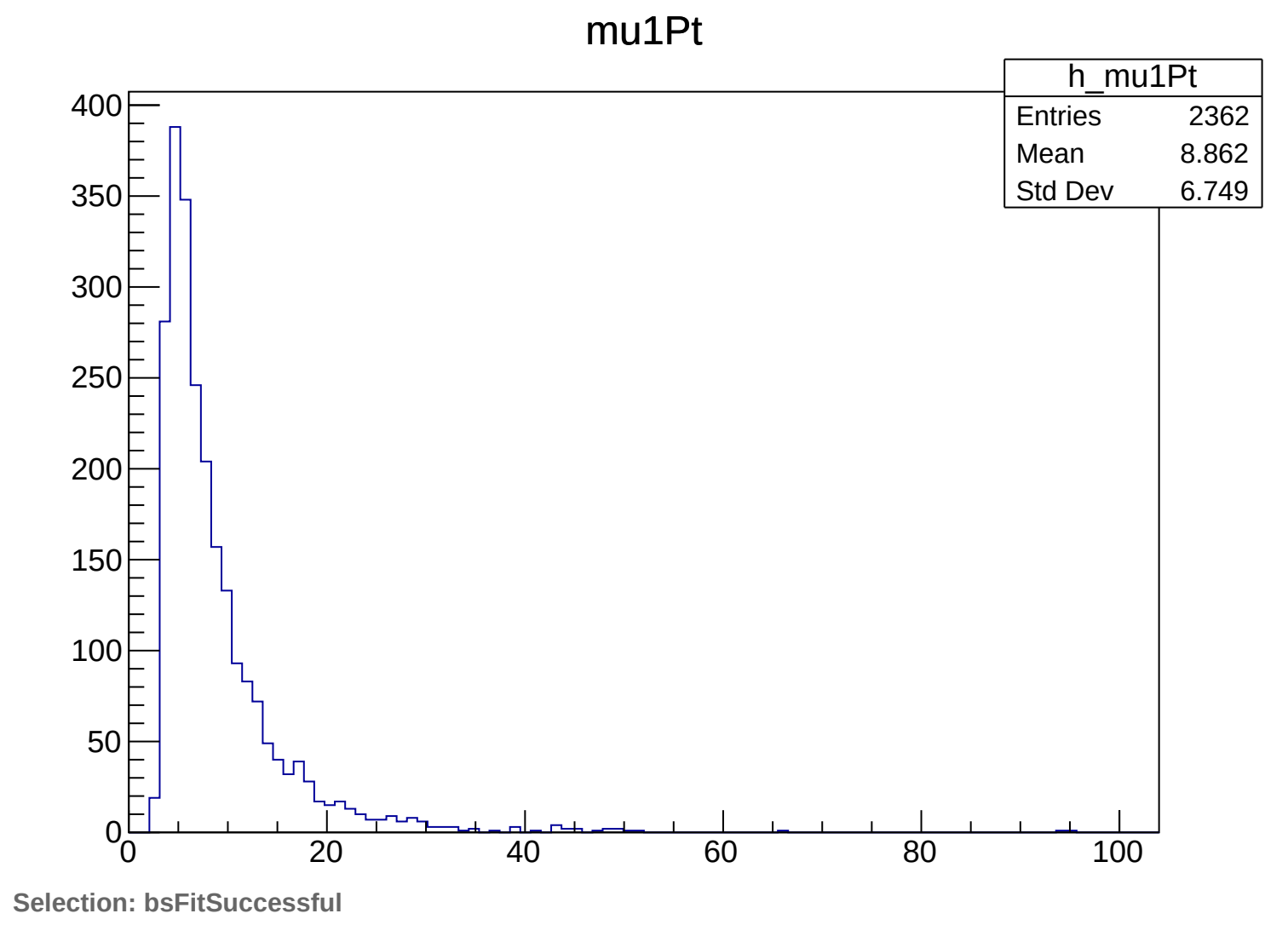

nMuons
h_nMuons
Entries 2362
Mean 2.267
Std Dev 0.568
1800
1600
1400
1200
1000
800
600
400
200
0
1
2
3
4
5
6
7
8
Selection: bsFitSuccessful

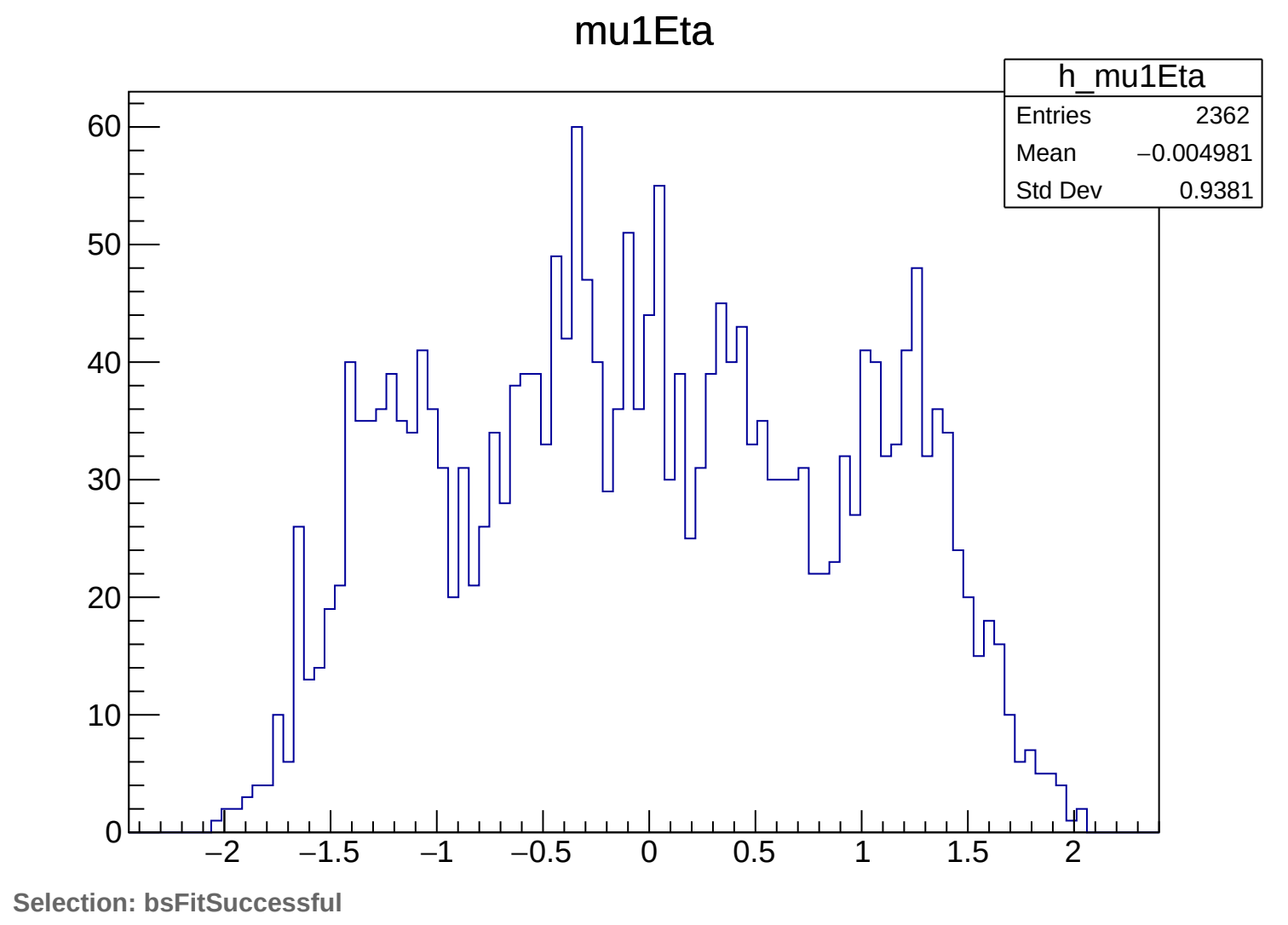

nPhotons
h_nPhotons
Entries 2362
Mean 1.196
Std Dev 0.4789
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
1
2
3
4
5
6
Selection: bsFitSuccessful

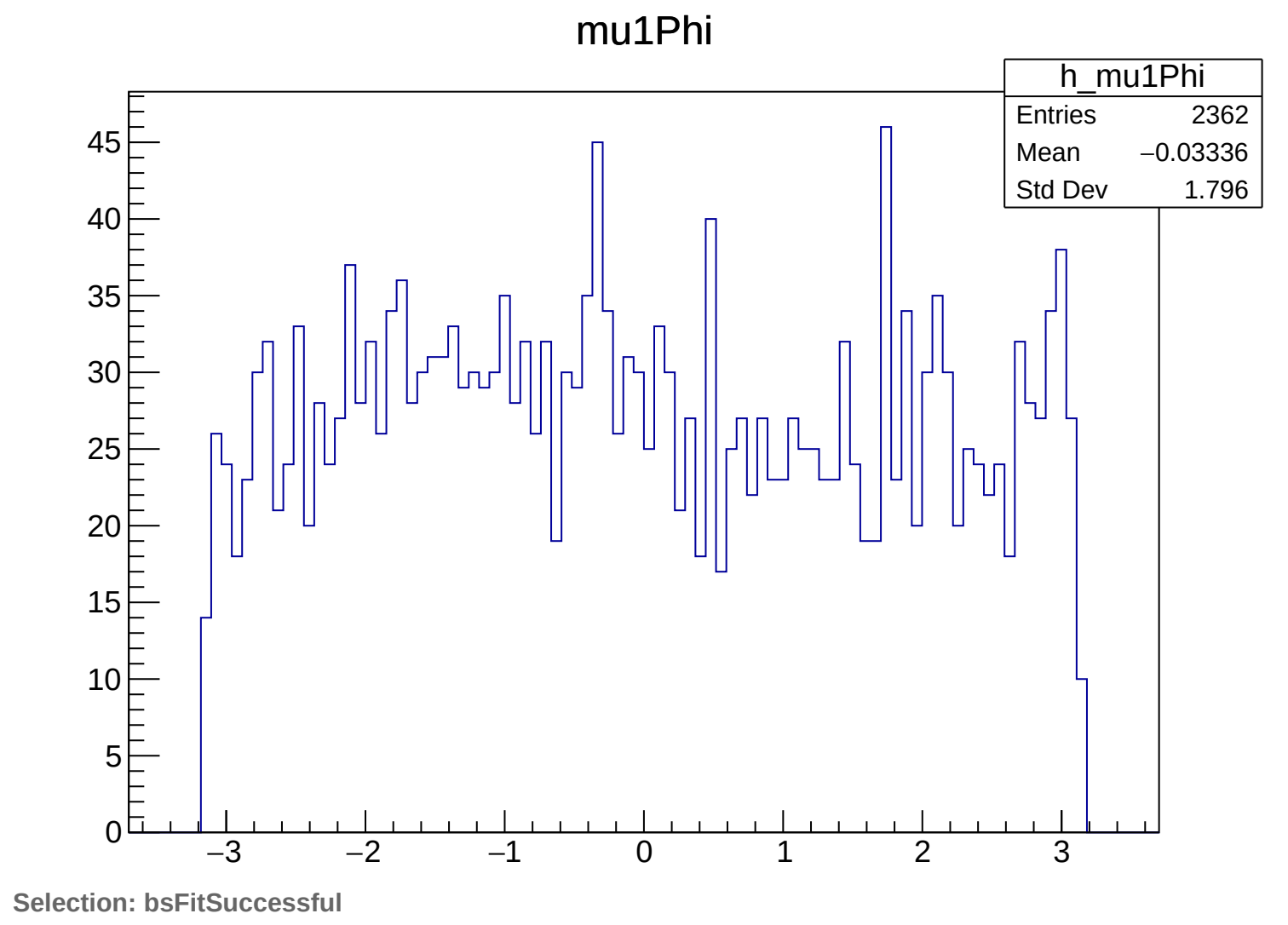

nTracks
h_nTracks
Entries 2362
Mean 99.52
Std Dev 43.3
70
60
50
40
30
20
10
0
0
50
100
150
200
250
300
Selection: bsFitSuccessful

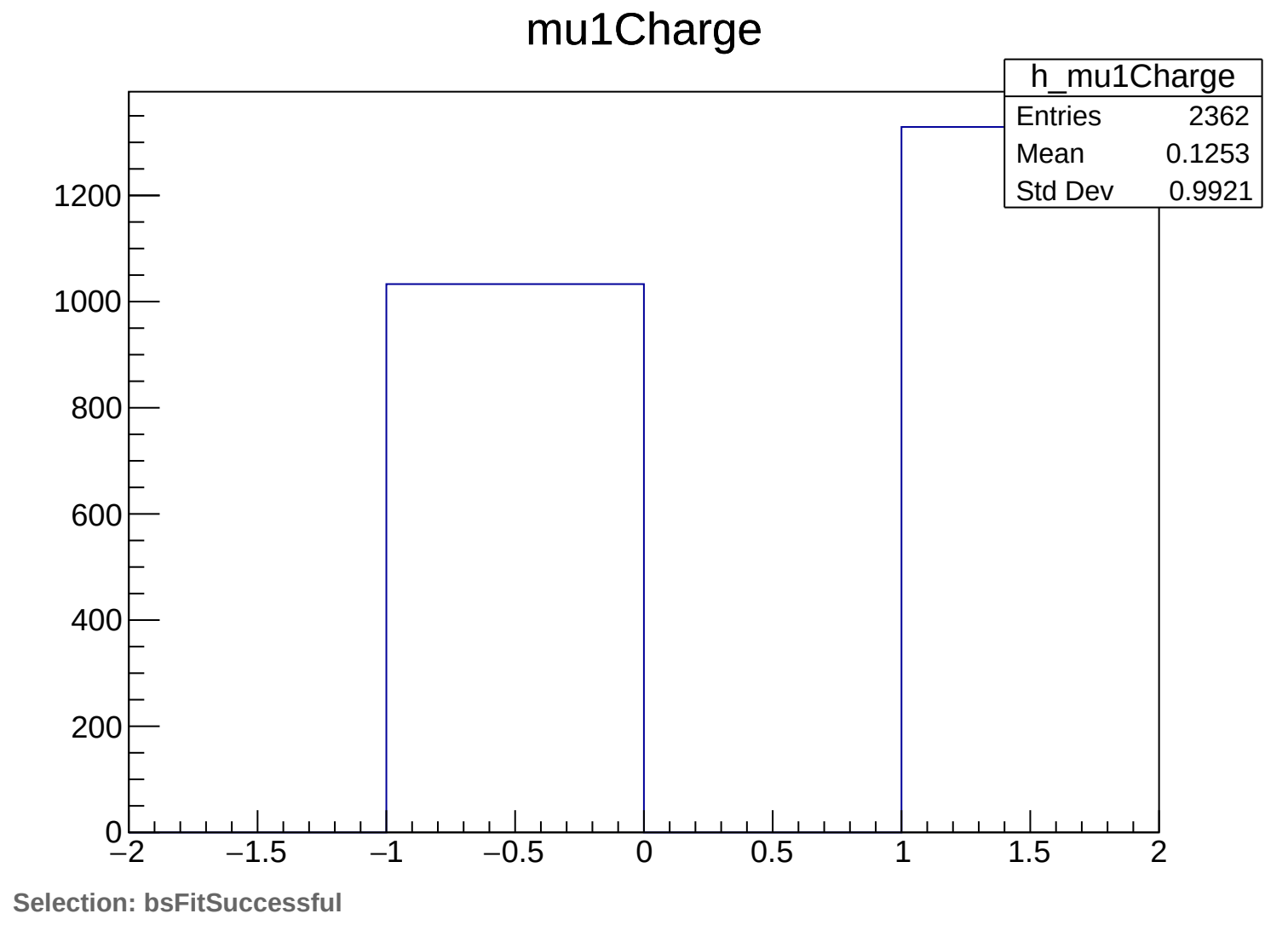

mu1MVAScore
h_mu1MVAScore
Entries 2362
Mean 0.8146
Std Dev 0.1311
240
220
200
180
160
140
120
100
80
60
40
20
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

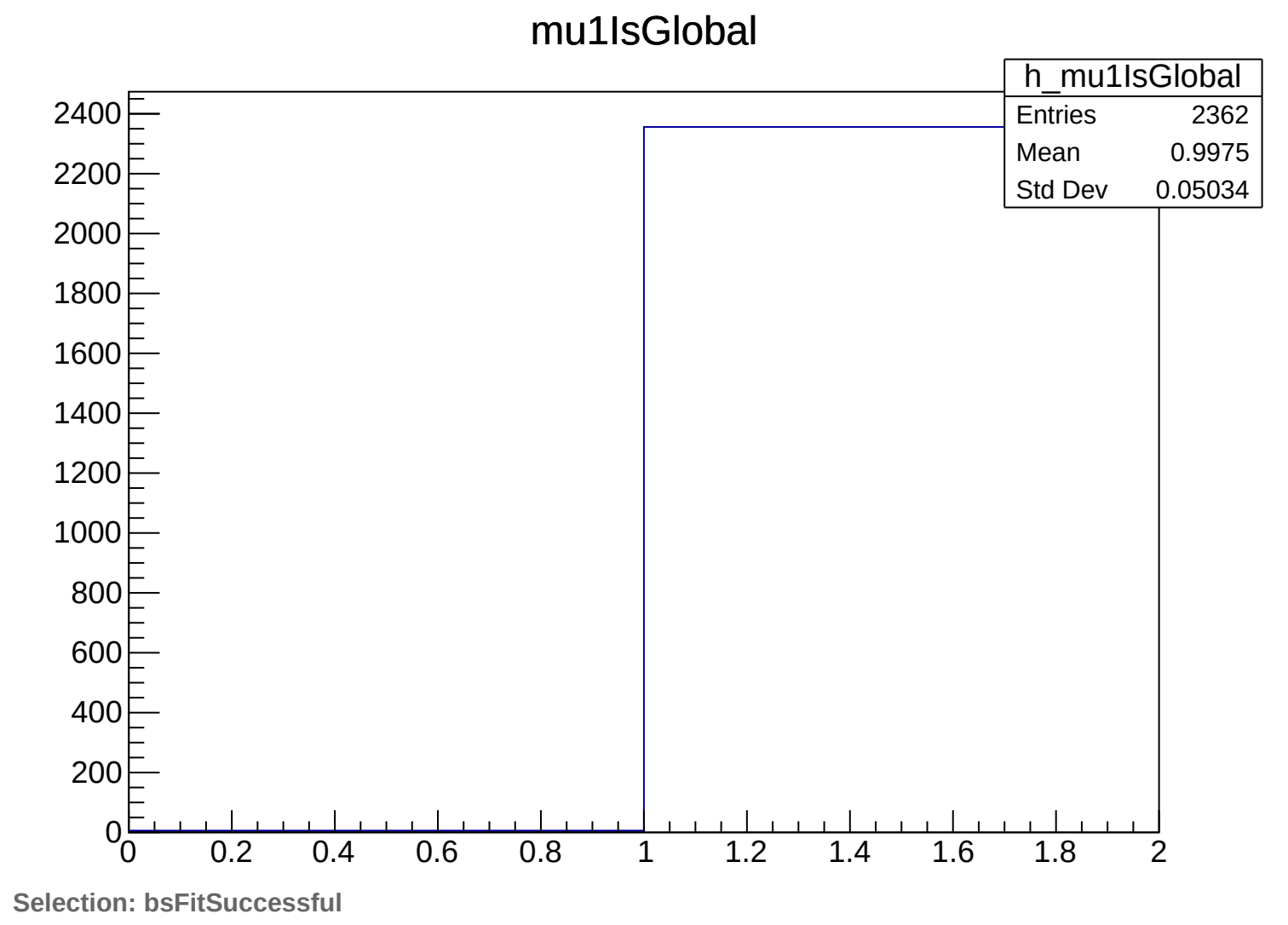

mu2MVAScore
h_mu2MVAScore
Entries 2362
Mean 0.7998
Std Dev 0.141
200
180
160
140
120
100
80
60
40
20
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

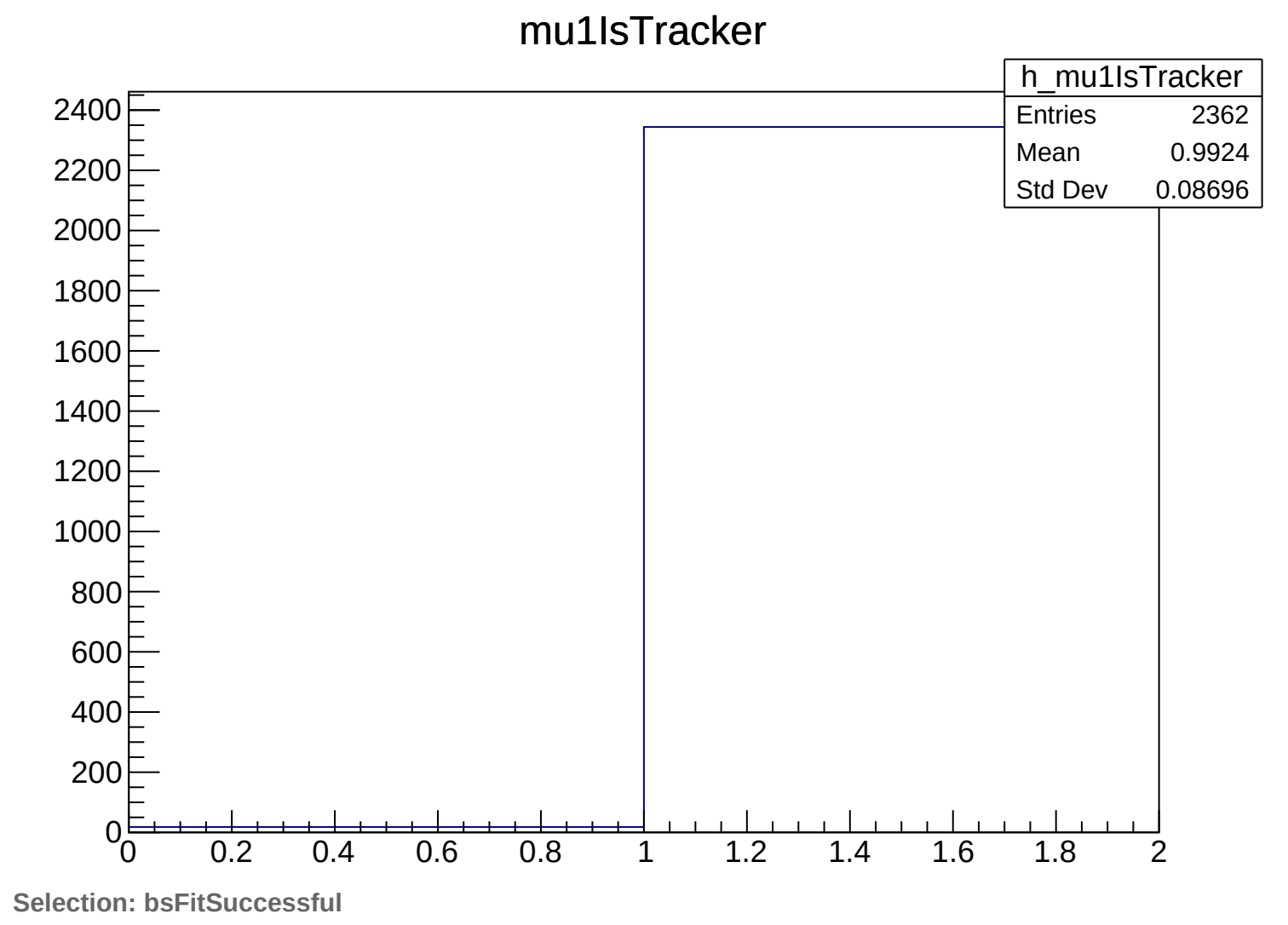

mu1Pt
h_mu1Pt
Entries 2362
Mean 8.862
Std Dev 6.749
400
350
300
250
200
150
100
50
0
0
20
40
60
80
100
Selection: bsFitSuccessful

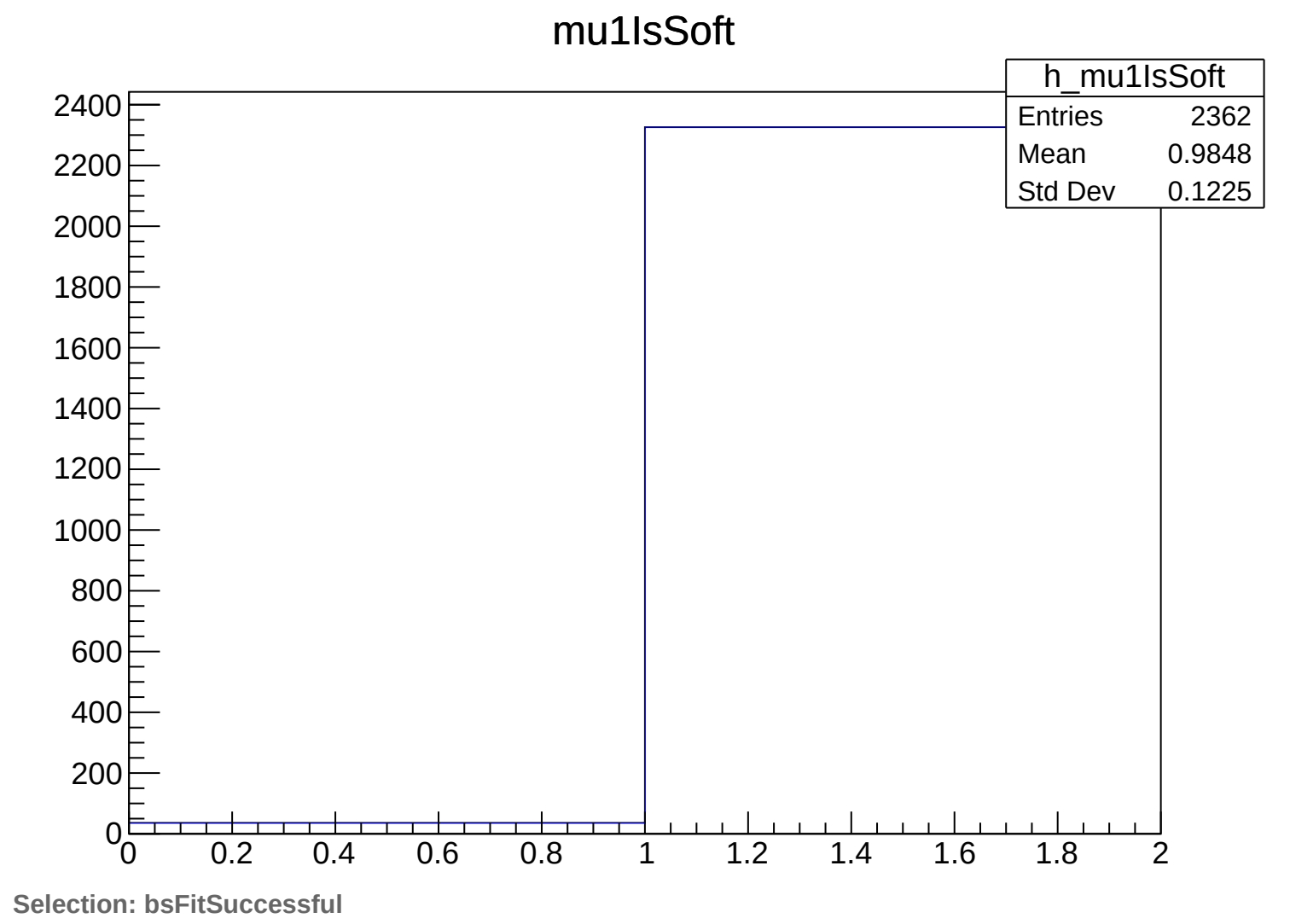

mu1Eta
h_mu1Eta
Entries 2362
Mean −0.004981
Std Dev 0.9381
60
50
40
30
20
10
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

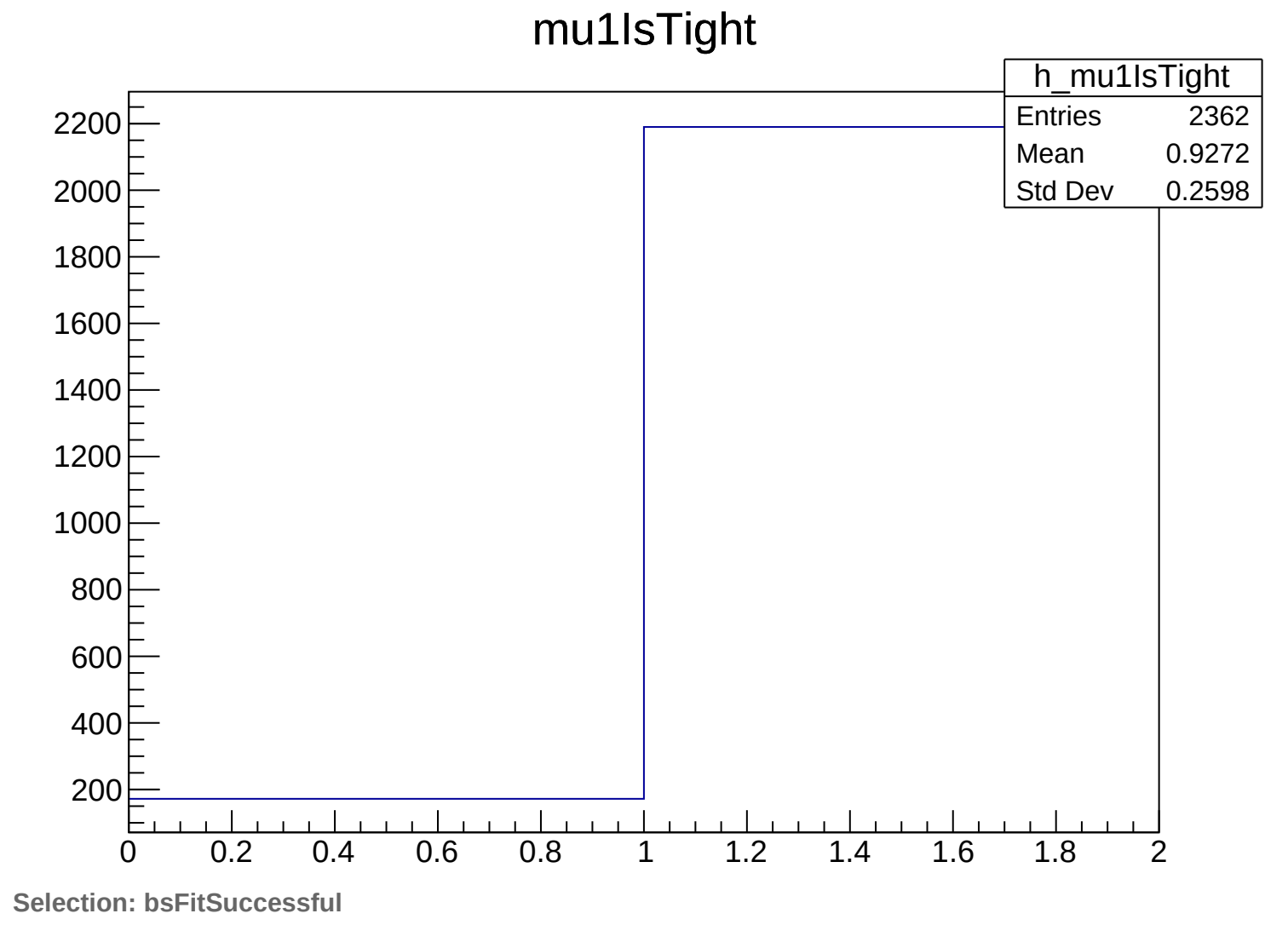

mu1Phi
h_mu1Phi
Entries 2362
Mean −0.03336
Std Dev 1.796
45
40
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

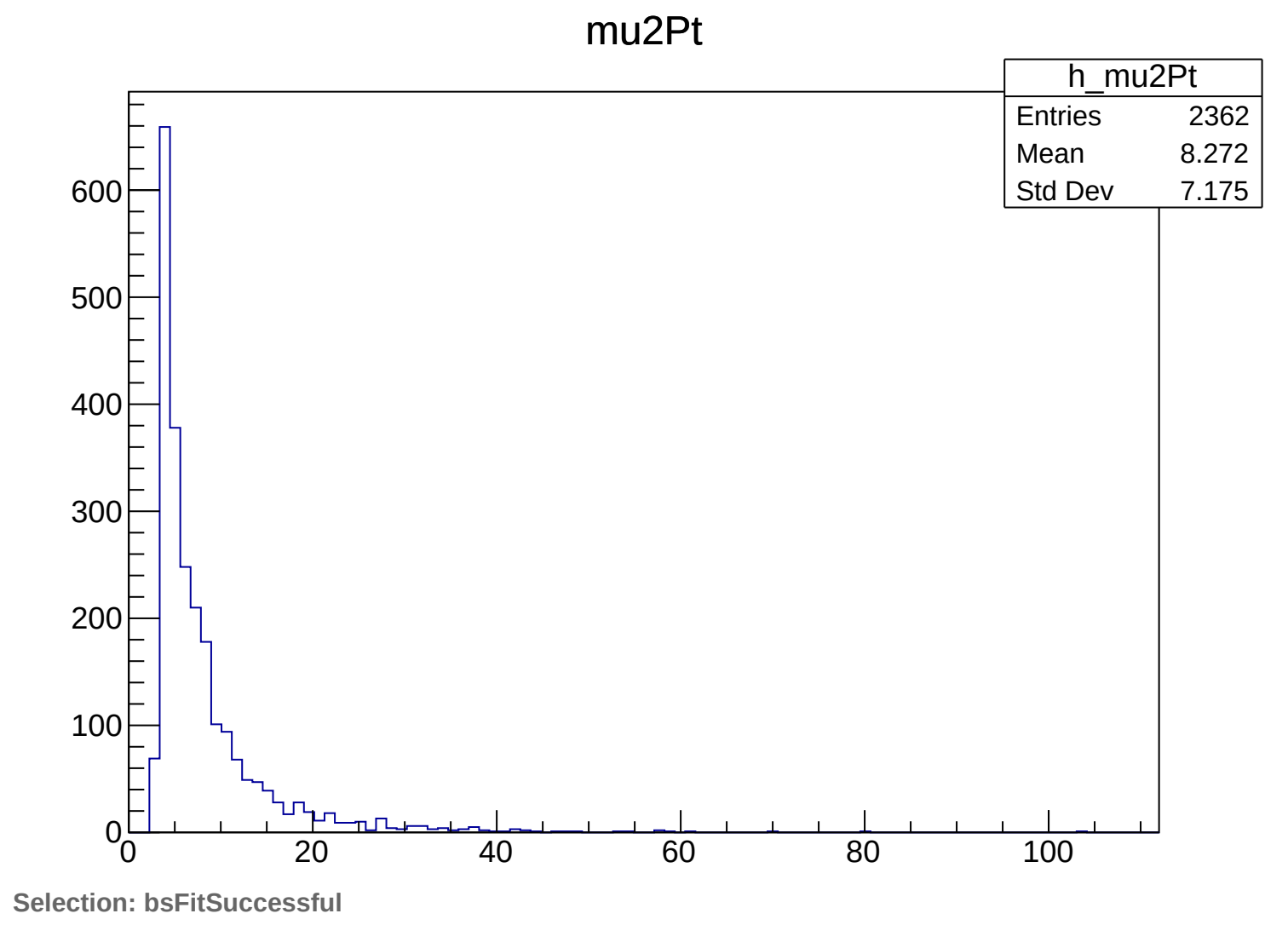

mu1Charge
h_mu1Charge
Entries 2362
Mean 0.1253
Std Dev 0.9921
1200
1000
800
600
400
200
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

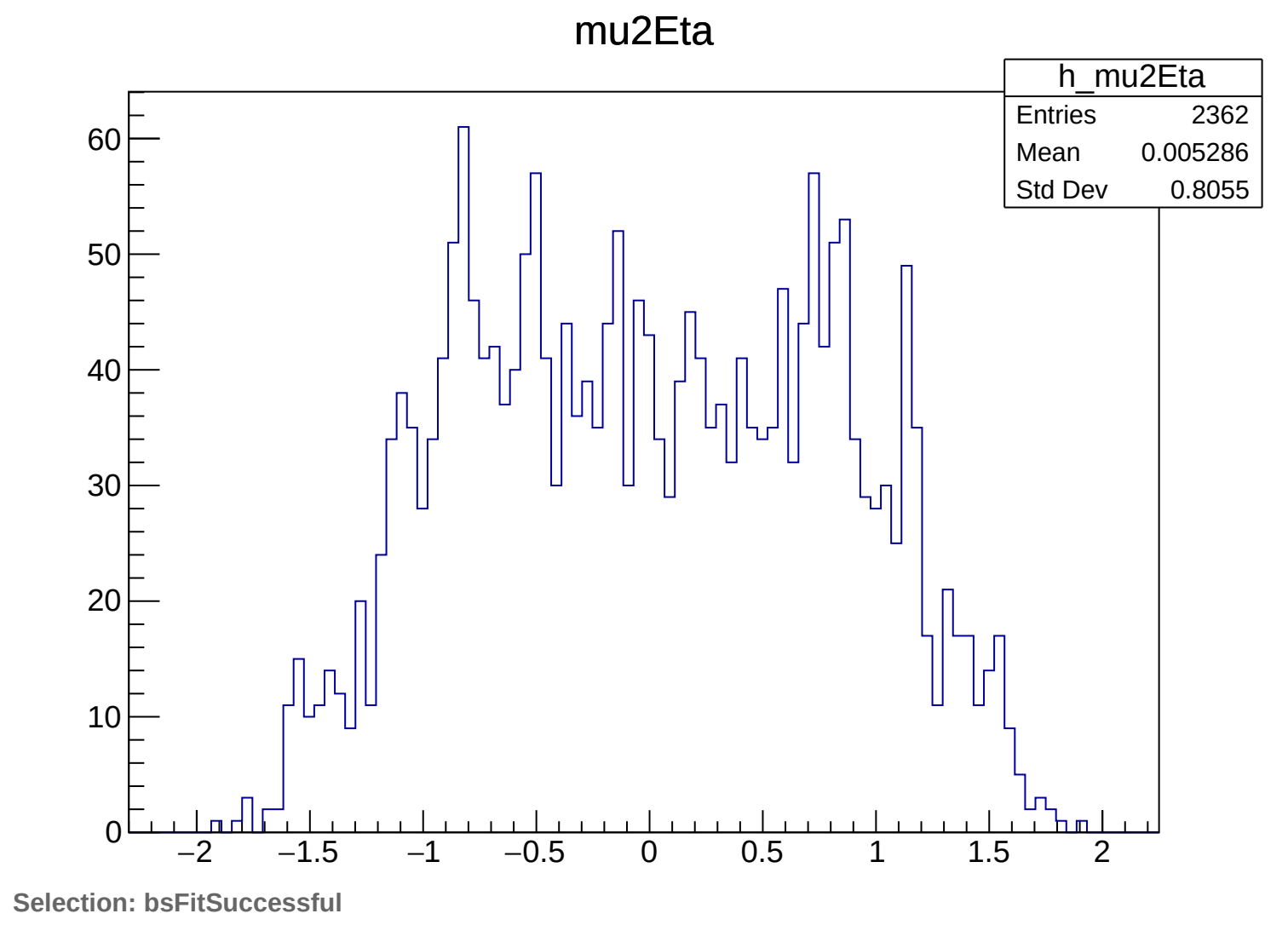

mu1IsGlobal
h_mu1IsGlobal
Entries 2362
Mean 0.9975
Std Dev 0.05034
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

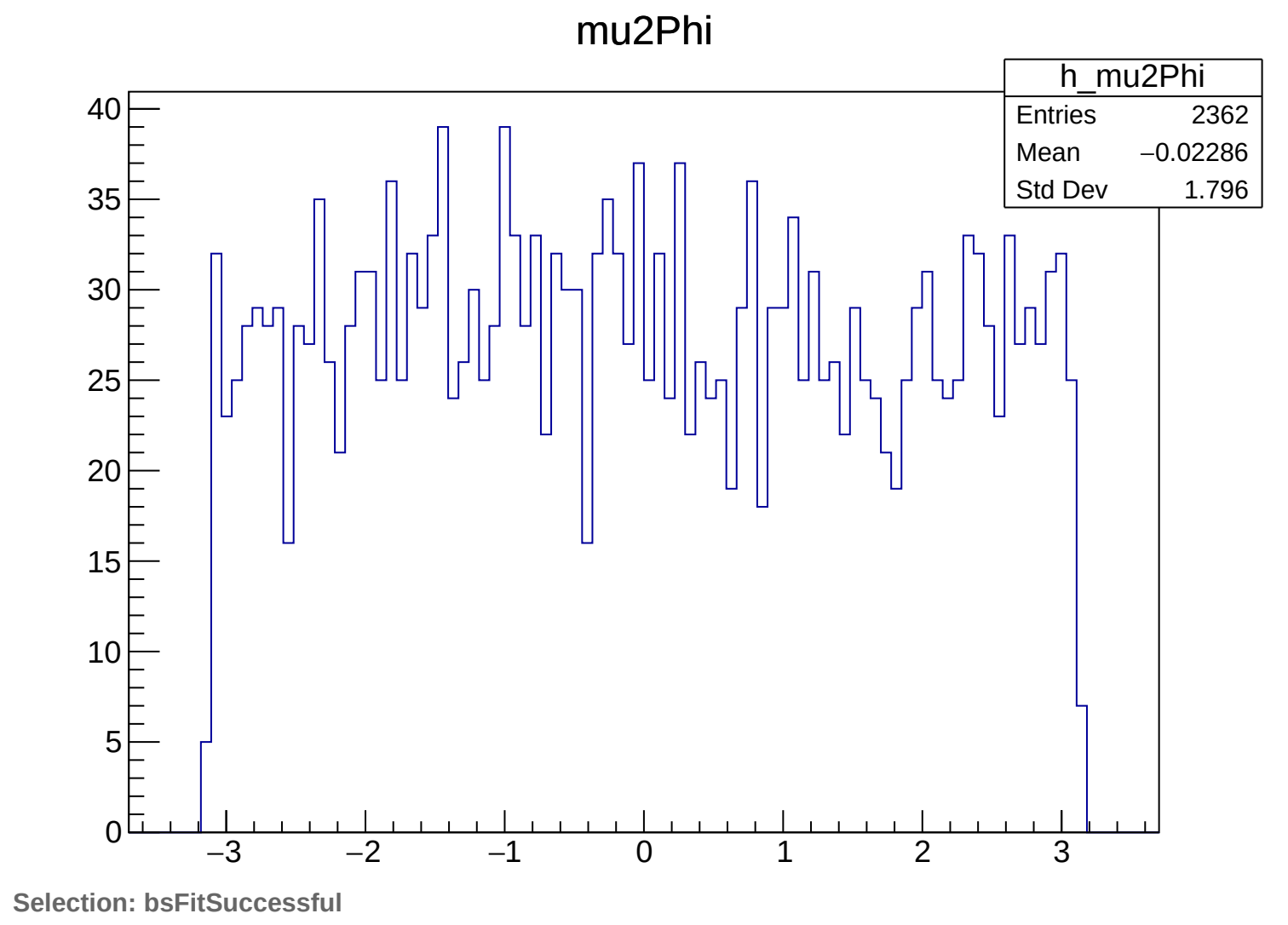

mu1IsTracker
h_mu1IsTracker
Entries 2362
Mean 0.9924
Std Dev 0.08696
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

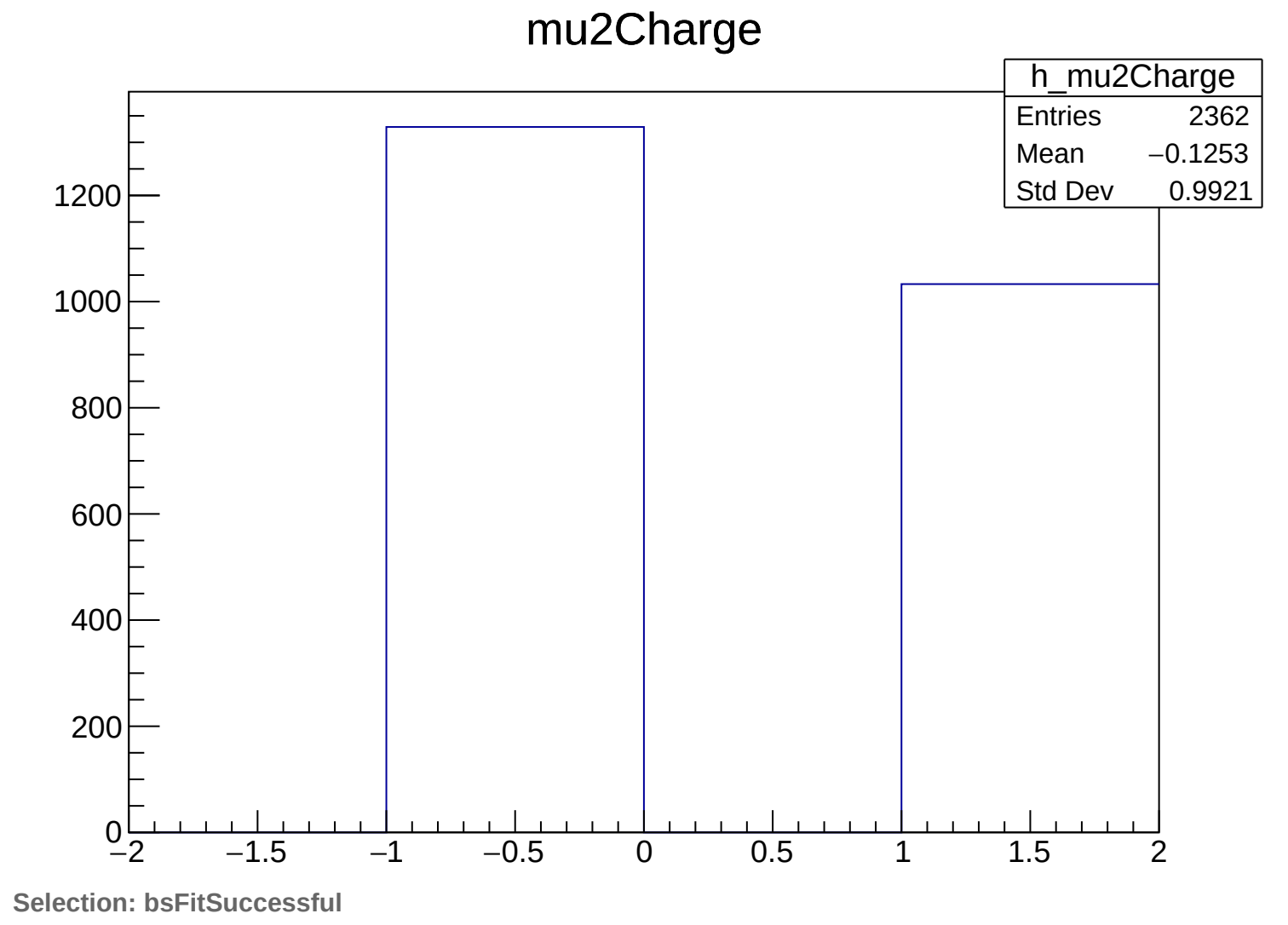

mu1IsSoft
h_mu1IsSoft
Entries 2362
Mean 0.9848
Std Dev 0.1225
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

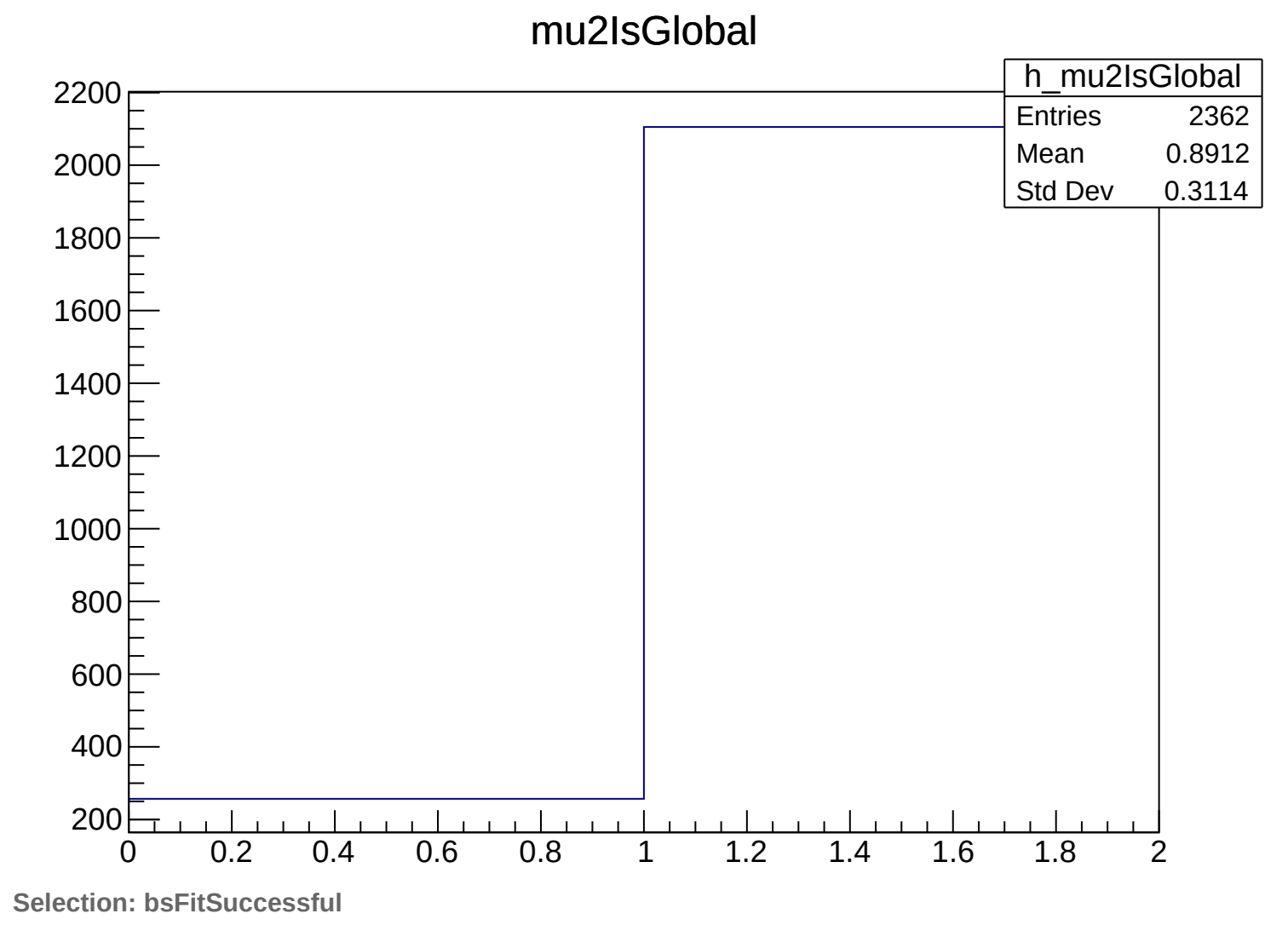

mu1IsTight
h_mu1IsTight
Entries 2362
Mean 0.9272
Std Dev 0.2598
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

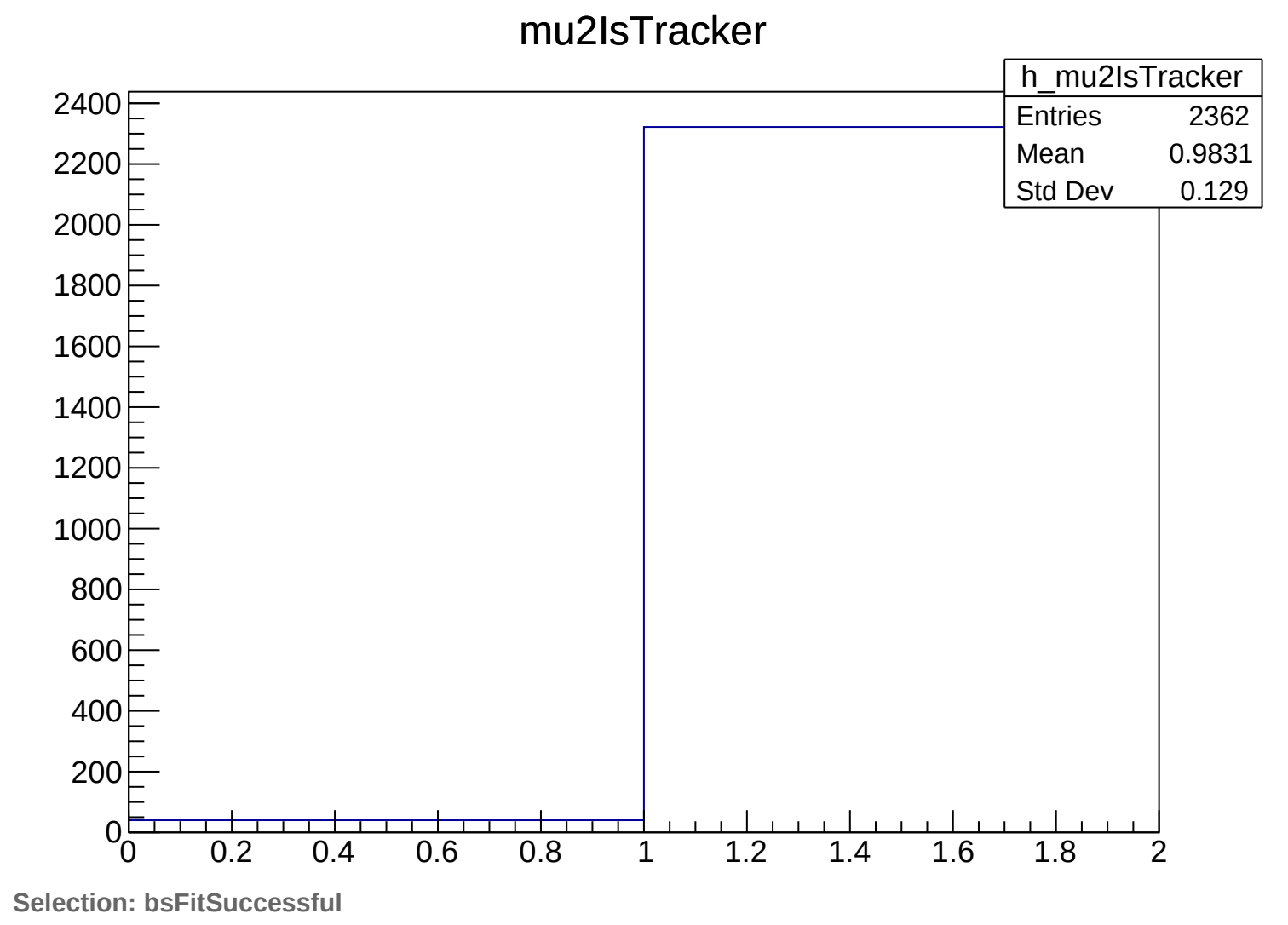

mu2Pt
h_mu2Pt
Entries 2362
Mean 8.272
Std Dev 7.175
600
500
400
300
200
100
0
0
20
40
60
80
100
Selection: bsFitSuccessful

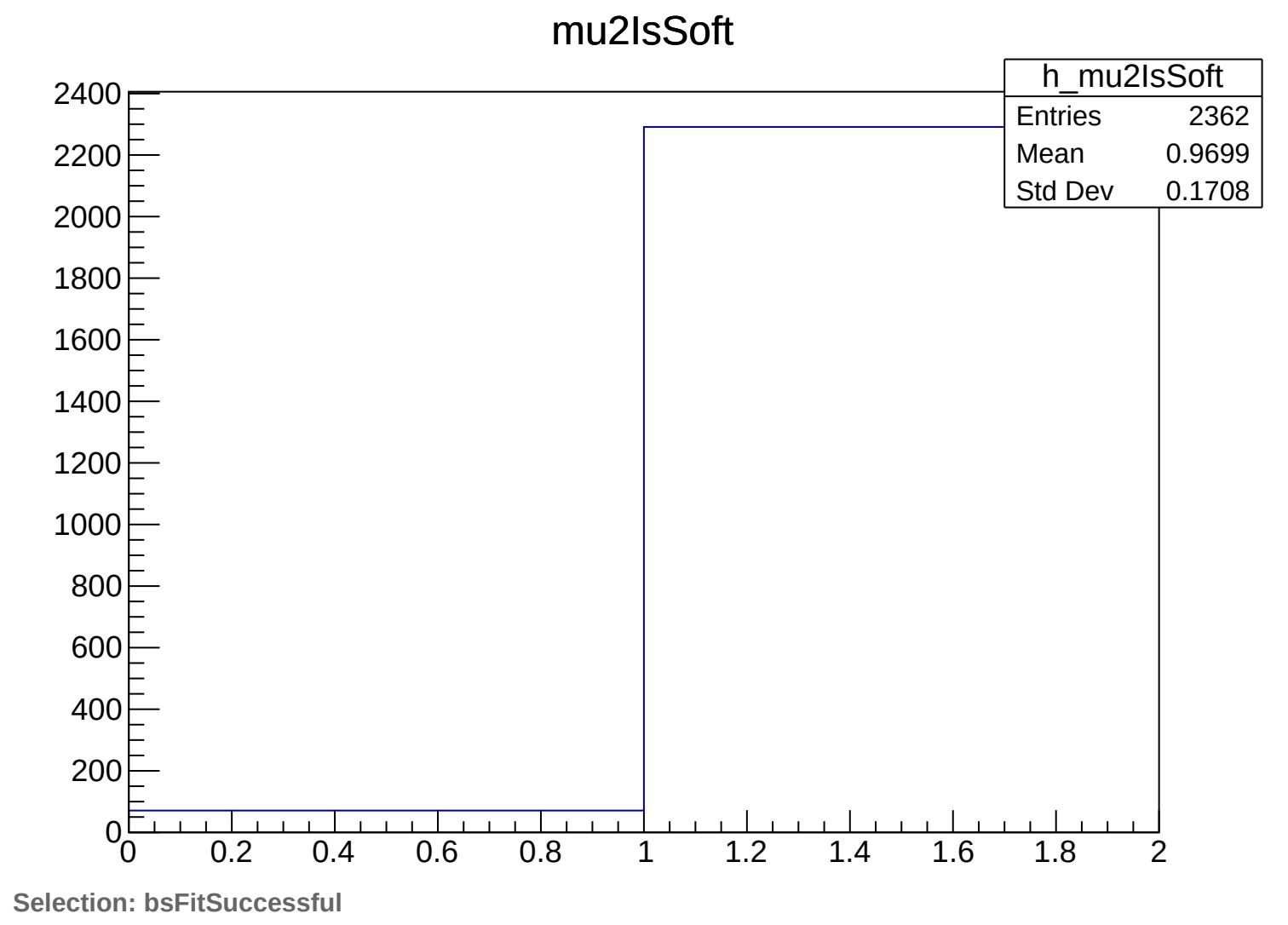

mu2Eta
h_mu2Eta
Entries 2362
Mean 0.005286
Std Dev 0.8055
60
50
40
30
20
10
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

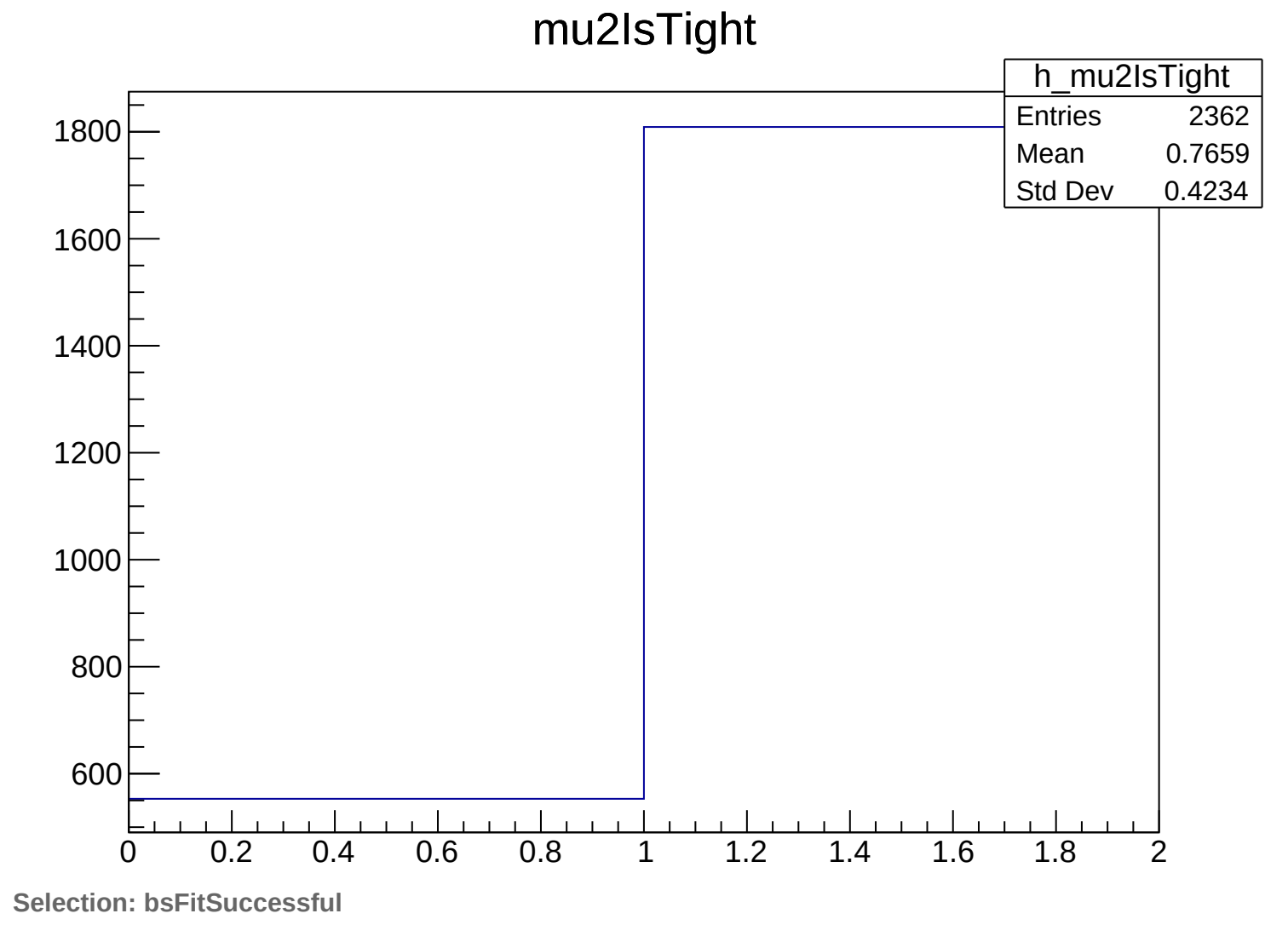

mu2Phi
h_mu2Phi
Entries 2362
Mean −0.02286
Std Dev 1.796
40
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

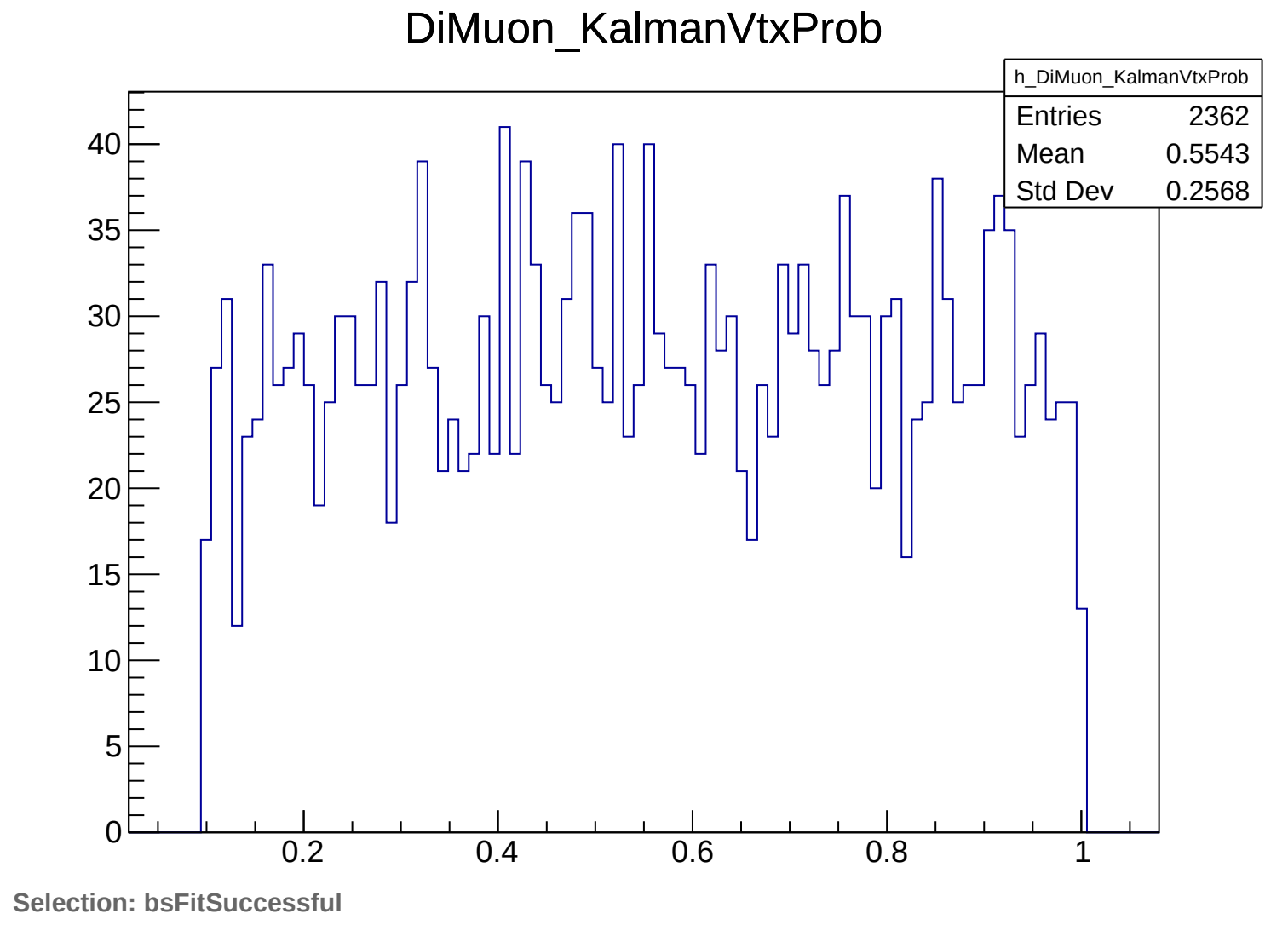

mu2Charge
h_mu2Charge
Entries 2362
Mean −0.1253
Std Dev 0.9921
1200
1000
800
600
400
200
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

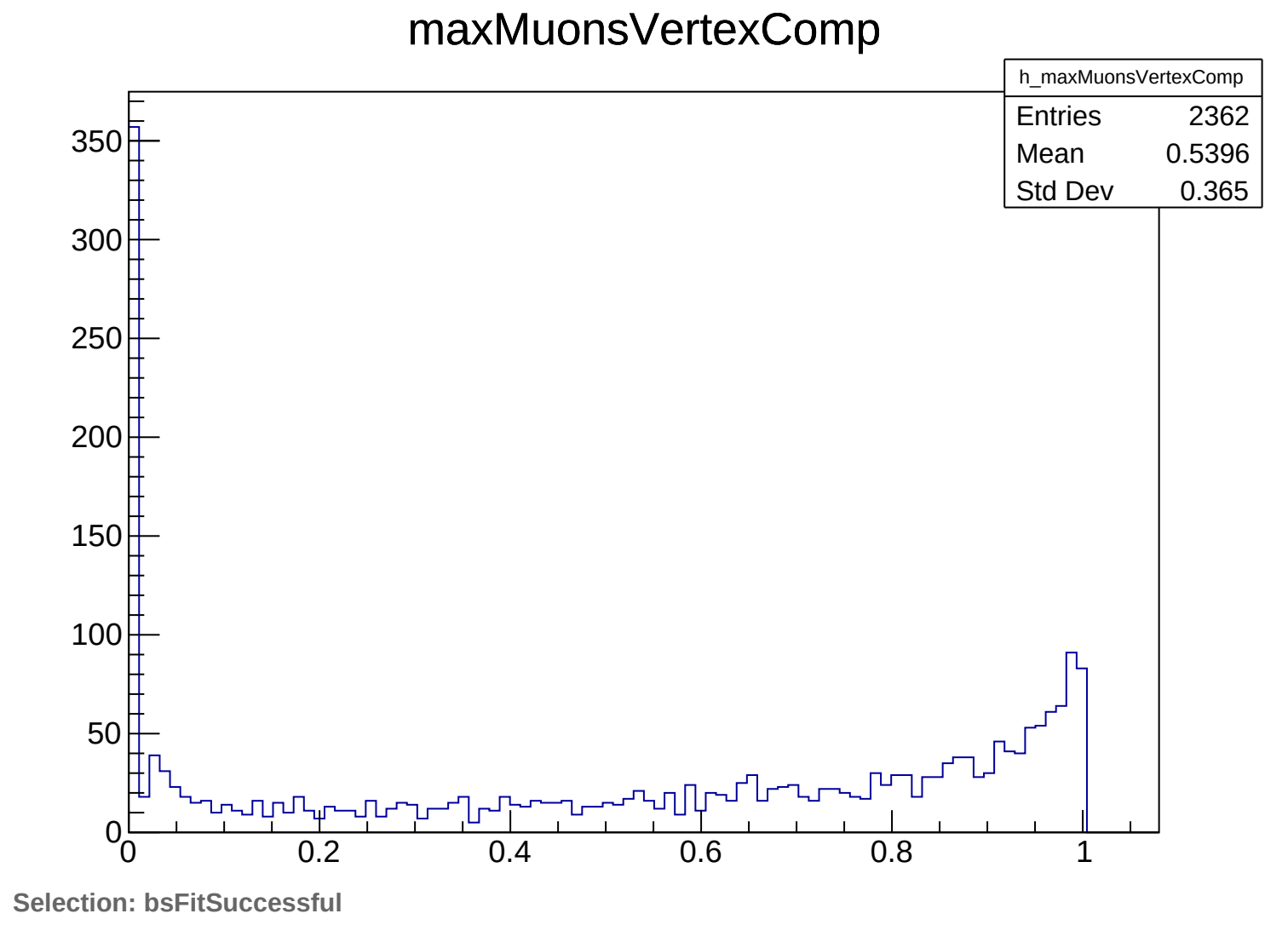

mu2IsGlobal
h_mu2IsGlobal
Entries 2362
Mean 0.8912
Std Dev 0.3114
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

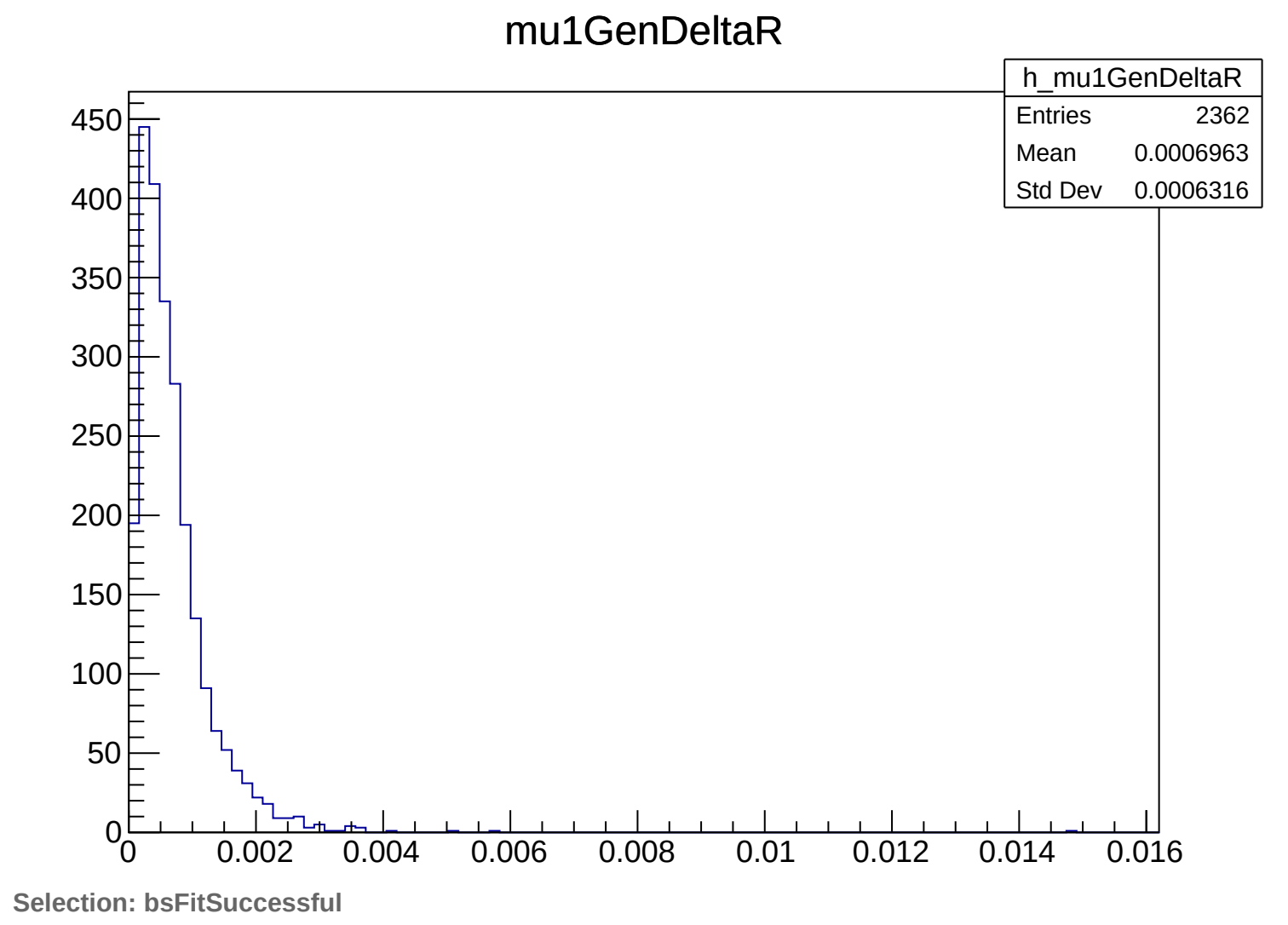

mu2IsTracker
h_mu2IsTracker
Entries 2362
Mean 0.9831
Std Dev 0.129
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

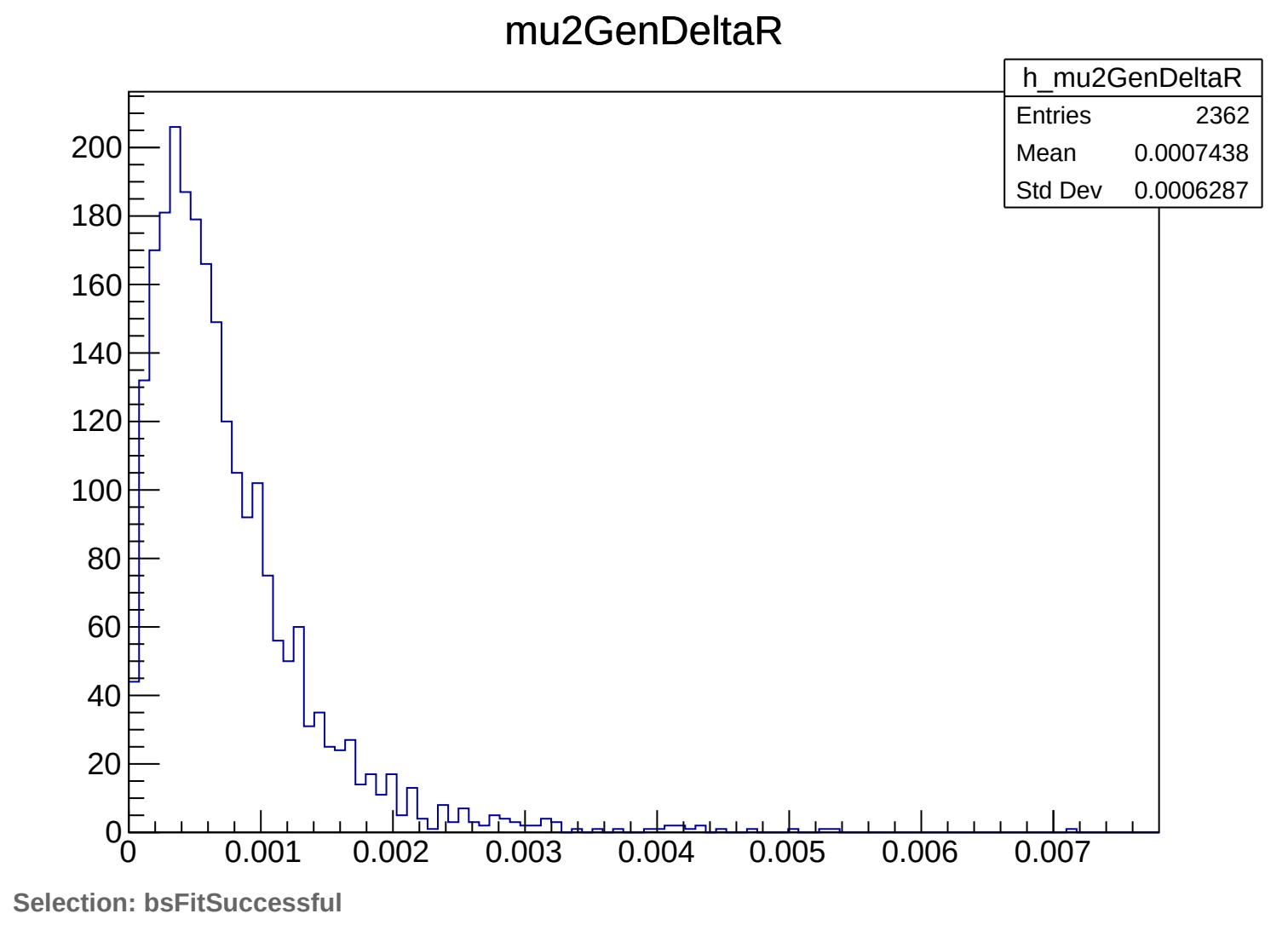

mu2IsSoft
h_mu2IsSoft
Entries 2362
Mean 0.9699
Std Dev 0.1708
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

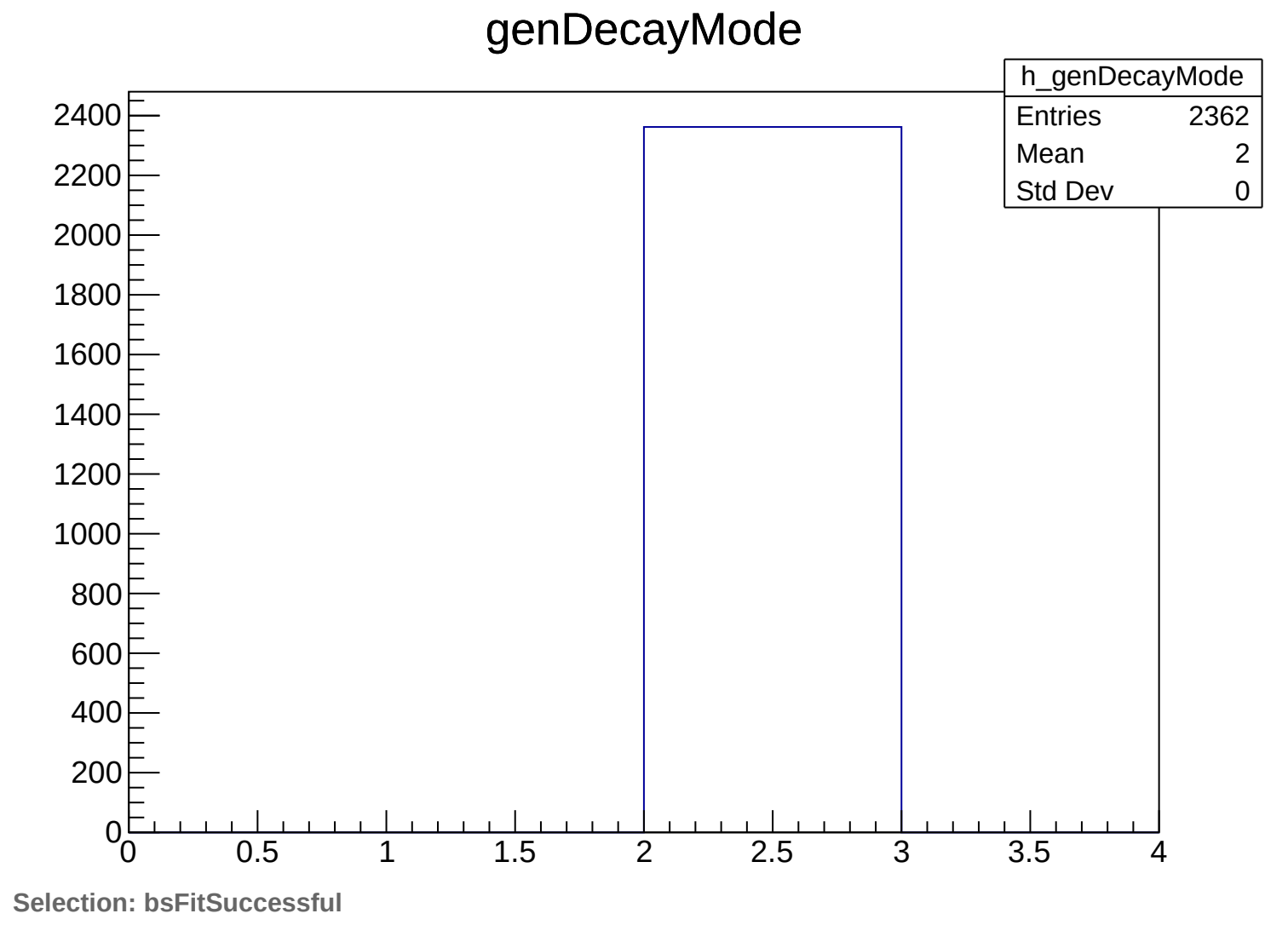

mu2IsTight
h_mu2IsTight
Entries 2362
Mean 0.7659
Std Dev 0.4234
1800
1600
1400
1200
1000
800
600
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

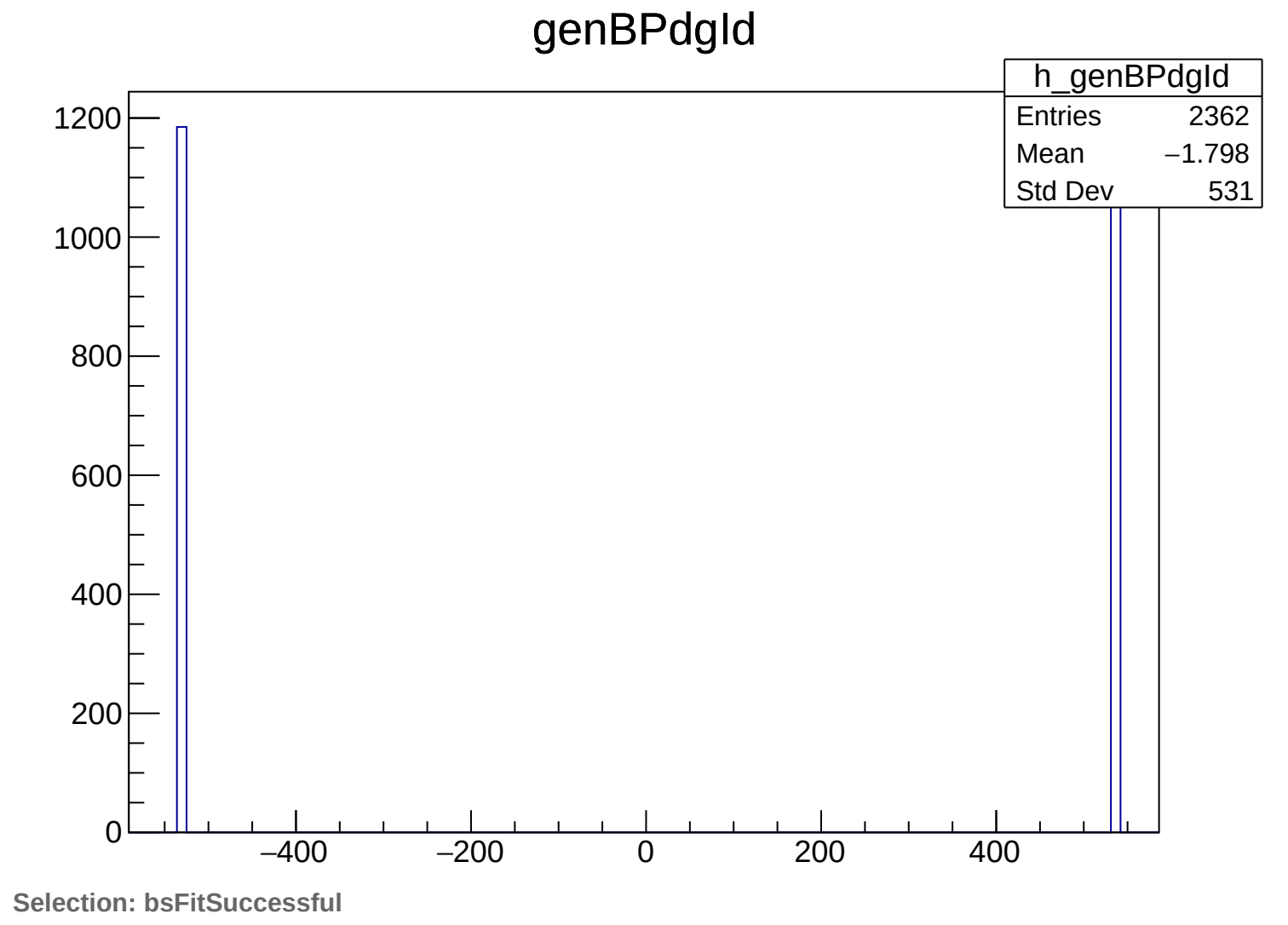

DiMuon_KalmanVtxProb
h_DiMuon_KalmanVtxProb
Entries 2362
Mean 0.5543
Std Dev 0.2568
0
5
10
15
20
25
30
35
40
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

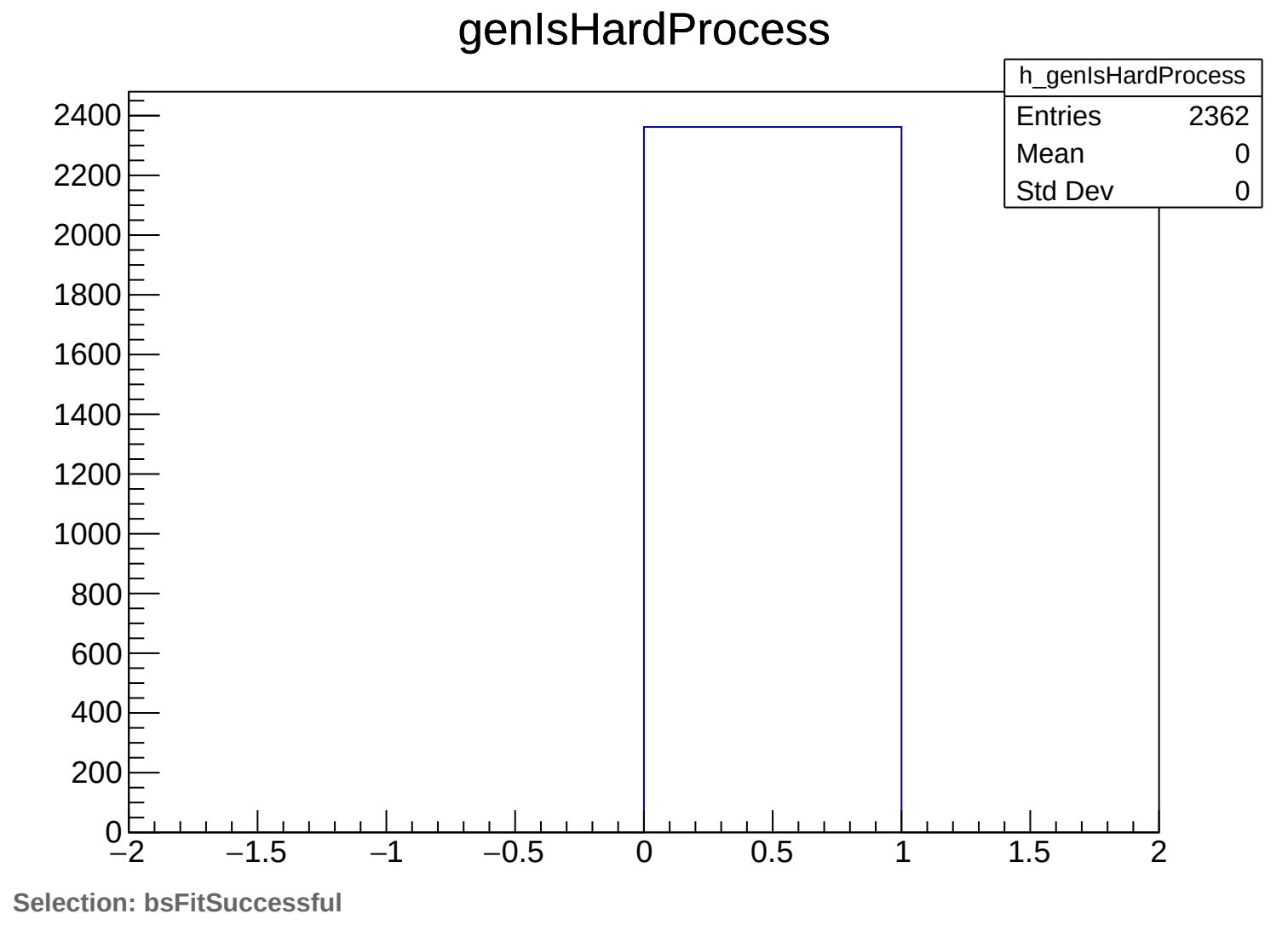

maxMuonsVertexComp
h_maxMuonsVertexComp
Entries 2362
Mean 0.5396
Std Dev 0.365
350
300
250
200
150
100
50
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

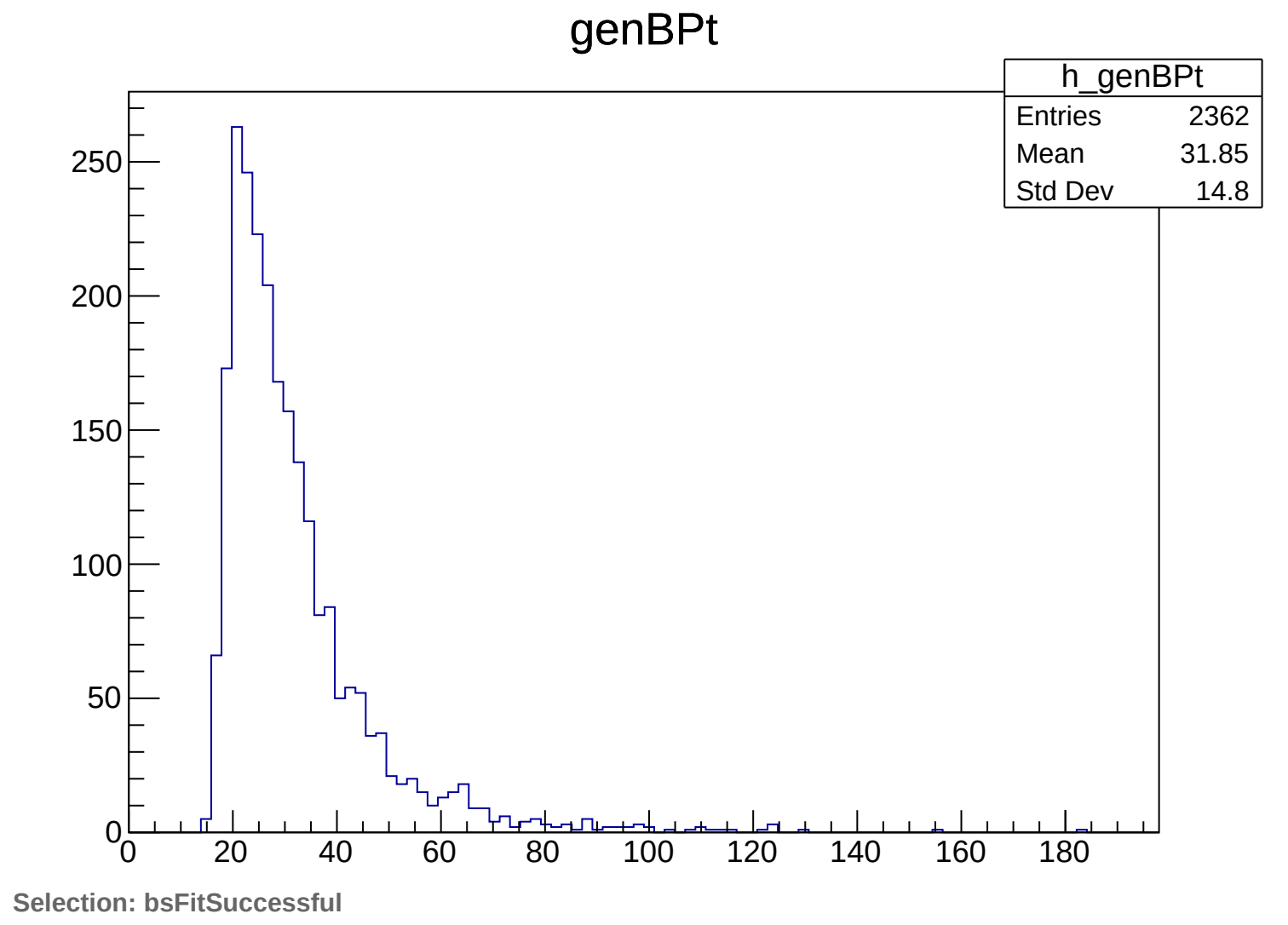

mu1GenDeltaR
h_mu1GenDeltaR
Entries 2362
Mean 0.0006963
Std Dev 0.0006316
450
400
350
300
250
200
150
100
50
0
0
0.002
0.004
0.006
0.008
0.01
0.012
0.014
0.016
Selection: bsFitSuccessful

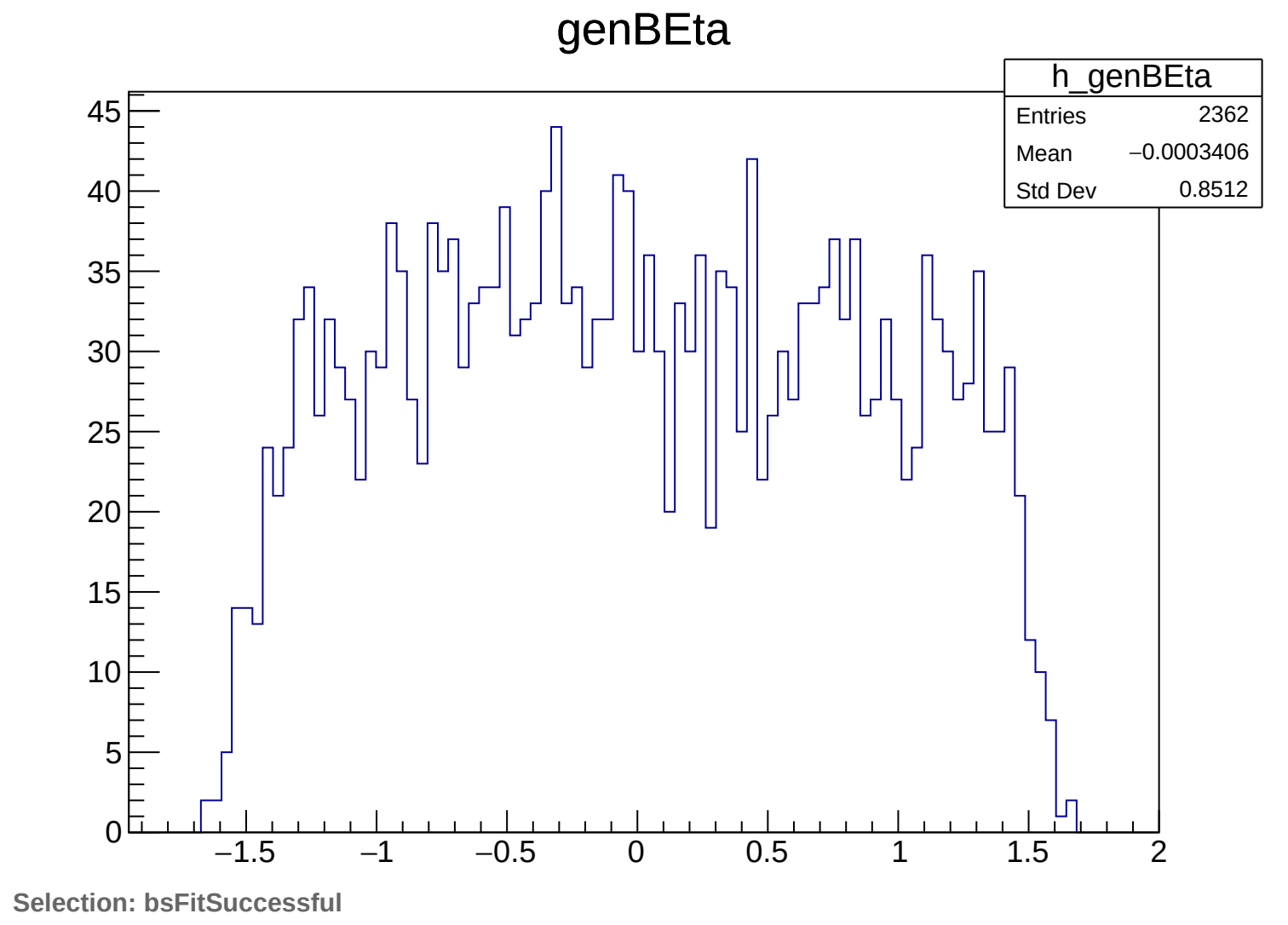

mu2GenDeltaR
h_mu2GenDeltaR
Entries 2362
Mean 0.0007438
Std Dev 0.0006287
200
180
160
140
120
100
80
60
40
20
0
0
0.001
0.002
0.003
0.004
0.005
0.006
0.007
Selection: bsFitSuccessful

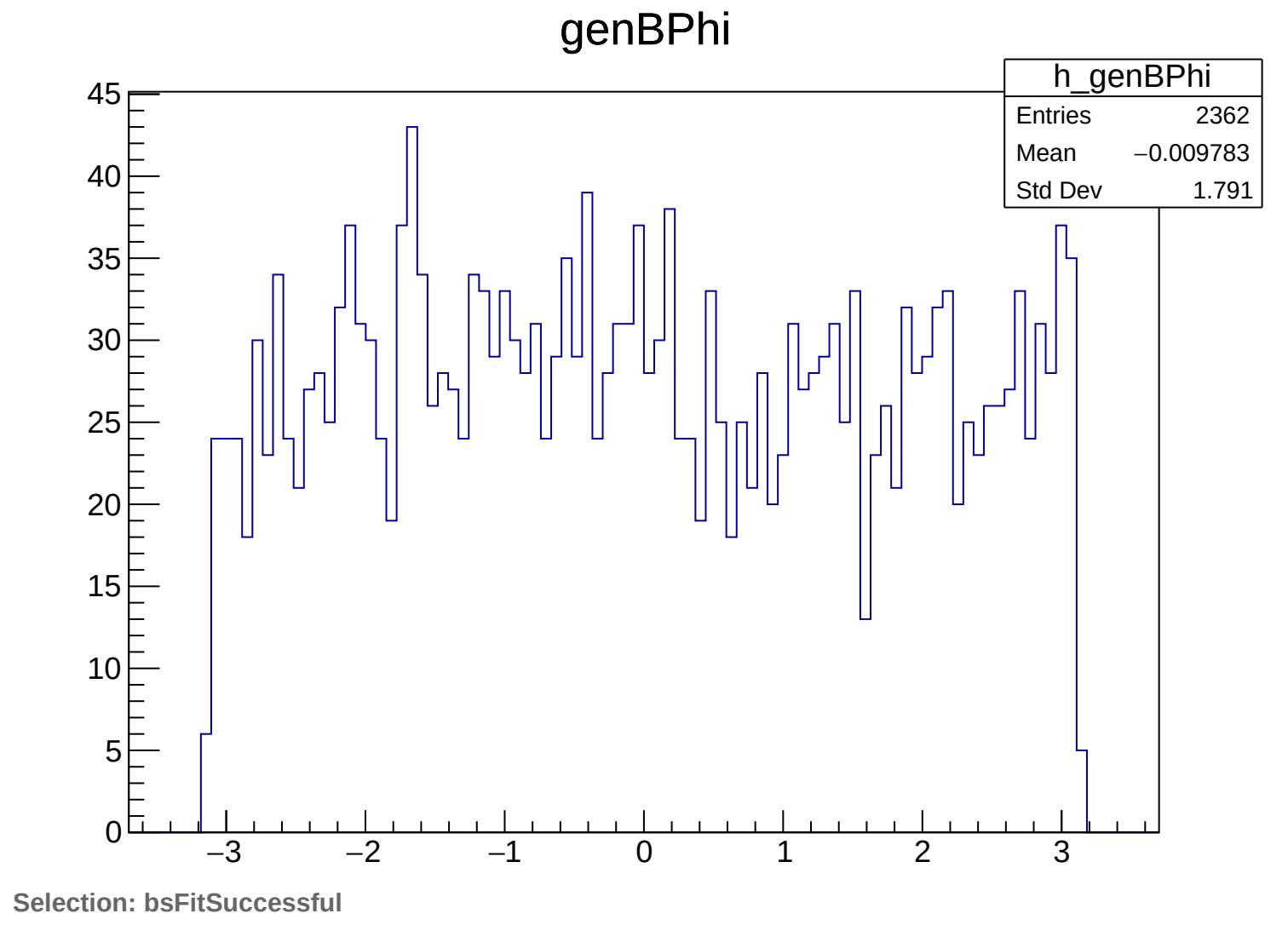

genDecayMode
h_genDecayMode
Entries 2362
Mean 2
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.5
1
1.5
2
2.5
3
3.5
4
Selection: bsFitSuccessful

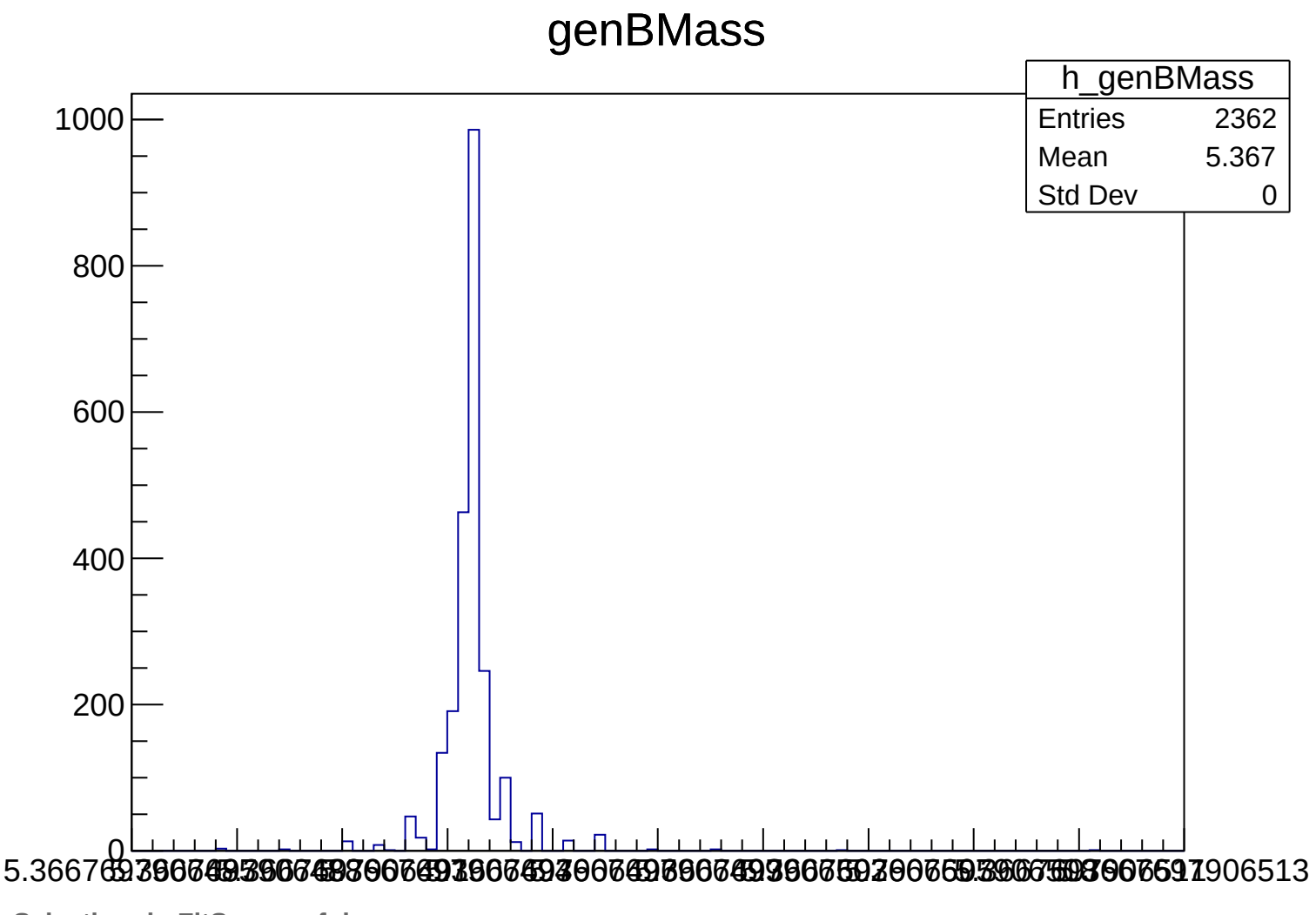

genBPdgId
h_genBPdgId
Entries 2362
Mean −1.798
Std Dev 531
1200
1000
800
600
400
200
0
−400
−200
0
200
400
Selection: bsFitSuccessful

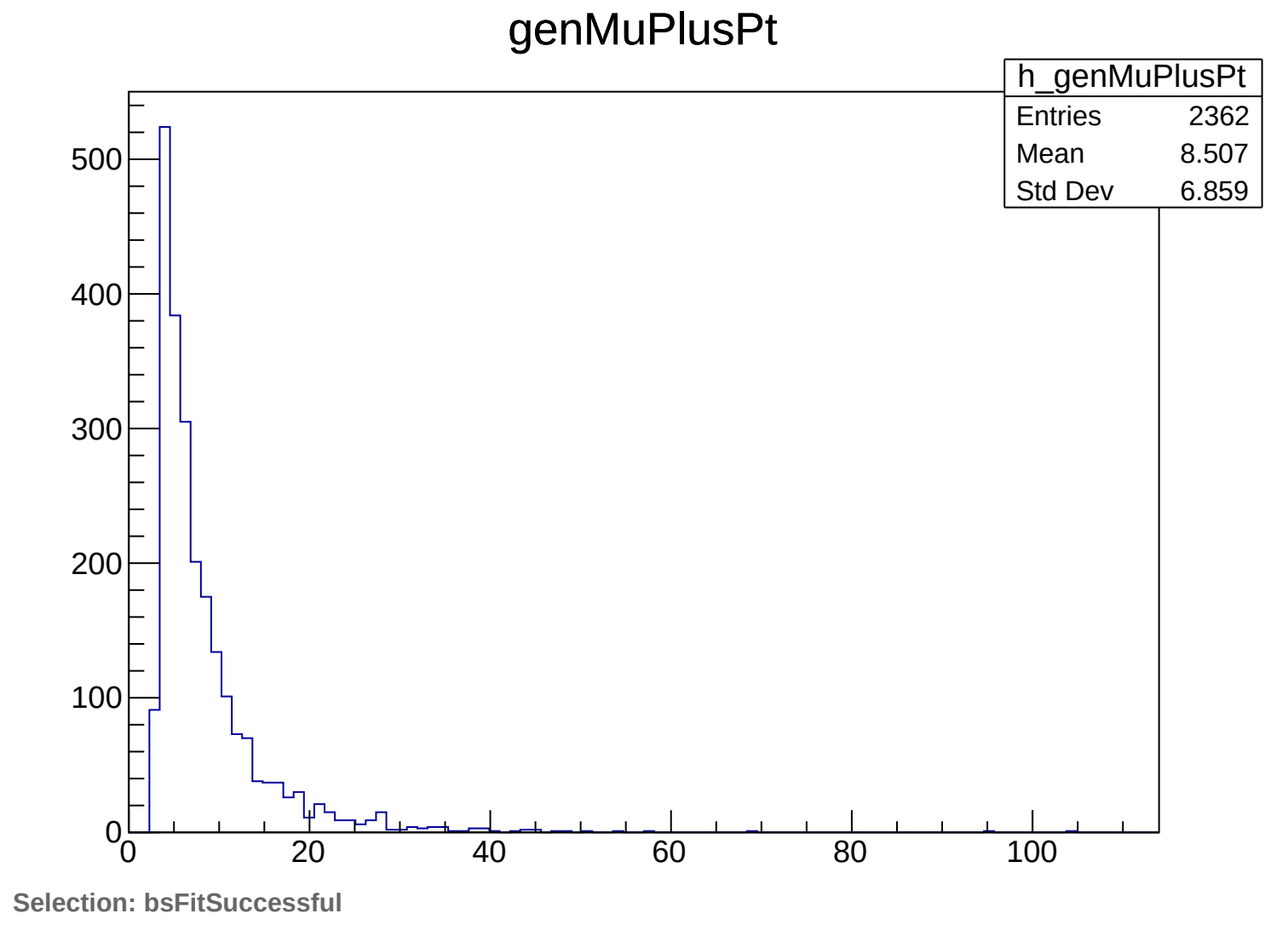

genIsHardProcess
h_genIsHardProcess
Entries 2362
Mean 0
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

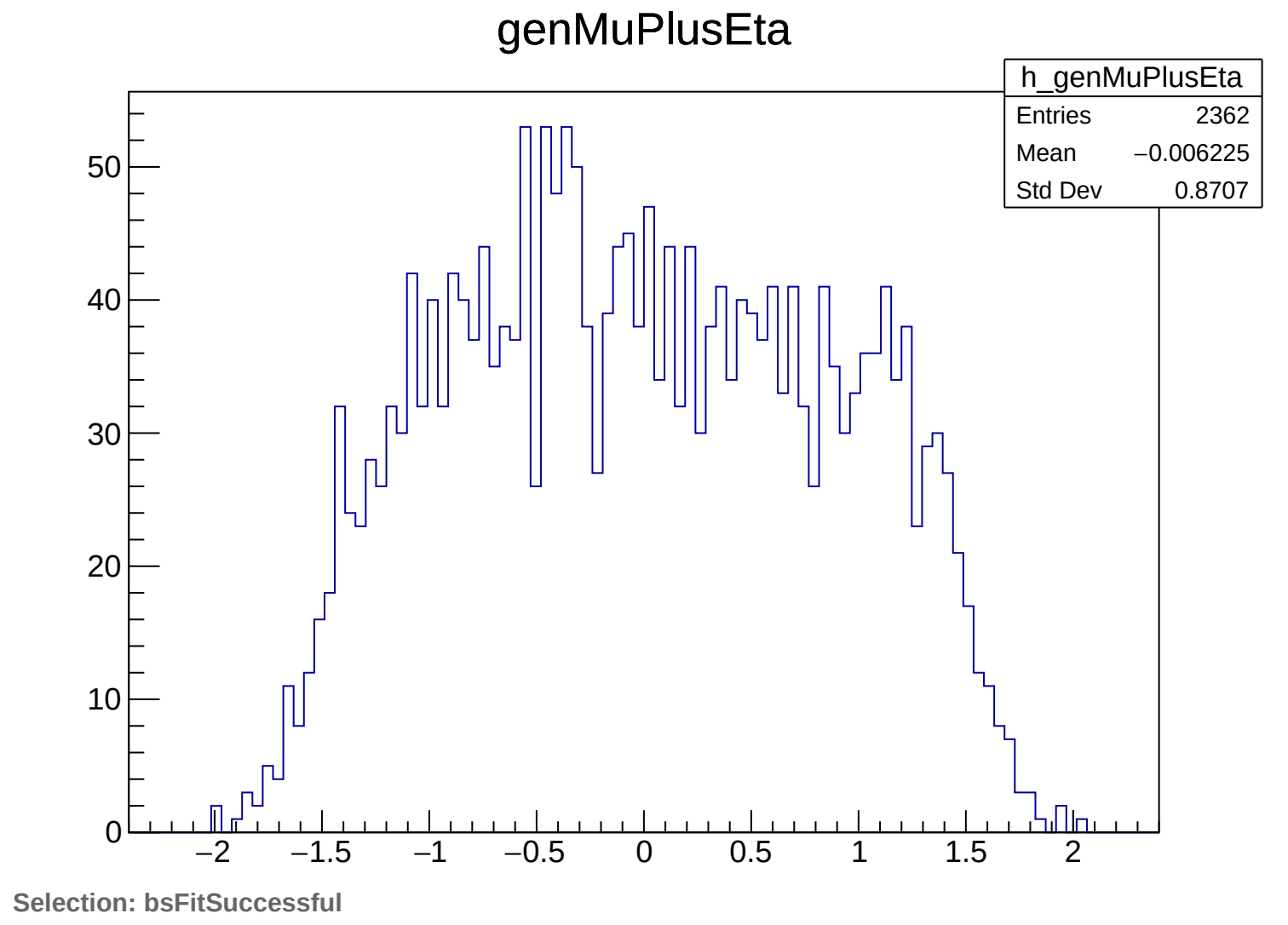

genBPt
h_genBPt
Entries 2362
Mean 31.85
Std Dev 14.8
250
200
150
100
50
0
0
20
40
60
80
100
120
140
160
180
Selection: bsFitSuccessful

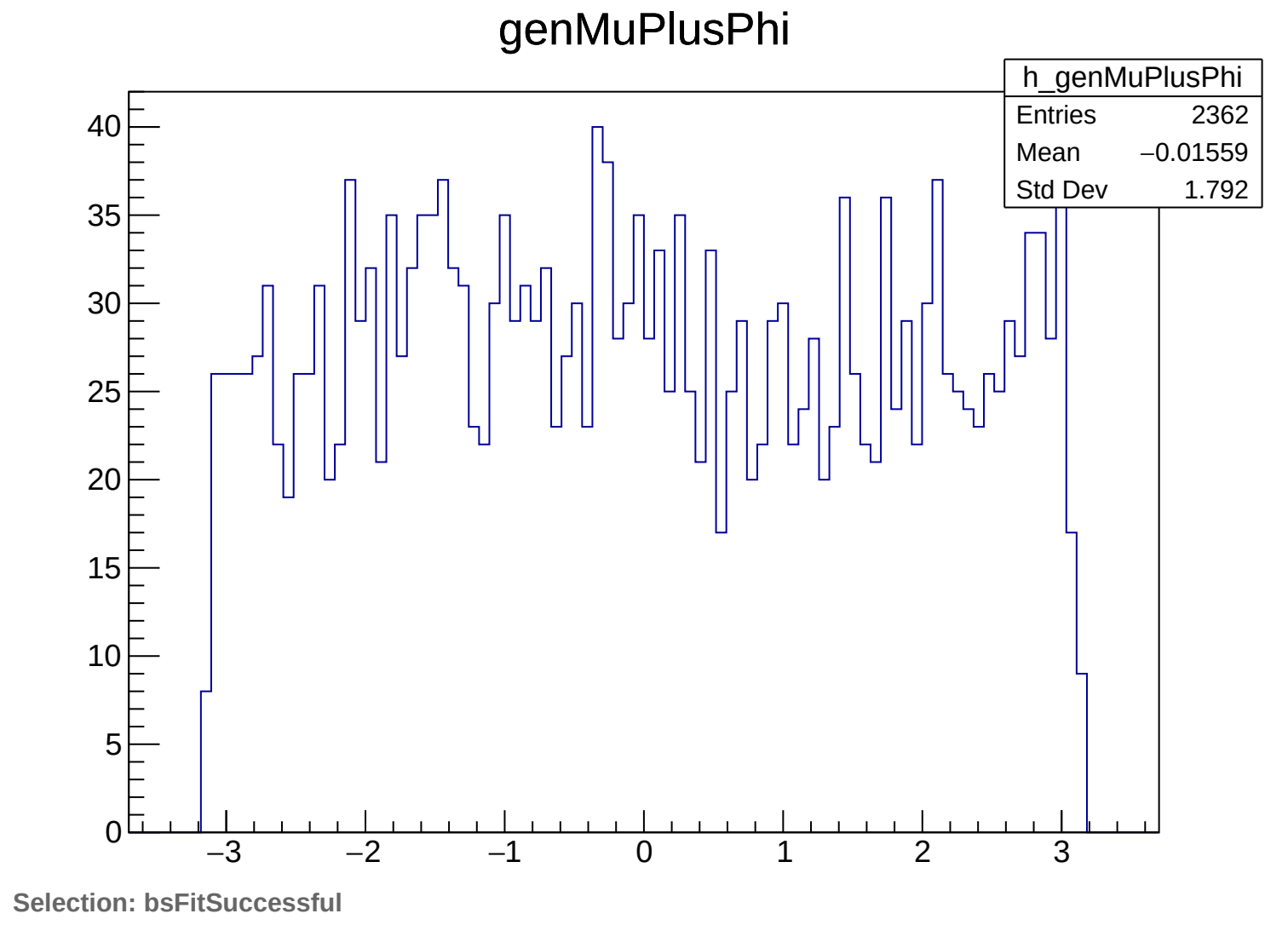

genBEta
h_genBEta
Entries 2362
Mean −0.0003406
Std Dev 0.8512
45
40
35
30
25
20
15
10
5
0
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

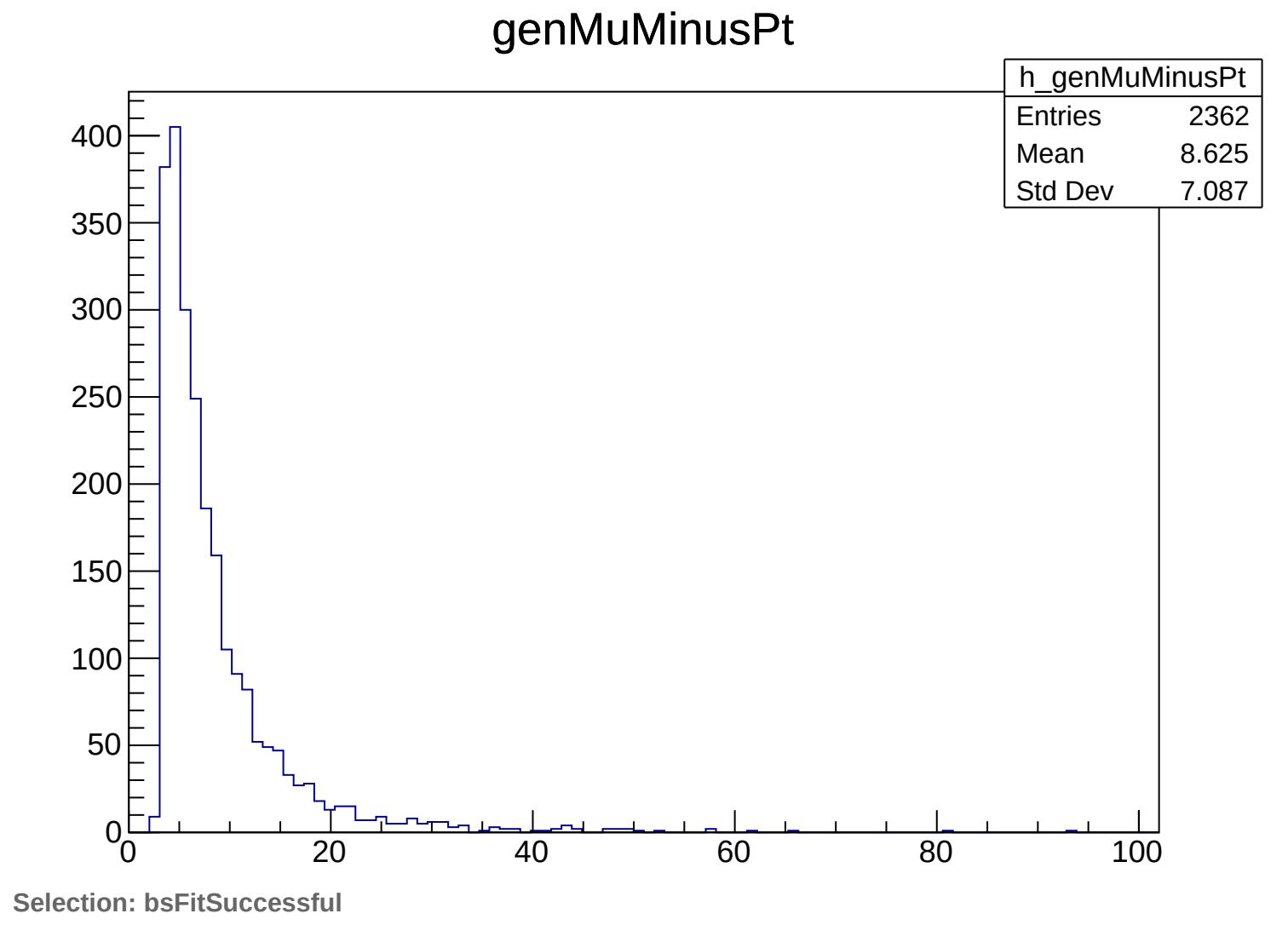

genBPhi
h_genBPhi
Entries 2362
Mean −0.009783
Std Dev 1.791
45
40
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

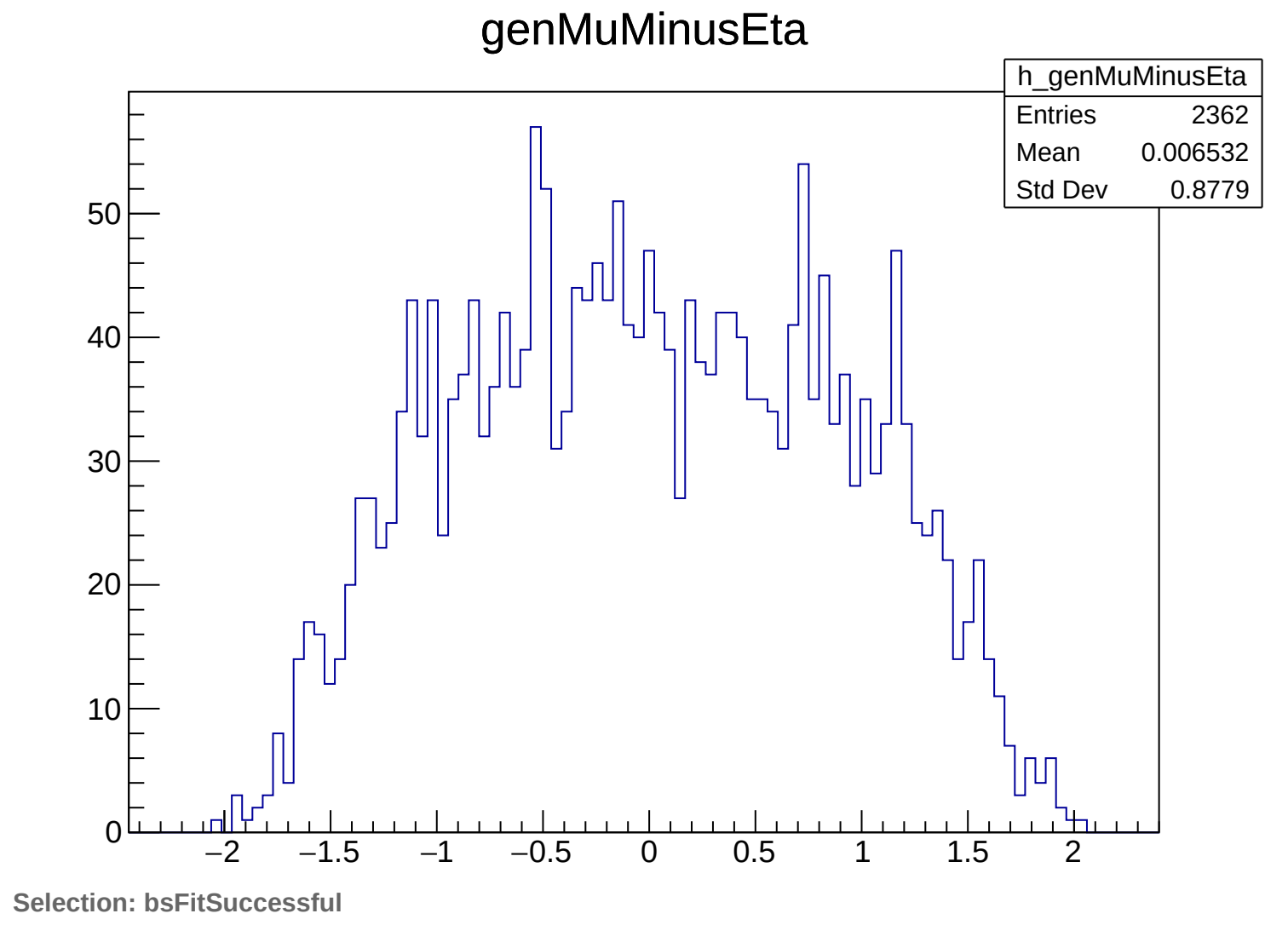

genBMass
h_genBMass
Entries 2362
Mean 5.367
Std Dev 0
1000
800
600
400
200
0
5.366769
5.367906513
Selection: bsFitSuccessful

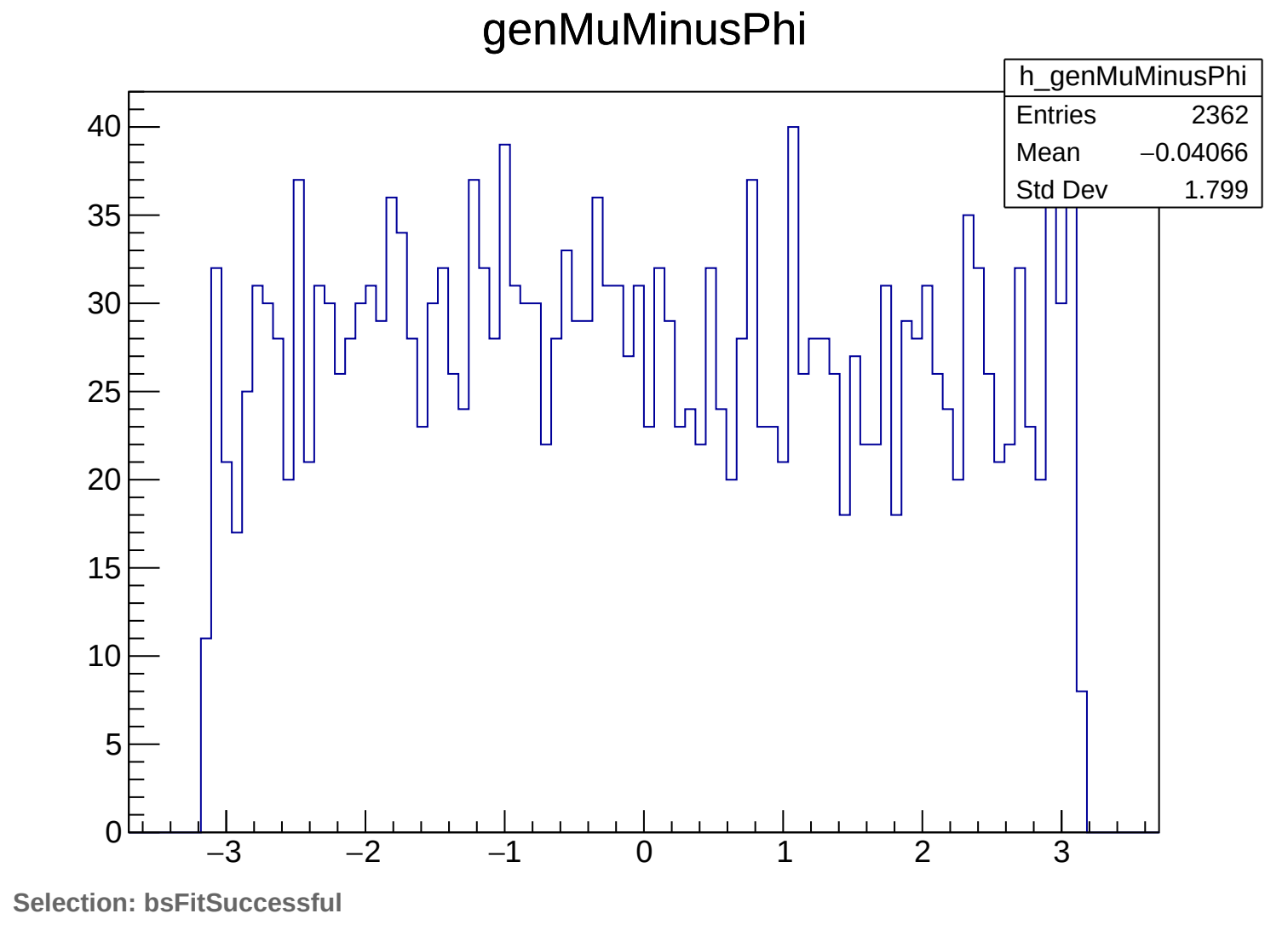

genMuPlusPt
h_genMuPlusPt
Entries 2362
Mean 8.507
Std Dev 6.859
500
400
300
200
100
0
0
20
40
60
80
100
Selection: bsFitSuccessful

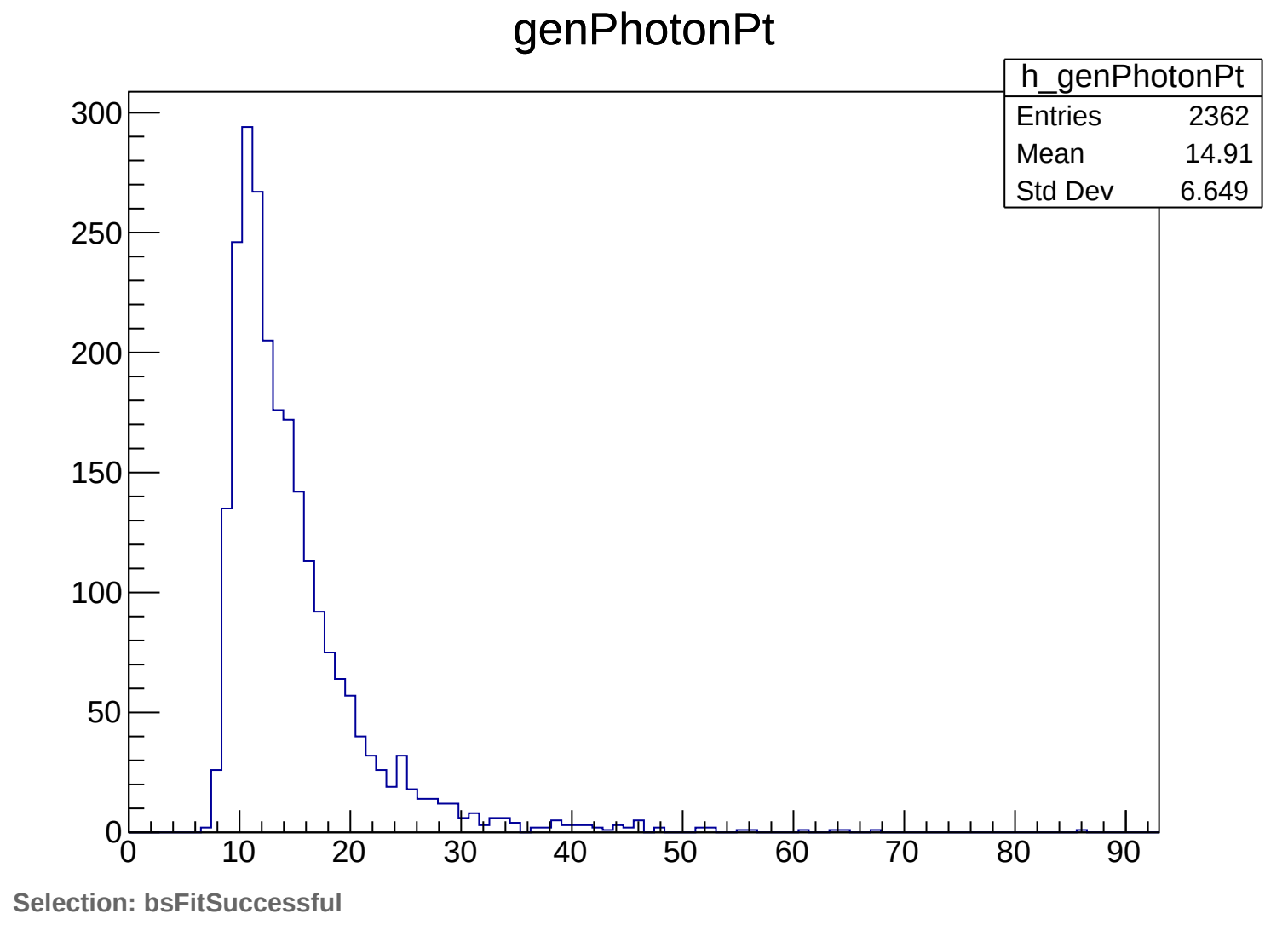

genMuPlusEta
h_genMuPlusEta
Entries 2362
Mean −0.006225
Std Dev 0.8707
50
40
30
20
10
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

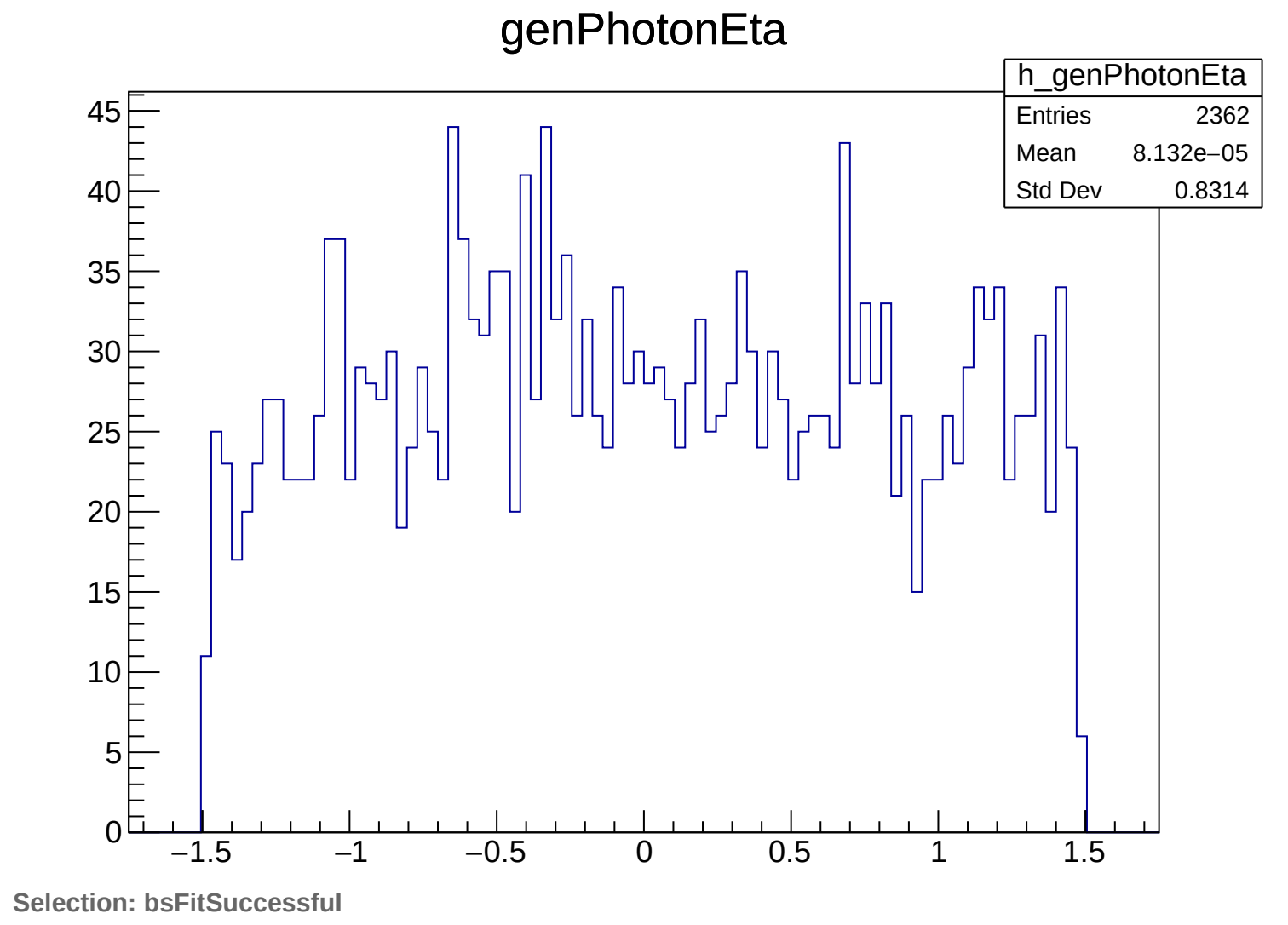

genMuPlusPhi
h_genMuPlusPhi
Entries 2362
Mean −0.01559
Std Dev 1.792
40
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

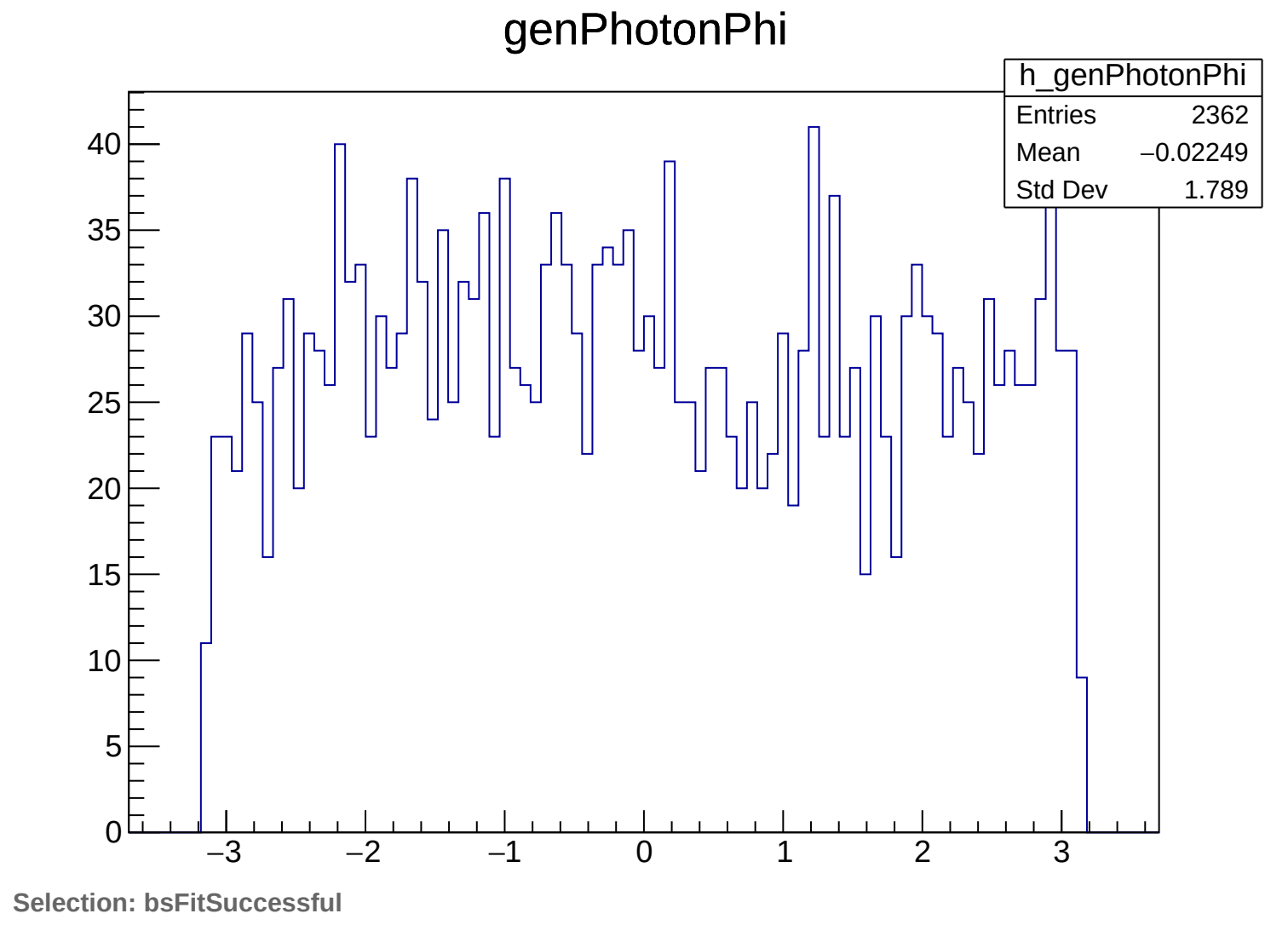

genMuMinusPt
h_genMuMinusPt
Entries 2362
Mean 8.625
Std Dev 7.087
400
350
300
250
200
150
100
50
0
0
20
40
60
80
100
Selection: bsFitSuccessful

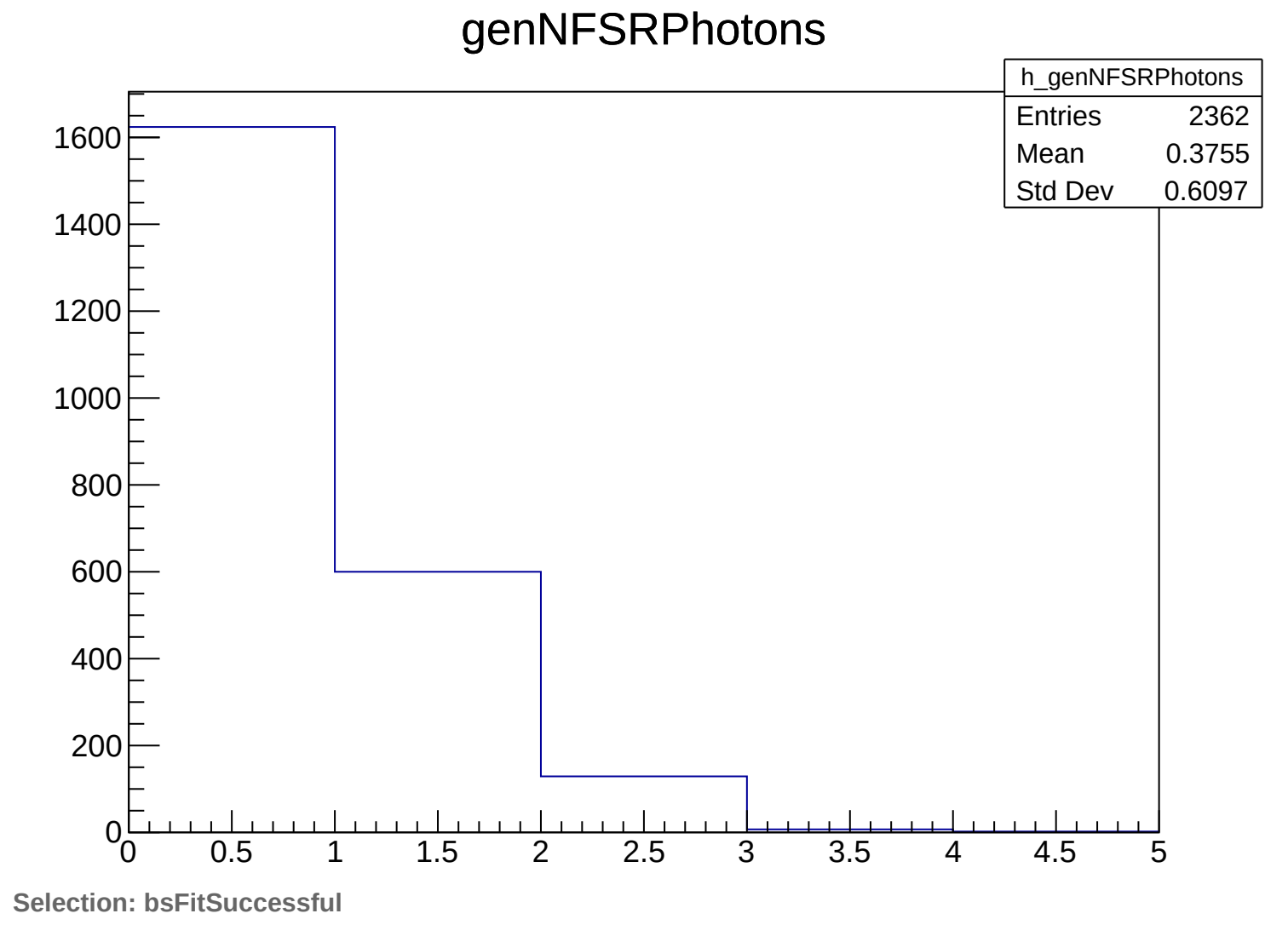

genMuMinusEta
h_genMuMinusEta
Entries 2362
Mean 0.006532
Std Dev 0.8779
50
40
30
20
10
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

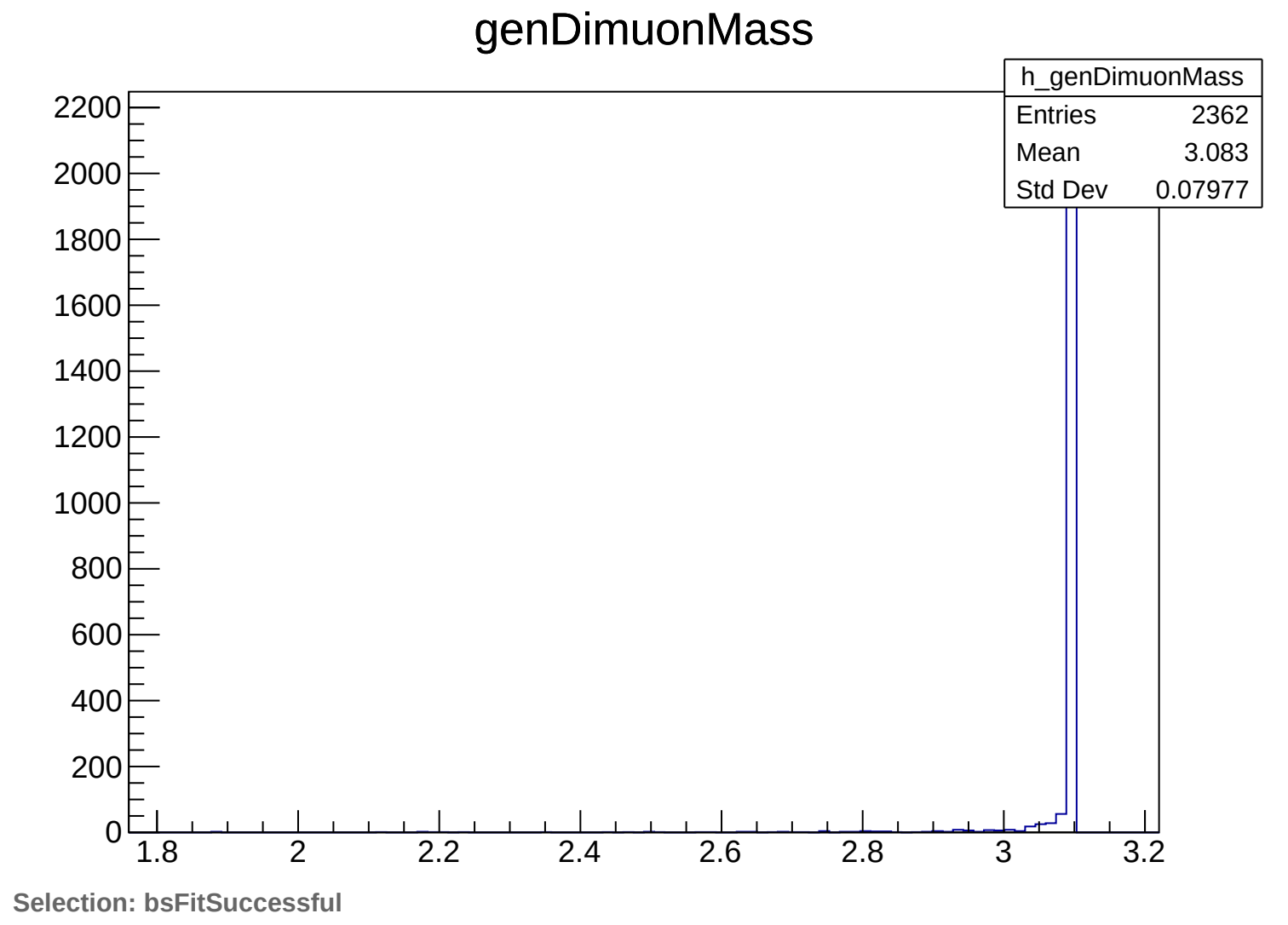

genMuMinusPhi
h_genMuMinusPhi
Entries 2362
Mean −0.04066
Std Dev 1.799
40
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

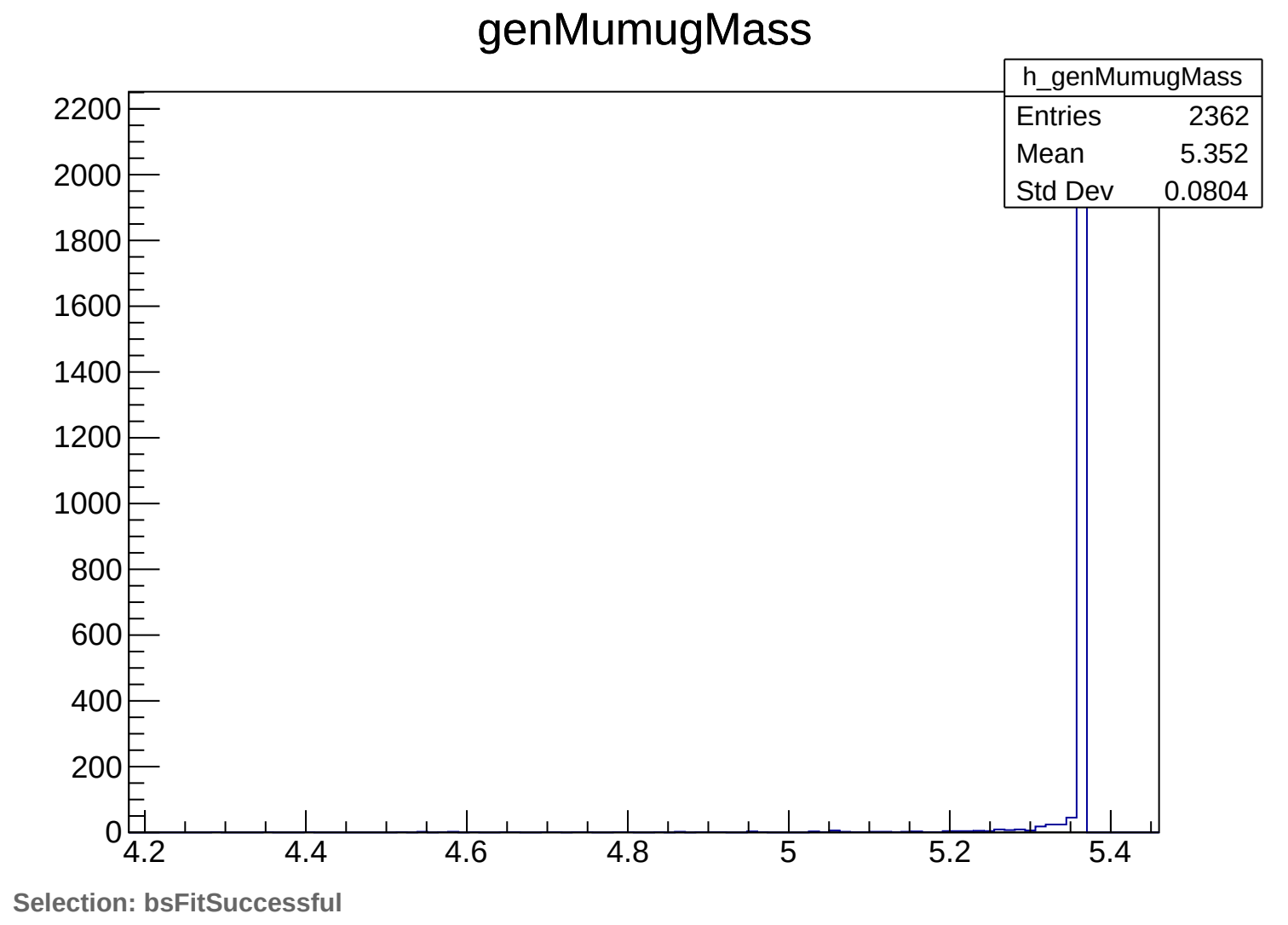

genPhotonPt
h_genPhotonPt
Entries 2362
Mean 14.91
Std Dev 6.649
300
250
200
150
100
50
0
0
10
20
30
40
50
60
70
80
90
Selection: bsFitSuccessful

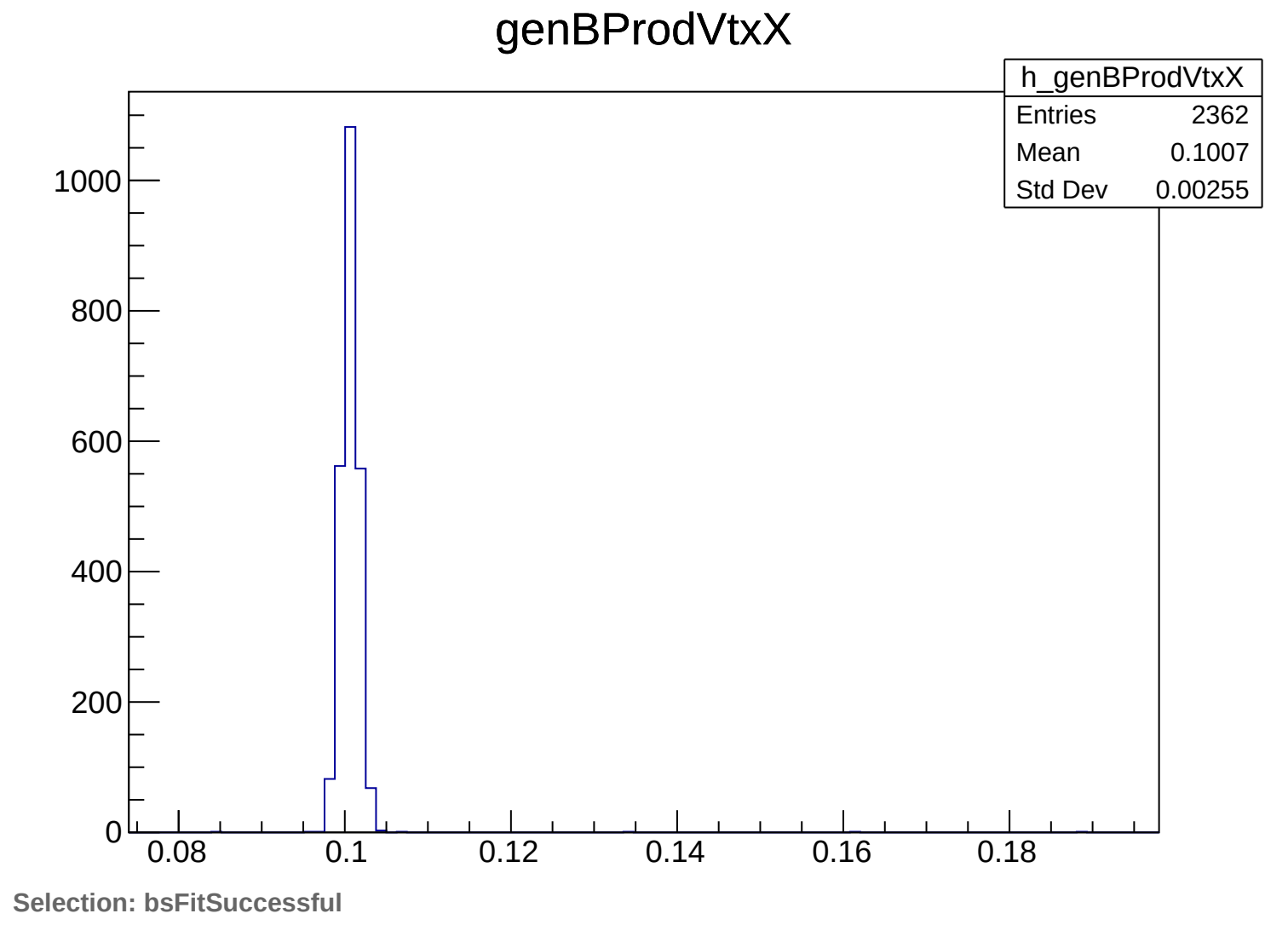

genPhotonEta
h_genPhotonEta
Entries 2362
Mean 8.132e−05
Std Dev 0.8314
45
40
35
30
25
20
15
10
5
0
−1.5
−1
−0.5
0
0.5
1
1.5
Selection: bsFitSuccessful

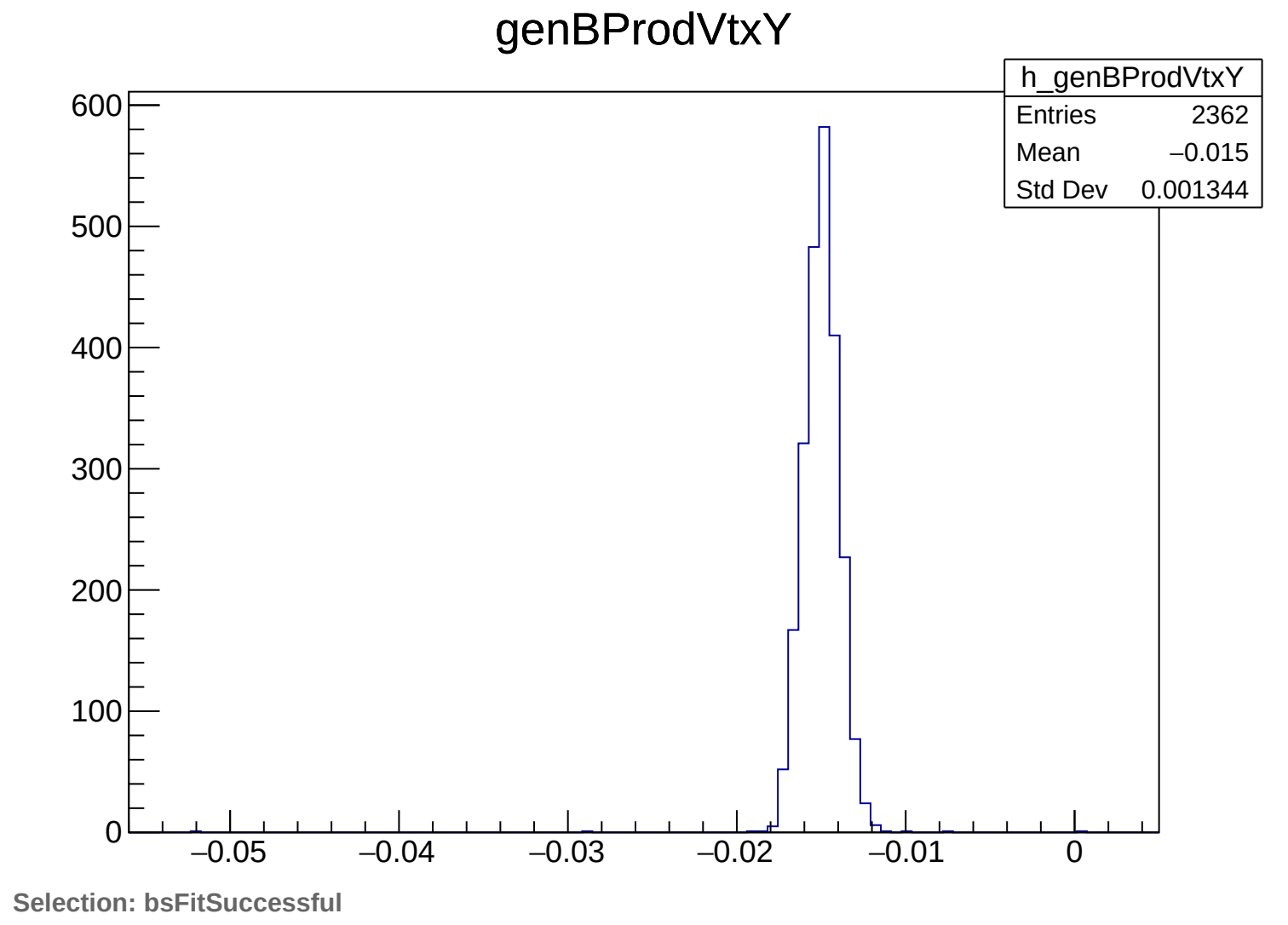

genPhotonPhi
h_genPhotonPhi
Entries 2362
Mean −0.02249
Std Dev 1.789
40
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

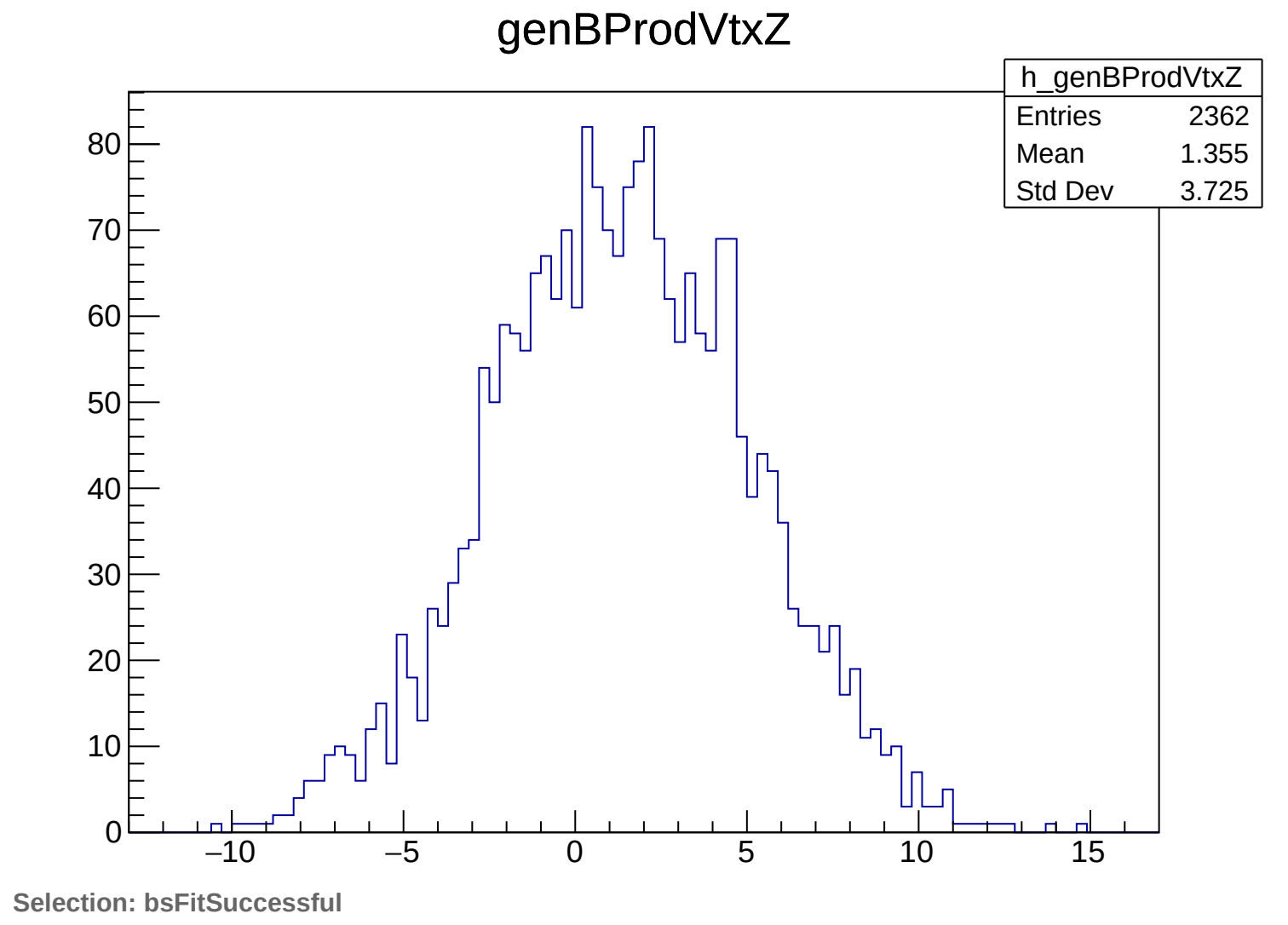

genNFSRPhotons
h_genNFSRPhotons
Entries 2362
Mean 0.3755
Std Dev 0.6097
1600
1400
1200
1000
800
600
400
200
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Selection: bsFitSuccessful

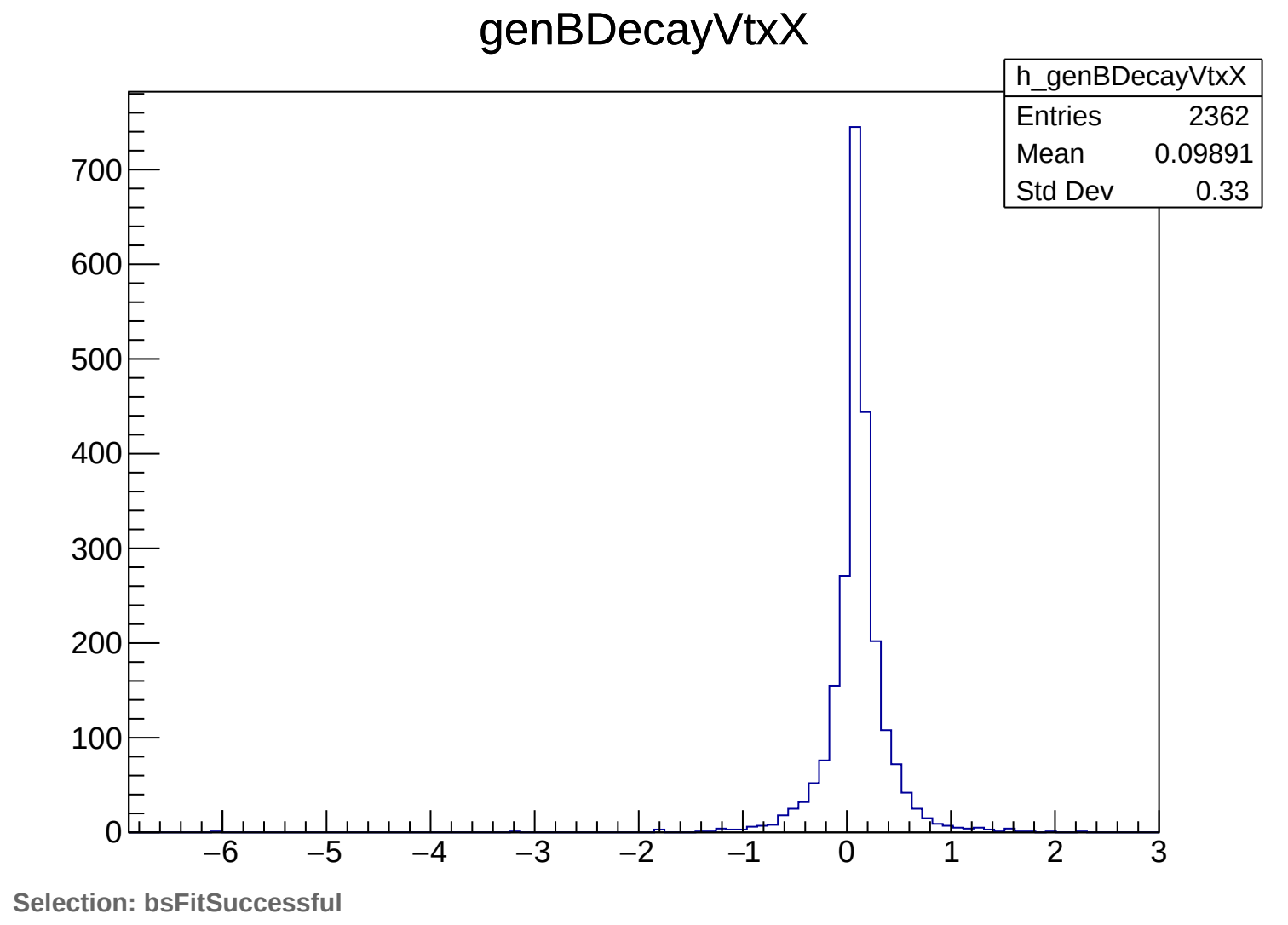

genDimuonMass
h_genDimuonMass
Entries 2362
Mean 3.083
Std Dev 0.07977
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
1.8
2
2.2
2.4
2.6
2.8
3
3.2
Selection: bsFitSuccessful

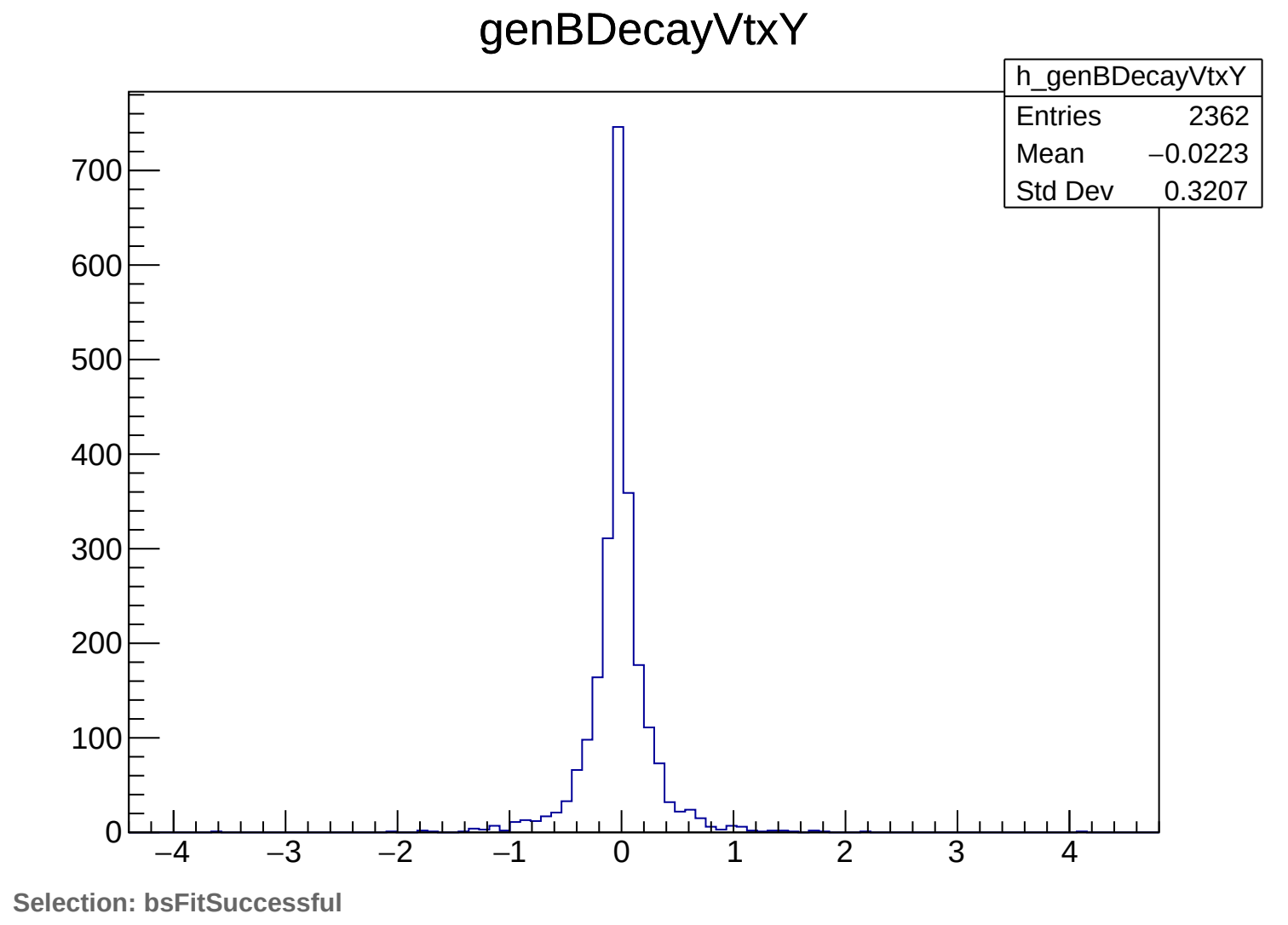

genMumugMass
h_genMumugMass
Entries 2362
Mean 5.352
Std Dev 0.0804
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
4.2
4.4
4.6
4.8
5
5.2
5.4
Selection: bsFitSuccessful

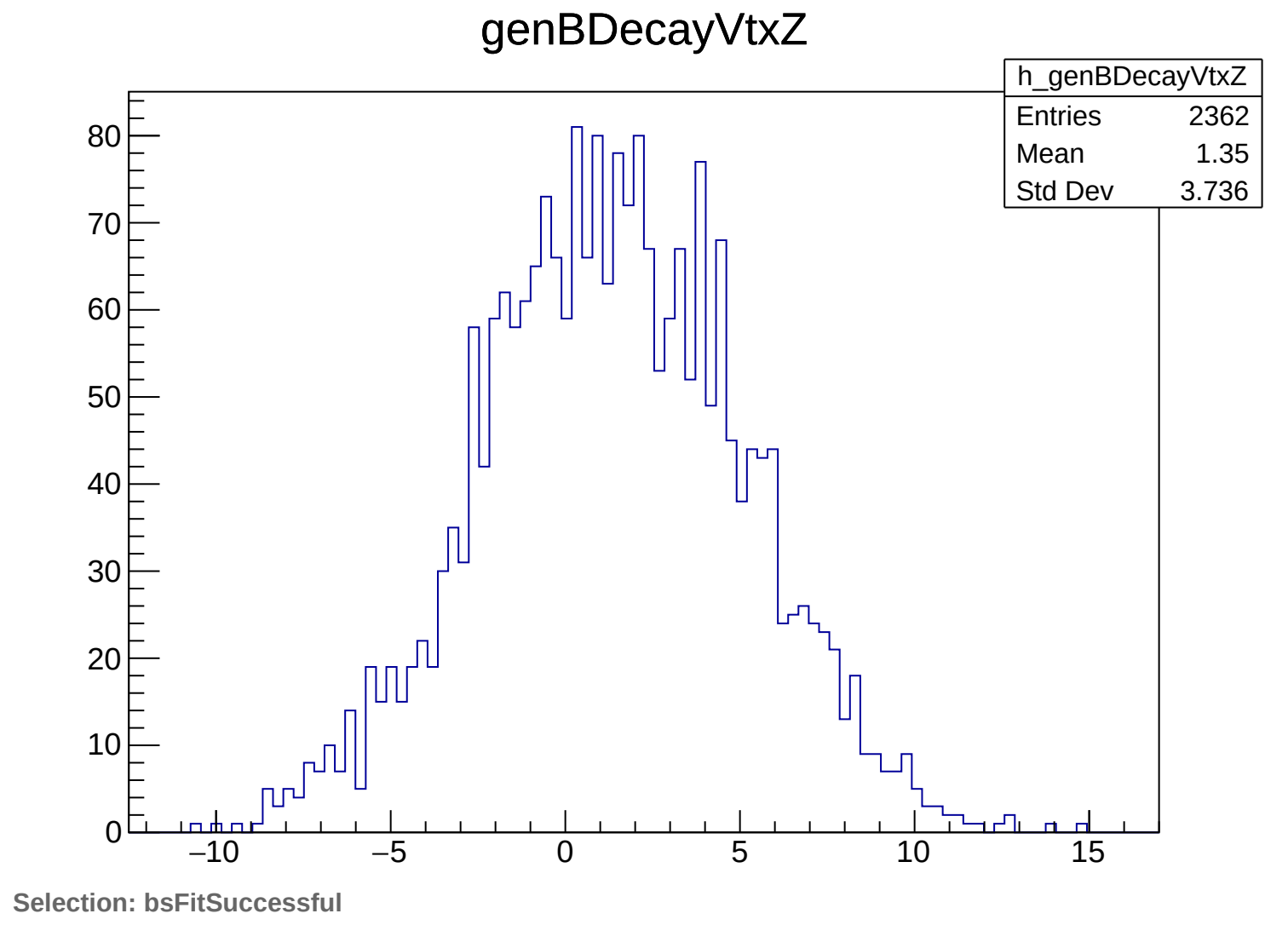

genBProdVtxX
h_genBProdVtxX
Entries 2362
Mean 0.1007
Std Dev 0.00255
1000
800
600
400
200
0
0.08
0.1
0.12
0.14
0.16
0.18
Selection: bsFitSuccessful

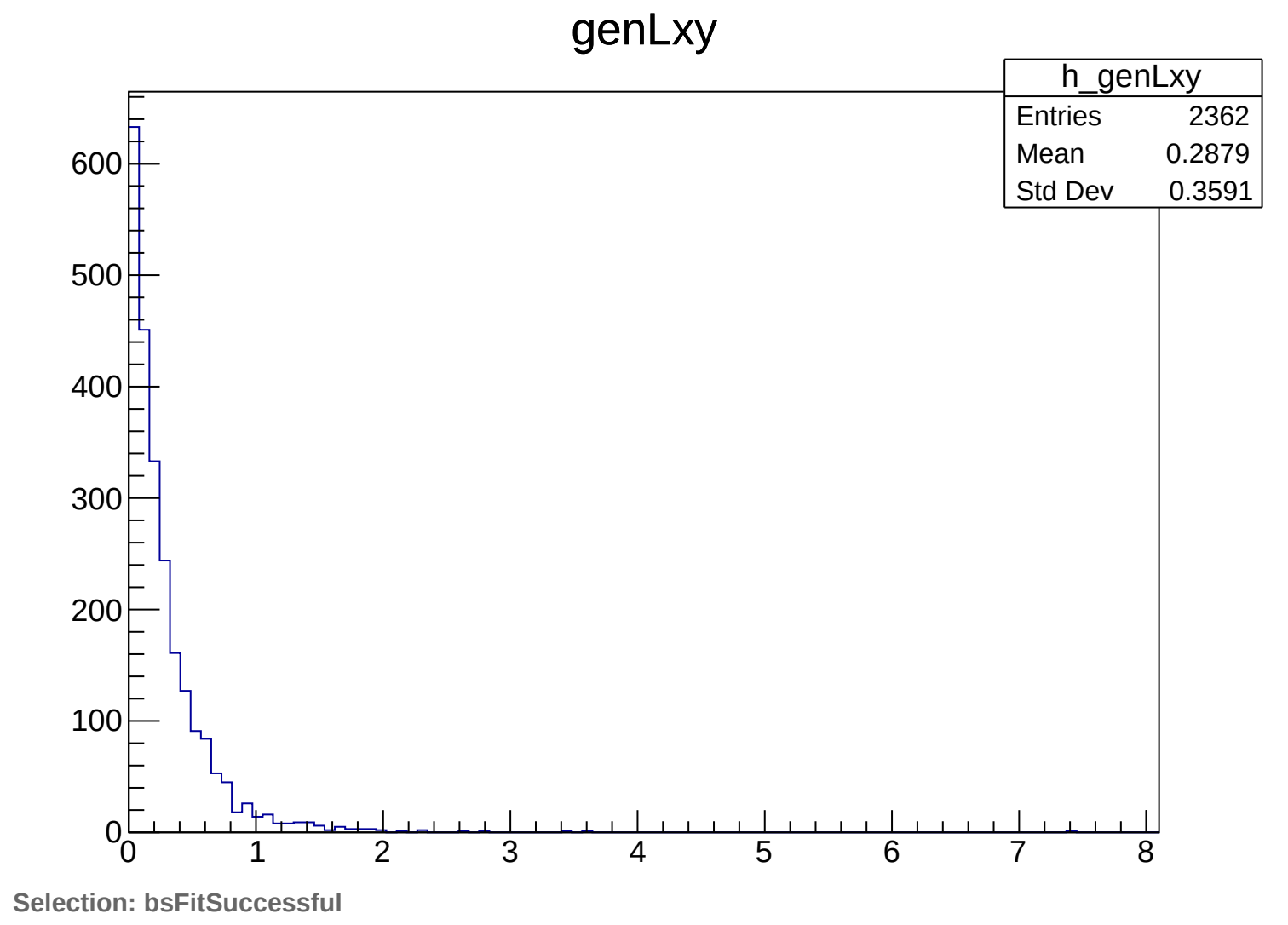

genBProdVtxY
h_genBProdVtxY
Entries 2362
Mean −0.015
Std Dev 0.001344
600
500
400
300
200
100
0
−0.05
−0.04
−0.03
−0.02
−0.01
0
Selection: bsFitSuccessful

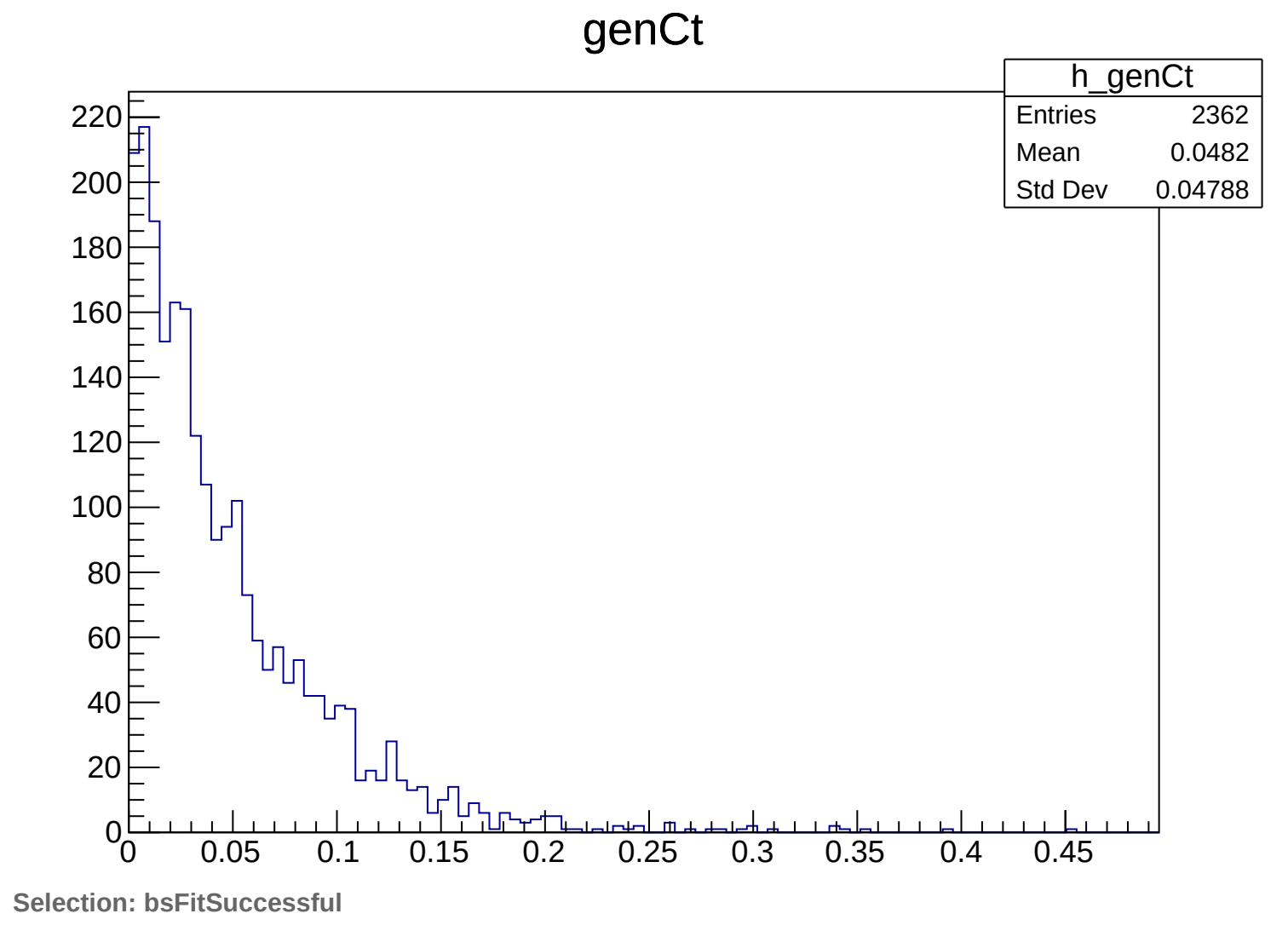

genBProdVtxZ
h_genBProdVtxZ
Entries 2362
Mean 1.355
Std Dev 3.725
80
70
60
50
40
30
20
10
0
−10
−5
0
5
10
15
Selection: bsFitSuccessful

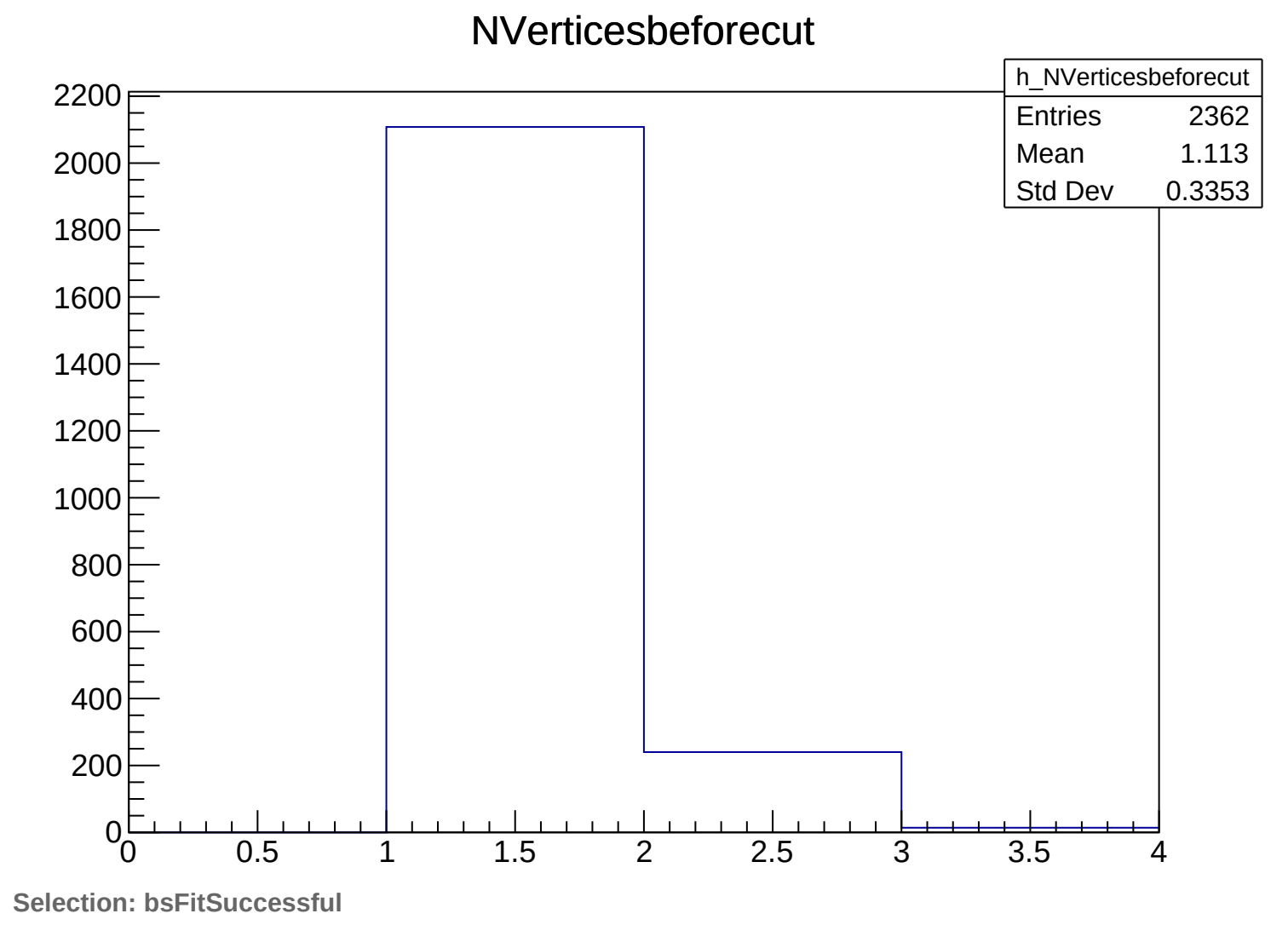

genBDecayVtxX
h_genBDecayVtxX
Entries 2362
Mean 0.09891
Std Dev 0.33
700
600
500
400
300
200
100
0
−6
−5
−4
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

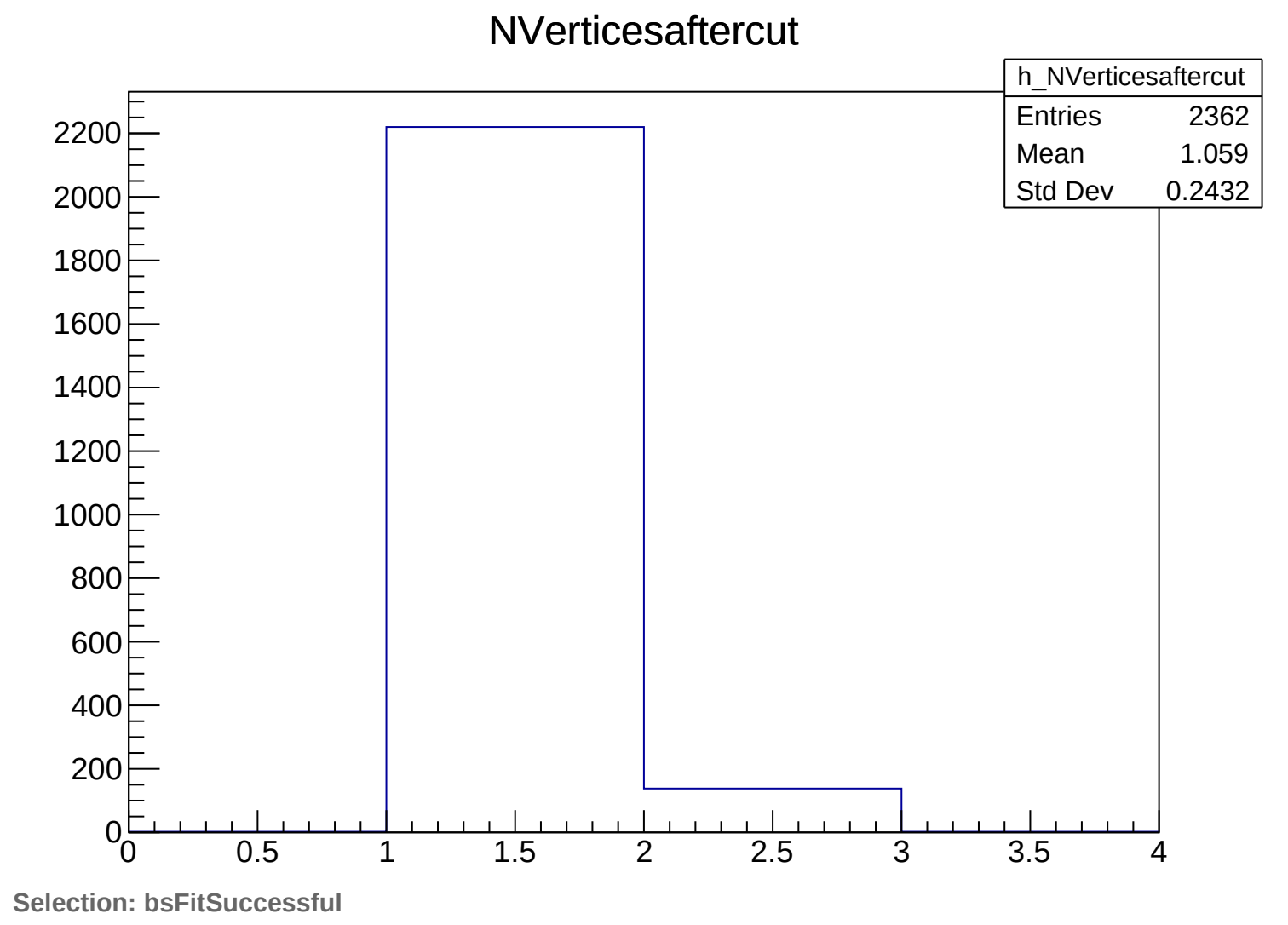

genBDecayVtxY
h_genBDecayVtxY
Entries 2362
Mean −0.0223
Std Dev 0.3207
700
600
500
400
300
200
100
0
−4
−3
−2
−1
0
1
2
3
4
Selection: bsFitSuccessful

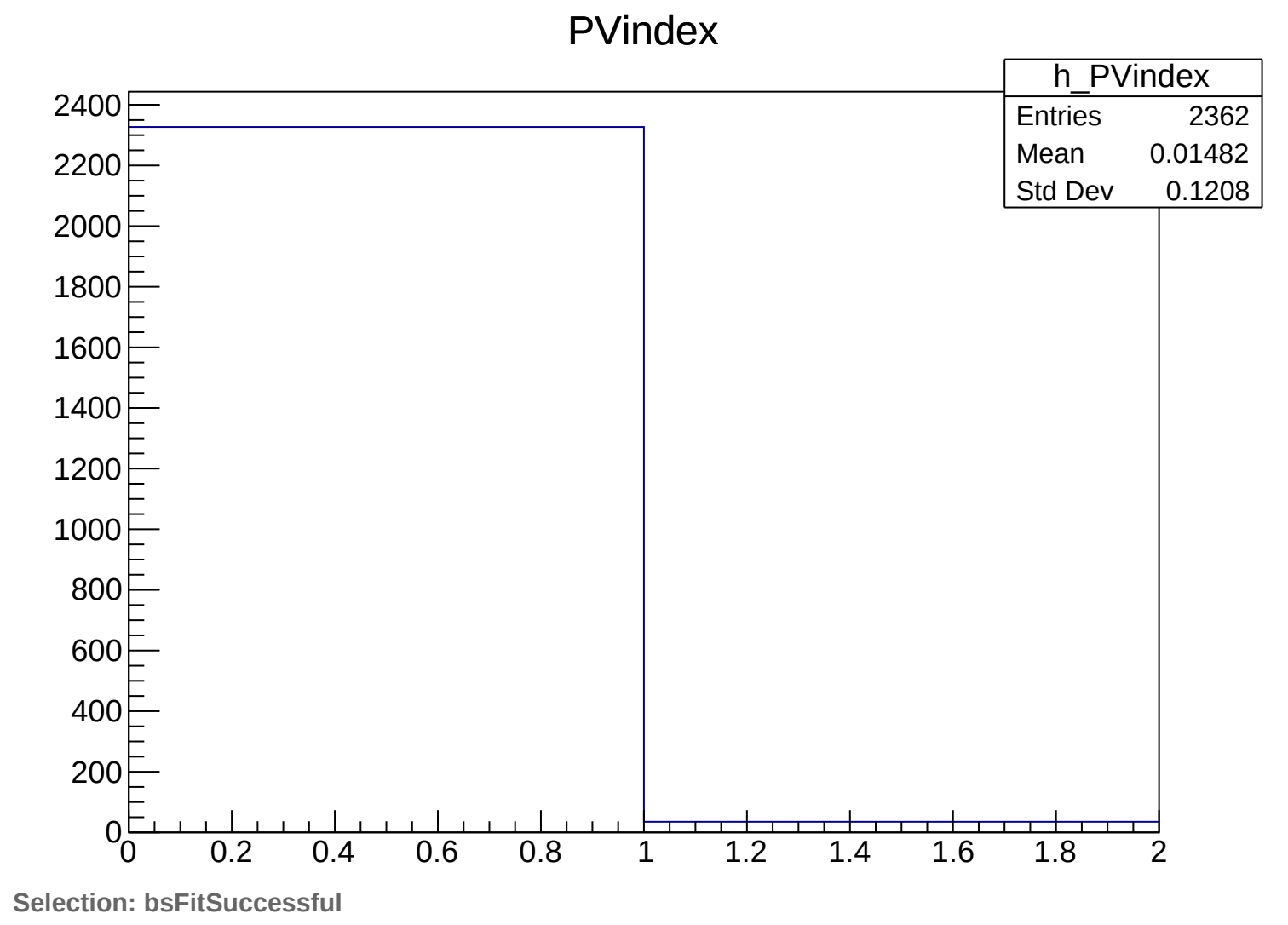

genBDecayVtxZ
h_genBDecayVtxZ
Entries 2362
Mean 1.35
Std Dev 3.736
80
70
60
50
40
30
20
10
0
−10
−5
0
5
10
15
Selection: bsFitSuccessful

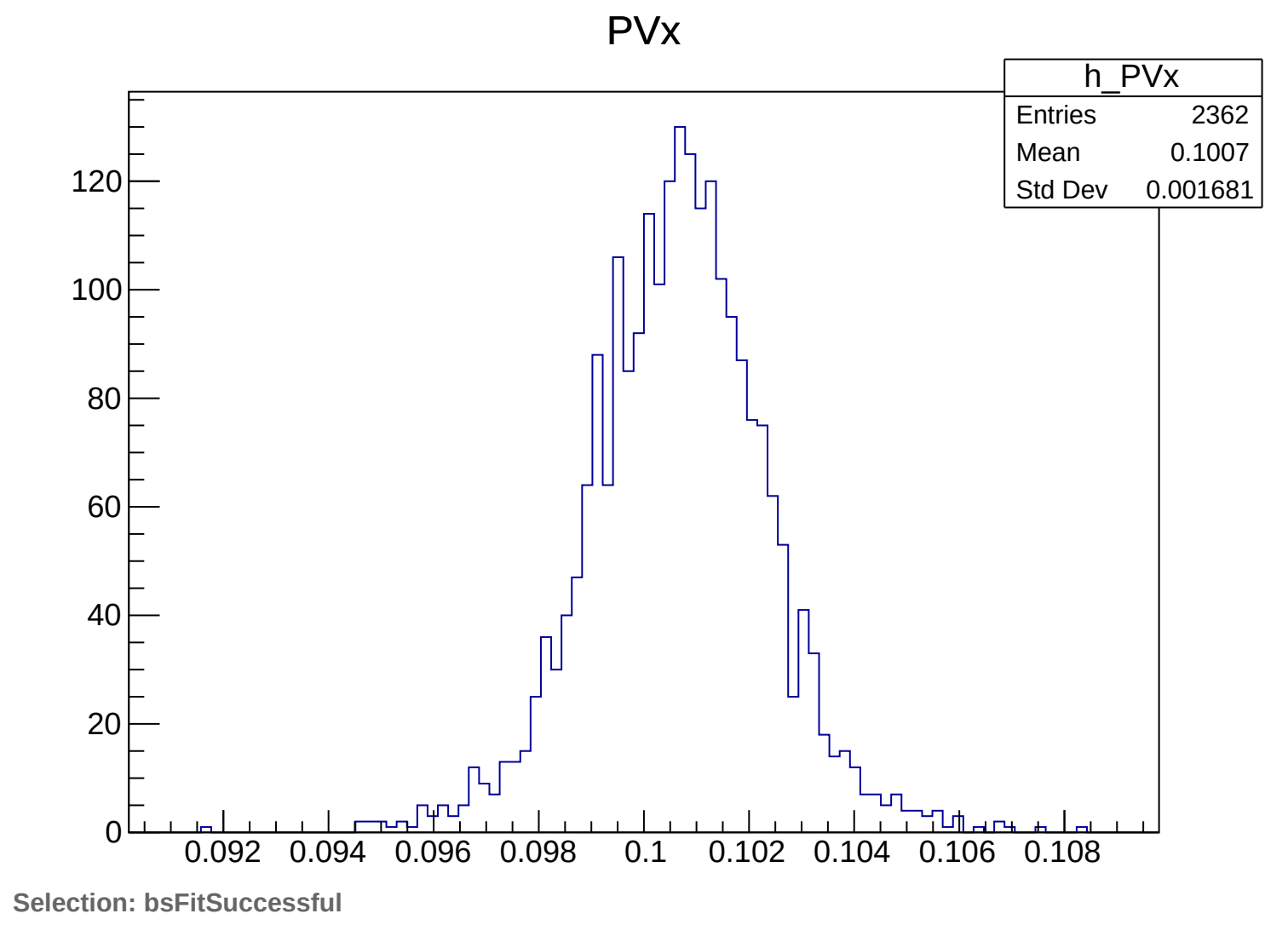

genLxy
h_genLxy
Entries 2362
Mean 0.2879
Std Dev 0.3591
600
500
400
300
200
100
0
0
1
2
3
4
5
6
7
8
Selection: bsFitSuccessful

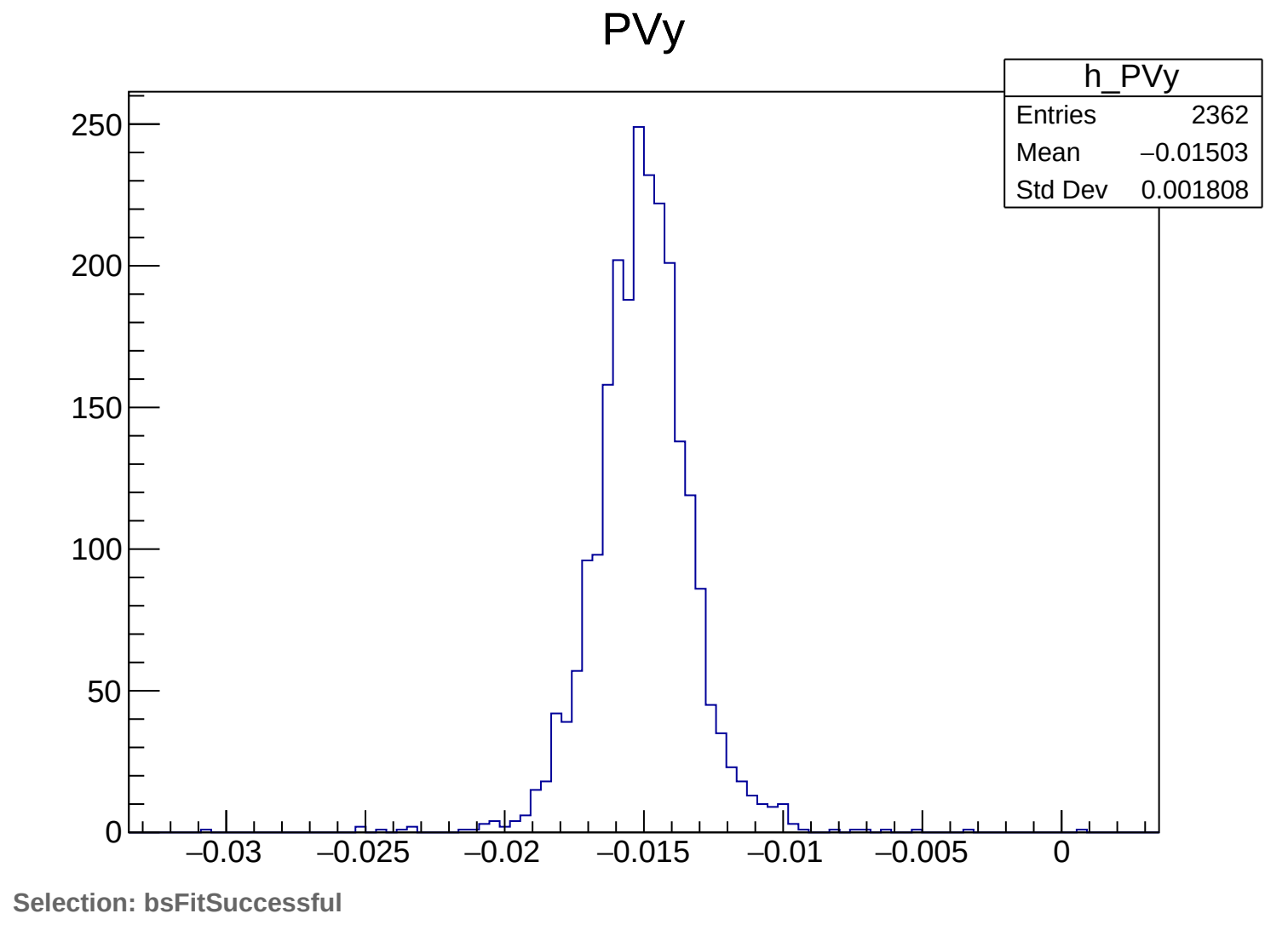

genCt
h_genCt
Entries 2362
Mean 0.0482
Std Dev 0.04788
220
200
180
160
140
120
100
80
60
40
20
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
Selection: bsFitSuccessful

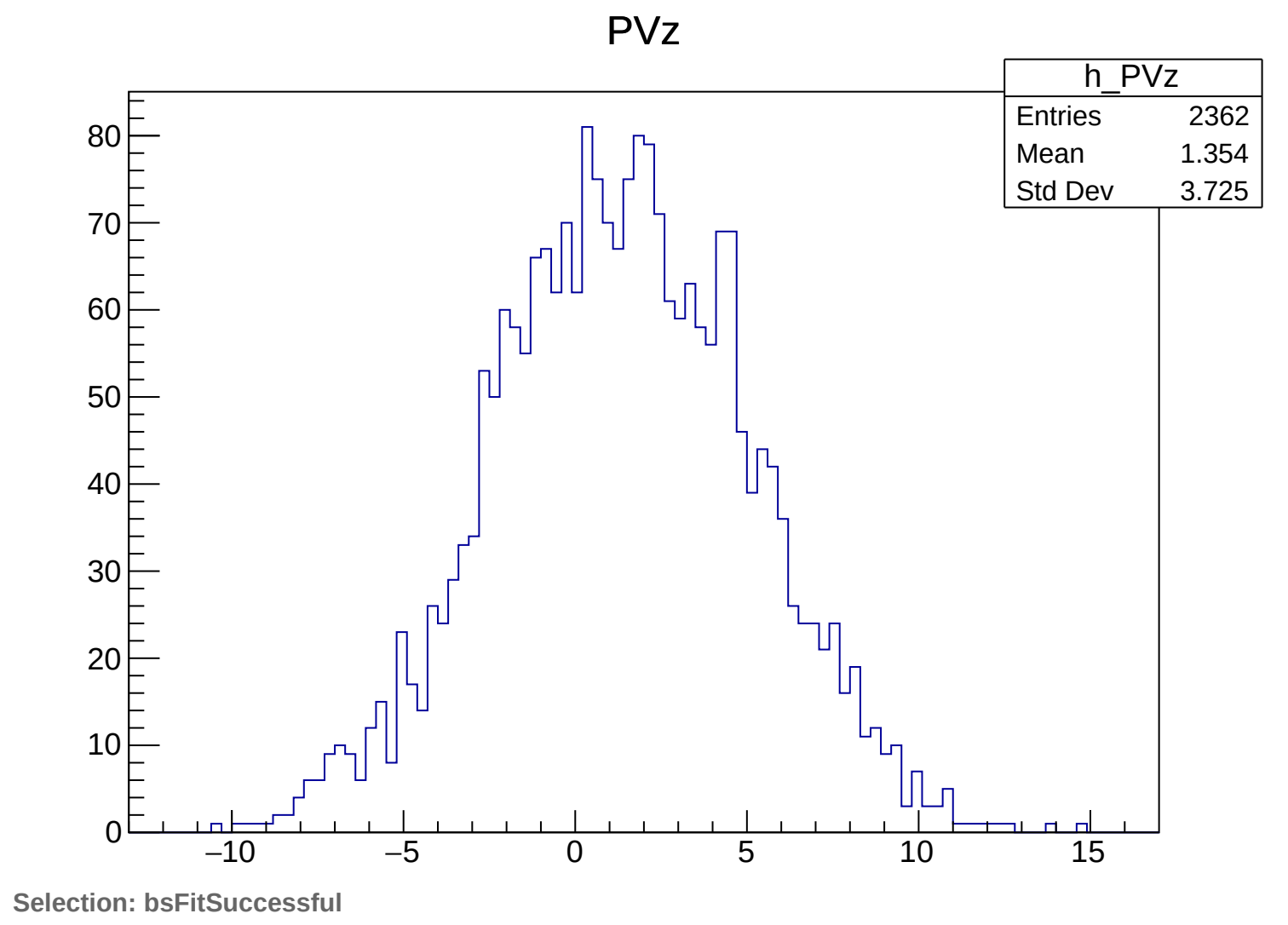

NVerticesbeforecut
h_NVerticesbeforecut
Entries 2362
Mean 1.113
Std Dev 0.3353
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.5
1
1.5
2
2.5
3
3.5
4
Selection: bsFitSuccessful

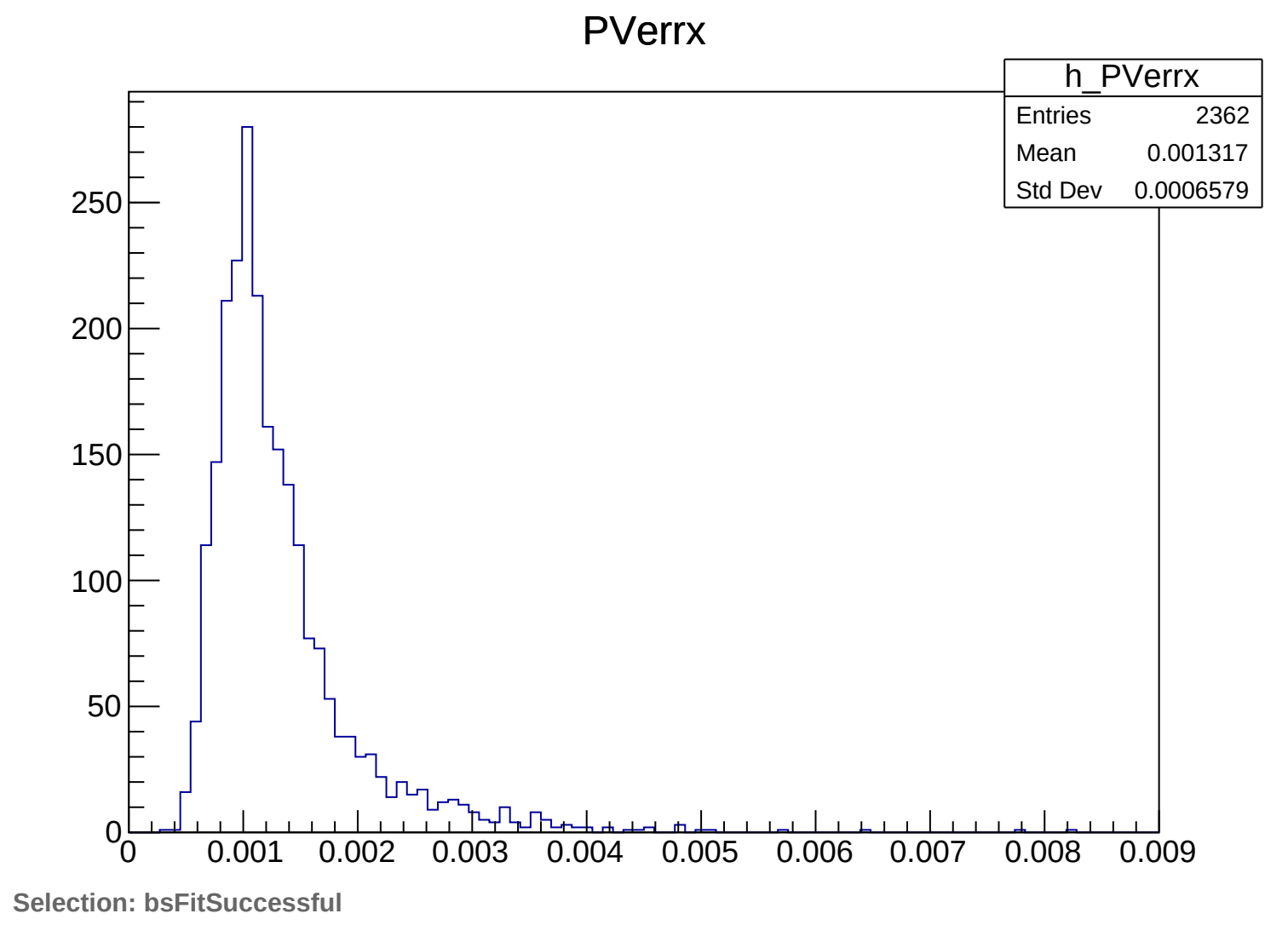

NVerticesaftercut
h_NVerticesaftercut
Entries 2362
Mean 1.059
Std Dev 0.2432
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.5
1
1.5
2
2.5
3
3.5
4
Selection: bsFitSuccessful

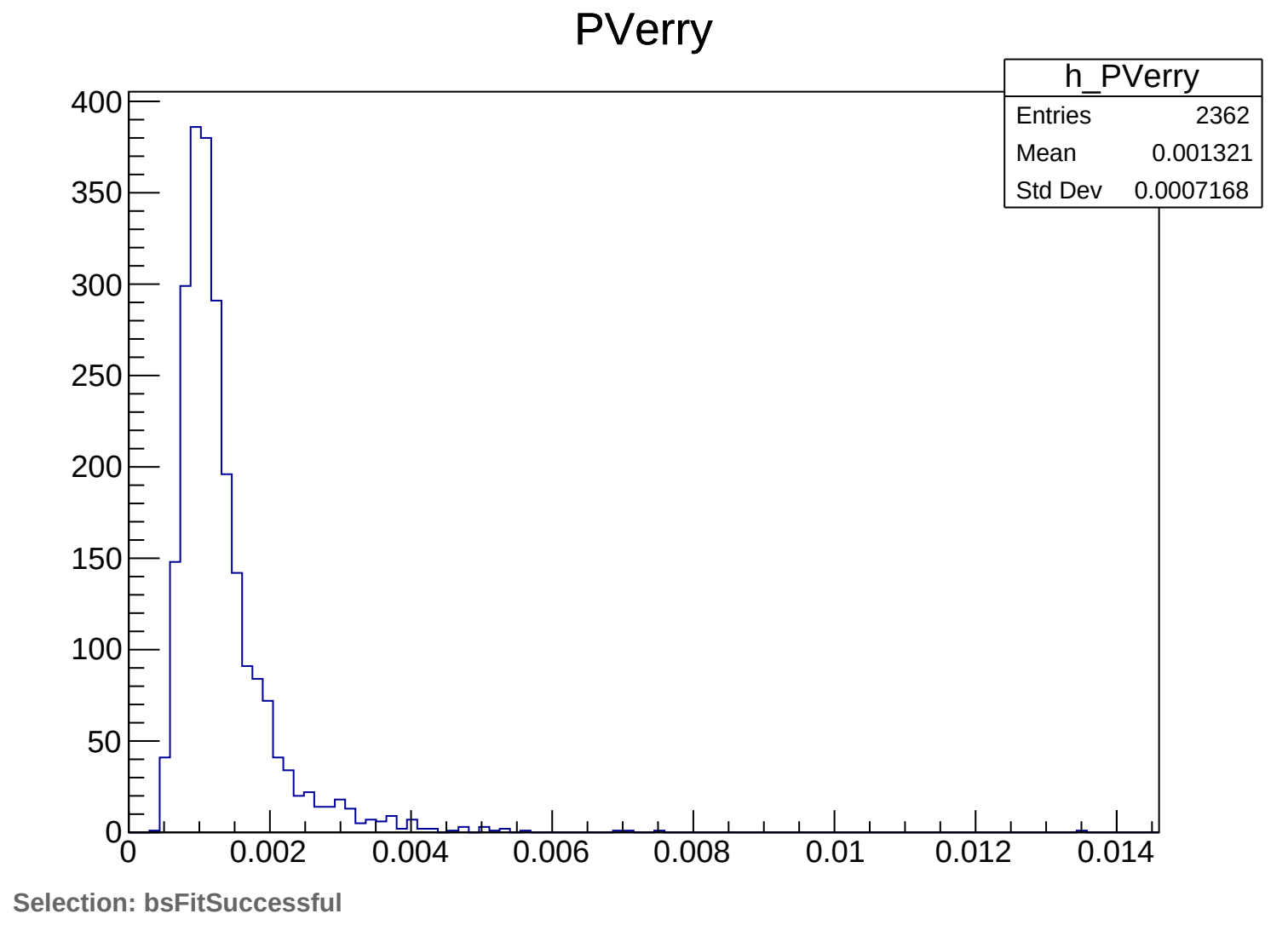

PVindex
h_PVindex
Entries 2362
Mean 0.01482
Std Dev 0.1208
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

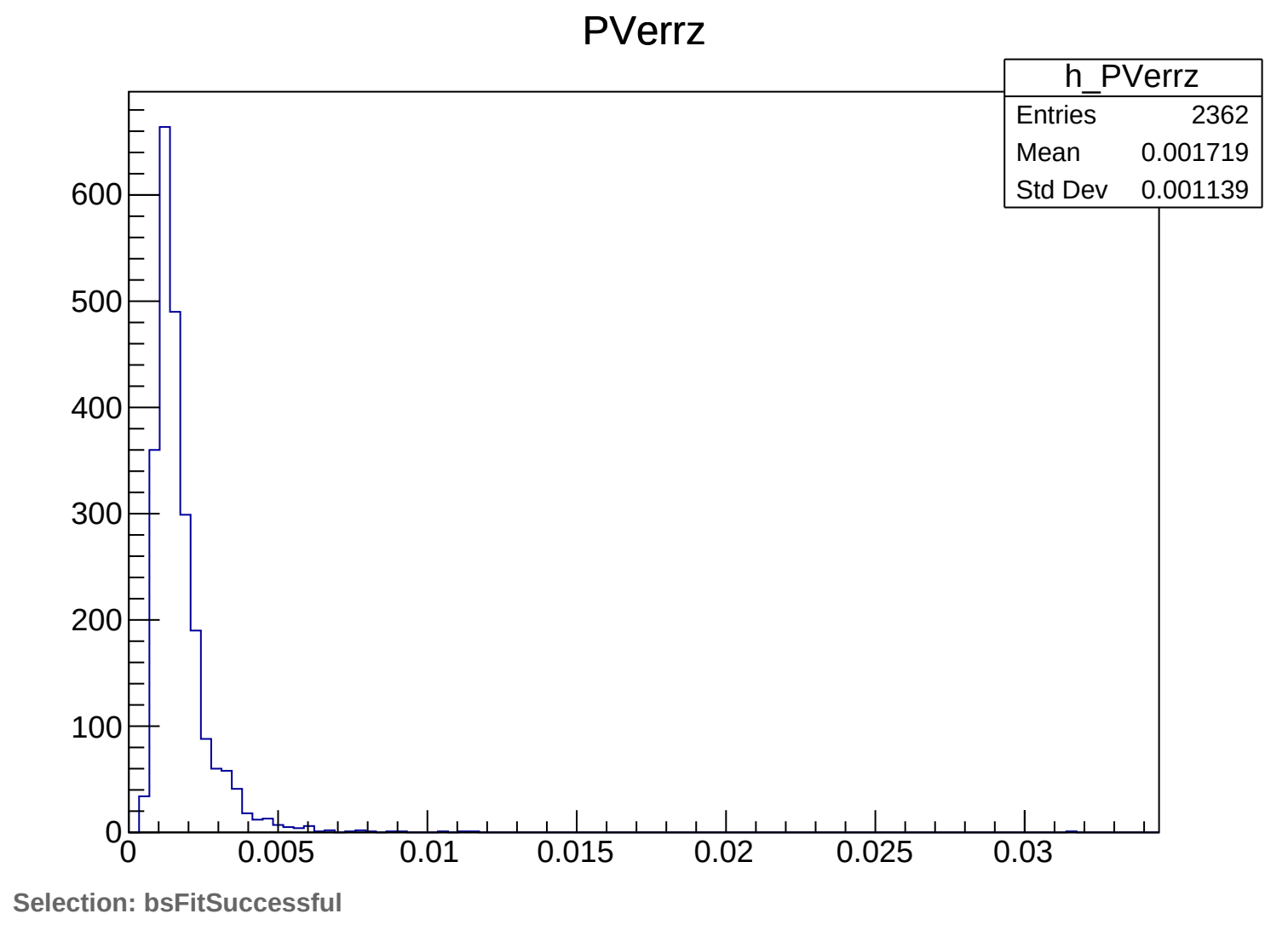

PVx
h_PVx
Entries 2362
Mean 0.1007
Std Dev 0.001681
120
100
80
60
40
20
0
0.092
0.094
0.096
0.098
0.1
0.102
0.104
0.106
0.108
Selection: bsFitSuccessful

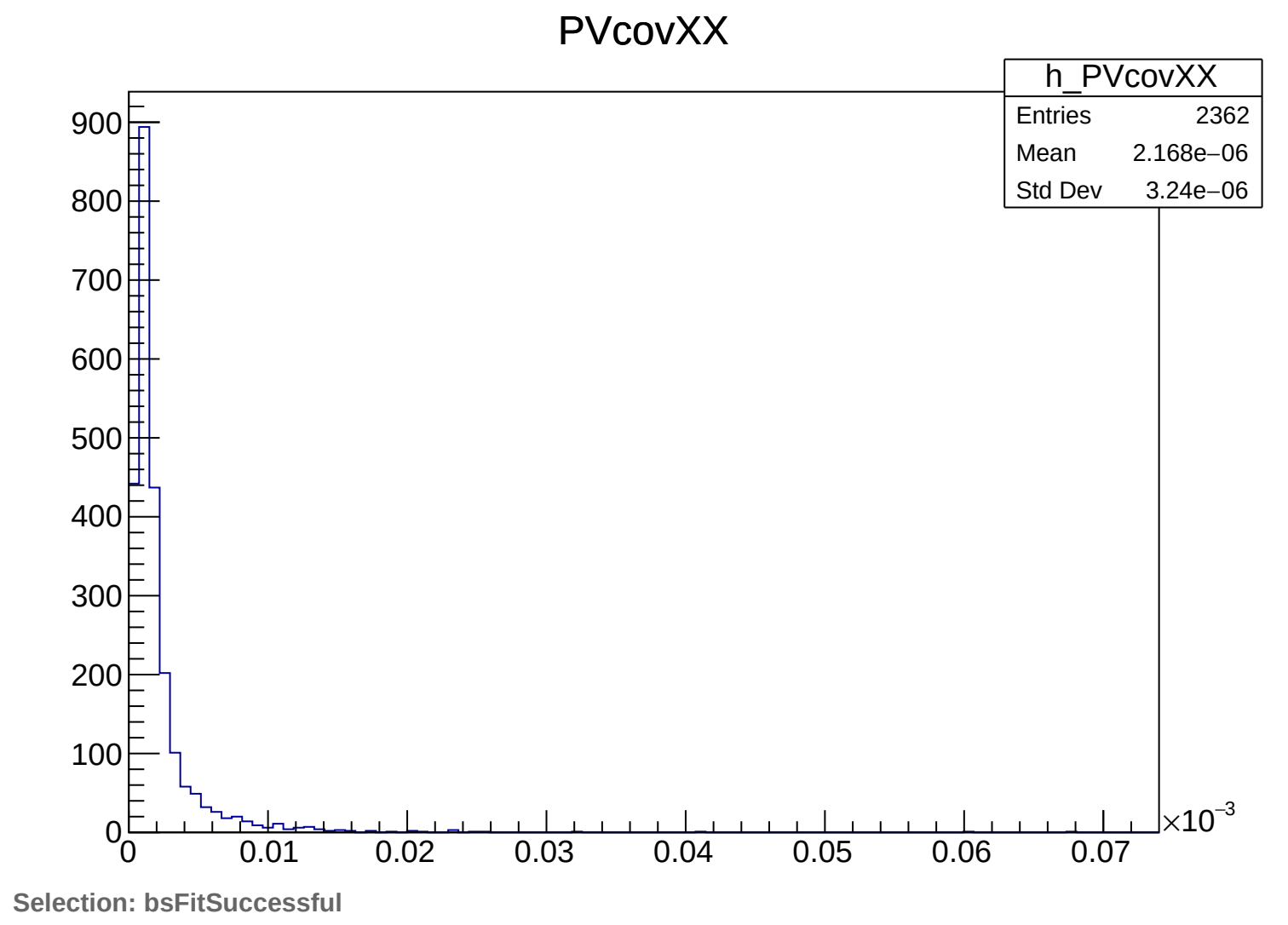

PVy
h_PVy
Entries 2362
Mean −0.01503
Std Dev 0.001808
250
200
150
100
50
0
−0.03
−0.025
−0.02
−0.015
−0.01
−0.005
0
Selection: bsFitSuccessful

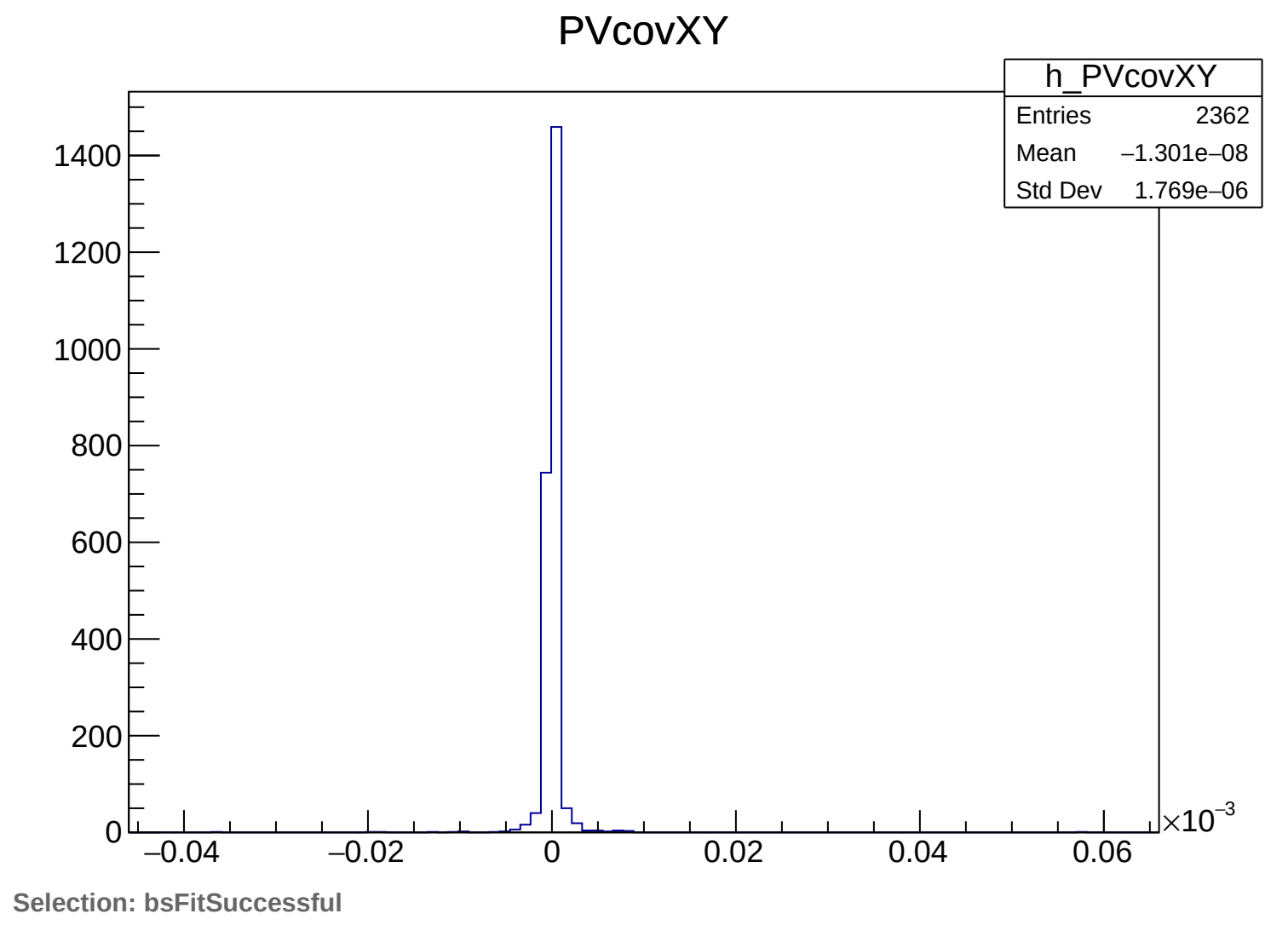

PVz
h_PVz
Entries 2362
Mean 1.354
Std Dev 3.725
80
70
60
50
40
30
20
10
0
−10
−5
0
5
10
15
Selection: bsFitSuccessful

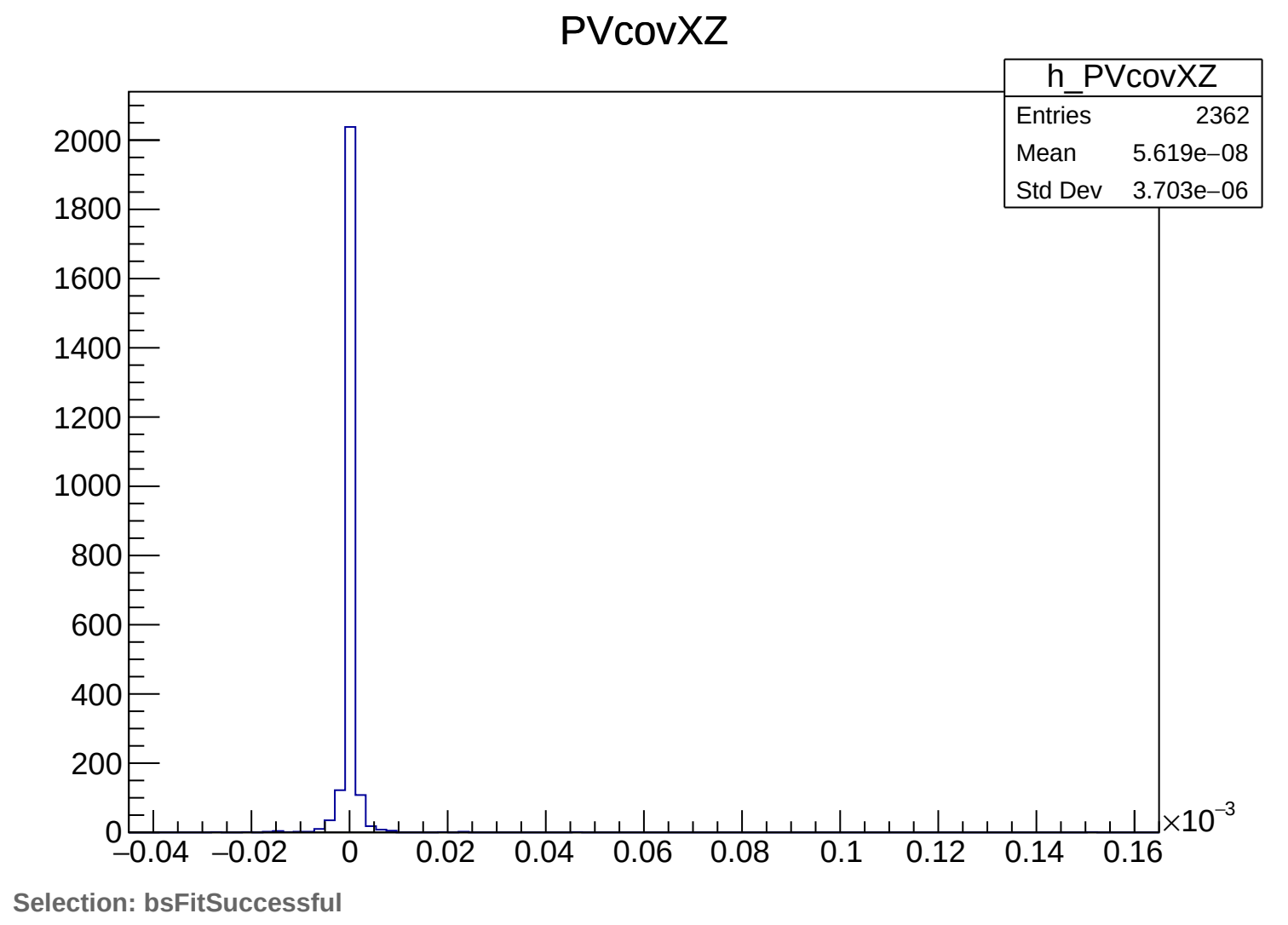

PVerrx
h_PVerrx
Entries 2362
Mean 0.001317
Std Dev 0.0006579
250
200
150
100
50
0
0
0.001
0.002
0.003
0.004
0.005
0.006
0.007
0.008
0.009
Selection: bsFitSuccessful

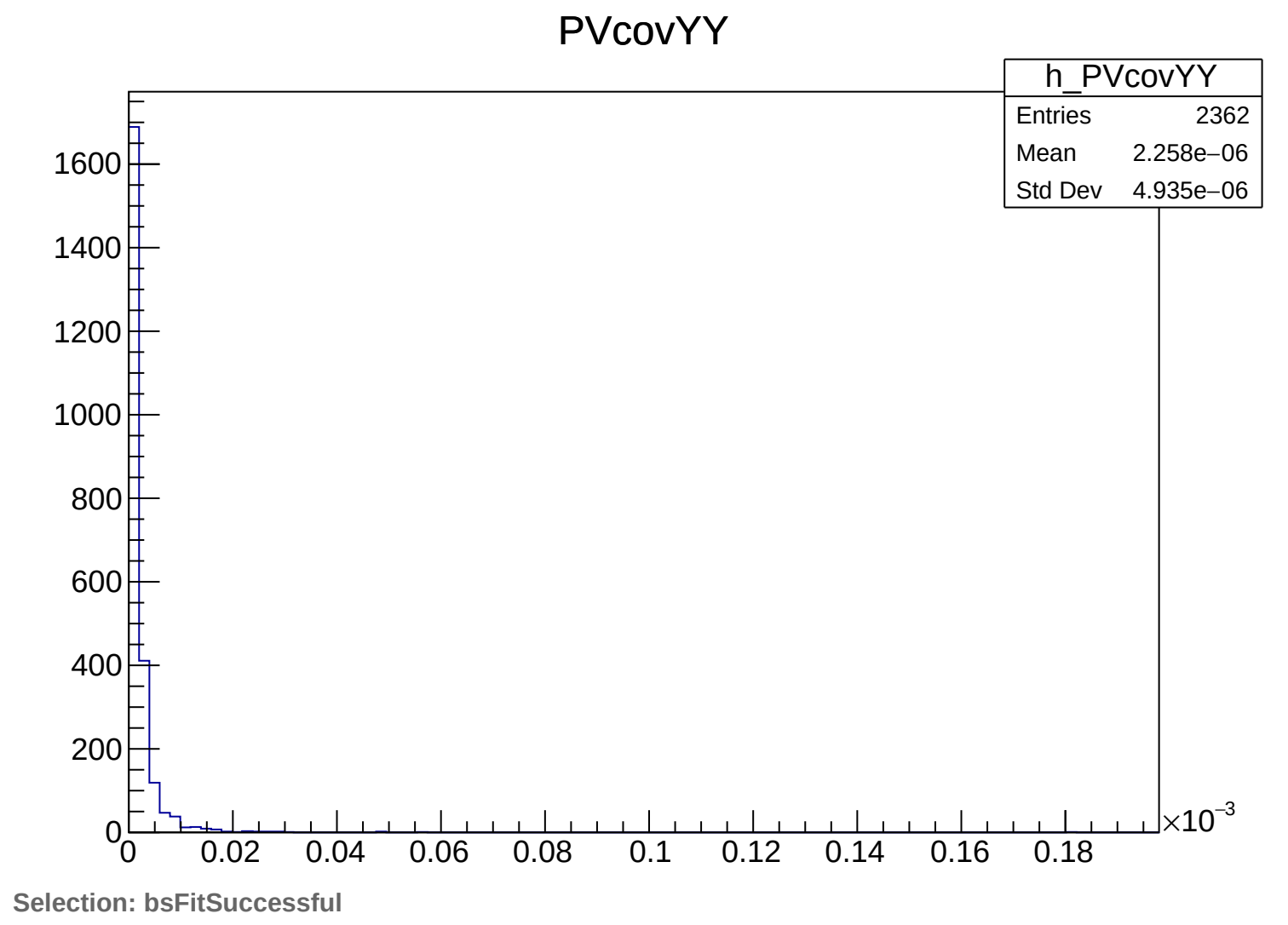

PVerry
h_PVerry
Entries 2362
Mean 0.001321
Std Dev 0.0007168
400
350
300
250
200
150
100
50
0
0
0.002
0.004
0.006
0.008
0.01
0.012
0.014
Selection: bsFitSuccessful

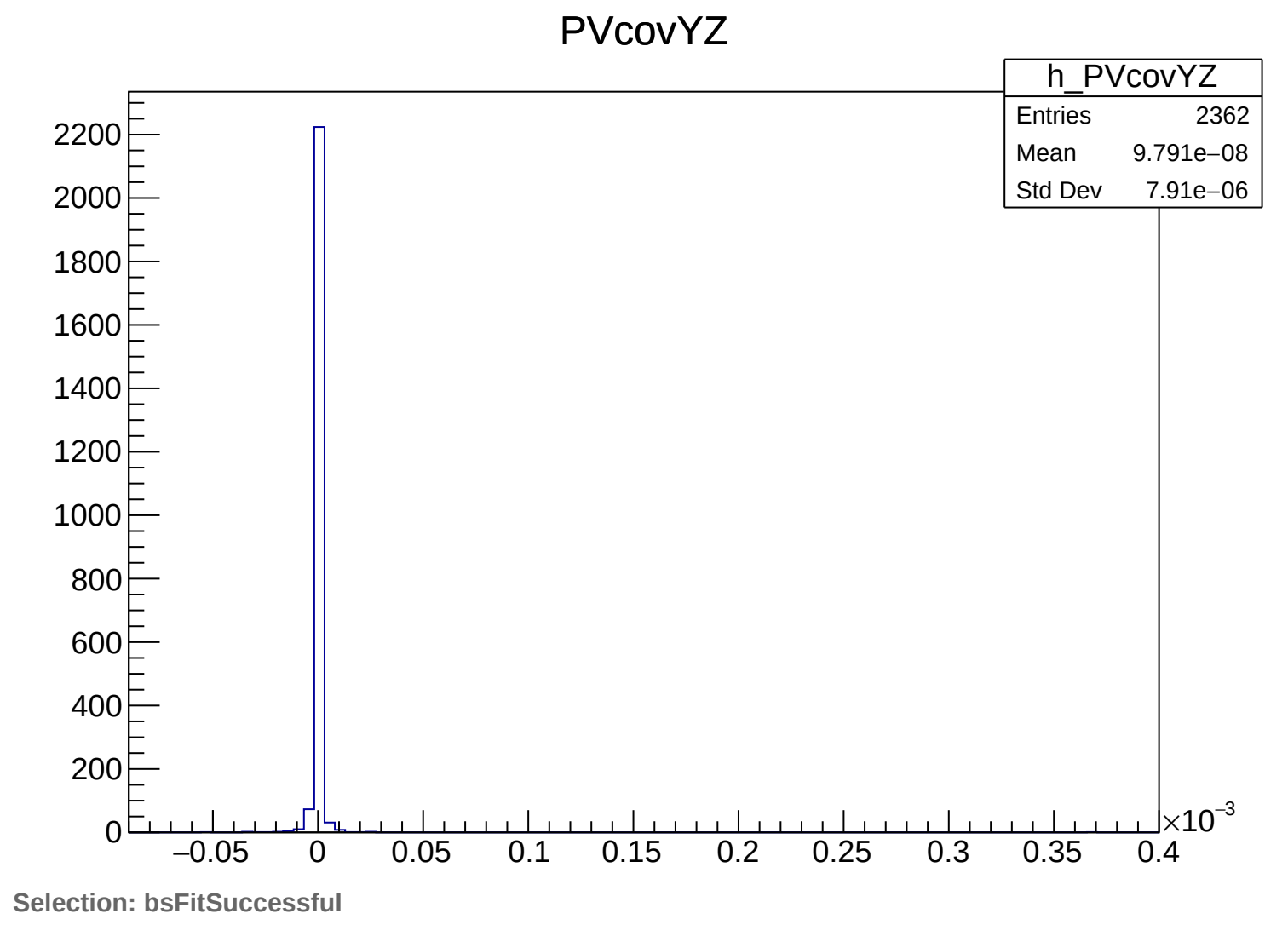

PVerrz
h_PVerrz
Entries 2362
Mean 0.001719
Std Dev 0.001139
600
500
400
300
200
100
0
0
0.005
0.01
0.015
0.02
0.025
0.03
Selection: bsFitSuccessful


PVcovXX
h_PVcovXX
Entries 2362
Mean 2.168e−06
Std Dev 3.24e−06
900
800
700
600
500
400
300
200
100
0
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07
×10⁻³
Selection: bsFitSuccessful

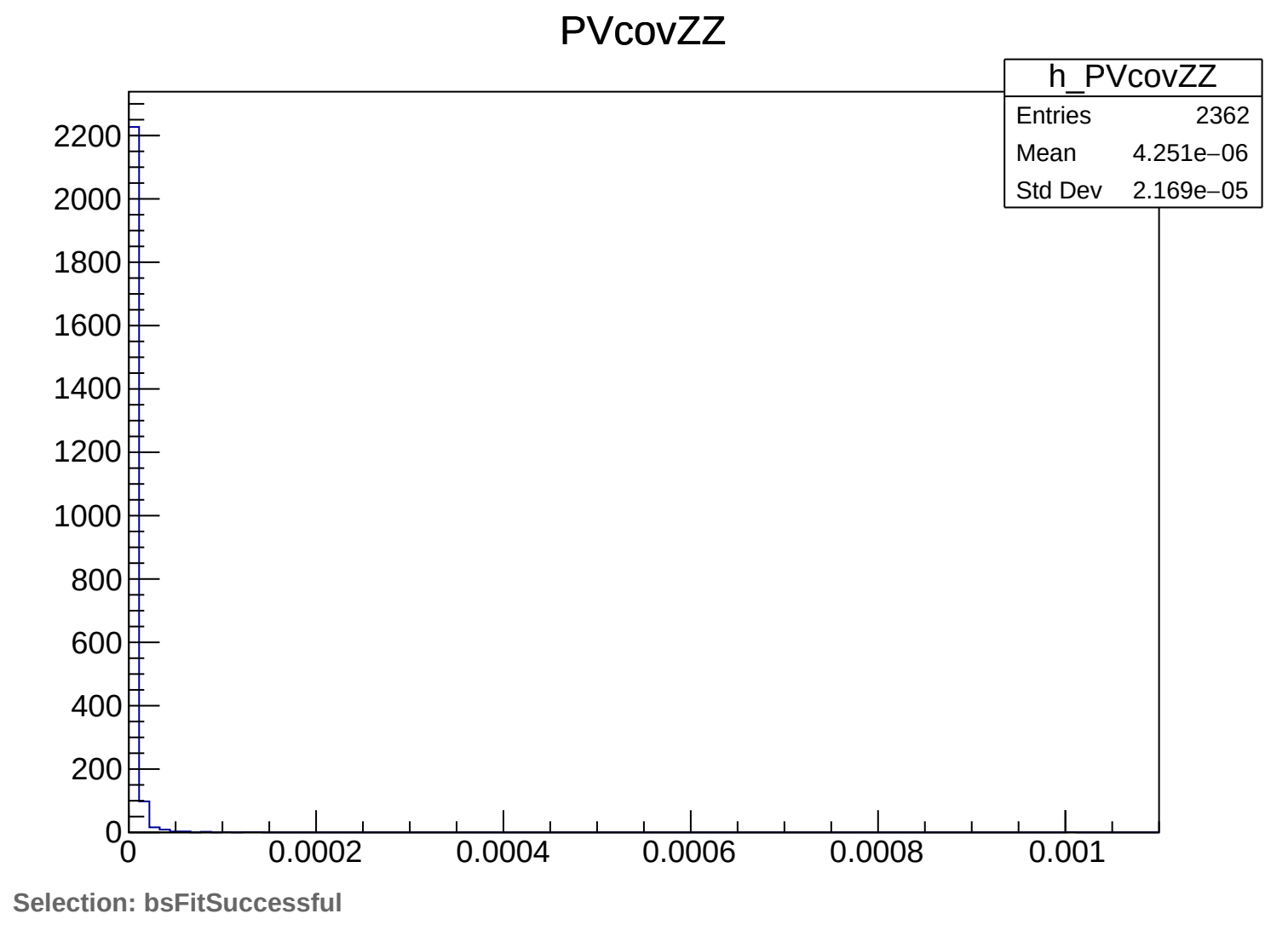

PVcovXY
h_PVcovXY
Entries 2362
Mean –1.301e–08
Std Dev 1.769e–06
1400
1200
1000
800
600
400
200
0
–0.04
–0.02
0
0.02
0.04
0.06
×10^–3
Selection: bsFitSuccessful

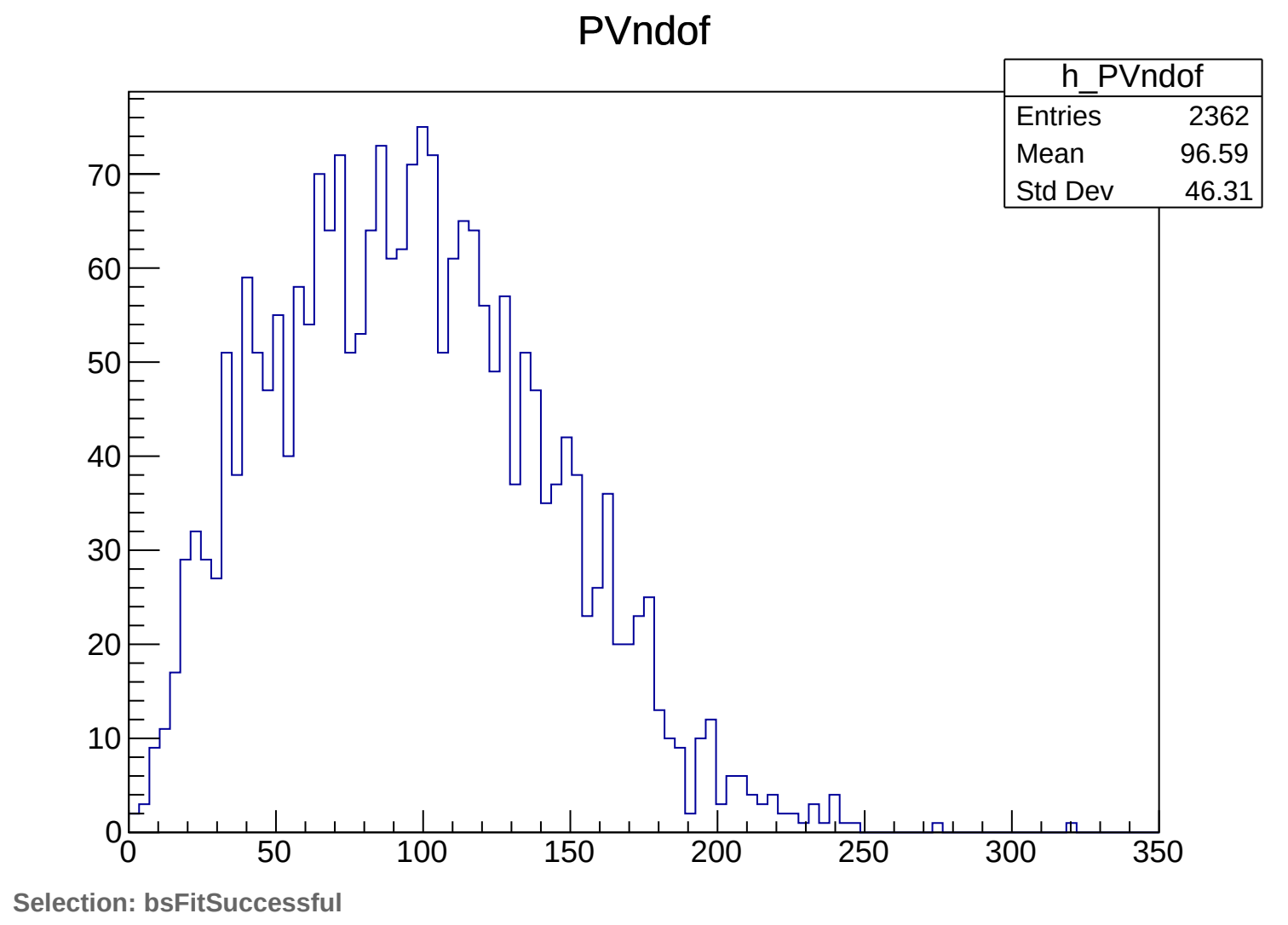

PVcovXZ
h_PVcovXZ
Entries 2362
Mean 5.619e−08
Std Dev 3.703e−06
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−0.04
−0.02
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
×10^−3
Selection: bsFitSuccessful

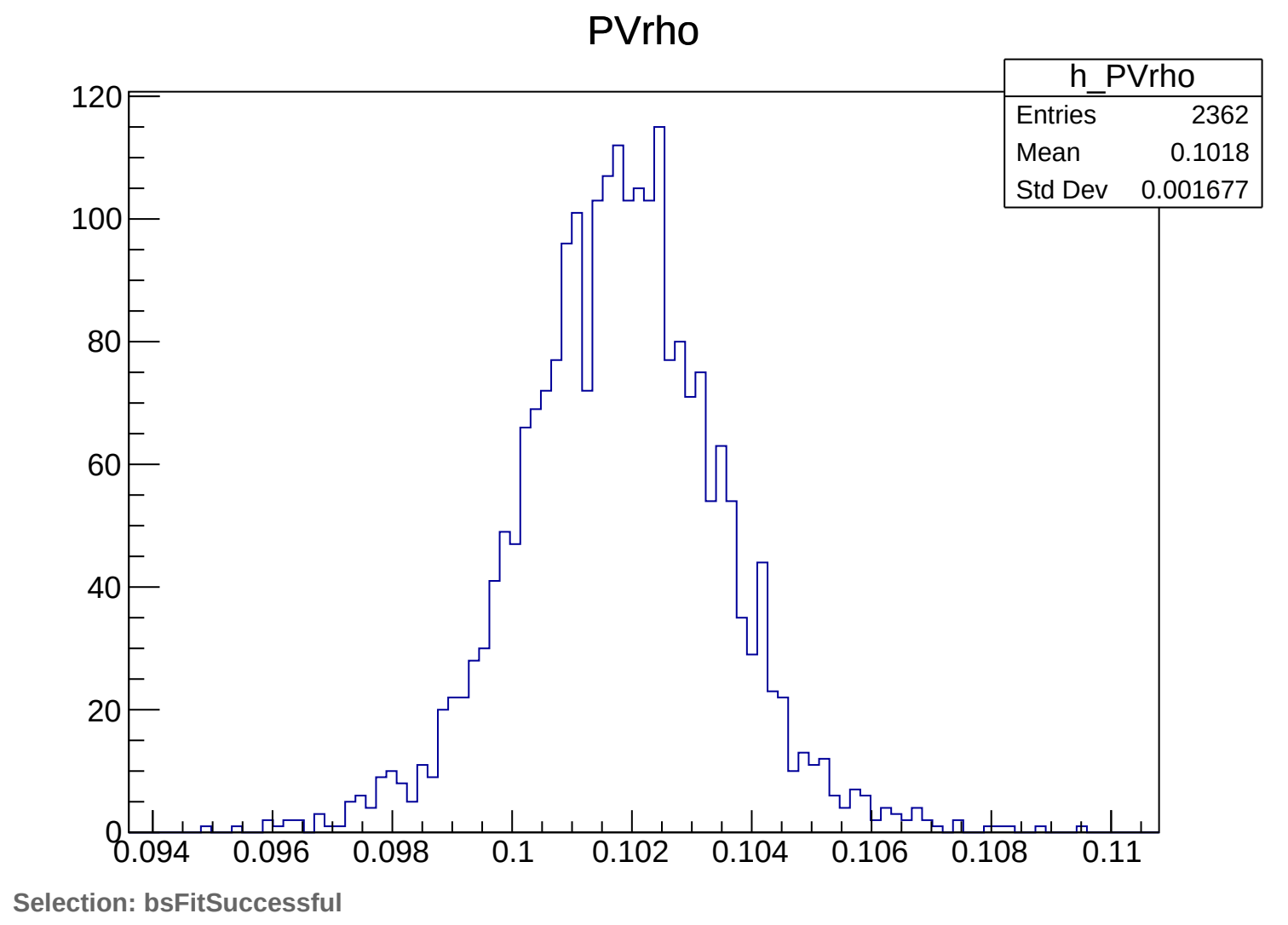

PVcovYY
h_PVcovYY
Entries 2362
Mean 2.258e−06
Std Dev 4.935e−06
1600
1400
1200
1000
800
600
400
200
0
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
×10⁻³
Selection: bsFitSuccessful

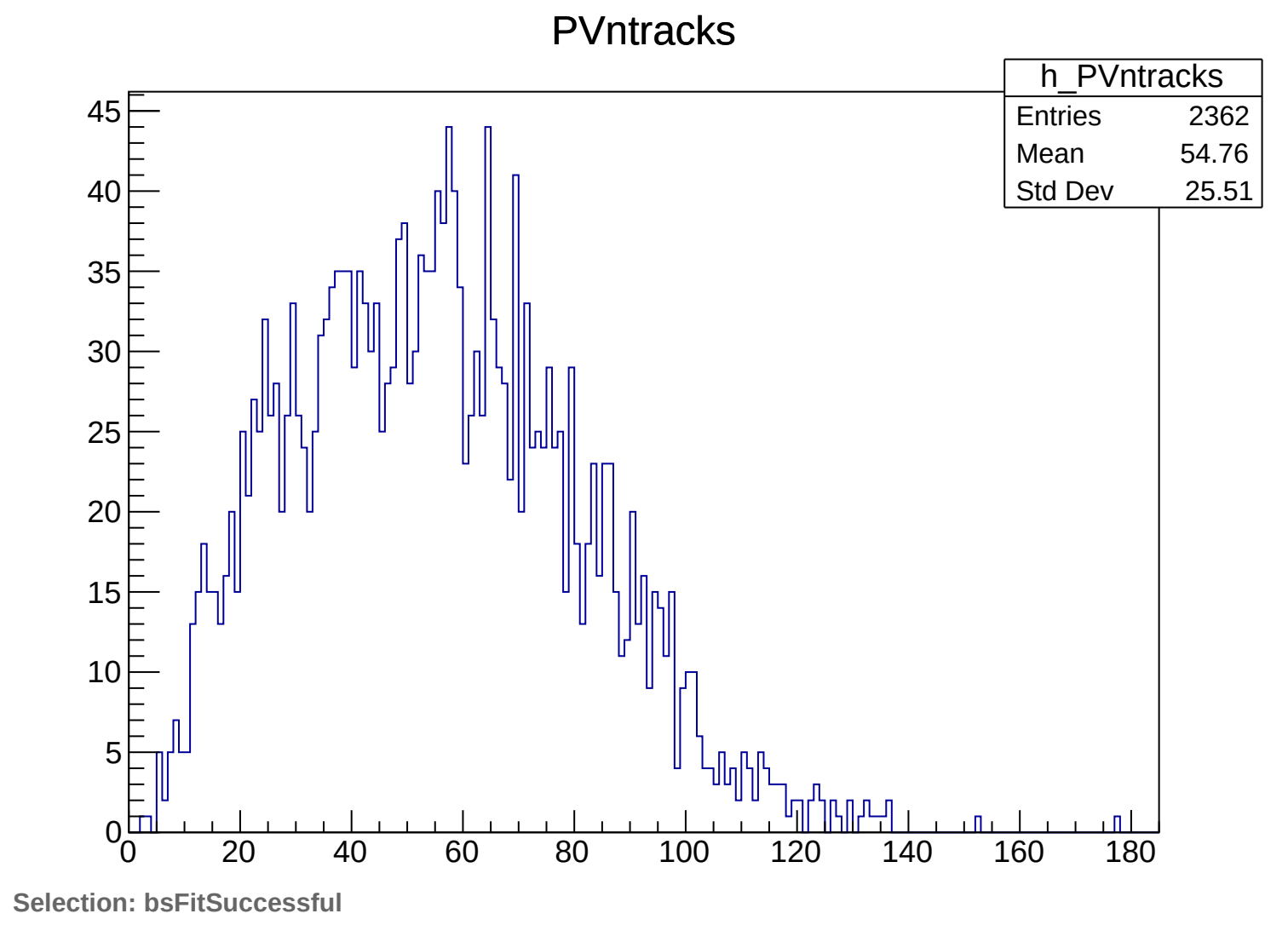


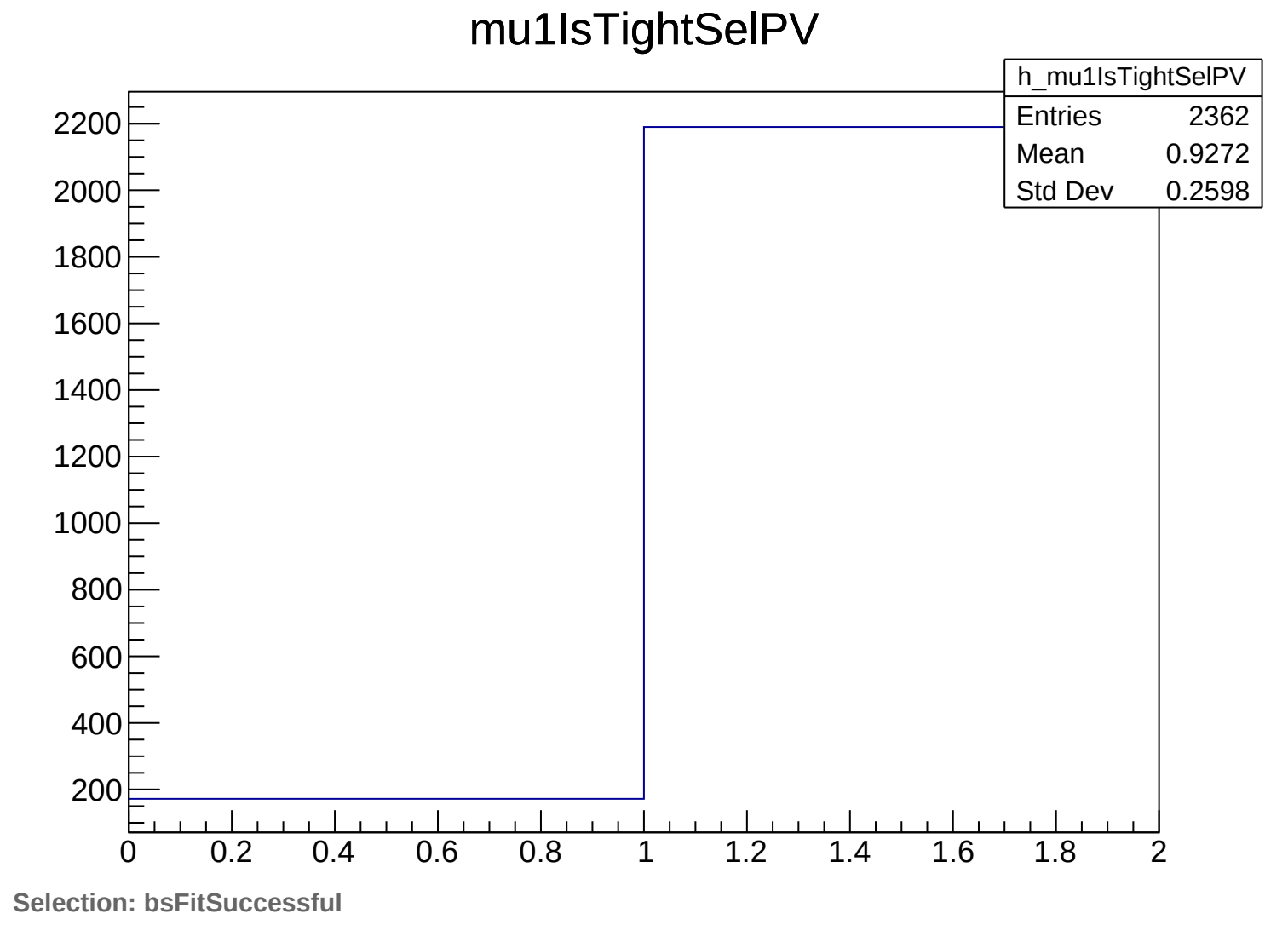
Selection: bsFitSuccessful

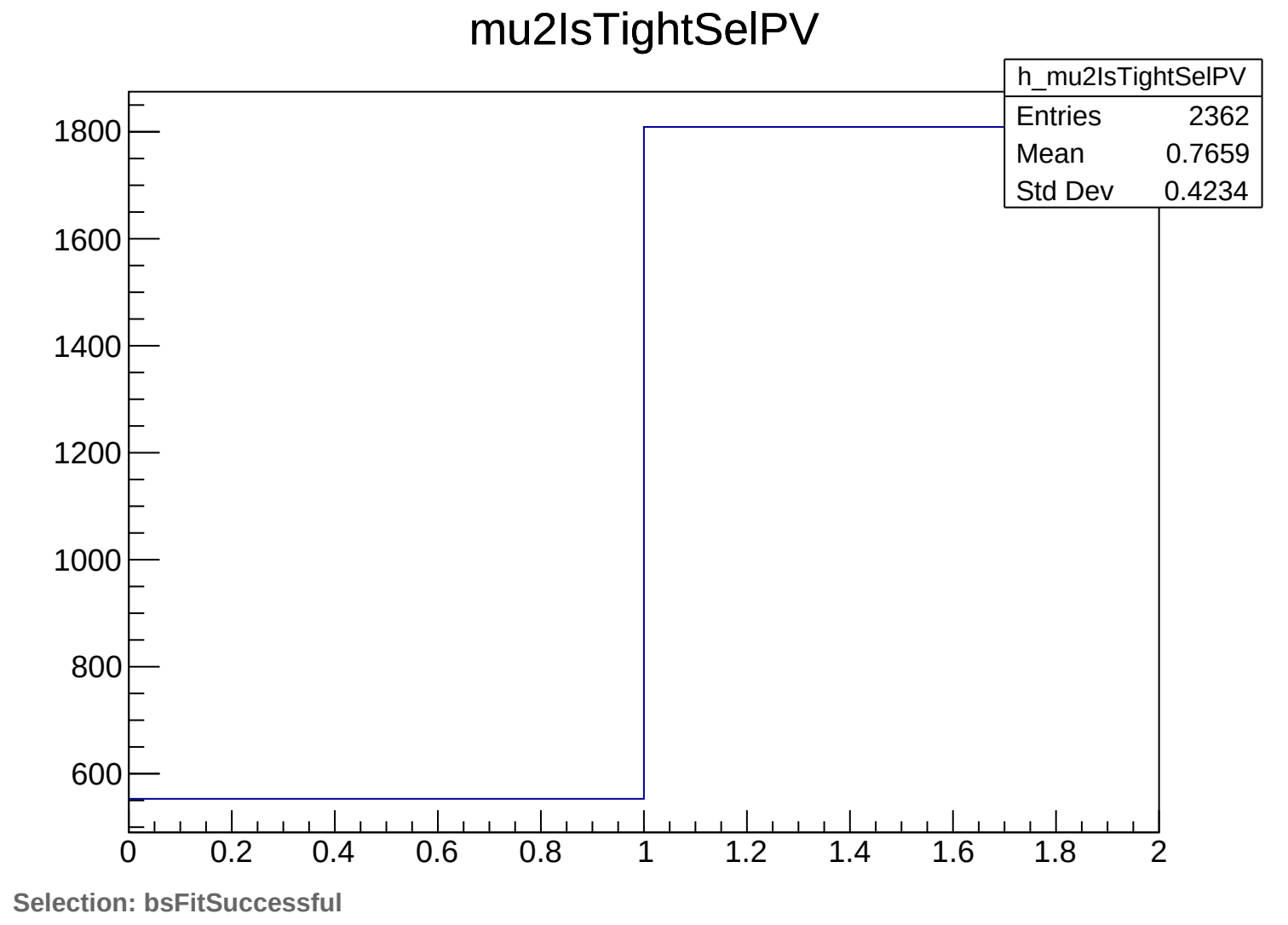

PVcovZZ
h_PVcovZZ
Entries 2362
Mean 4.251e−06
Std Dev 2.169e−05
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.0002
0.0004
0.0006
0.0008
0.001
Selection: bsFitSuccessful

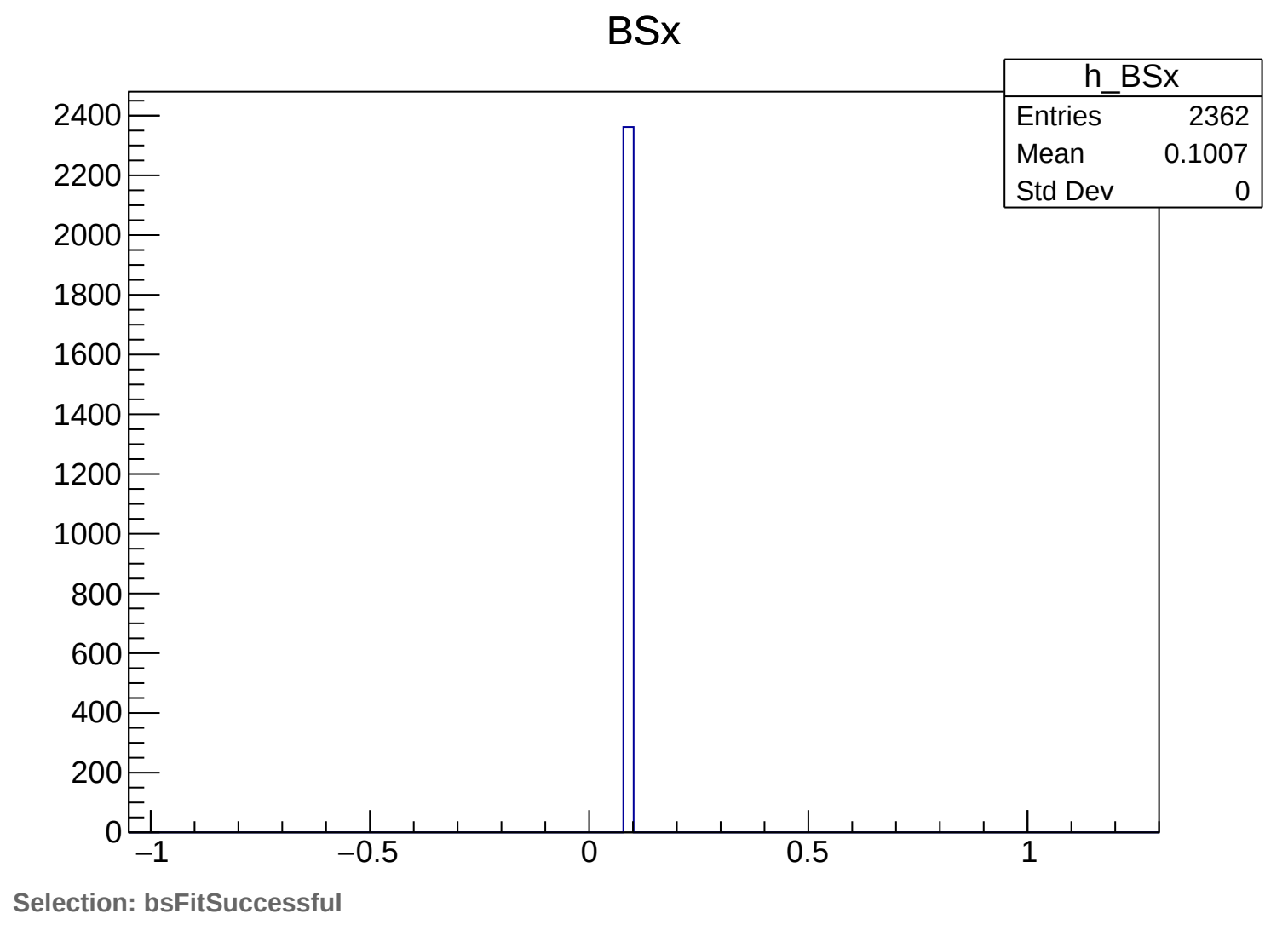

PVndof
h_PVndof
Entries 2362
Mean 96.59
Std Dev 46.31
0
10
20
30
40
50
60
70
0
50
100
150
200
250
300
350
Selection: bsFitSuccessful

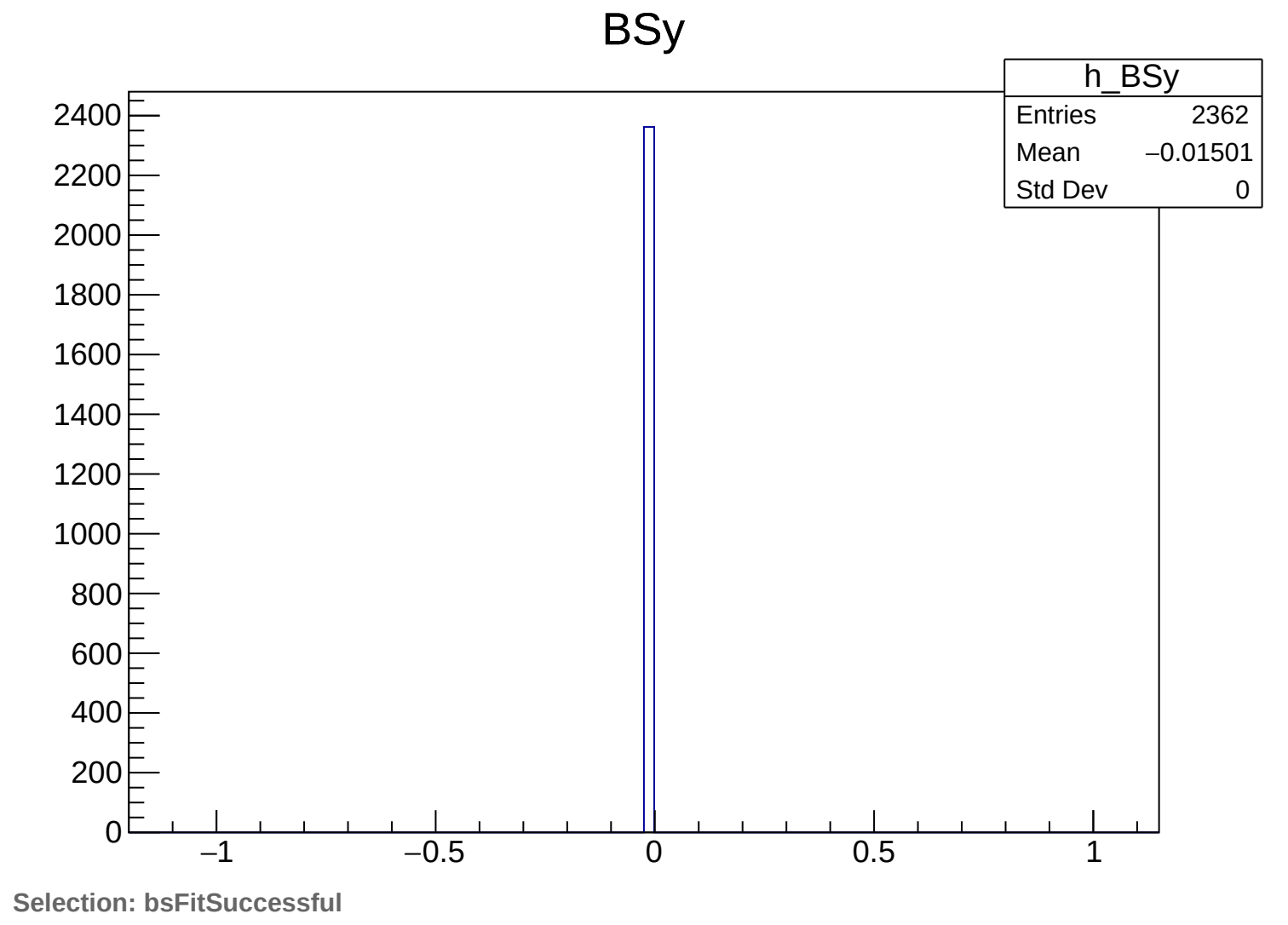

PVrho
h_PVrho
Entries 2362
Mean 0.1018
Std Dev 0.001677
120
100
80
60
40
20
0
0.094
0.096
0.098
0.1
0.102
0.104
0.106
0.108
0.11
Selection: bsFitSuccessful

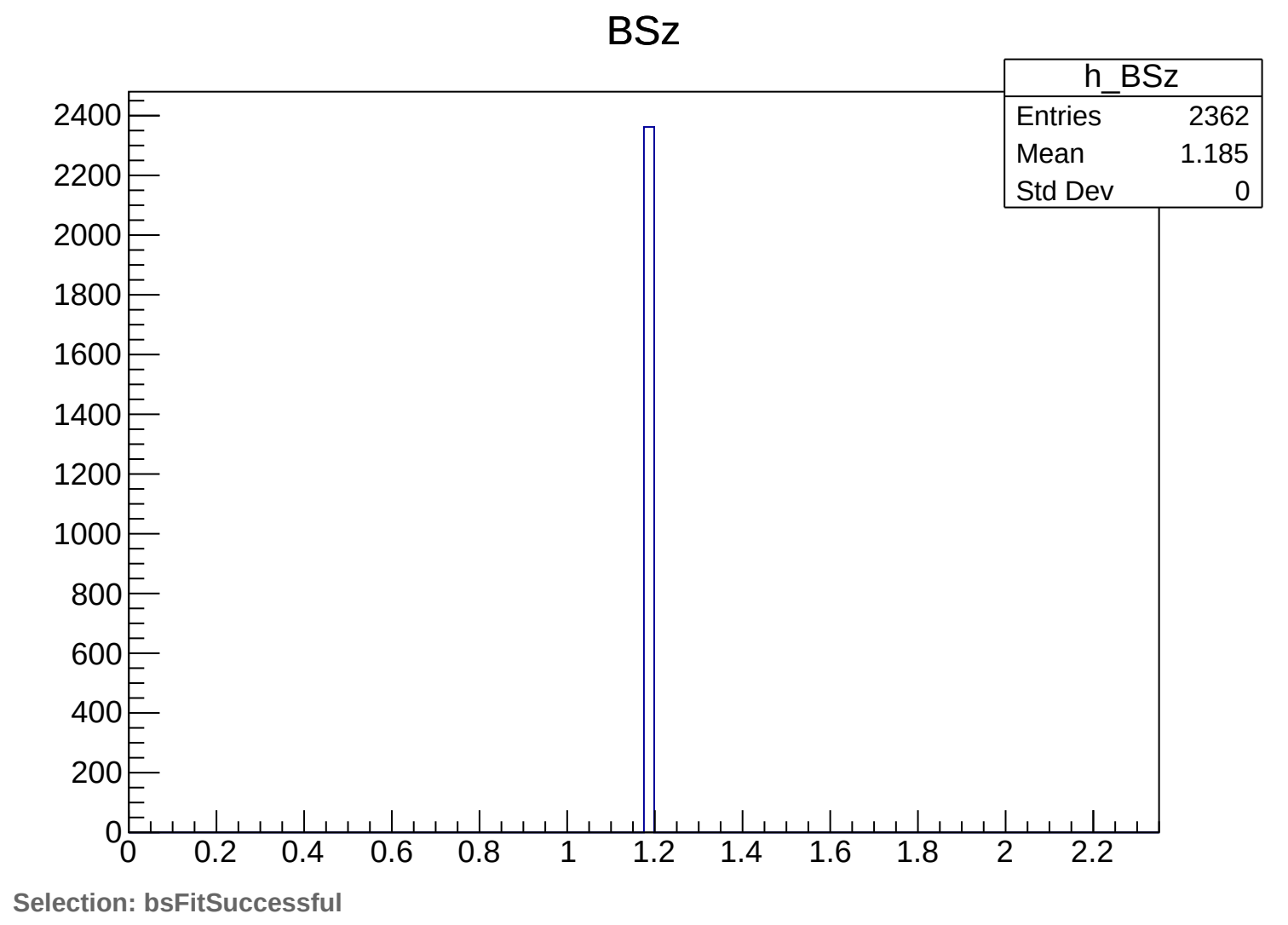

PVntracks
h_PVntracks
Entries 2362
Mean 54.76
Std Dev 25.51
0
5
10
15
20
25
30
35
40
45
0
20
40
60
80
100
120
140
160
180
Selection: bsFitSuccessful

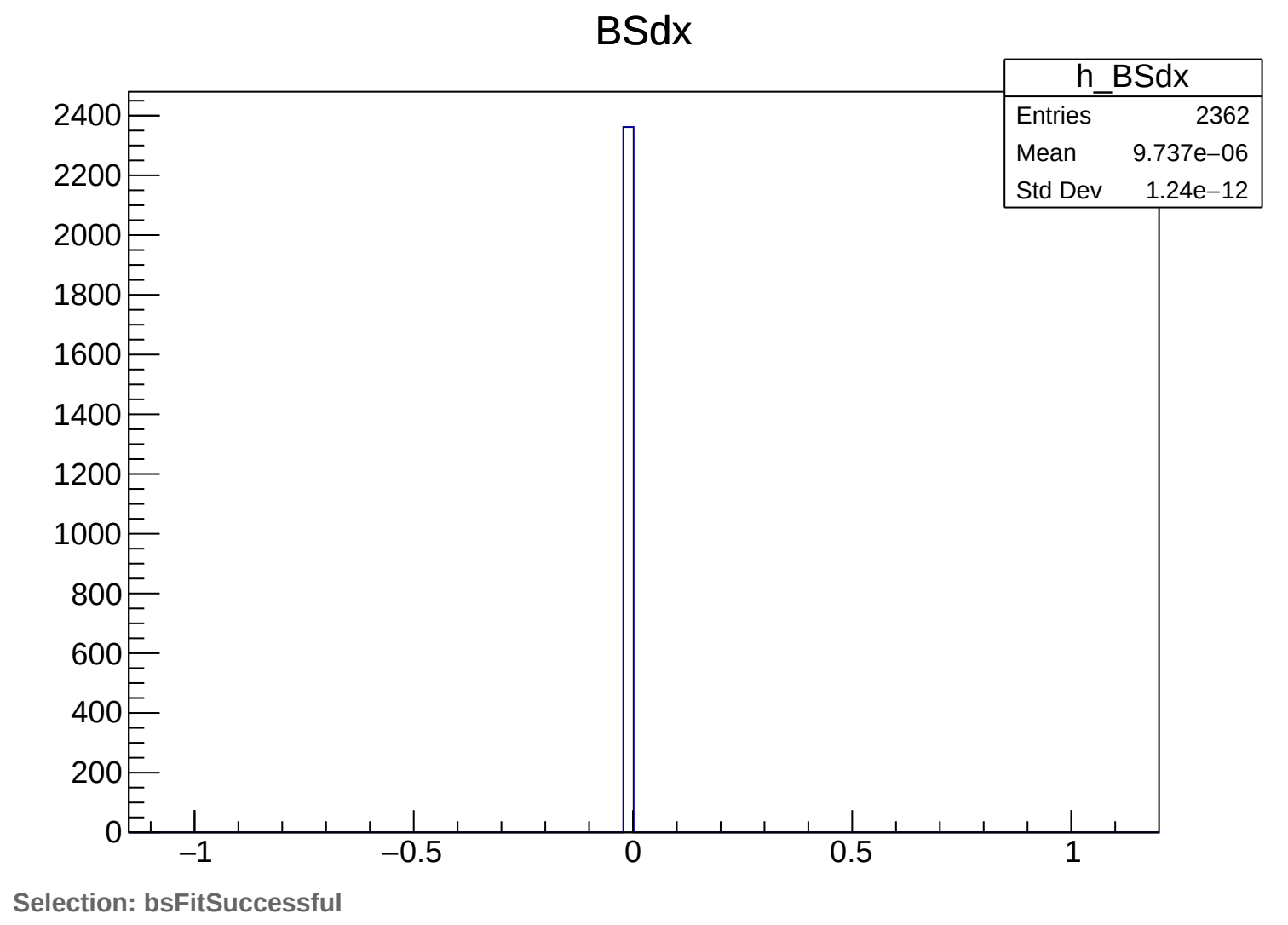

mu1IsTightSelPV
h_mu1IsTightSelPV
Entries 2362
Mean 0.9272
Std Dev 0.2598
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

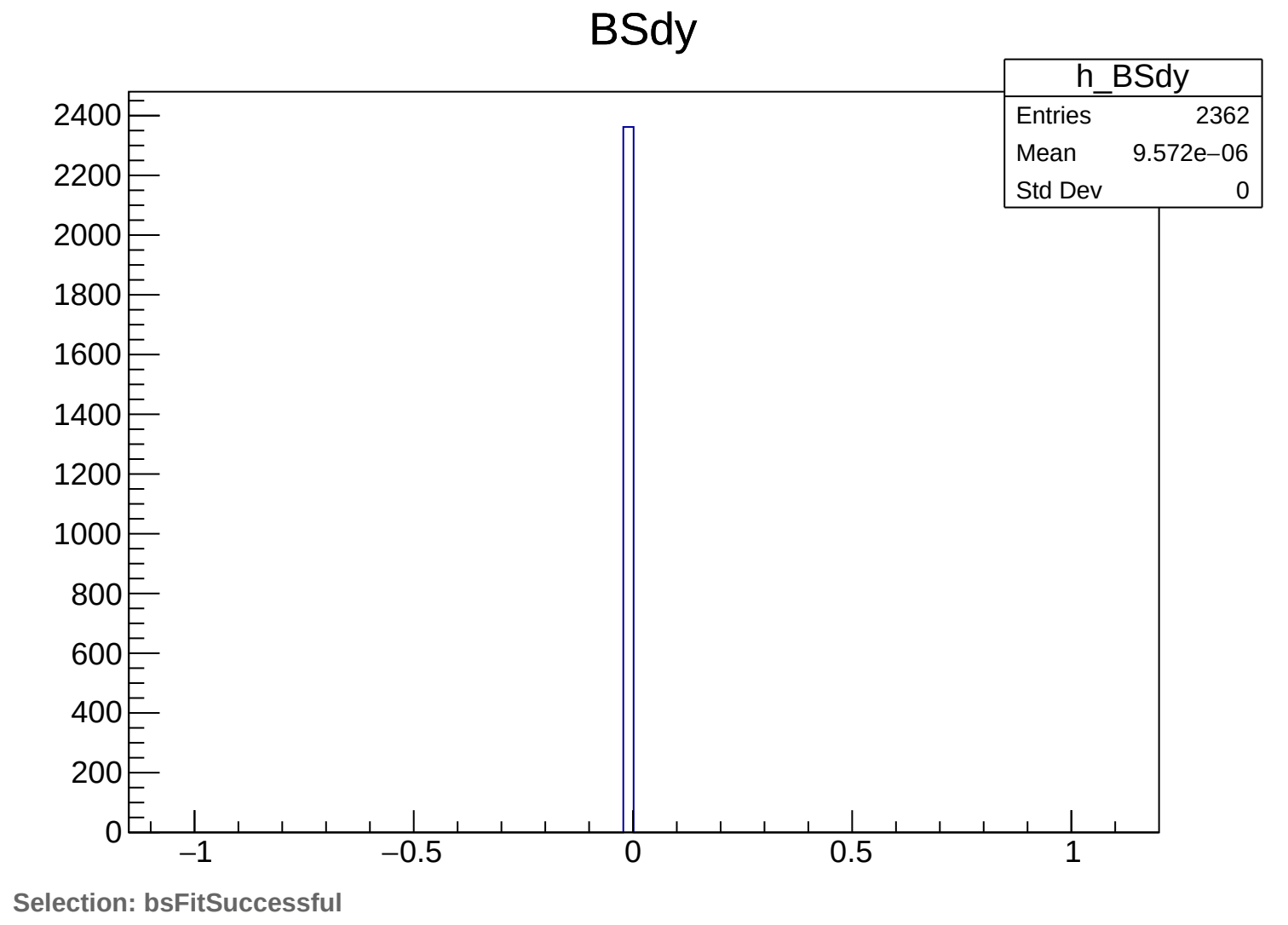

mu2IsTightSelPV
h_mu2IsTightSelPV
Entries 2362
Mean 0.7659
Std Dev 0.4234
1800
1600
1400
1200
1000
800
600
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

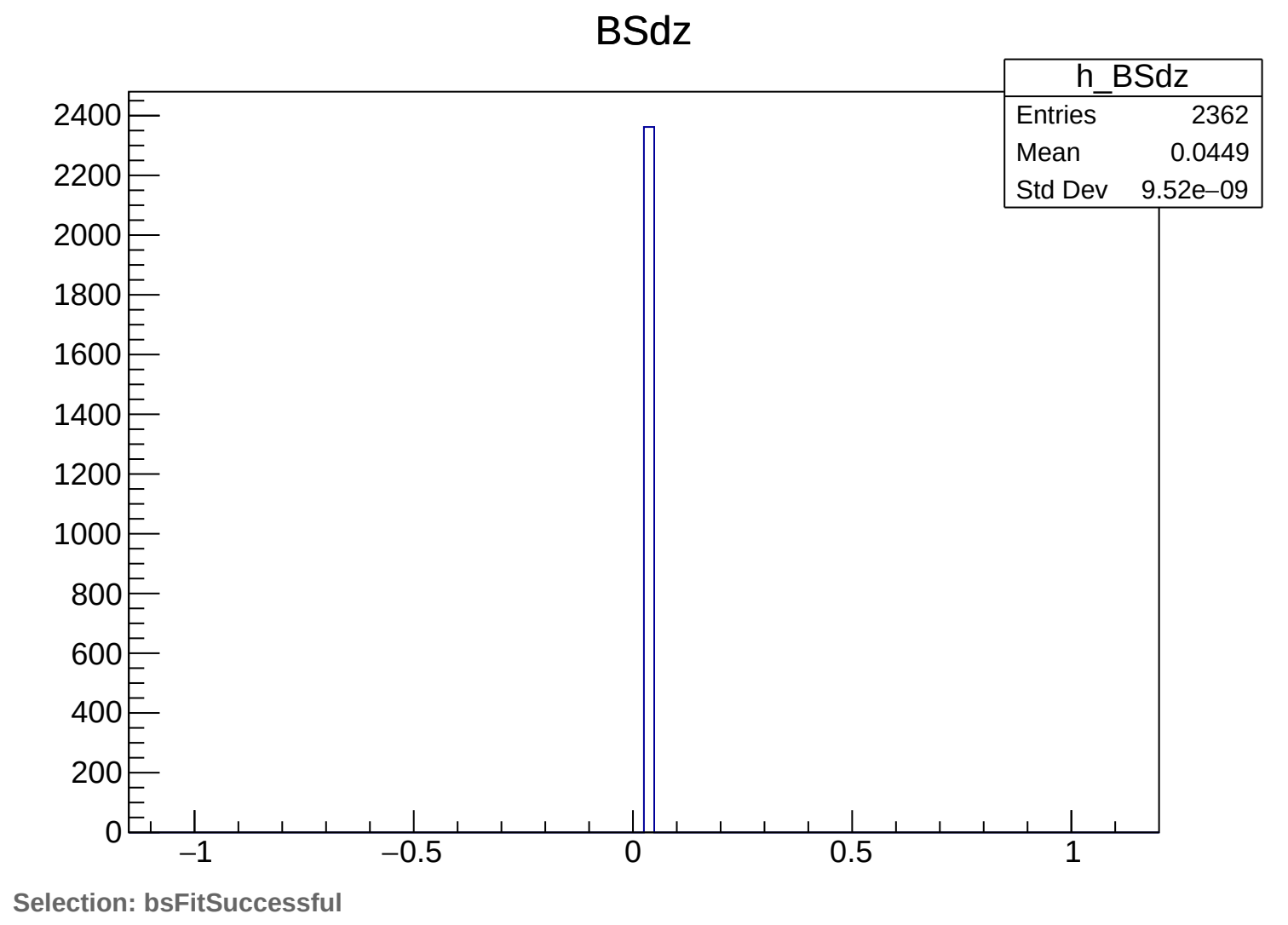

BSx
h_BSx
Entries 2362
Mean 0.1007
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

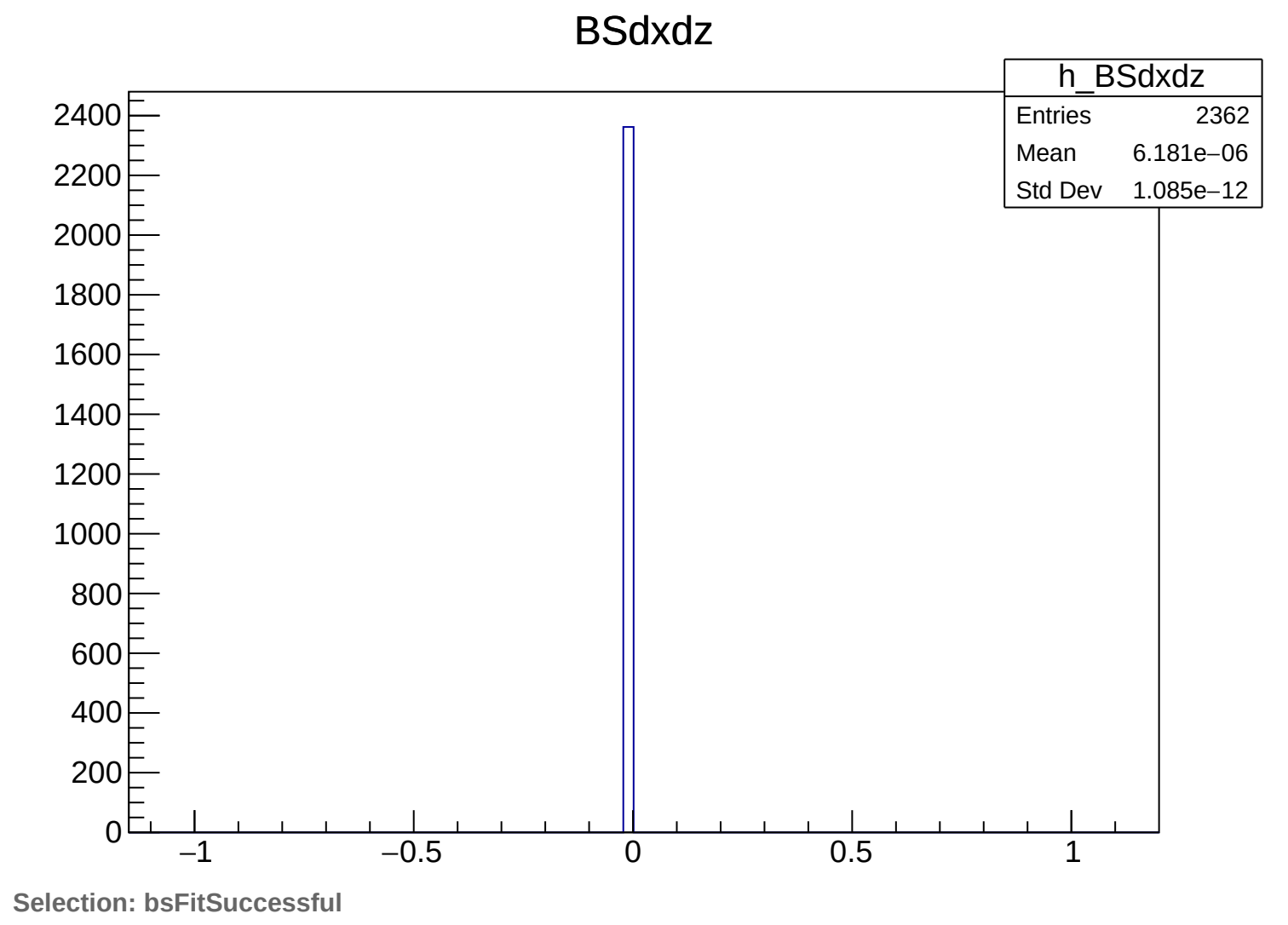

BSy
h_BSy
Entries 2362
Mean −0.01501
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

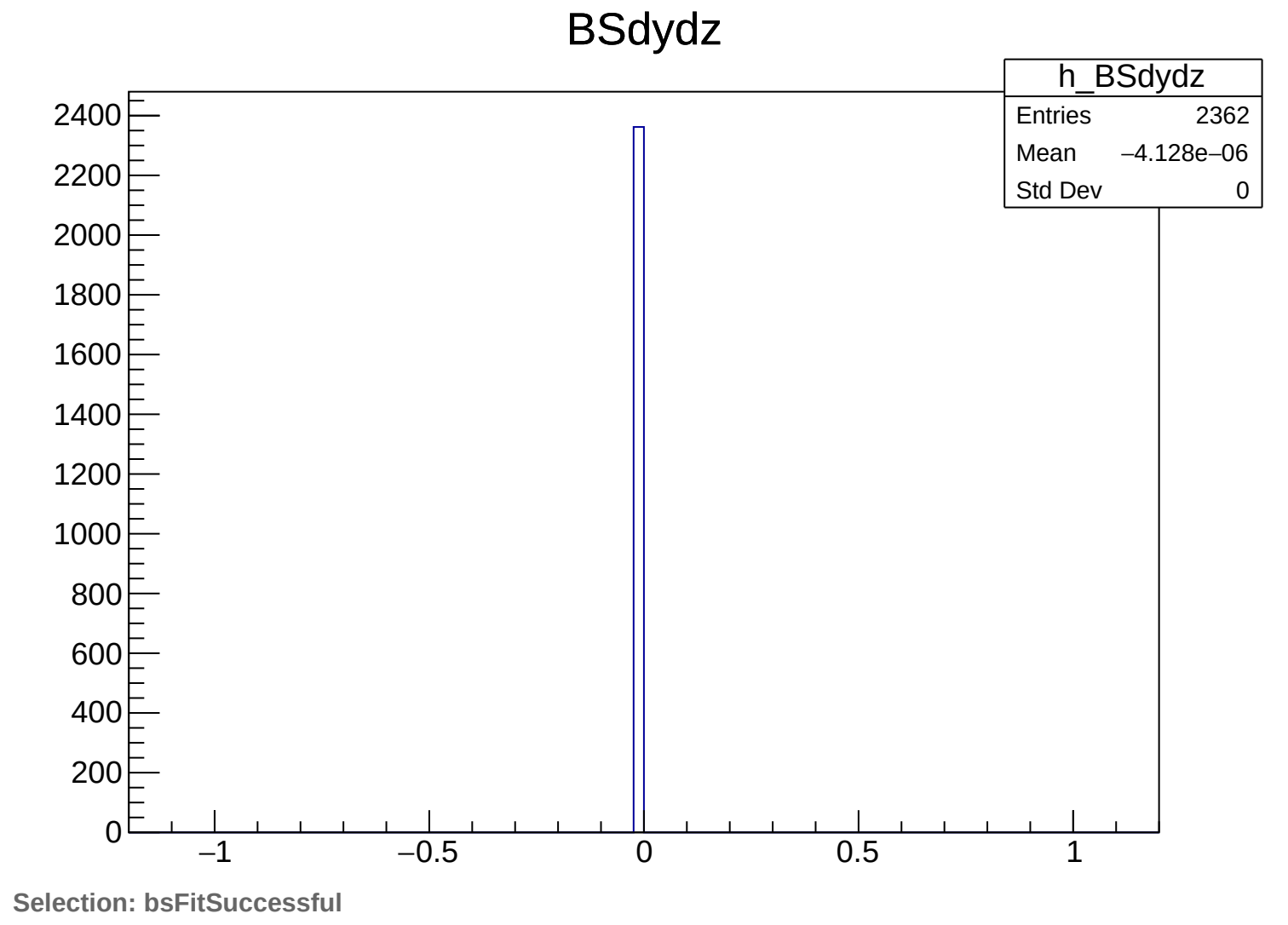

BSz
h_BSz
Entries 2362
Mean 1.185
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
2.2
Selection: bsFitSuccessful

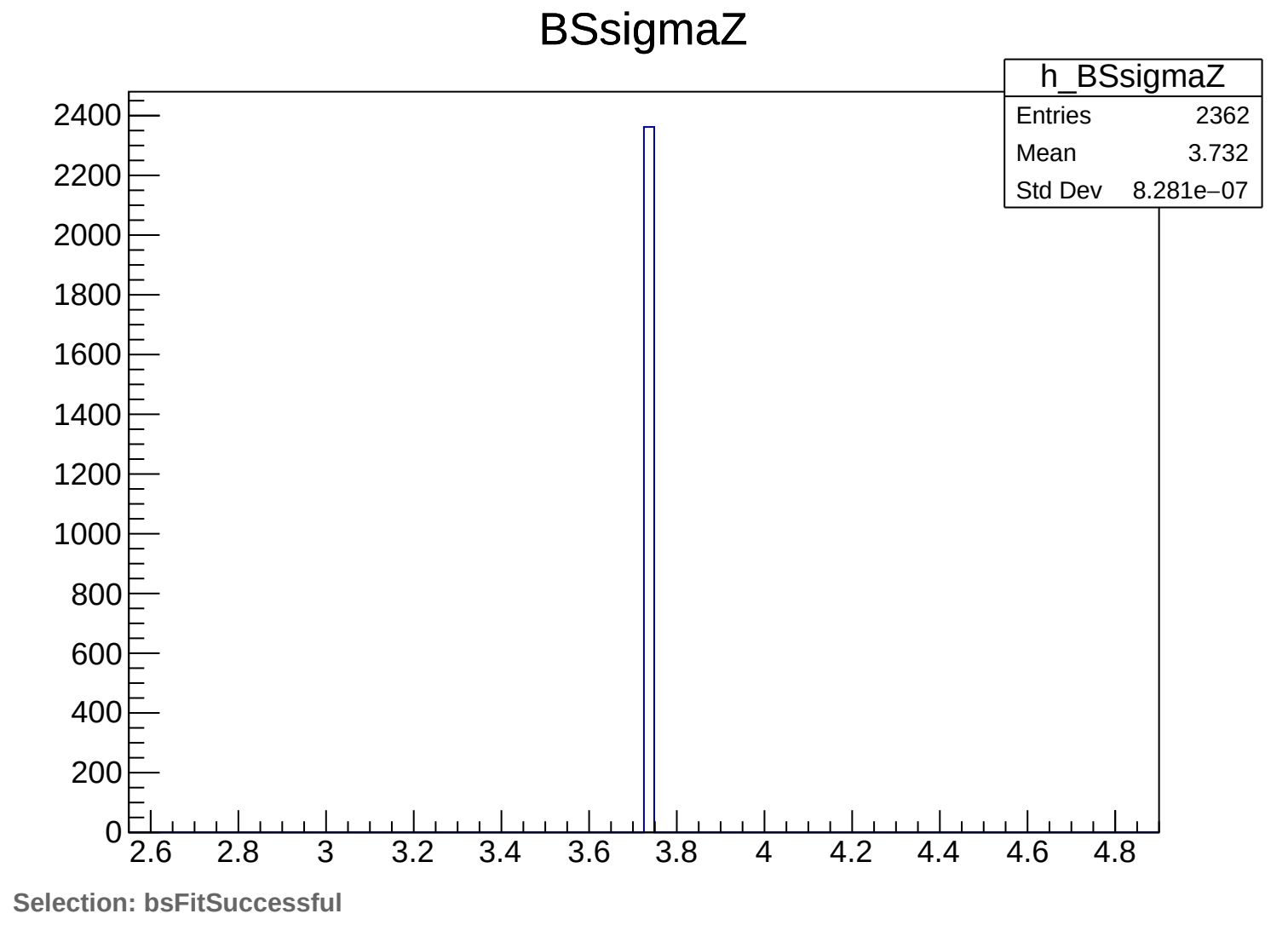

BSdx
h_BSdx
Entries 2362
Mean 9.737e−06
Std Dev 1.24e−12
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

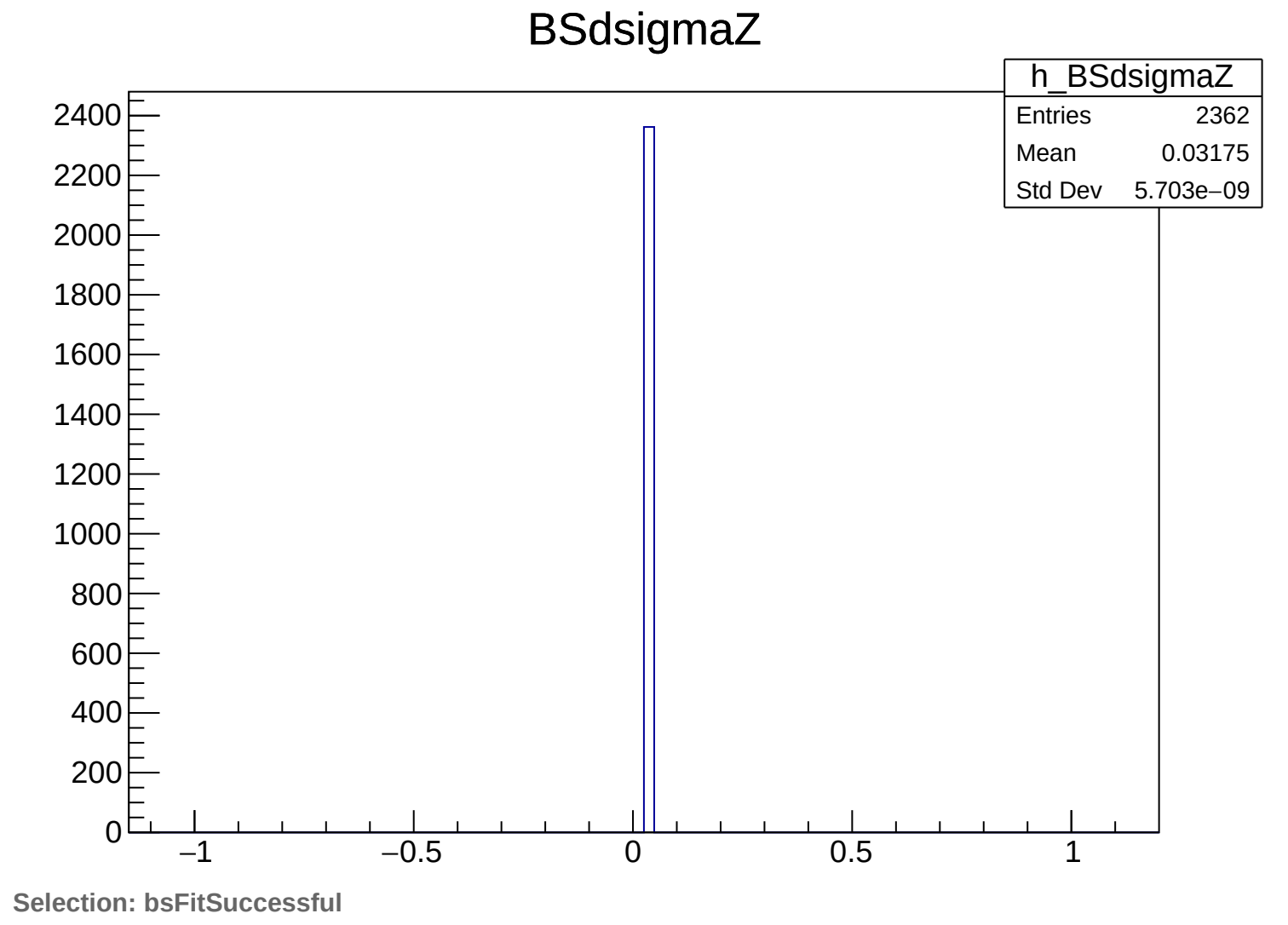

BSdy
h_BSdy
Entries 2362
Mean 9.572e−06
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

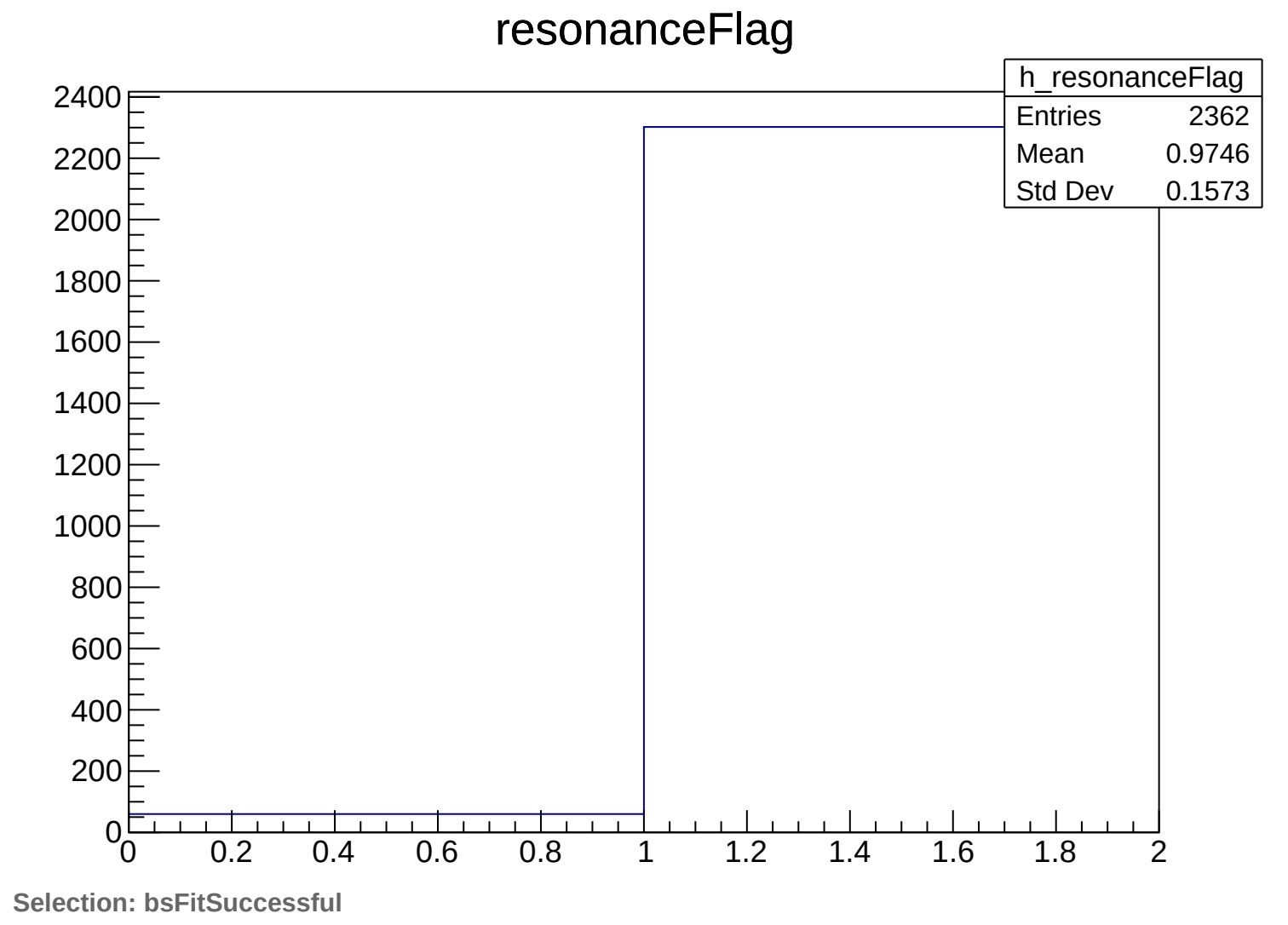

BSdz
h_BSdz
Entries 2362
Mean 0.0449
Std Dev 9.52e−09
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

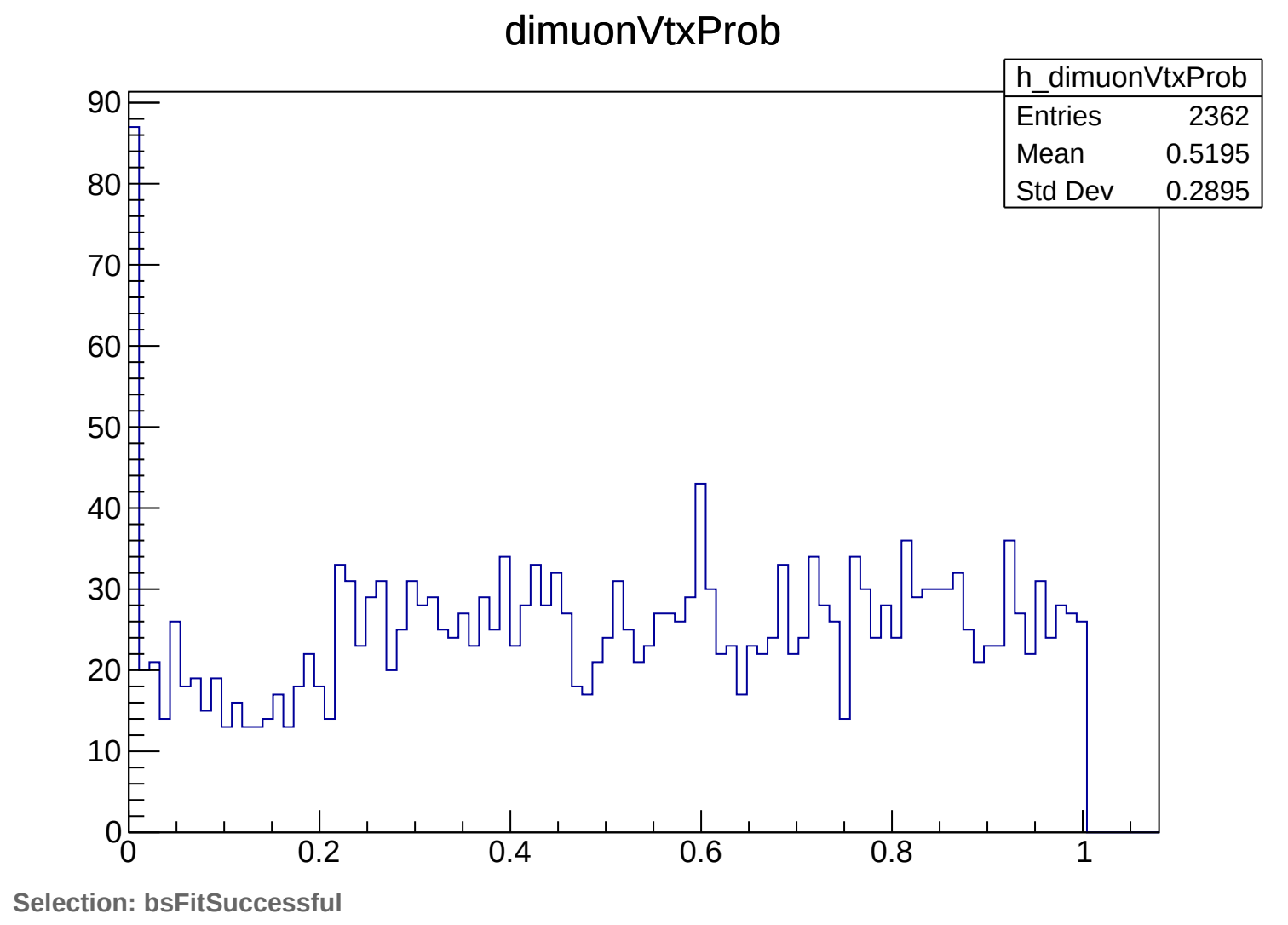

BSdxdz
h_BSdxdz
Entries 2362
Mean 6.181e−06
Std Dev 1.085e−12
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

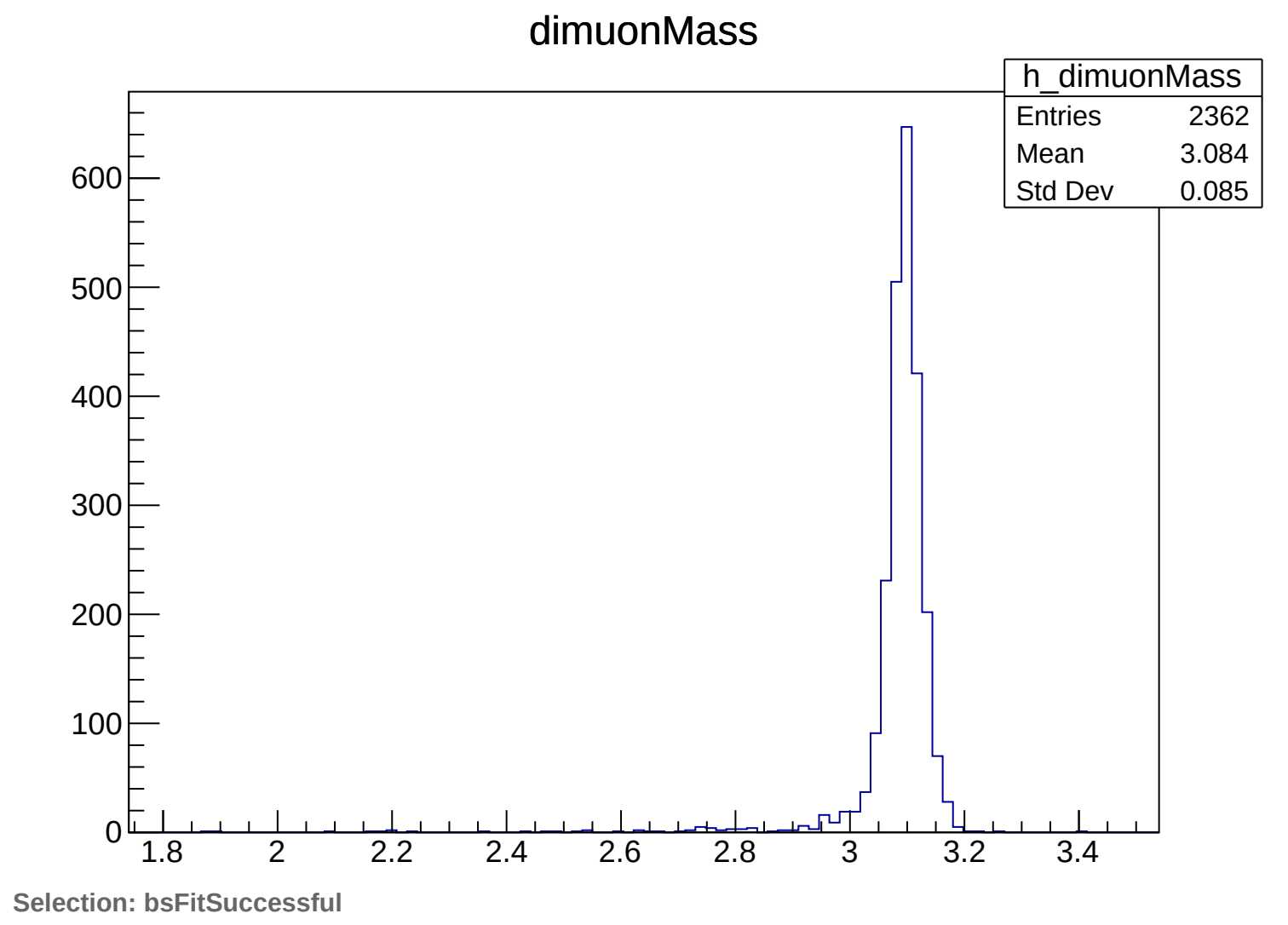

BSdydz
h_BSdydz
Entries 2362
Mean –4.128e–06
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
–1
–0.5
0
0.5
1
Selection: bsFitSuccessful

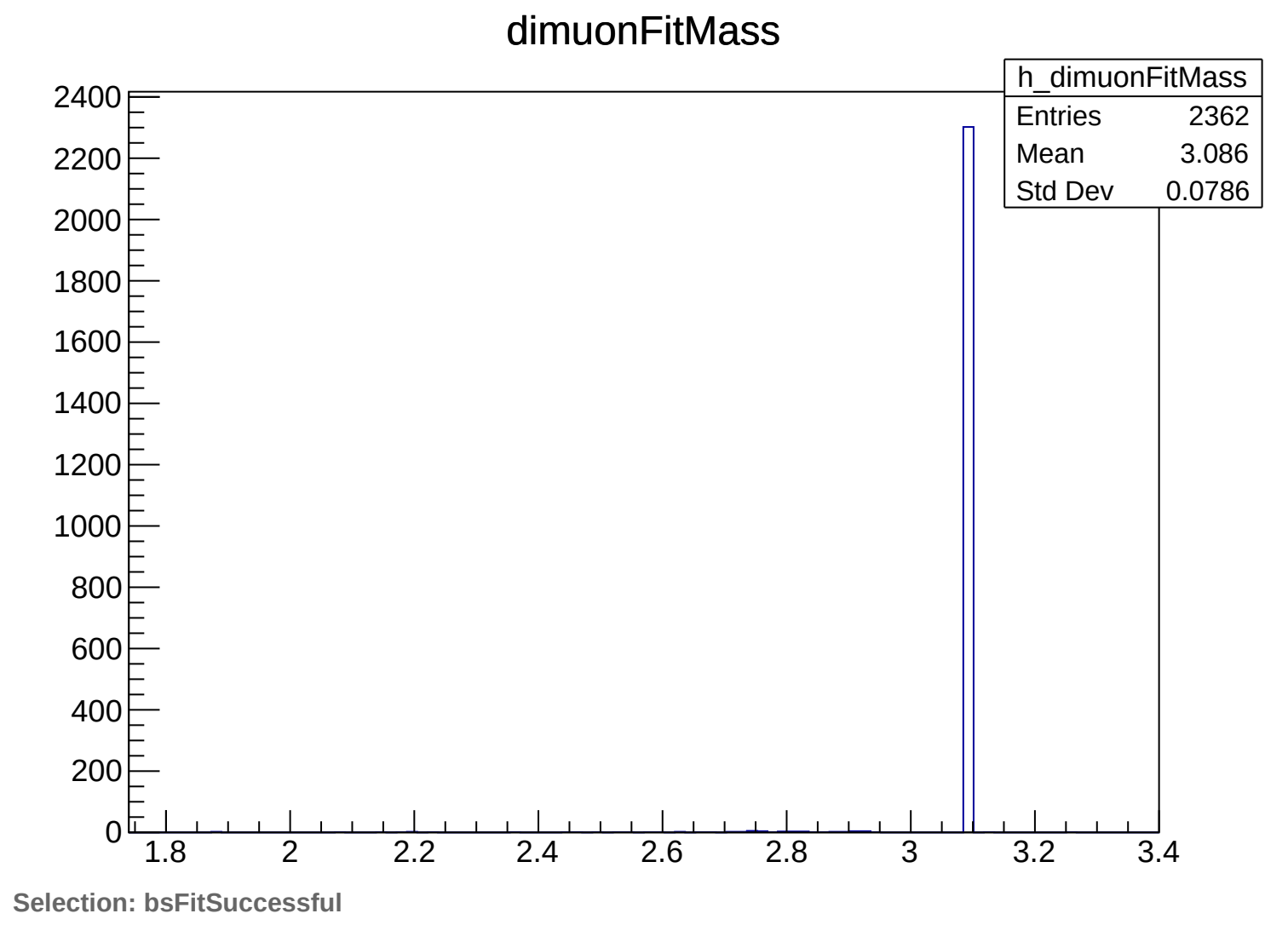

BSsigmaZ
h_BSsigmaZ
Entries 2362
Mean 3.732
Std Dev 8.281e−07
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
2.6
2.8
3
3.2
3.4
3.6
3.8
4
4.2
4.4
4.6
4.8
Selection: bsFitSuccessful

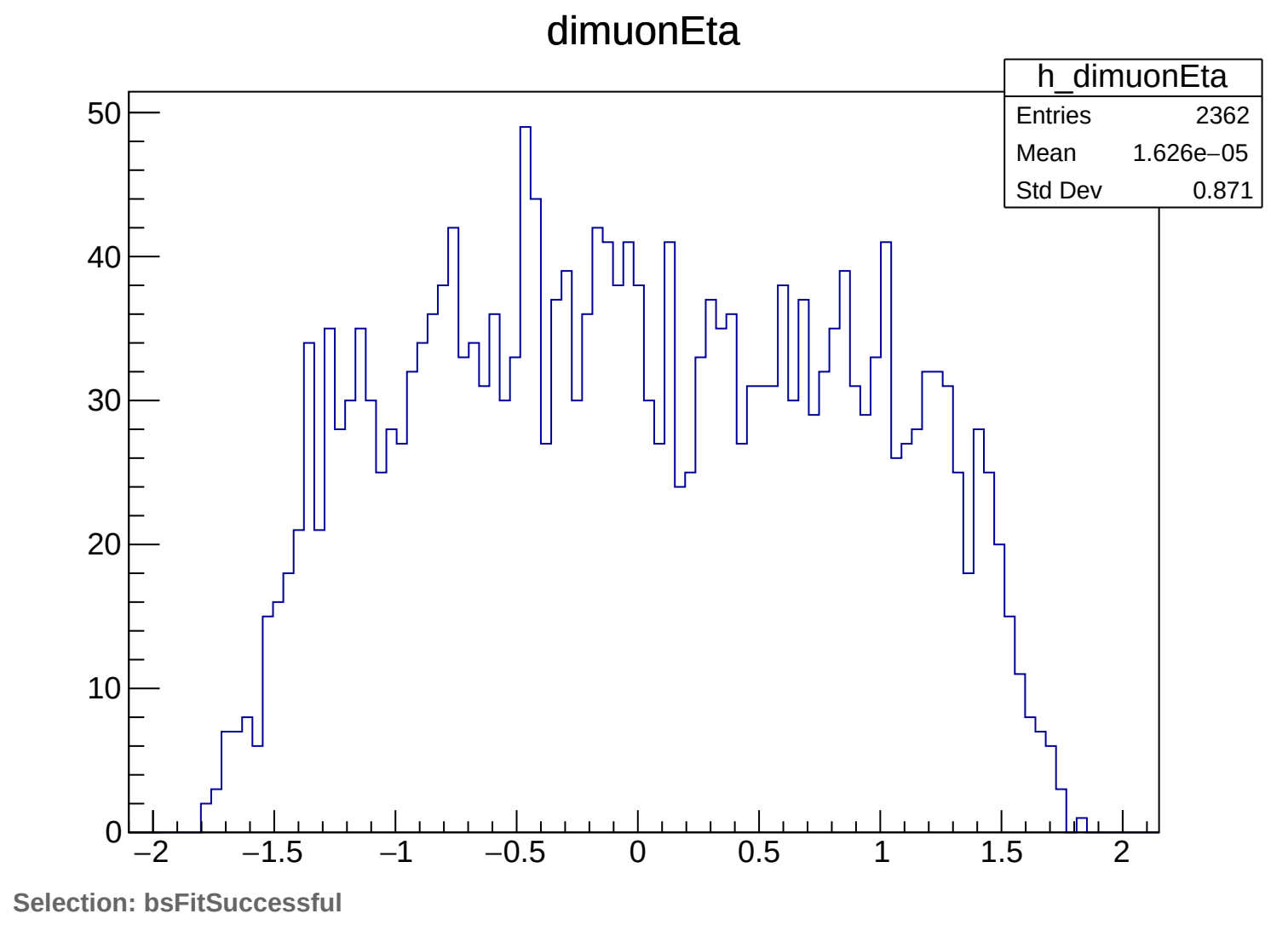

BSdsigmaZ
h_BSdsigmaZ
Entries 2362
Mean 0.03175
Std Dev 5.703e−09
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful

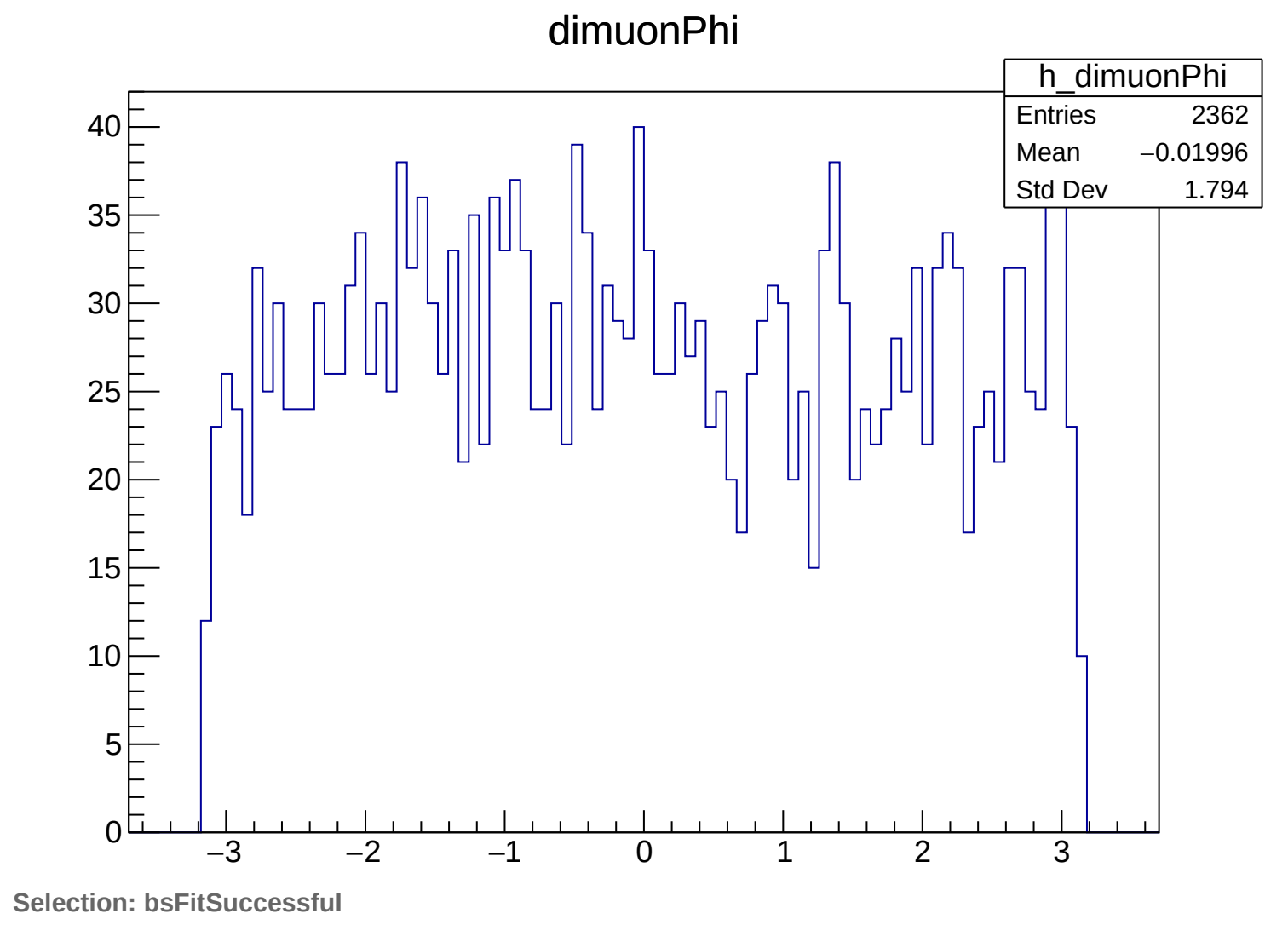

resonanceFlag
h_resonanceFlag
Entries 2362
Mean 0.9746
Std Dev 0.1573
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
Selection: bsFitSuccessful

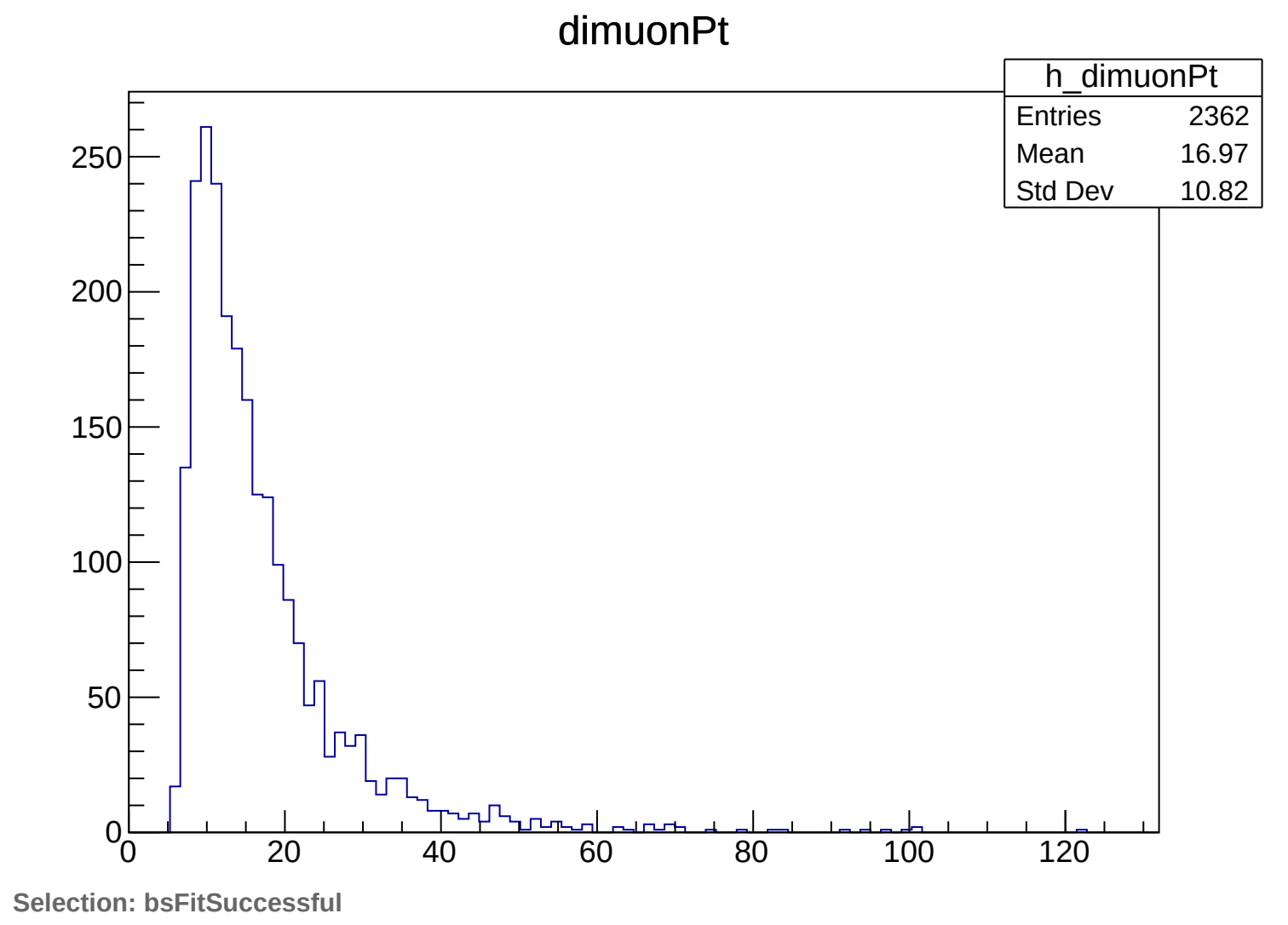

dimuonVtxProb
h_dimuonVtxProb
Entries 2362
Mean 0.5195
Std Dev 0.2895
90
80
70
60
50
40
30
20
10
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

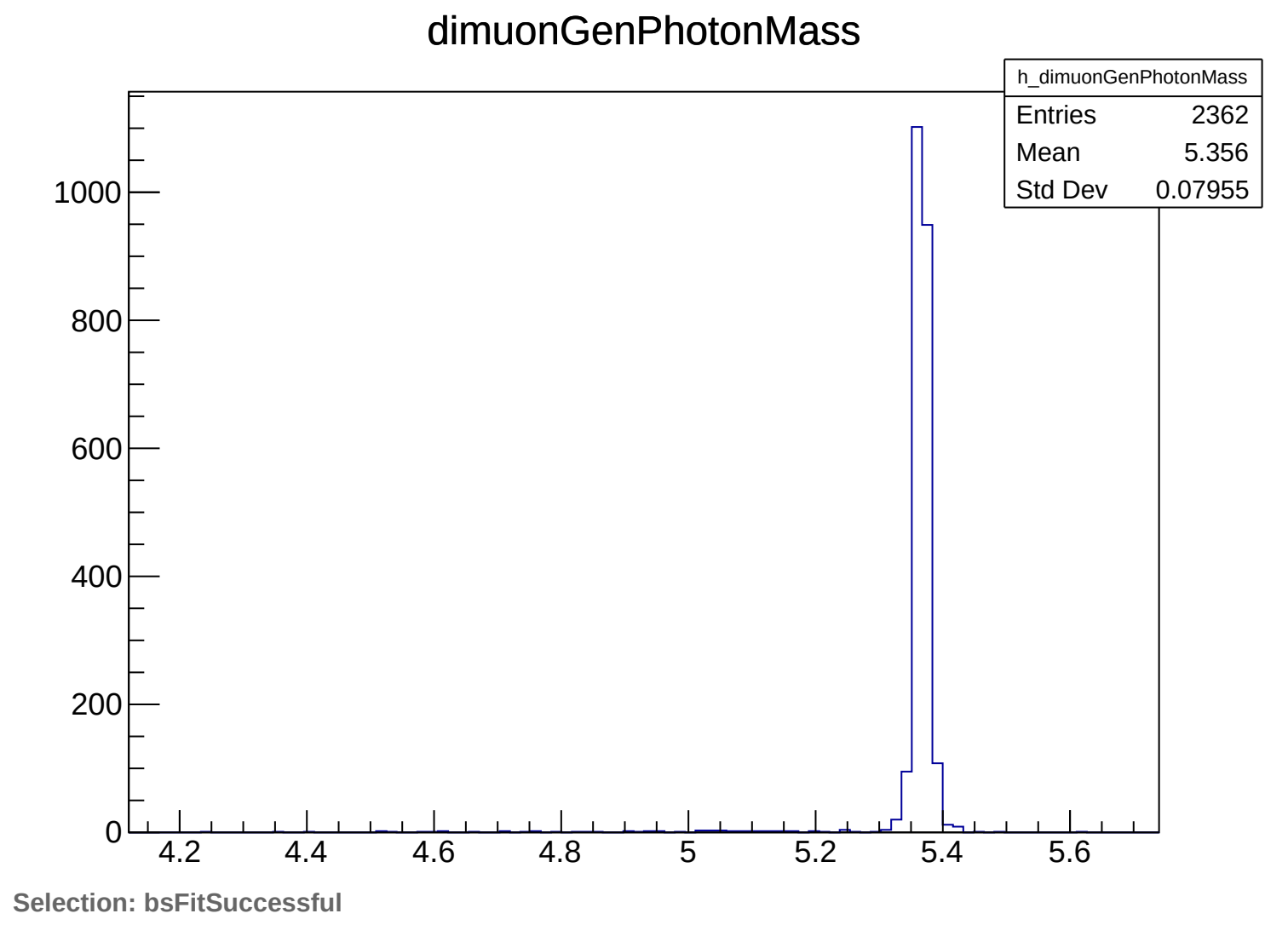

dimuonMass
h_dimuonMass
Entries 2362
Mean 3.084
Std Dev 0.085
600
500
400
300
200
100
0
1.8
2
2.2
2.4
2.6
2.8
3
3.2
3.4
Selection: bsFitSuccessful

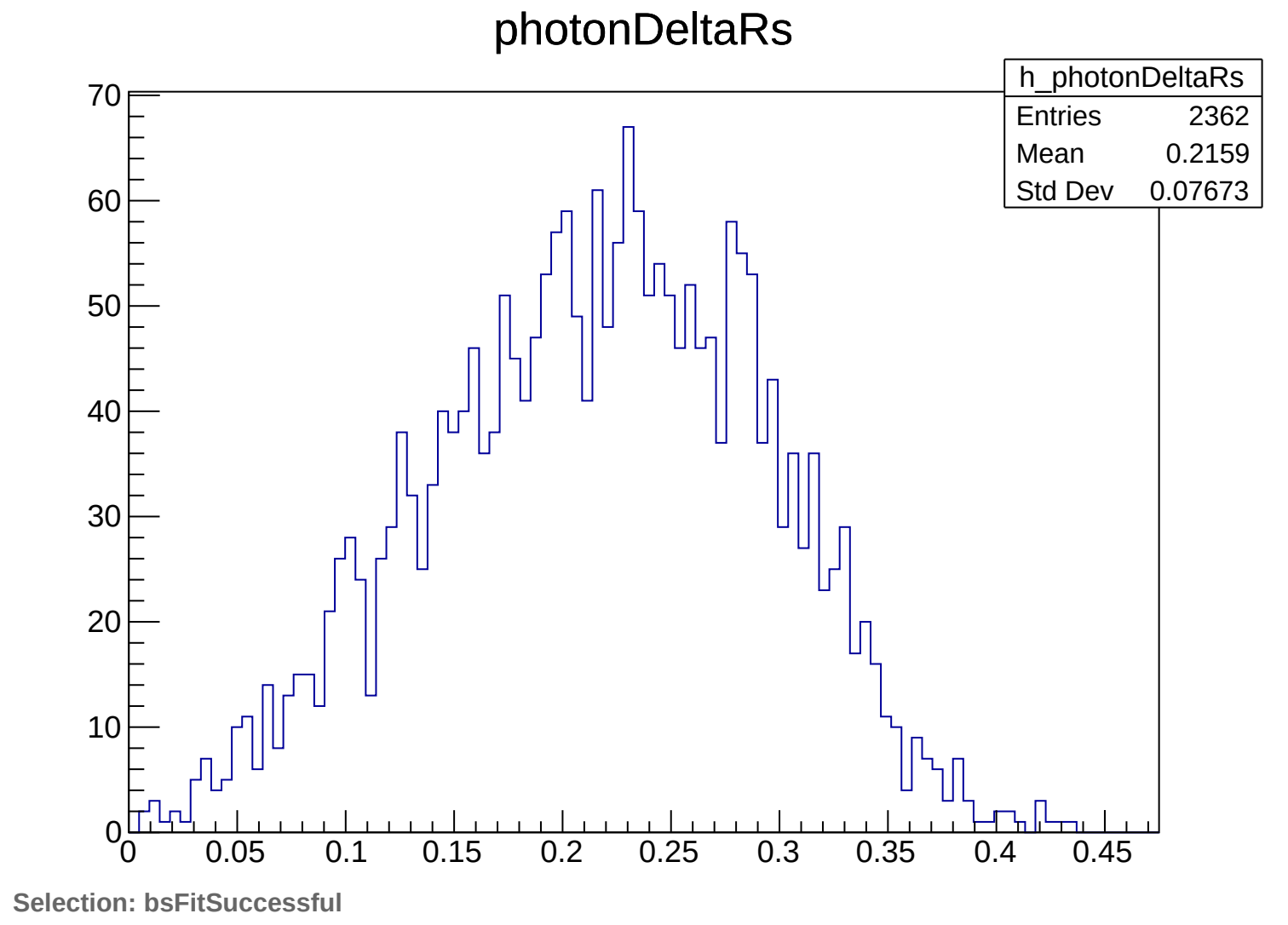

dimuonFitMass
h_dimuonFitMass
Entries 2362
Mean 3.086
Std Dev 0.0786
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
1.8
2
2.2
2.4
2.6
2.8
3
3.2
3.4
Selection: bsFitSuccessful

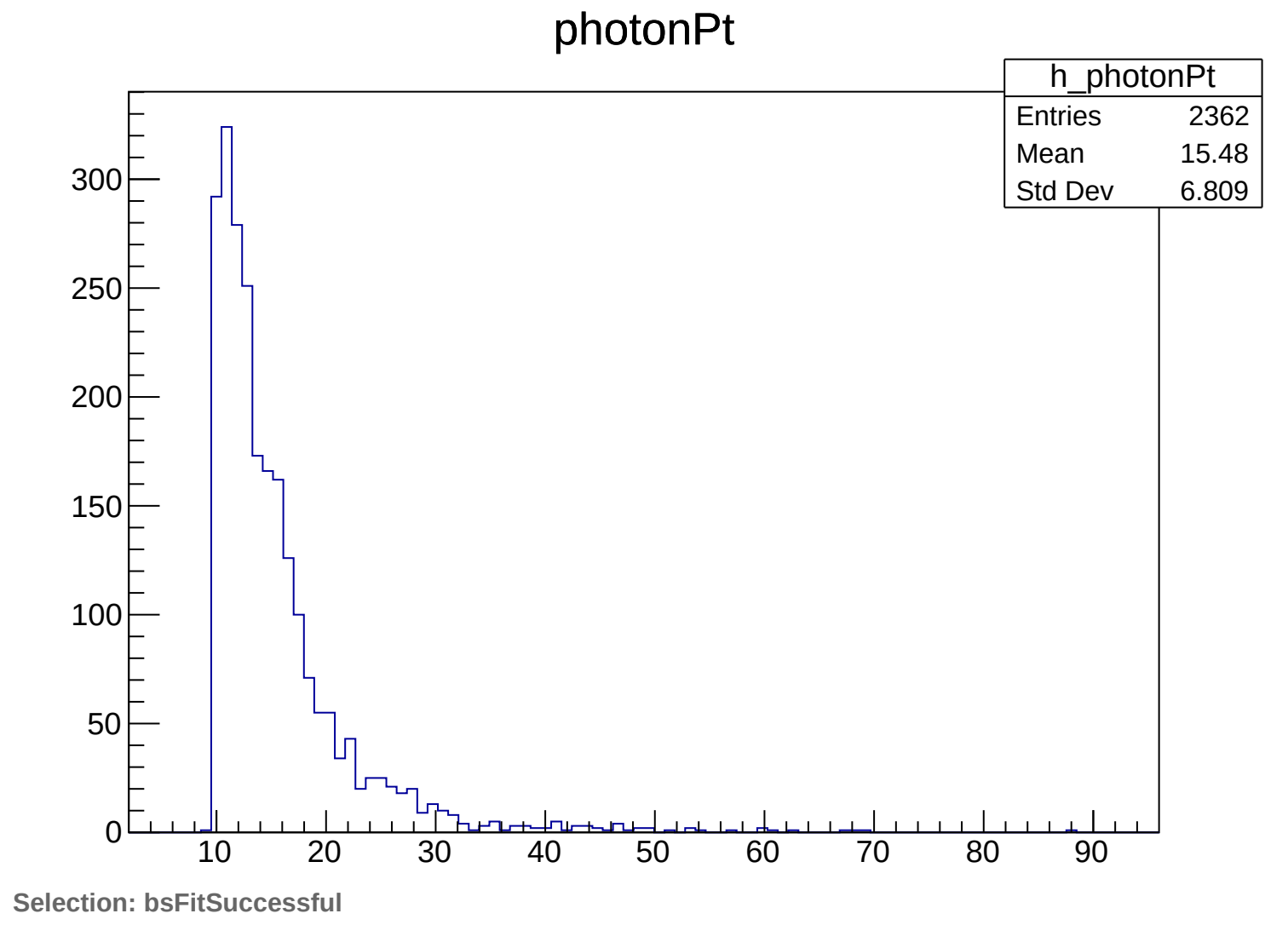

dimuonEta
h_dimuonEta
Entries 2362
Mean 1.626e−05
Std Dev 0.871
50
40
30
20
10
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
Selection: bsFitSuccessful

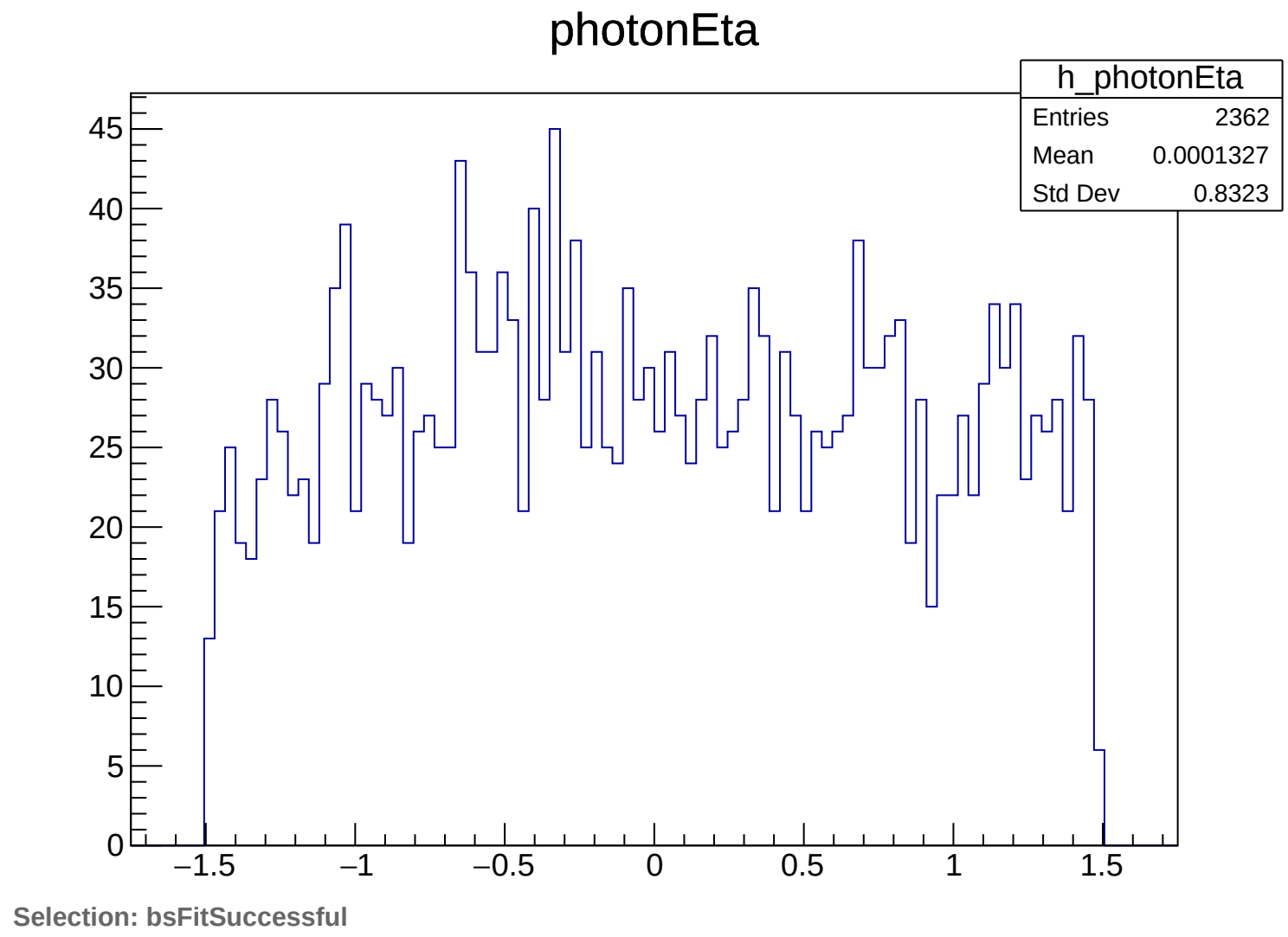

dimuonPhi
h_dimuonPhi
Entries 2362
Mean −0.01996
Std Dev 1.794
40
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

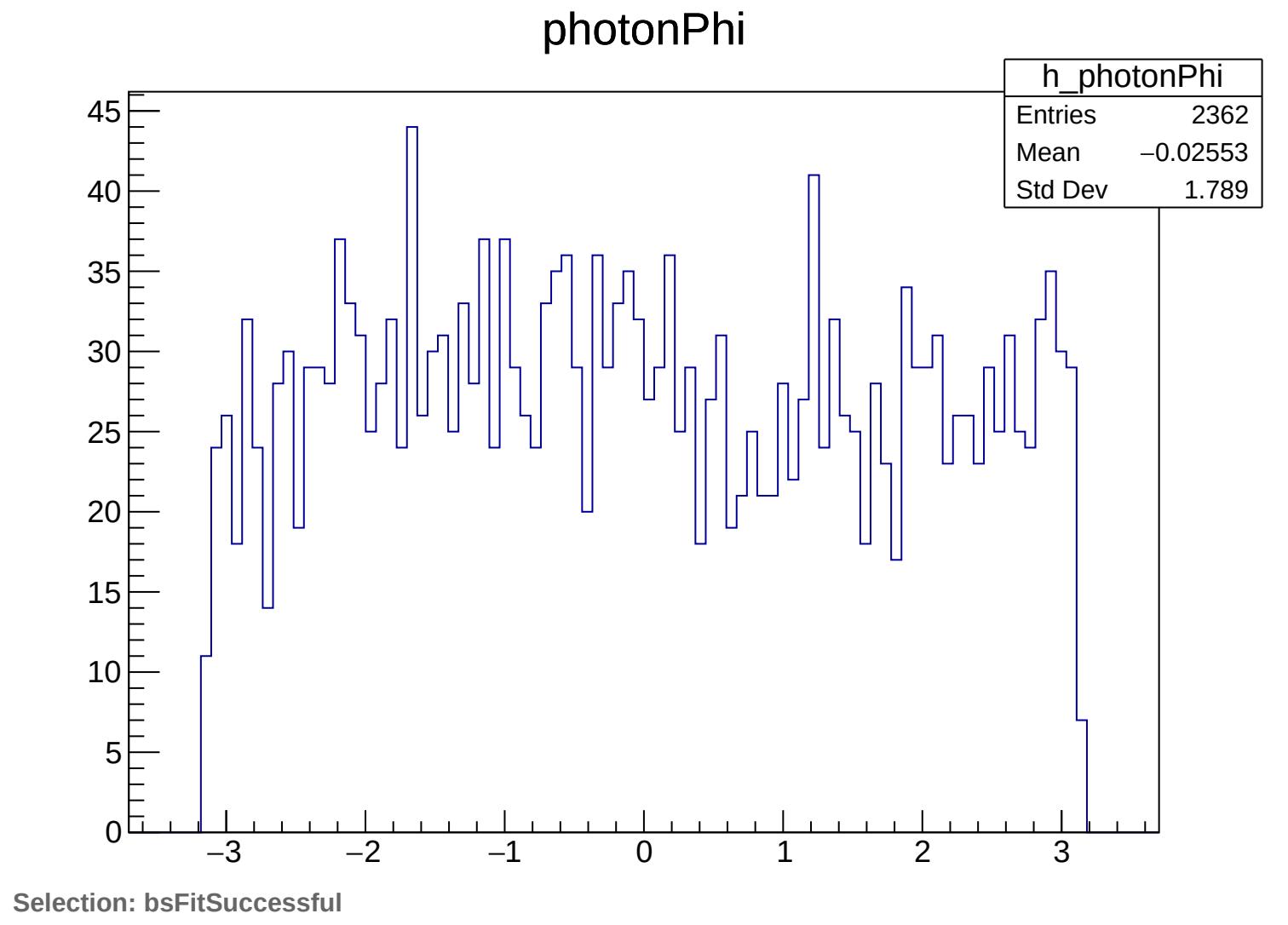

dimuonPt
h_dimuonPt
Entries 2362
Mean 16.97
Std Dev 10.82
250
200
150
100
50
0
0
20
40
60
80
100
120
Selection: bsFitSuccessful

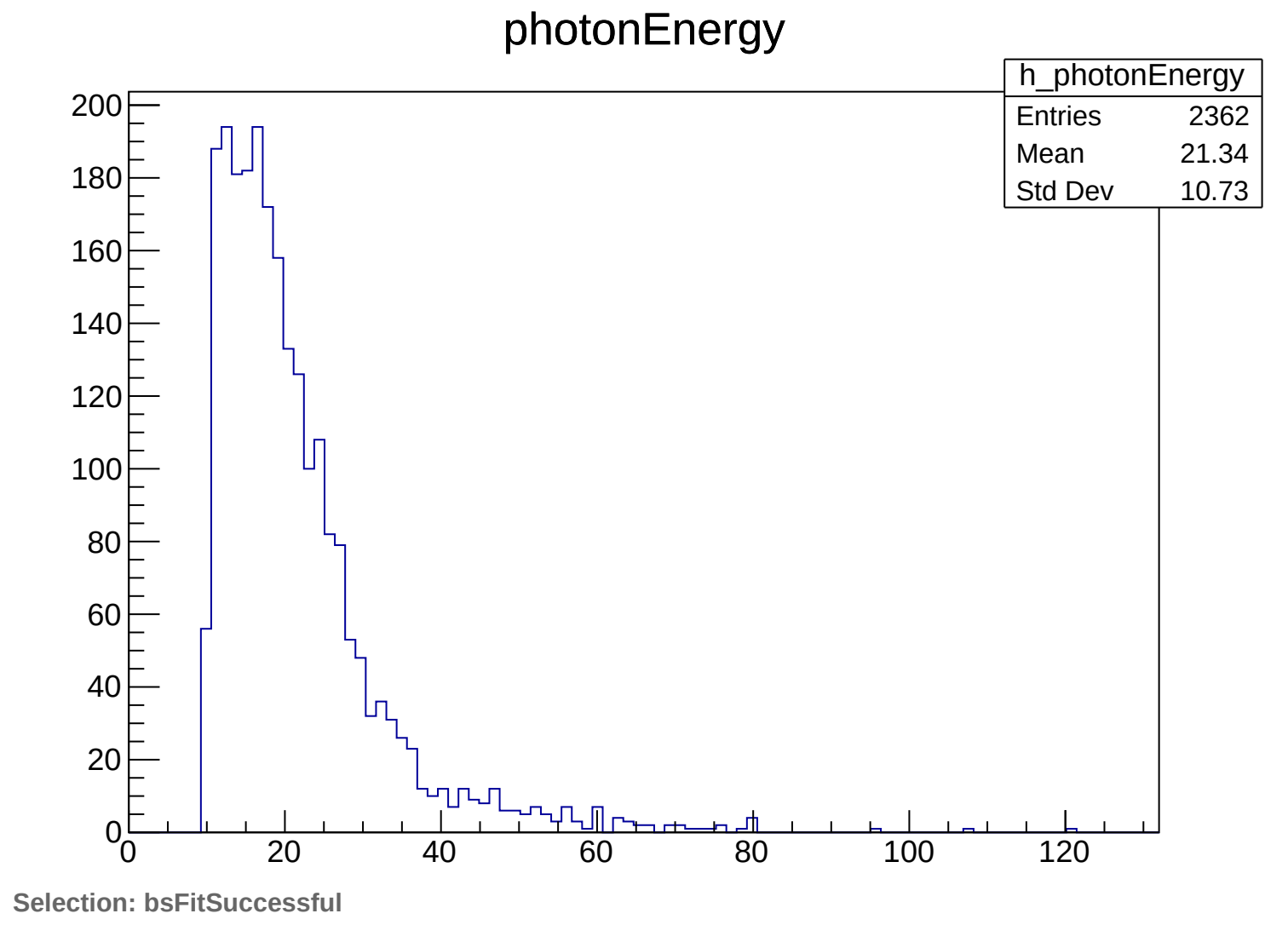

dimuonGenPhotonMass
h_dimuonGenPhotonMass
Entries 2362
Mean 5.356
Std Dev 0.07955
1000
800
600
400
200
0
4.2
4.4
4.6
4.8
5
5.2
5.4
5.6
Selection: bsFitSuccessful

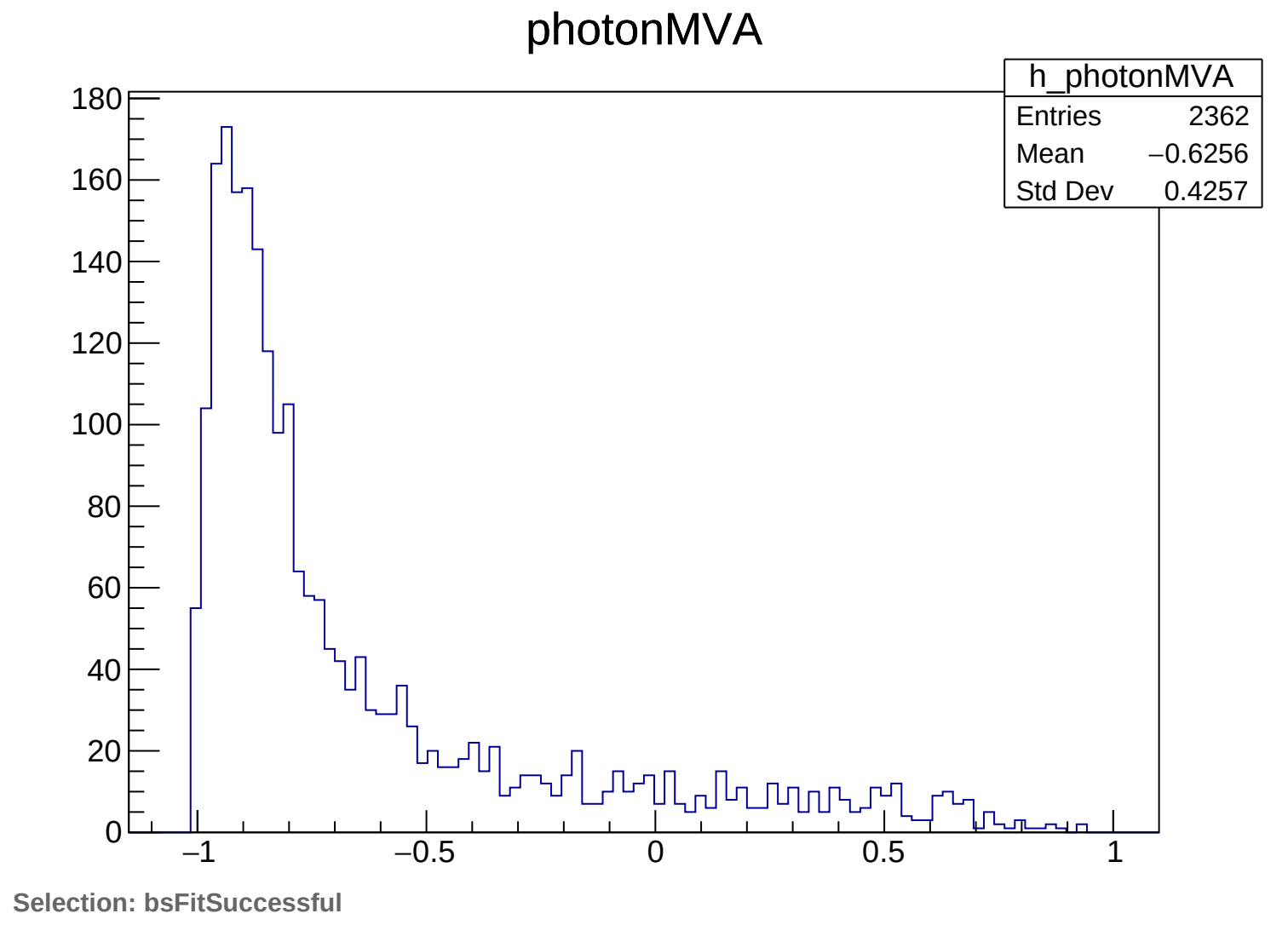

photonDeltaRs
h_photonDeltaRs
Entries 2362
Mean 0.2159
Std Dev 0.07673
70
60
50
40
30
20
10
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
Selection: bsFitSuccessful

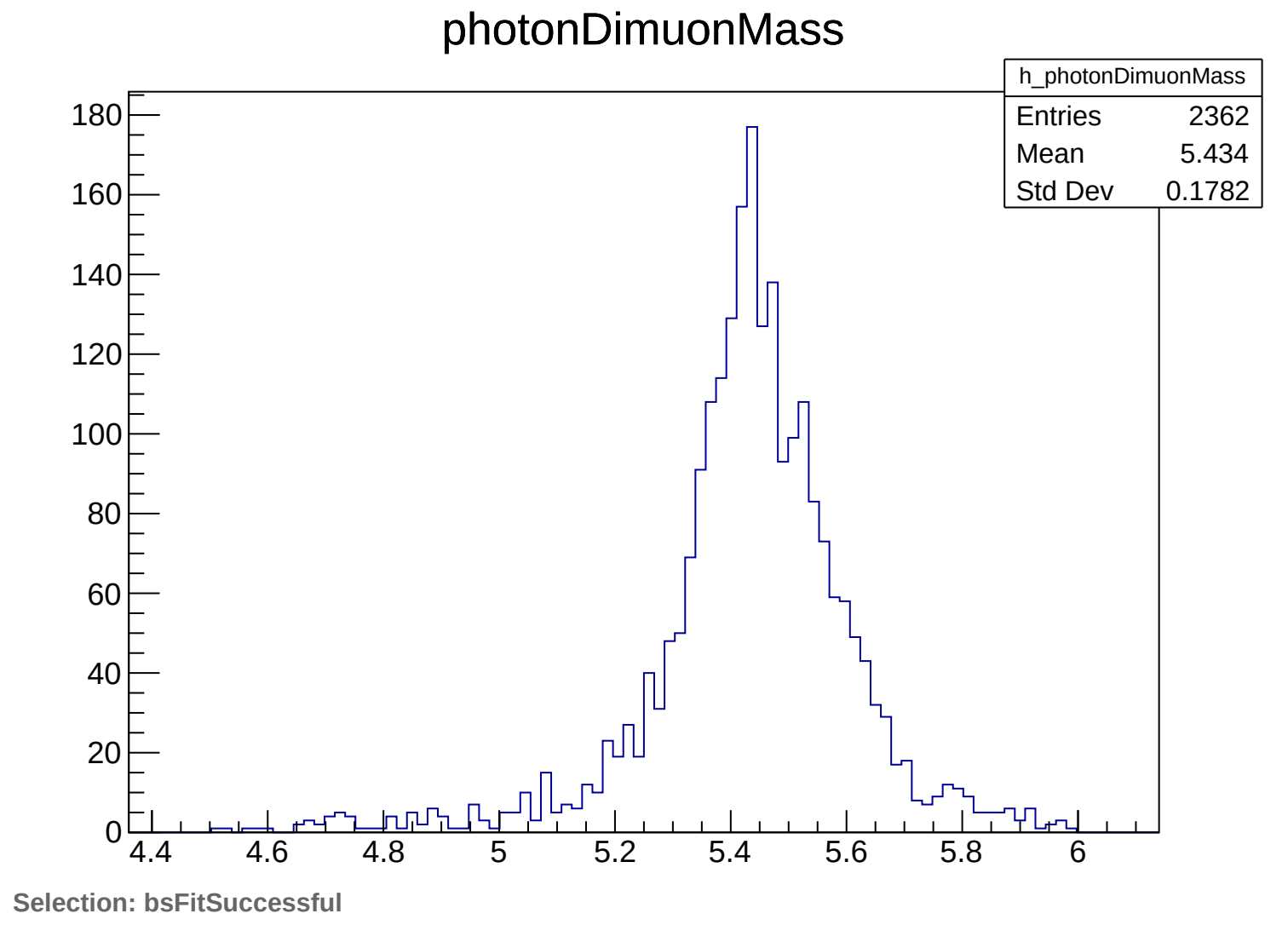

photonPt
h_photonPt
Entries 2362
Mean 15.48
Std Dev 6.809
300
250
200
150
100
50
0
10
20
30
40
50
60
70
80
90
Selection: bsFitSuccessful

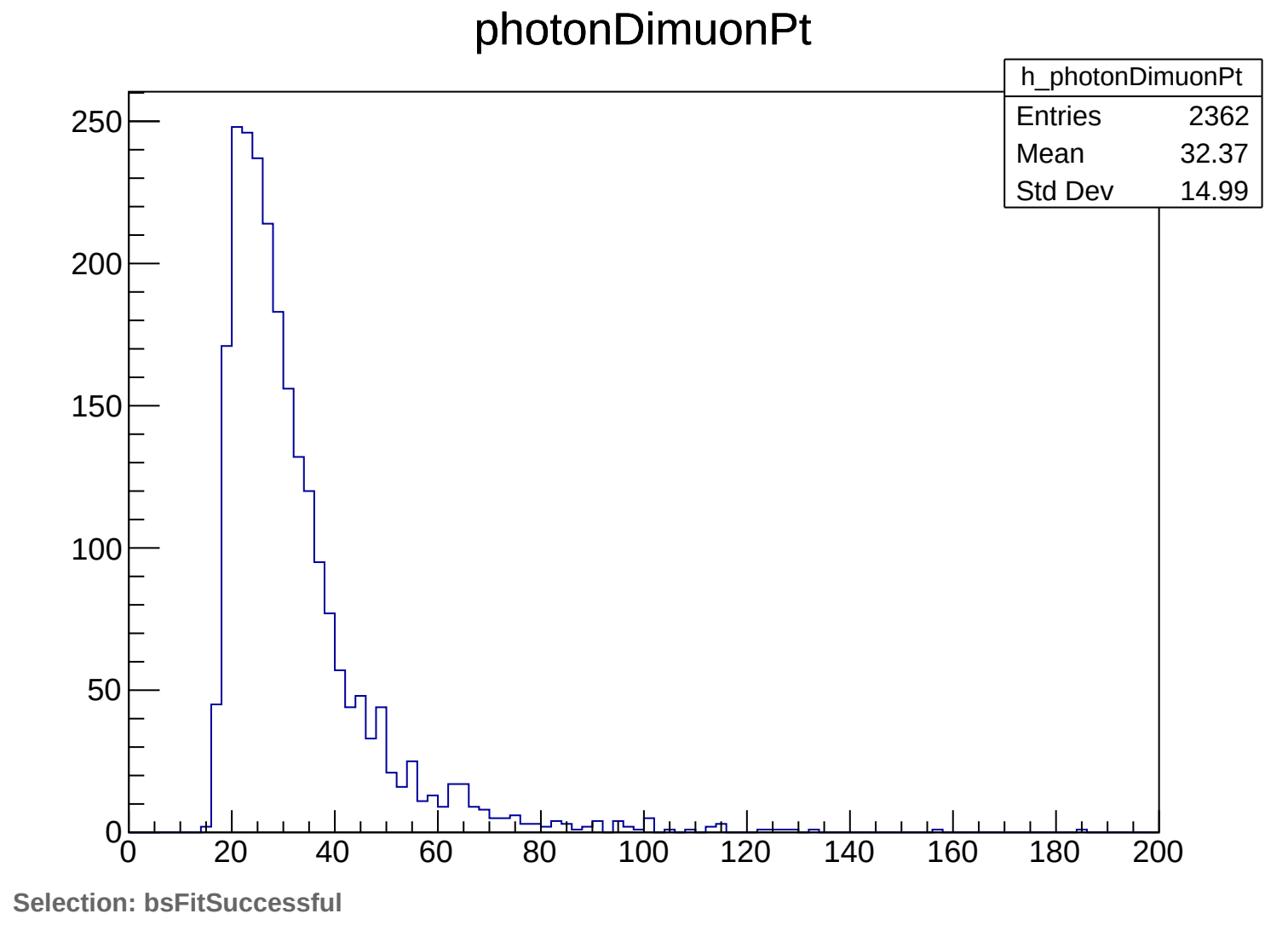

photonEta
h_photonEta
Entries 2362
Mean 0.0001327
Std Dev 0.8323
45
40
35
30
25
20
15
10
5
0
−1.5
−1
−0.5
0
0.5
1
1.5
Selection: bsFitSuccessful

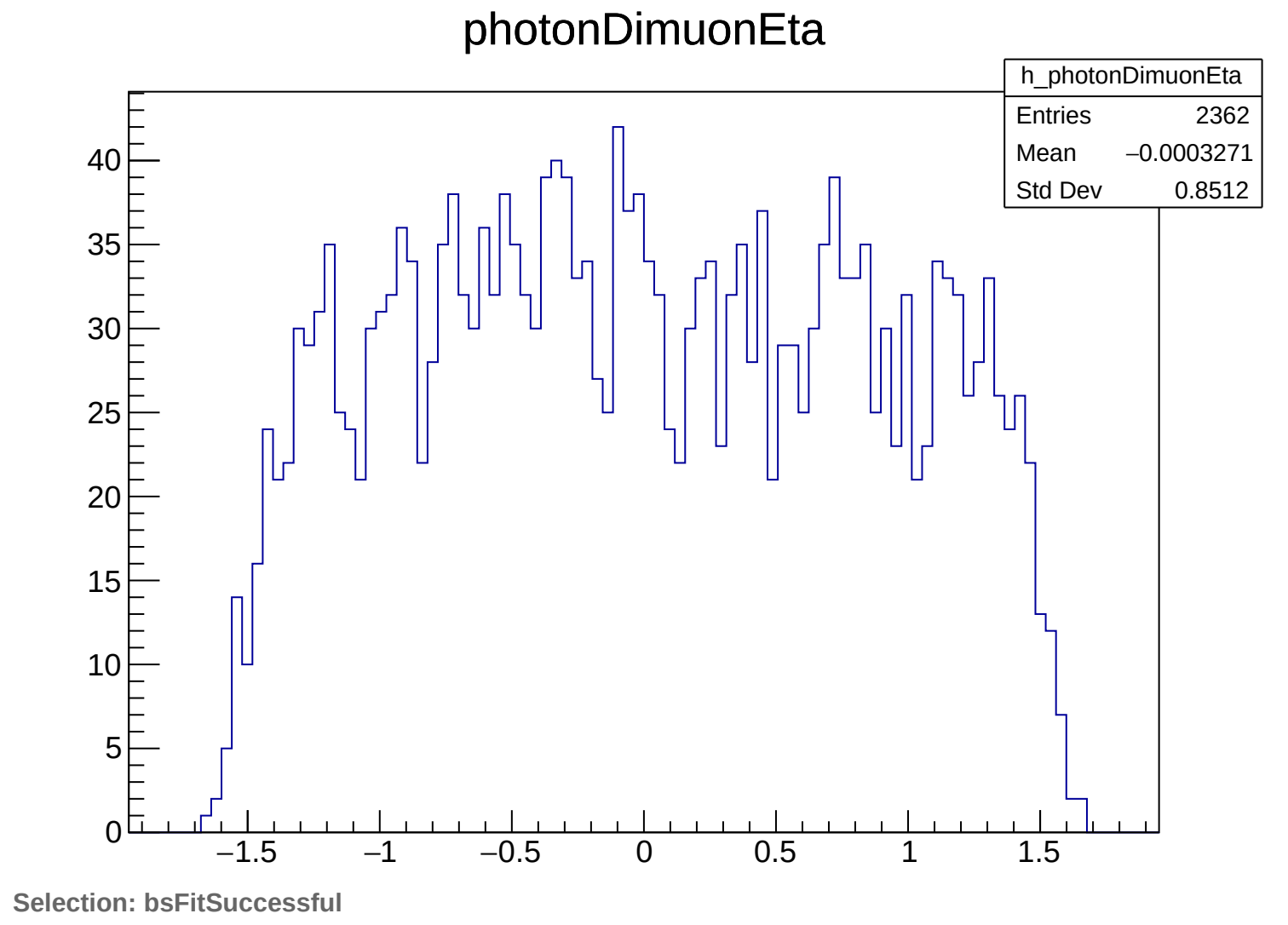

photonPhi
h_photonPhi
Entries 2362
Mean −0.02553
Std Dev 1.789
45
40
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

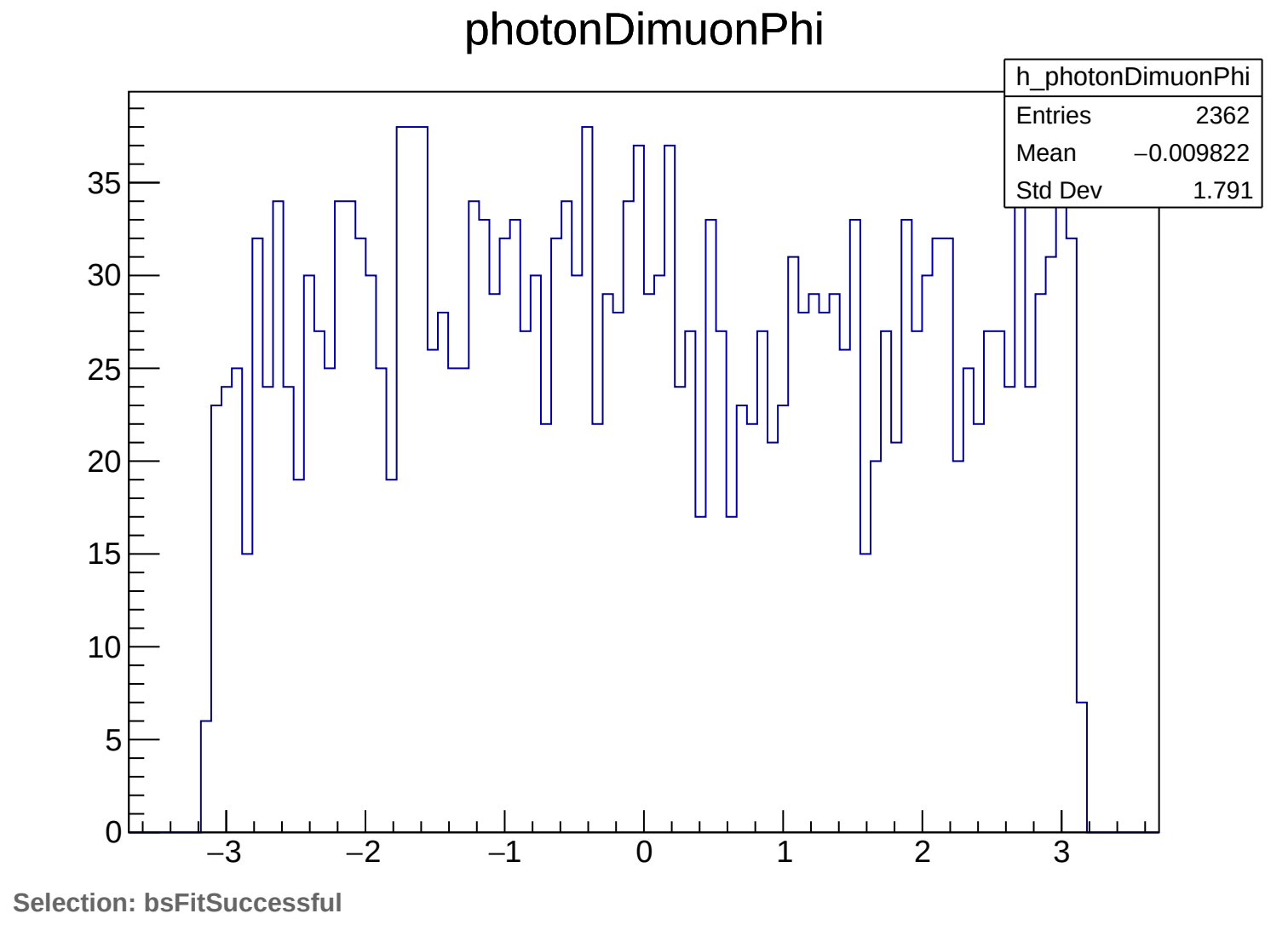

photonEnergy
h_photonEnergy
Entries 2362
Mean 21.34
Std Dev 10.73
200
180
160
140
120
100
80
60
40
20
0
0
20
40
60
80
100
120
Selection: bsFitSuccessful

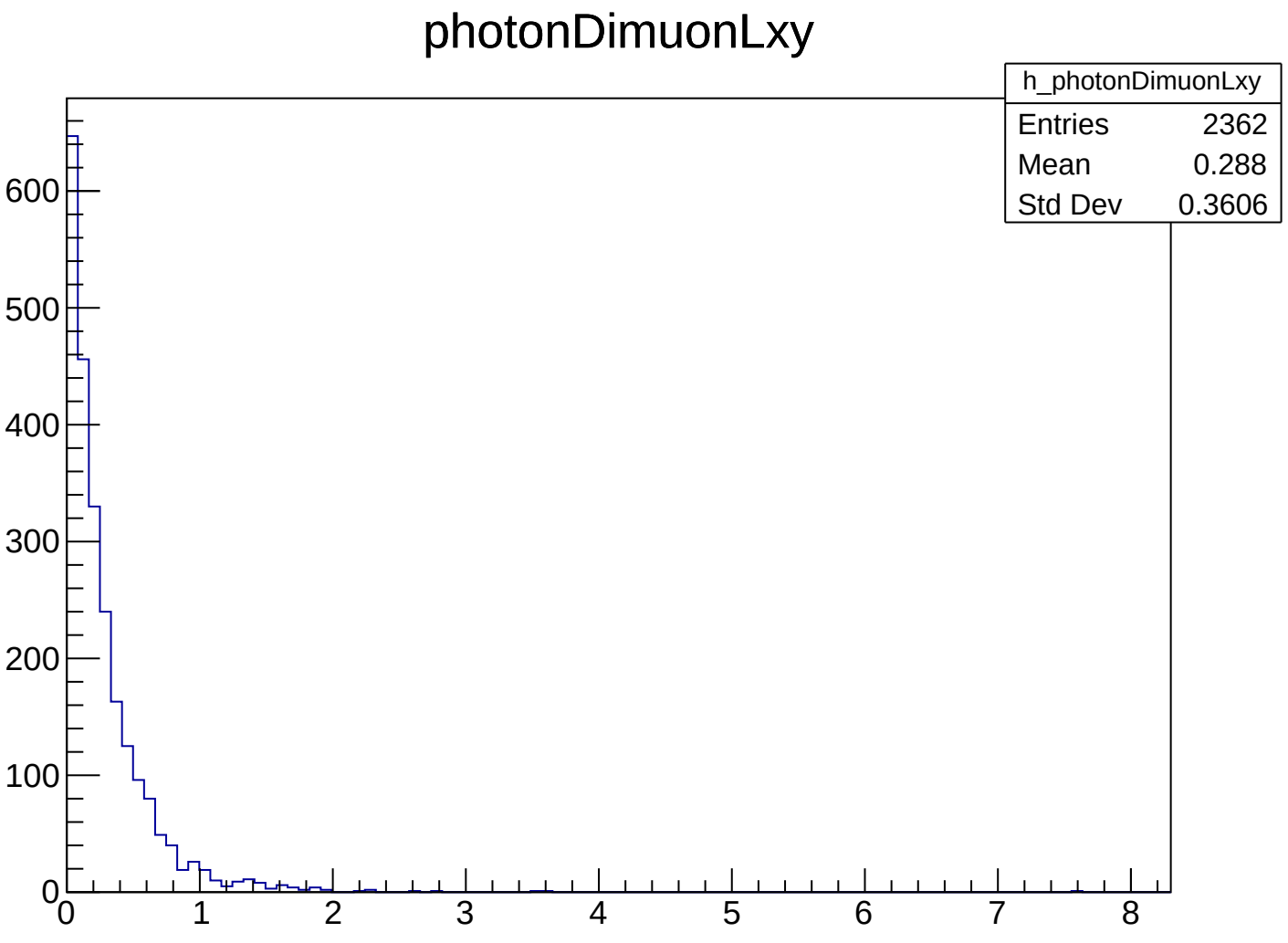

photonMVA
h_photonMVA
Entries 2362
Mean −0.6256
Std Dev 0.4257
180
160
140
120
100
80
60
40
20
0
−1
−0.5
0
0.5
1
Selection: bsFitSuccessful


photonDimuonMass
h_photonDimuonMass
Entries 2362
Mean 5.434
Std Dev 0.1782
180
160
140
120
100
80
60
40
20
0
4.4
4.6
4.8
5
5.2
5.4
5.6
5.8
6
Selection: bsFitSuccessful

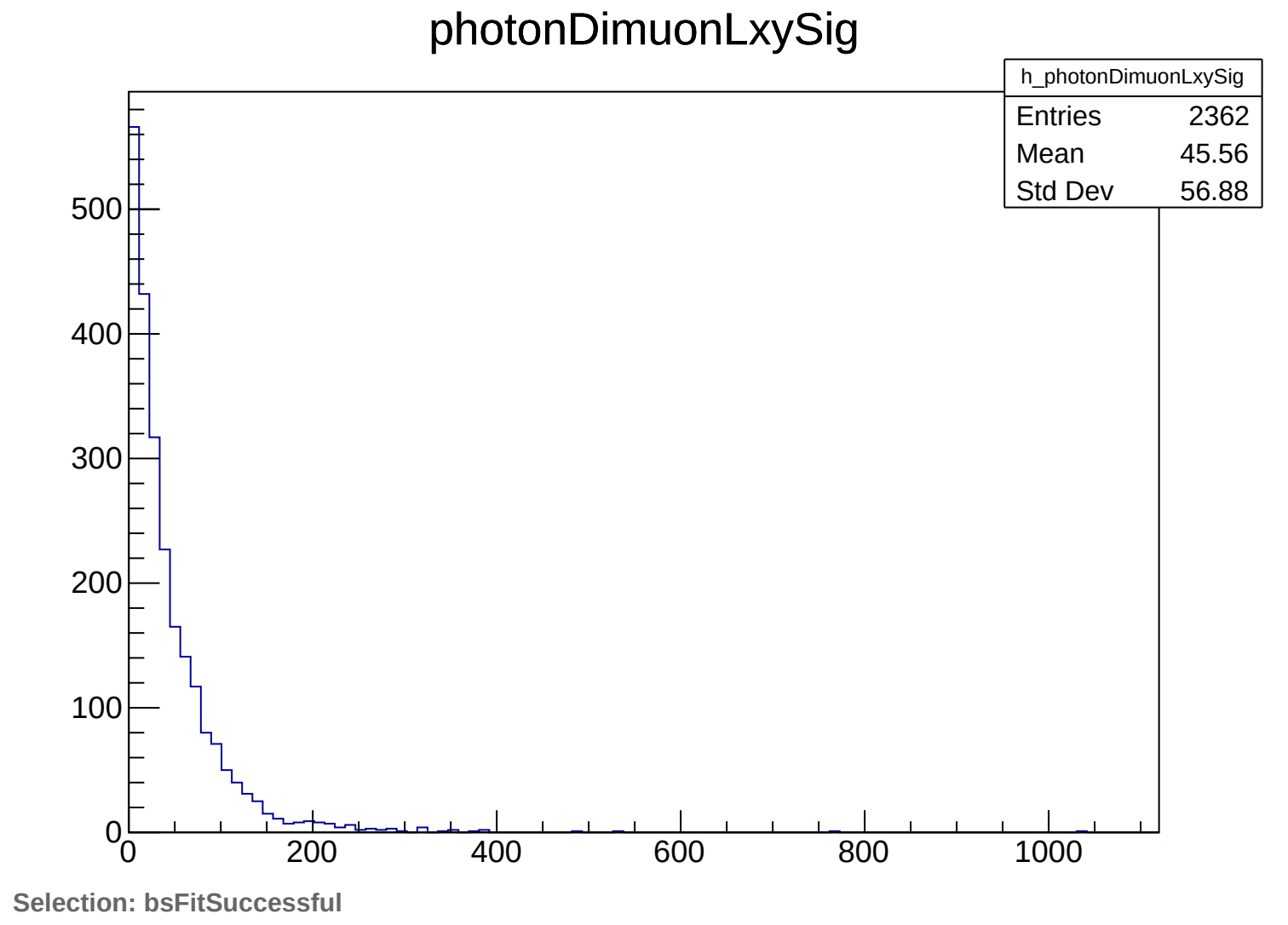

photonDimuonPt
h_photonDimuonPt
Entries 2362
Mean 32.37
Std Dev 14.99
250
200
150
100
50
0
0
20
40
60
80
100
120
140
160
180
200
Selection: bsFitSuccessful

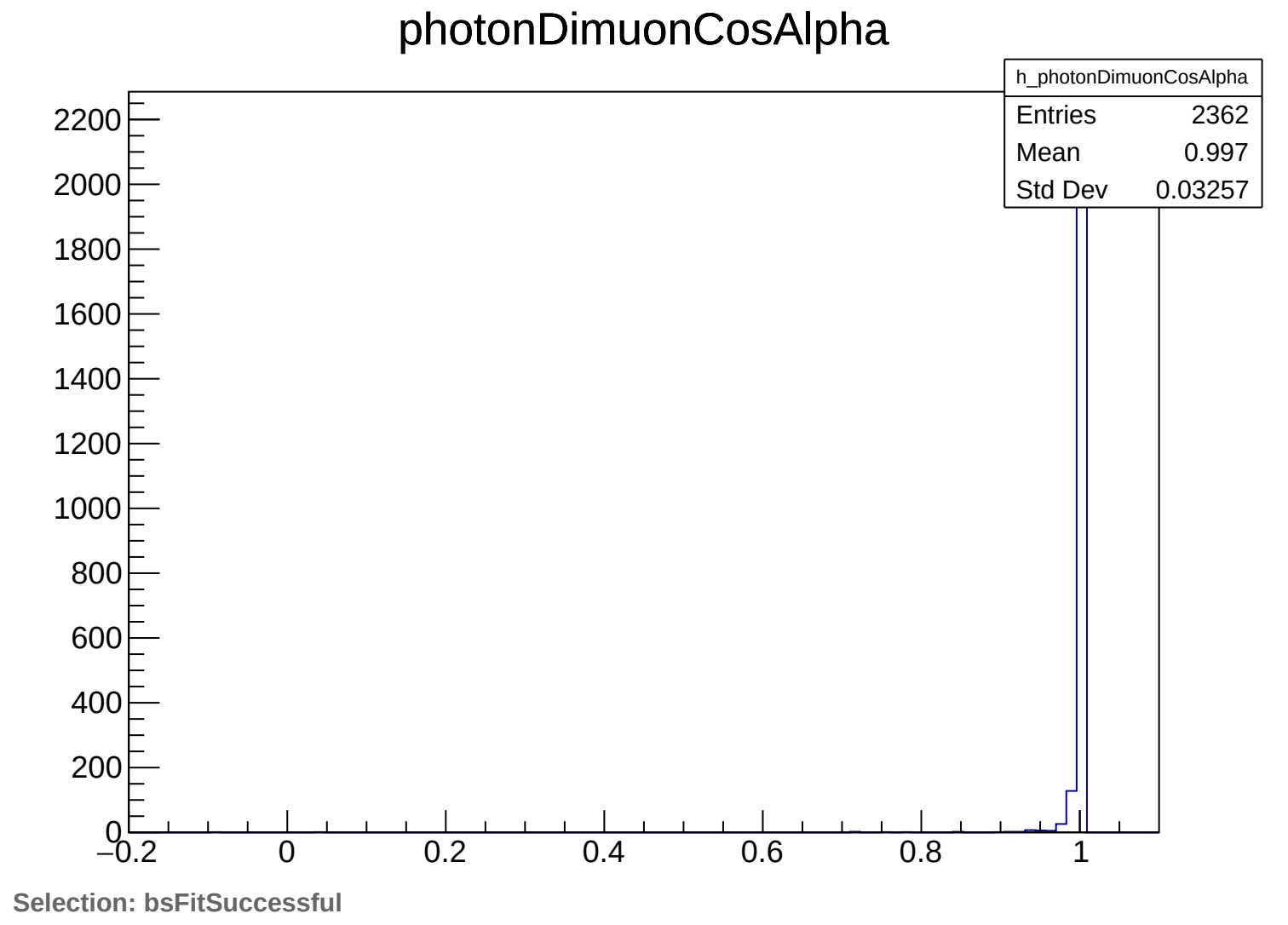

photonDimuonEta
h_photonDimuonEta
Entries 2362
Mean –0.0003271
Std Dev 0.8512
40
35
30
25
20
15
10
5
0
–1.5
–1
–0.5
0
0.5
1
1.5
Selection: bsFitSuccessful

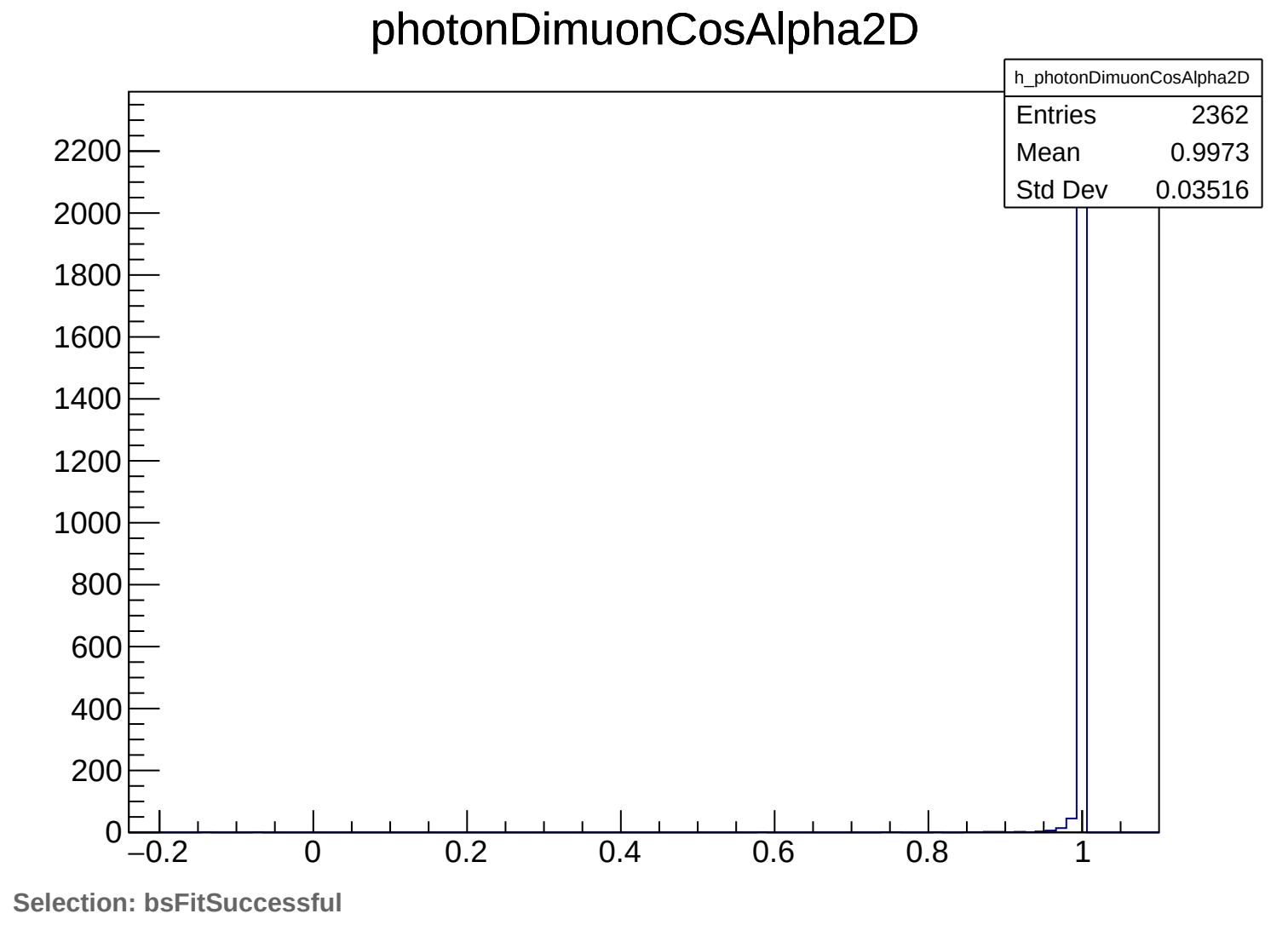

photonDimuonPhi
h_photonDimuonPhi
Entries 2362
Mean −0.009822
Std Dev 1.791
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

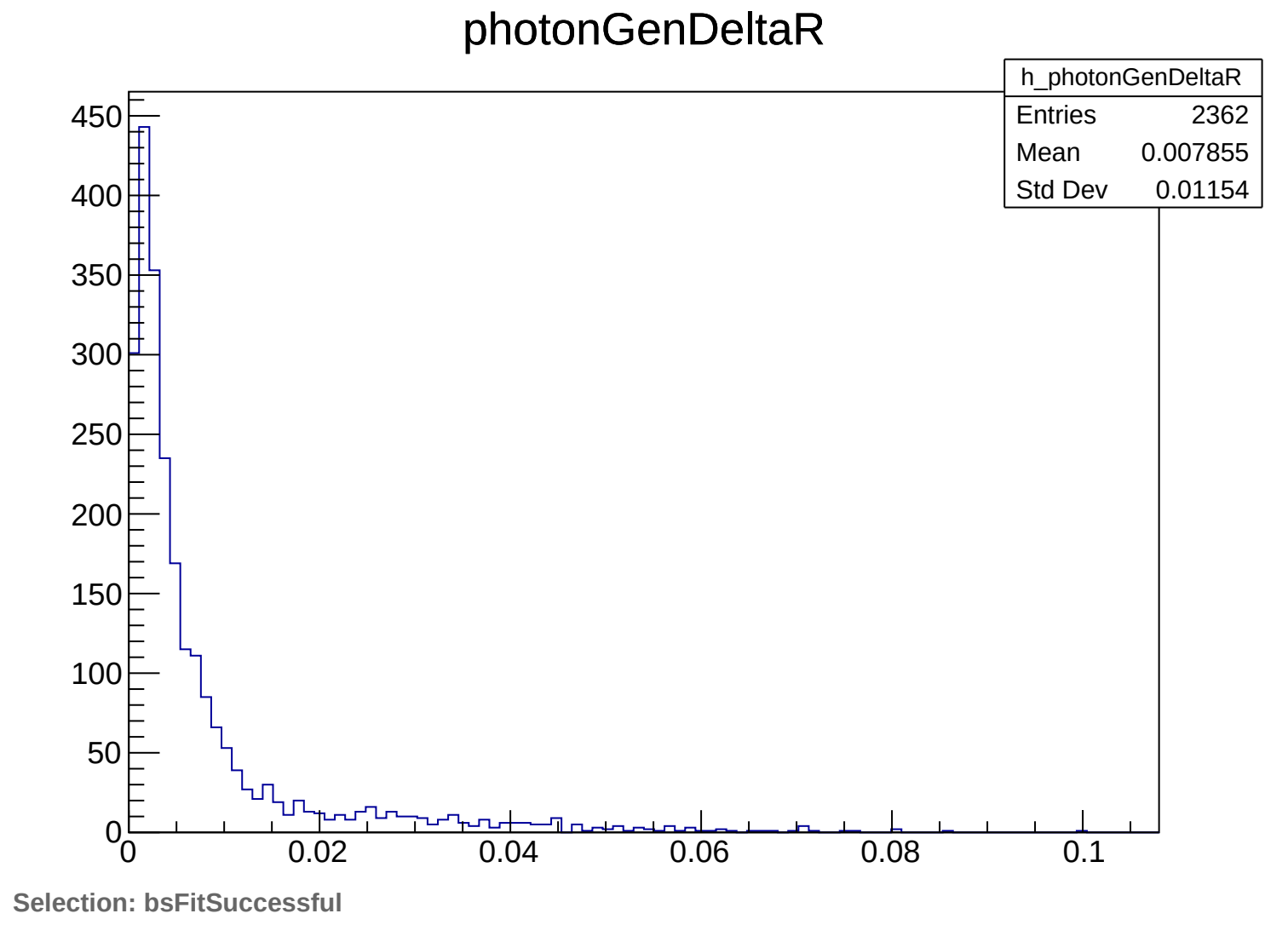


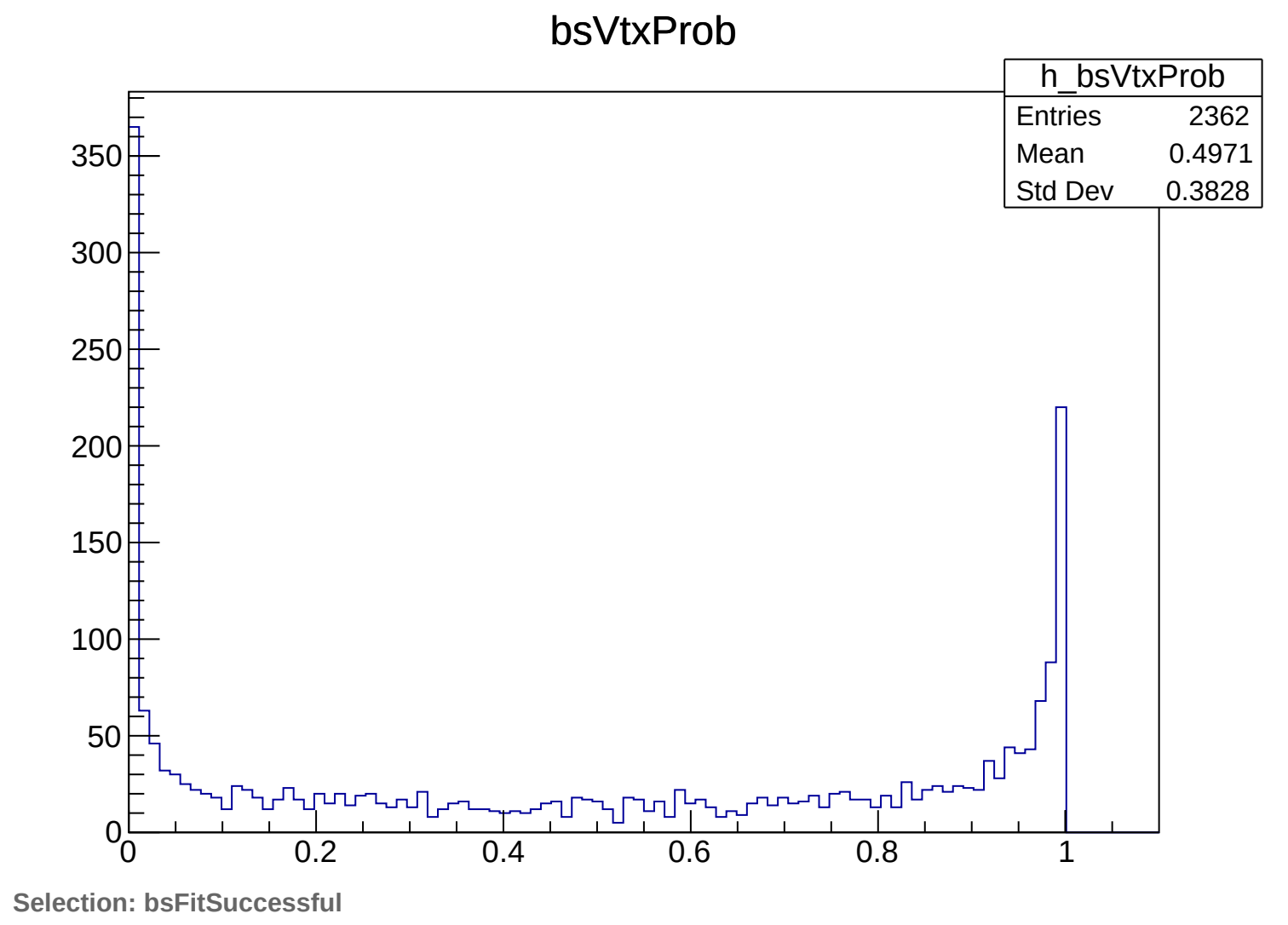
Selection: bsFitSuccessful

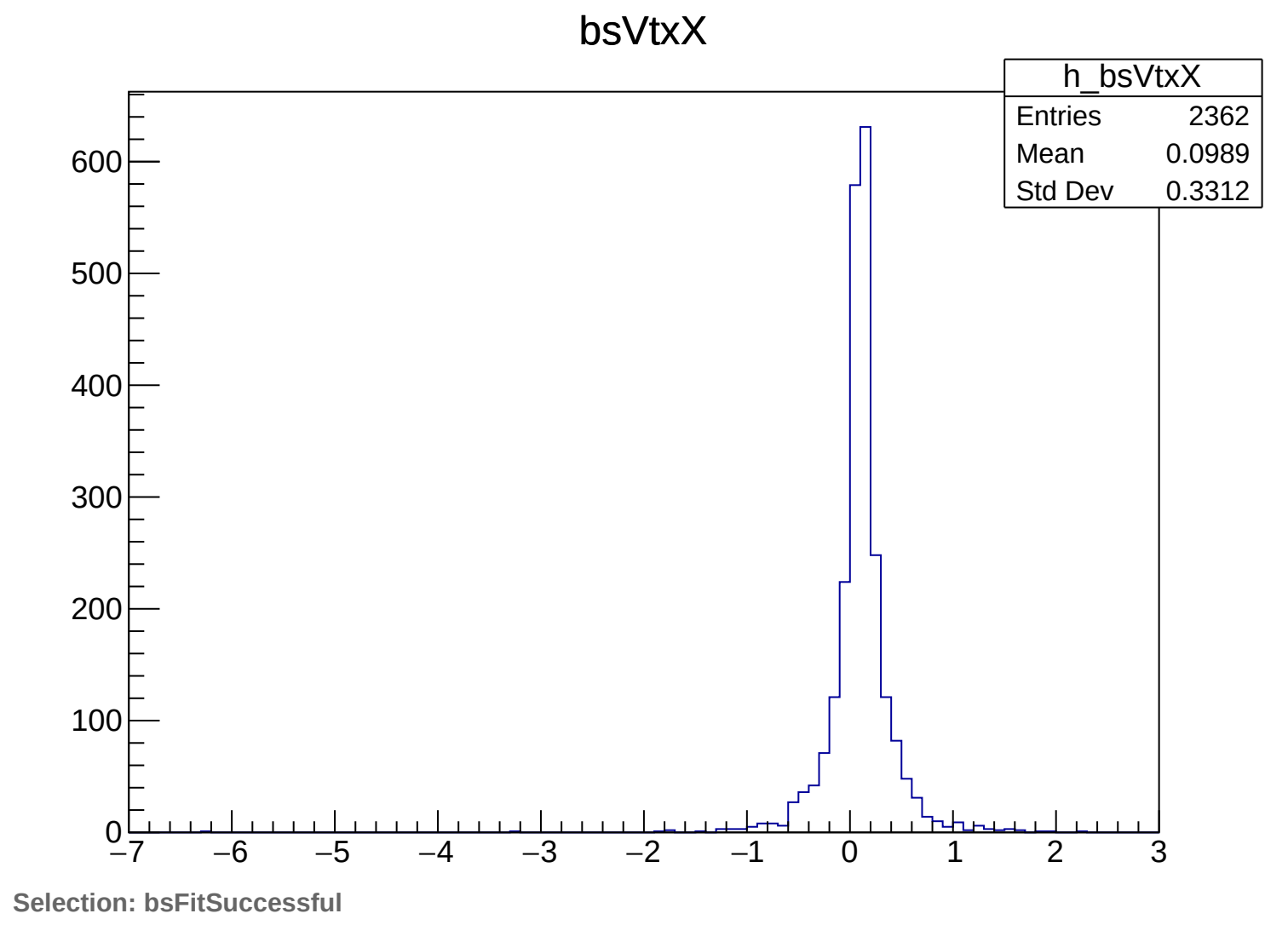

photonDimuonLxySig
h_photonDimuonLxySig
Entries 2362
Mean 45.56
Std Dev 56.88
500
400
300
200
100
0
0
200
400
600
800
1000
Selection: bsFitSuccessful

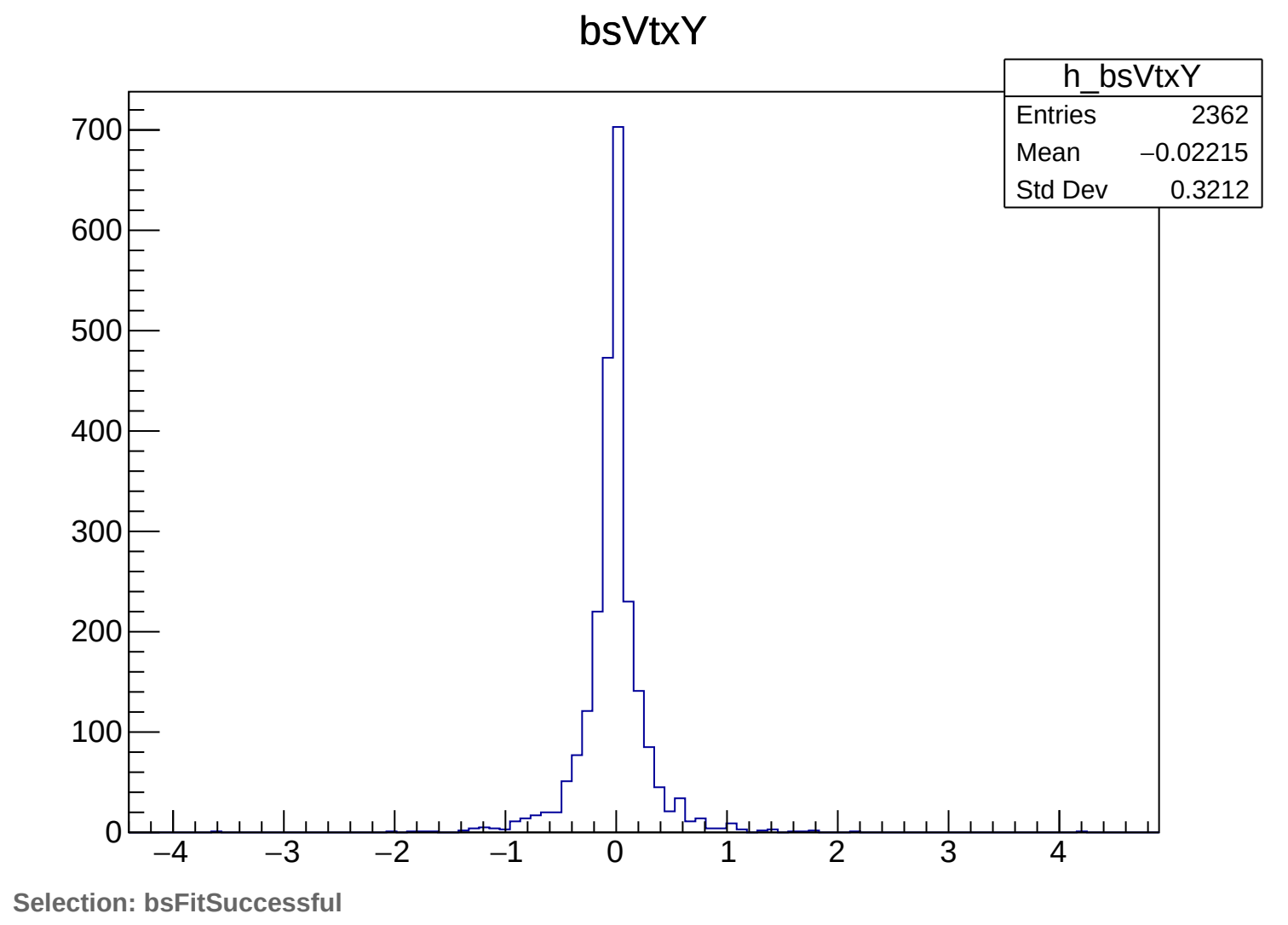

photonDimuonCosAlpha
h_photonDimuonCosAlpha
Entries 2362
Mean 0.997
Std Dev 0.03257
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
–0.2
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

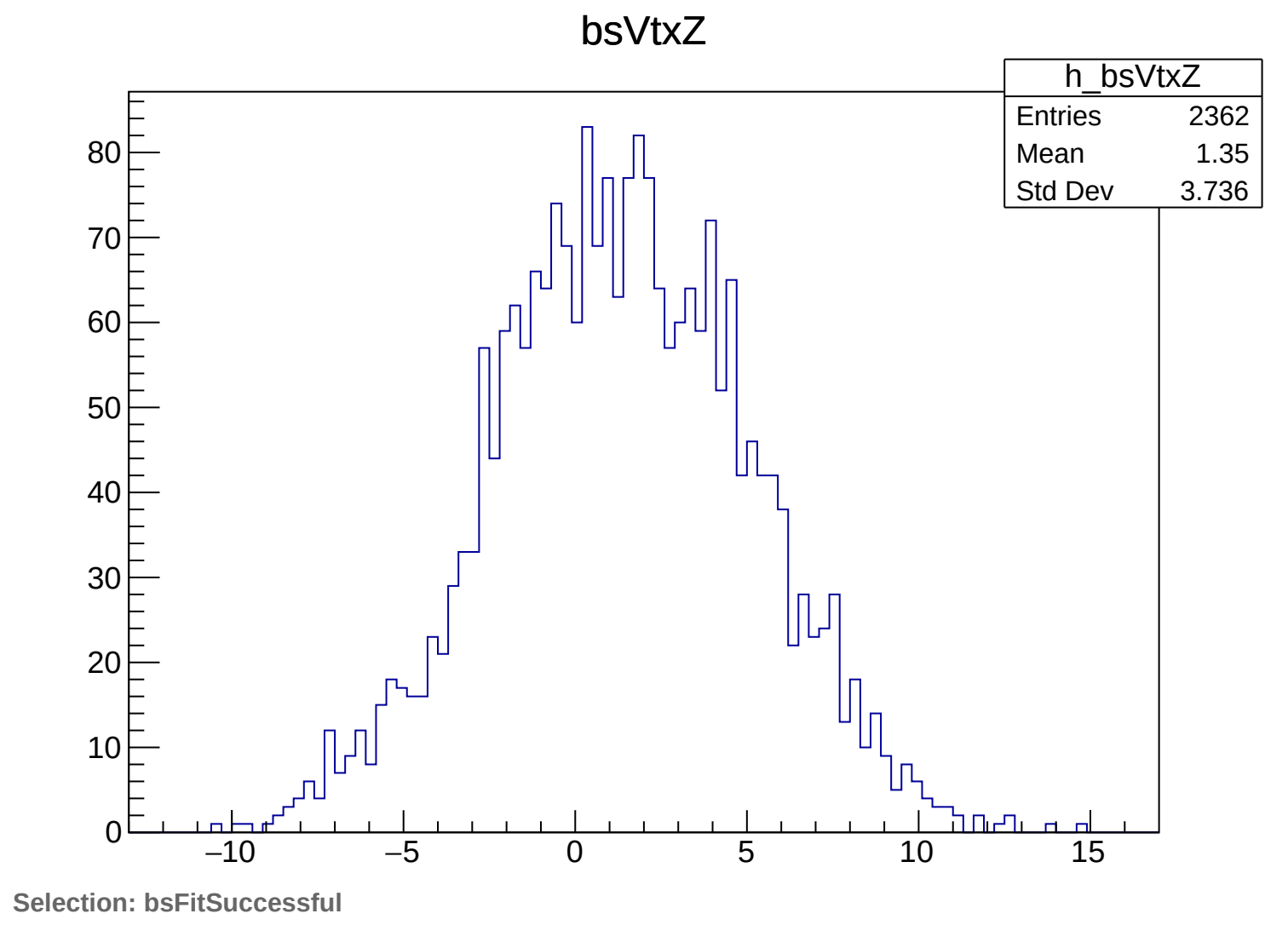

photonDimuonCosAlpha2D
h_photonDimuonCosAlpha2D
Entries 2362
Mean 0.9973
Std Dev 0.03516
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−0.2
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

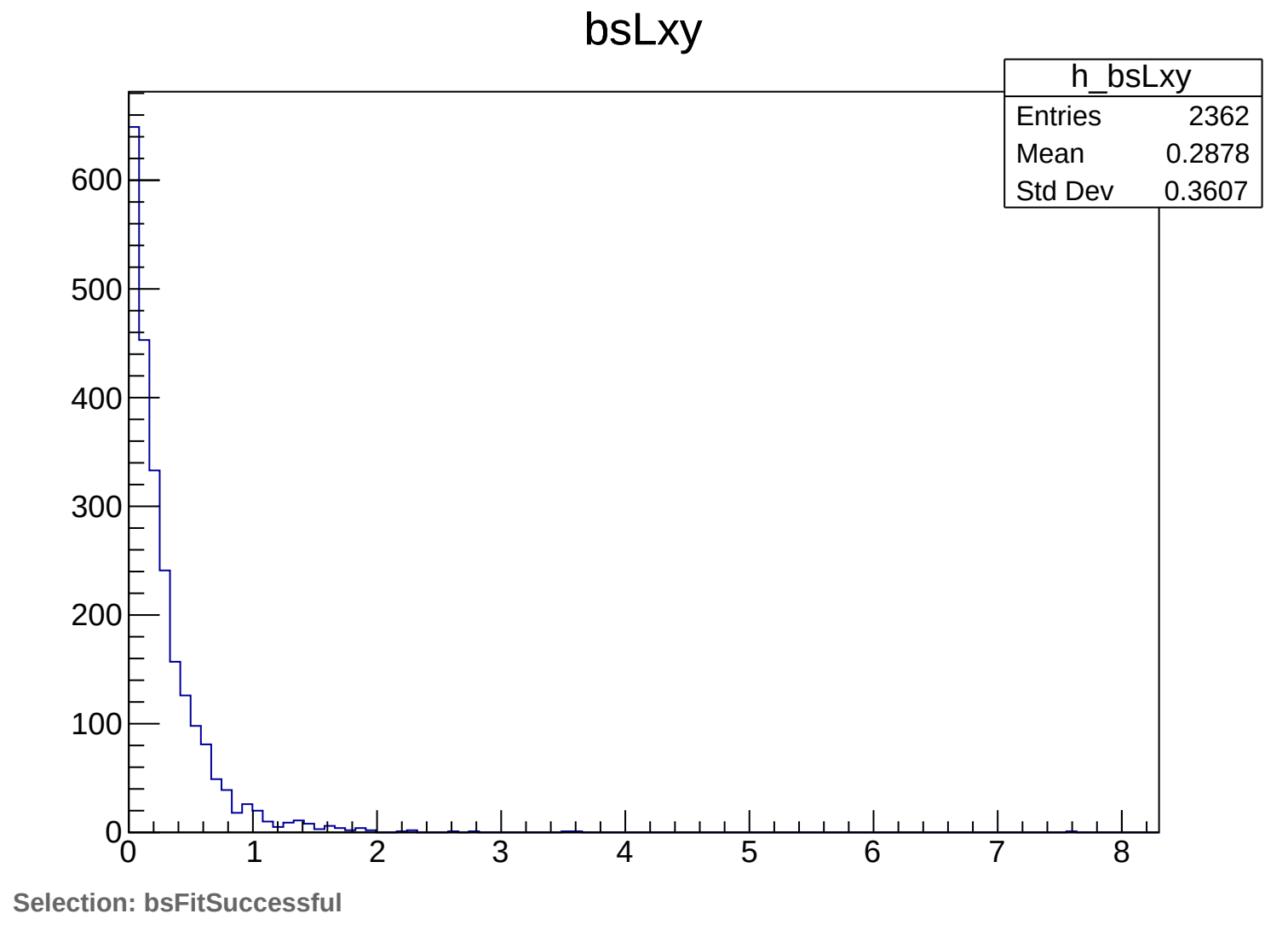

photonGenDeltaR
h_photonGenDeltaR
Entries 2362
Mean 0.007855
Std Dev 0.01154
450
400
350
300
250
200
150
100
50
0
0
0.02
0.04
0.06
0.08
0.1
Selection: bsFitSuccessful

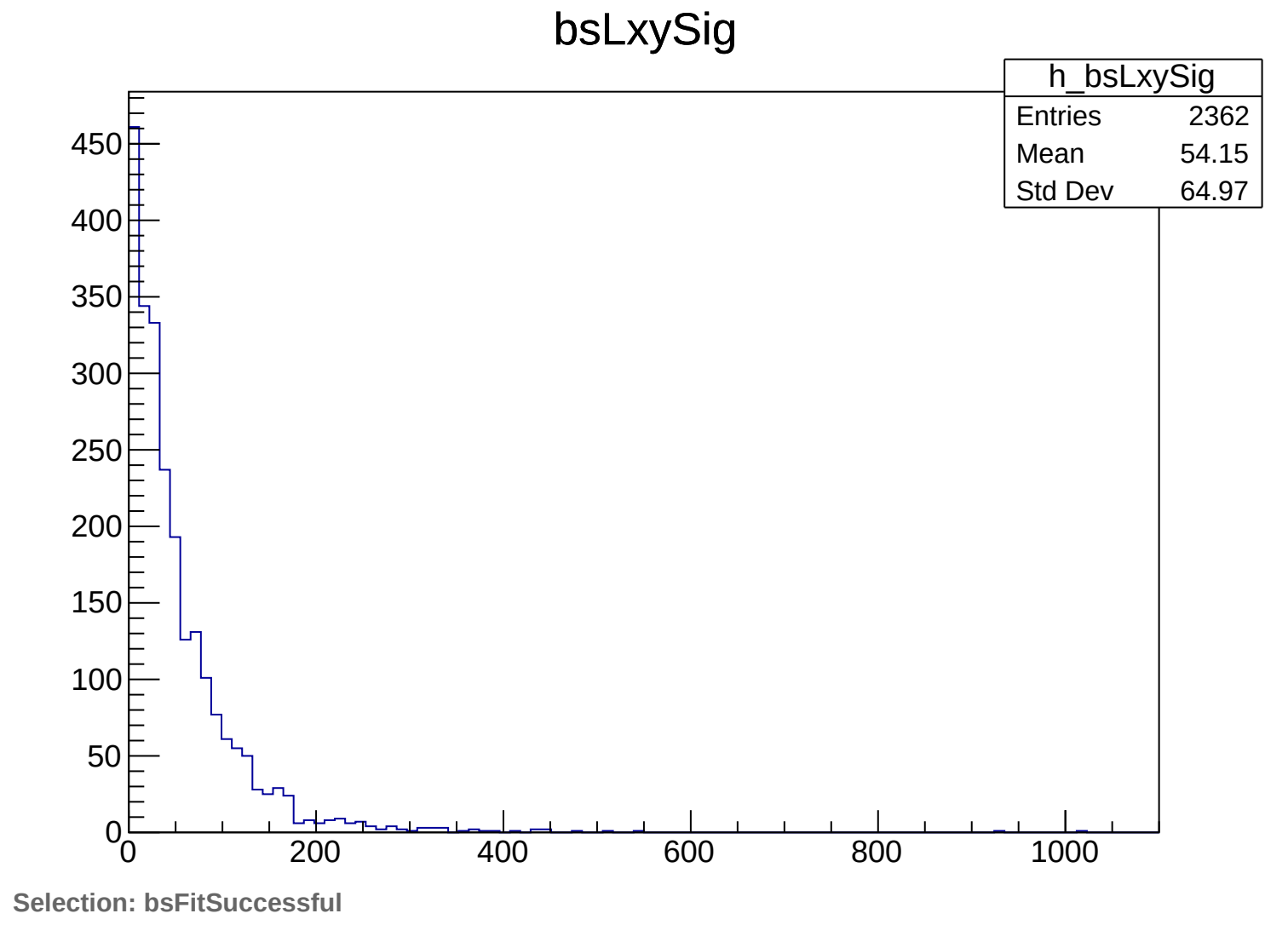

bsVtxProb
h_bsVtxProb
Entries 2362
Mean 0.4971
Std Dev 0.3828
350
300
250
200
150
100
50
0
0
0.2
0.4
0.6
0.8
1
Selection: bsFitSuccessful

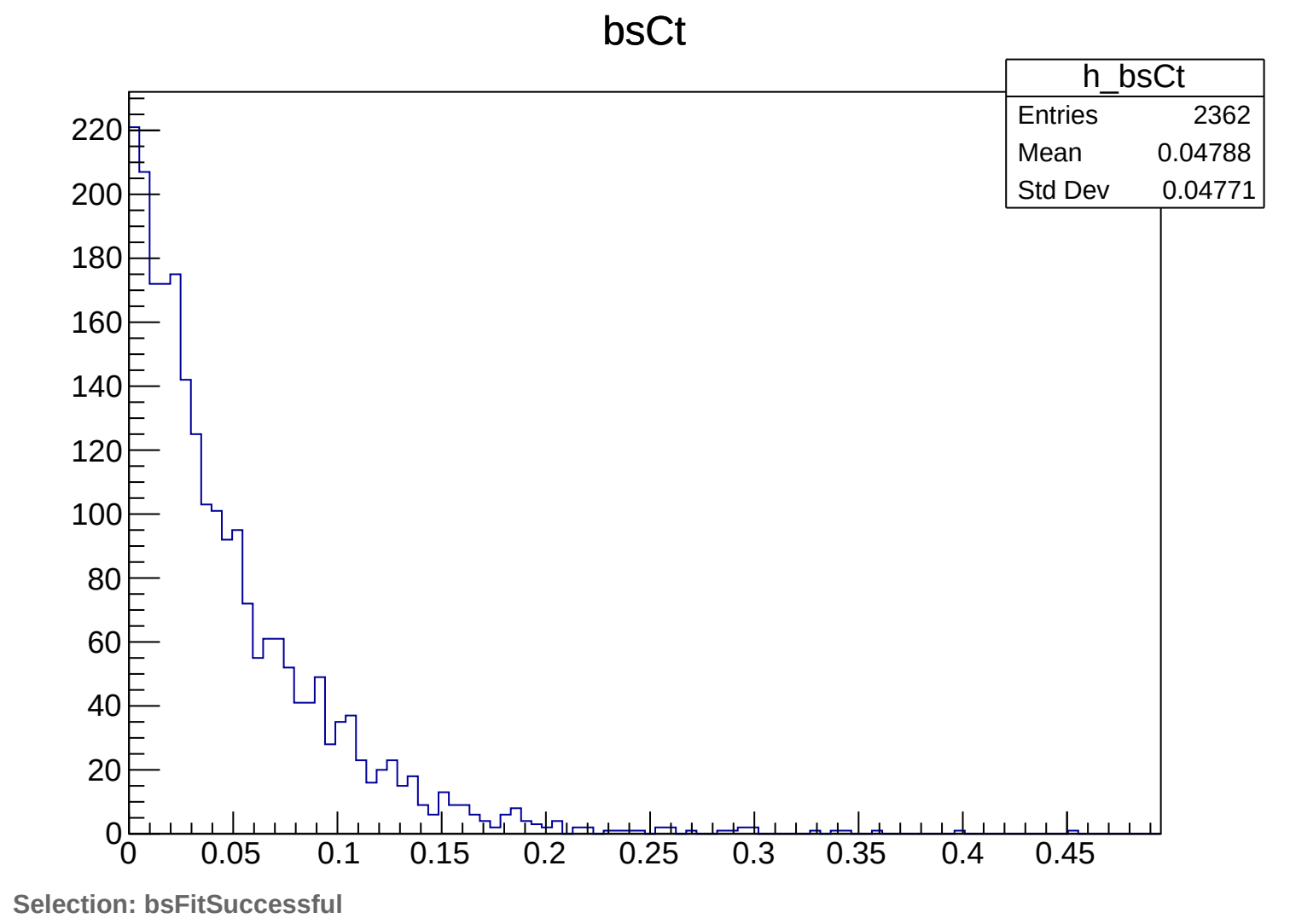

bsVtxX
h_bsVtxX
Entries 2362
Mean 0.0989
Std Dev 0.3312
600
500
400
300
200
100
0
−7
−6
−5
−4
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful

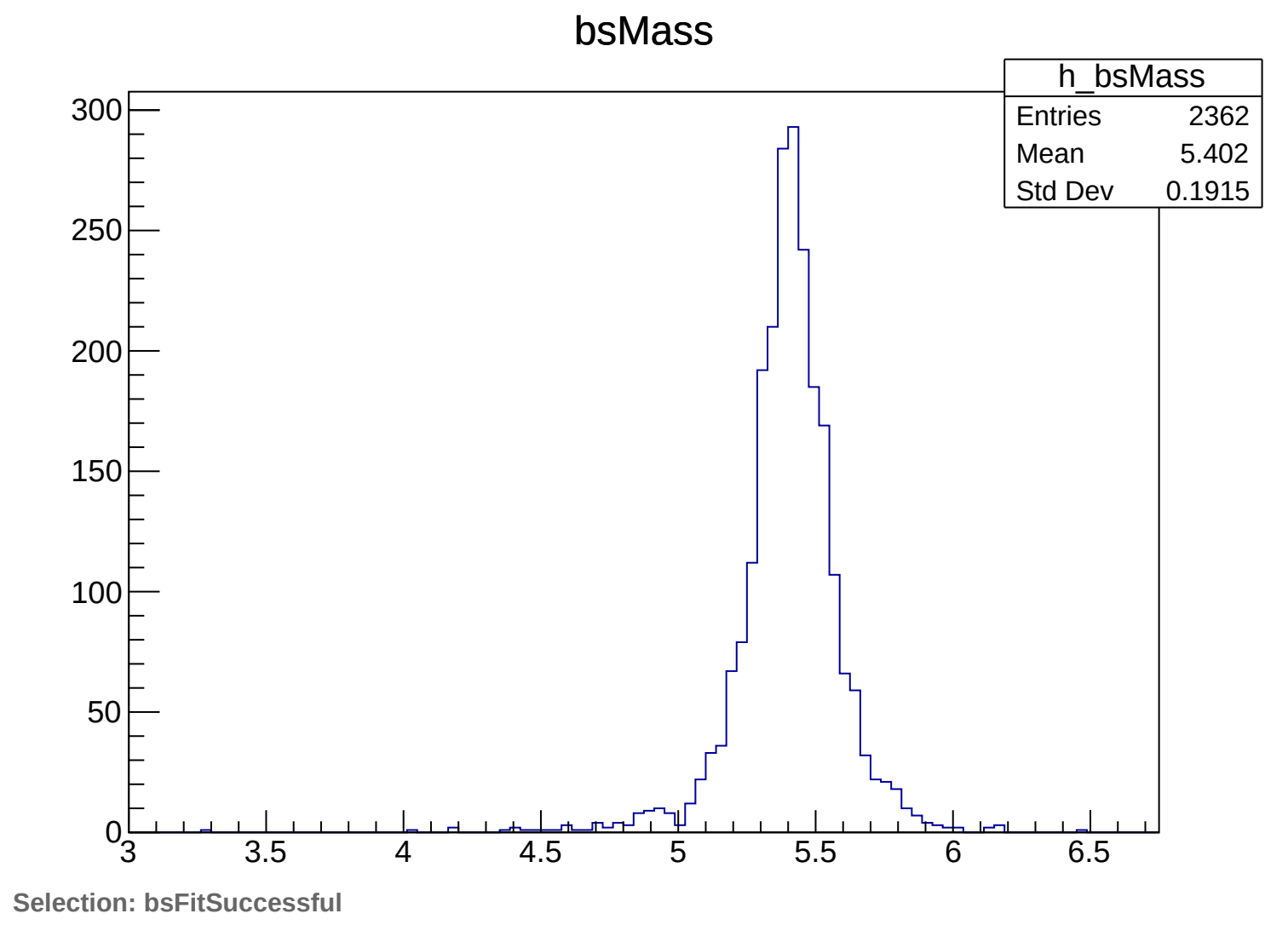

bsVtxY
h_bsVtxY
Entries 2362
Mean −0.02215
Std Dev 0.3212
700
600
500
400
300
200
100
0
−4
−3
−2
−1
0
1
2
3
4
Selection: bsFitSuccessful

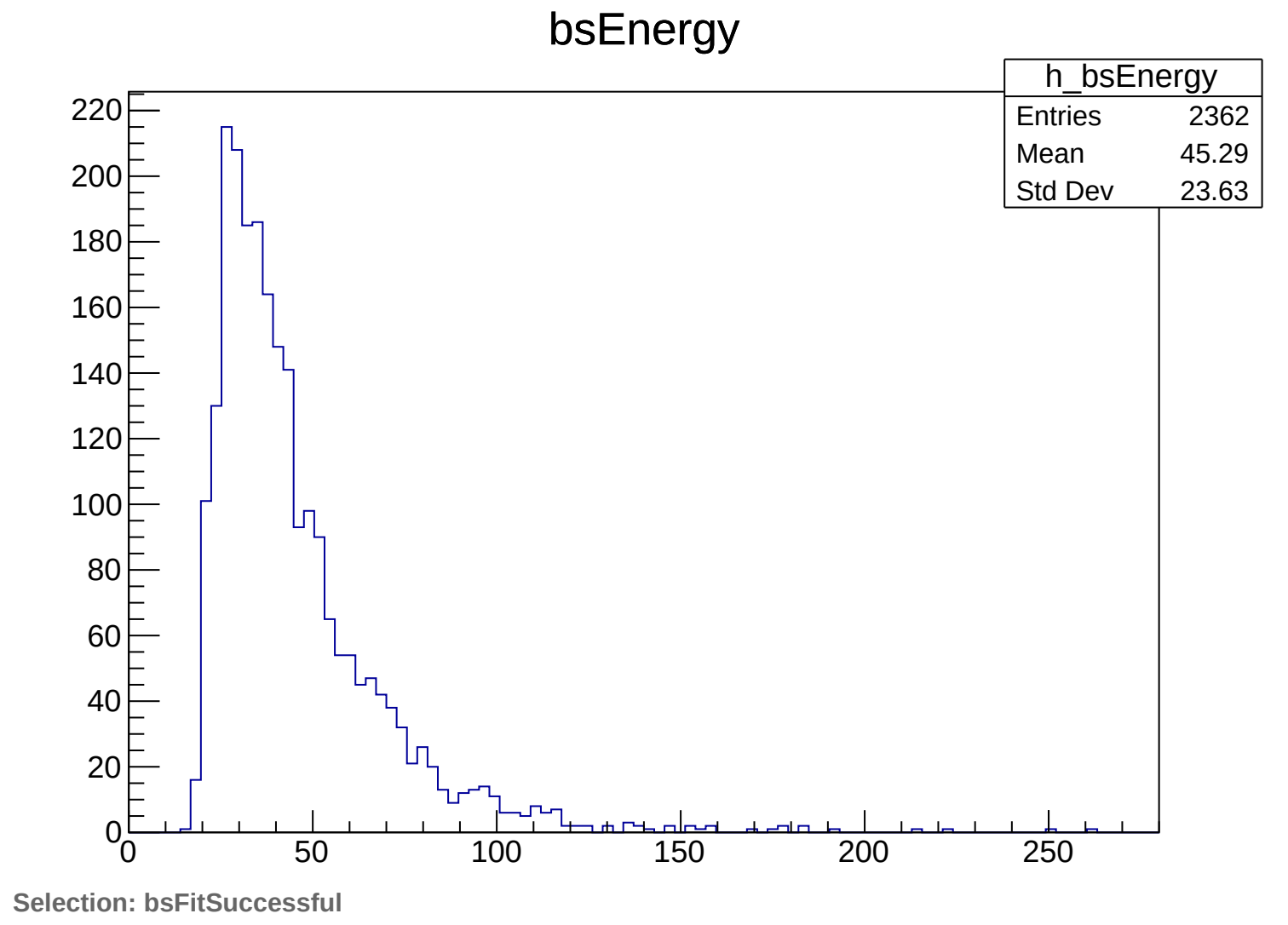

bsVtxZ
h_bsVtxZ
Entries 2362
Mean 1.35
Std Dev 3.736
80
70
60
50
40
30
20
10
0
−10
−5
0
5
10
15
Selection: bsFitSuccessful

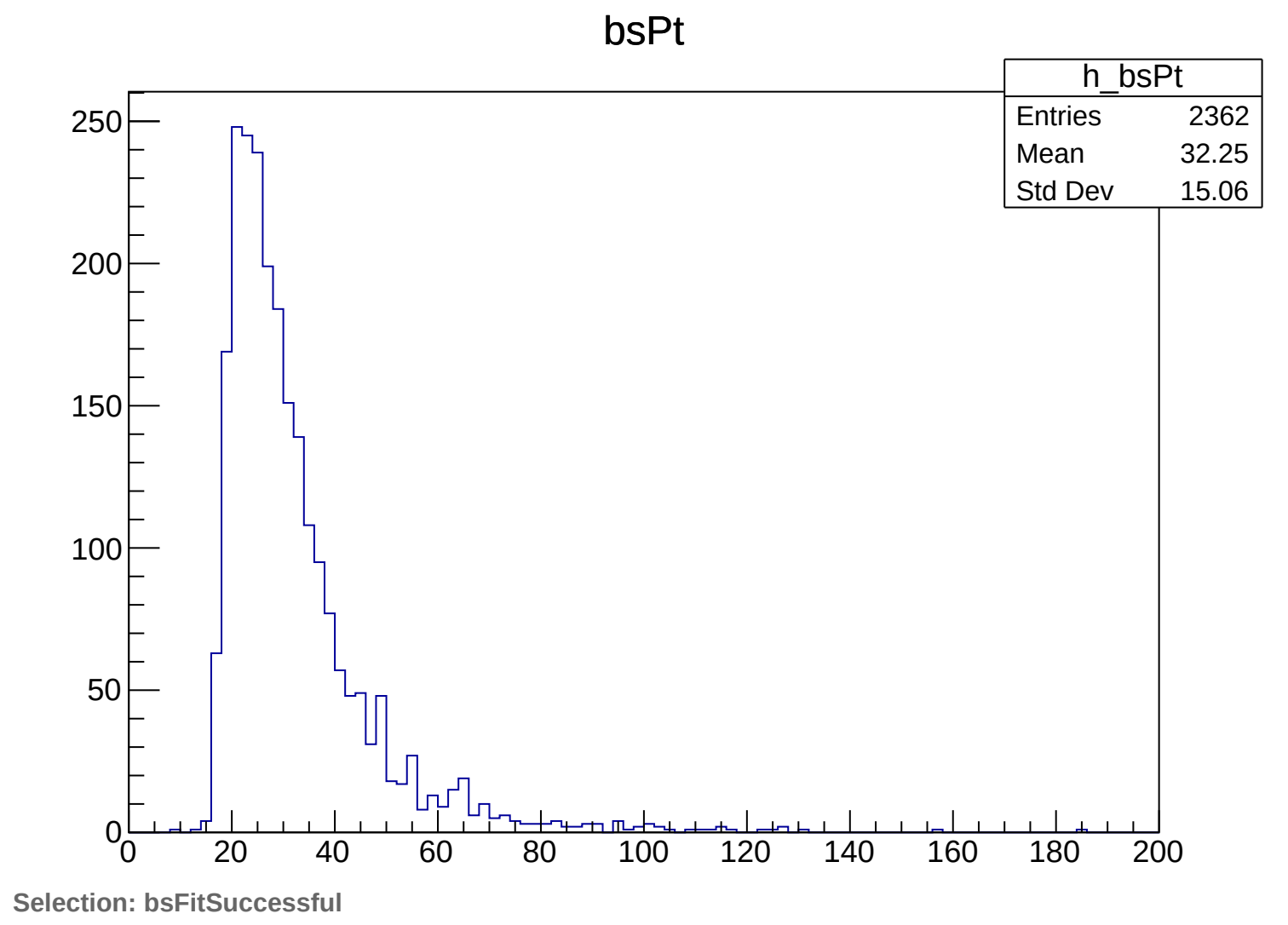

bsLxy
h_bsLxy
Entries 2362
Mean 0.2878
Std Dev 0.3607
600
500
400
300
200
100
0
0
1
2
3
4
5
6
7
8
Selection: bsFitSuccessful

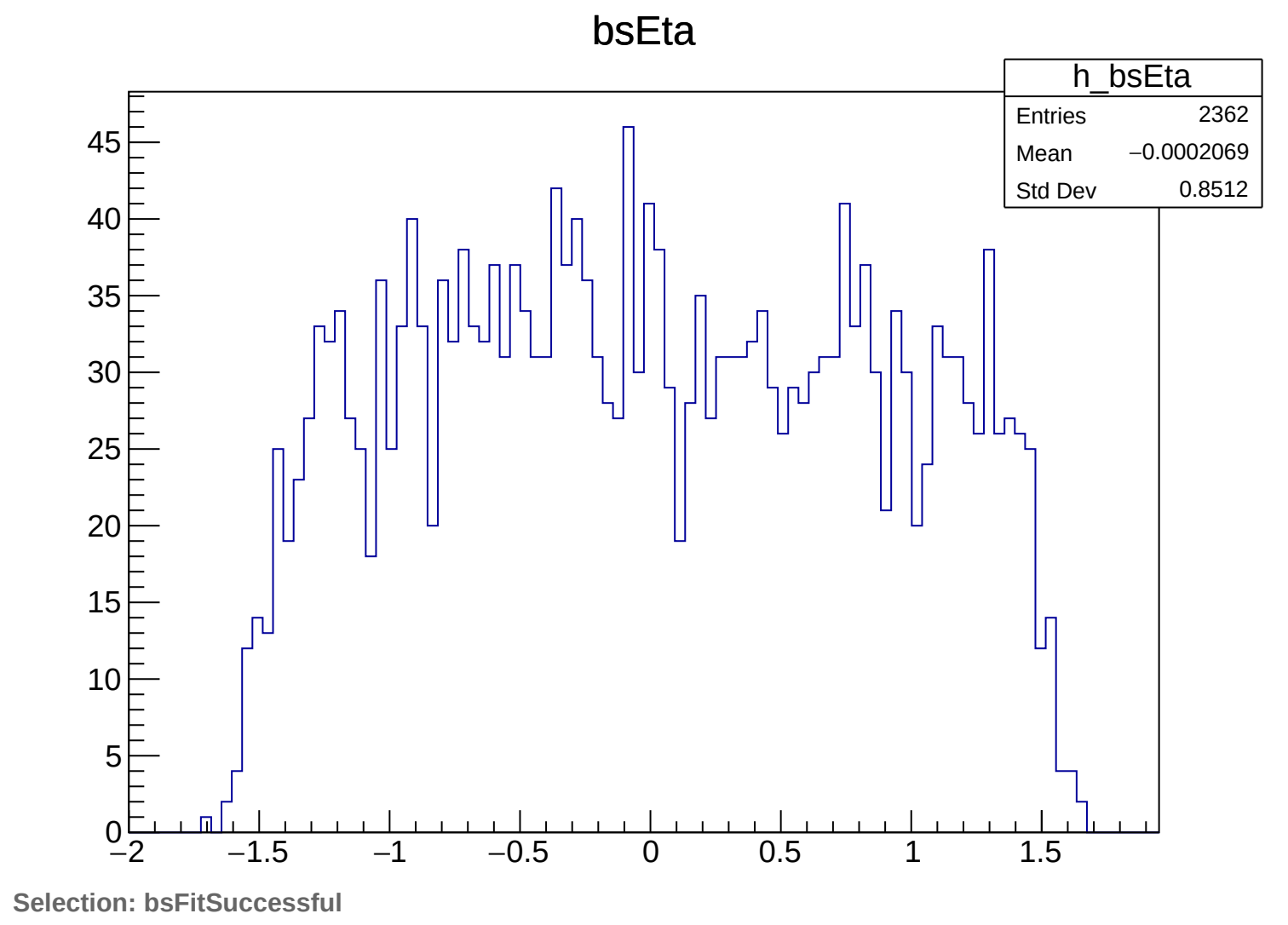

bsLxySig
h_bsLxySig
Entries 2362
Mean 54.15
Std Dev 64.97
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
Selection: bsFitSuccessful

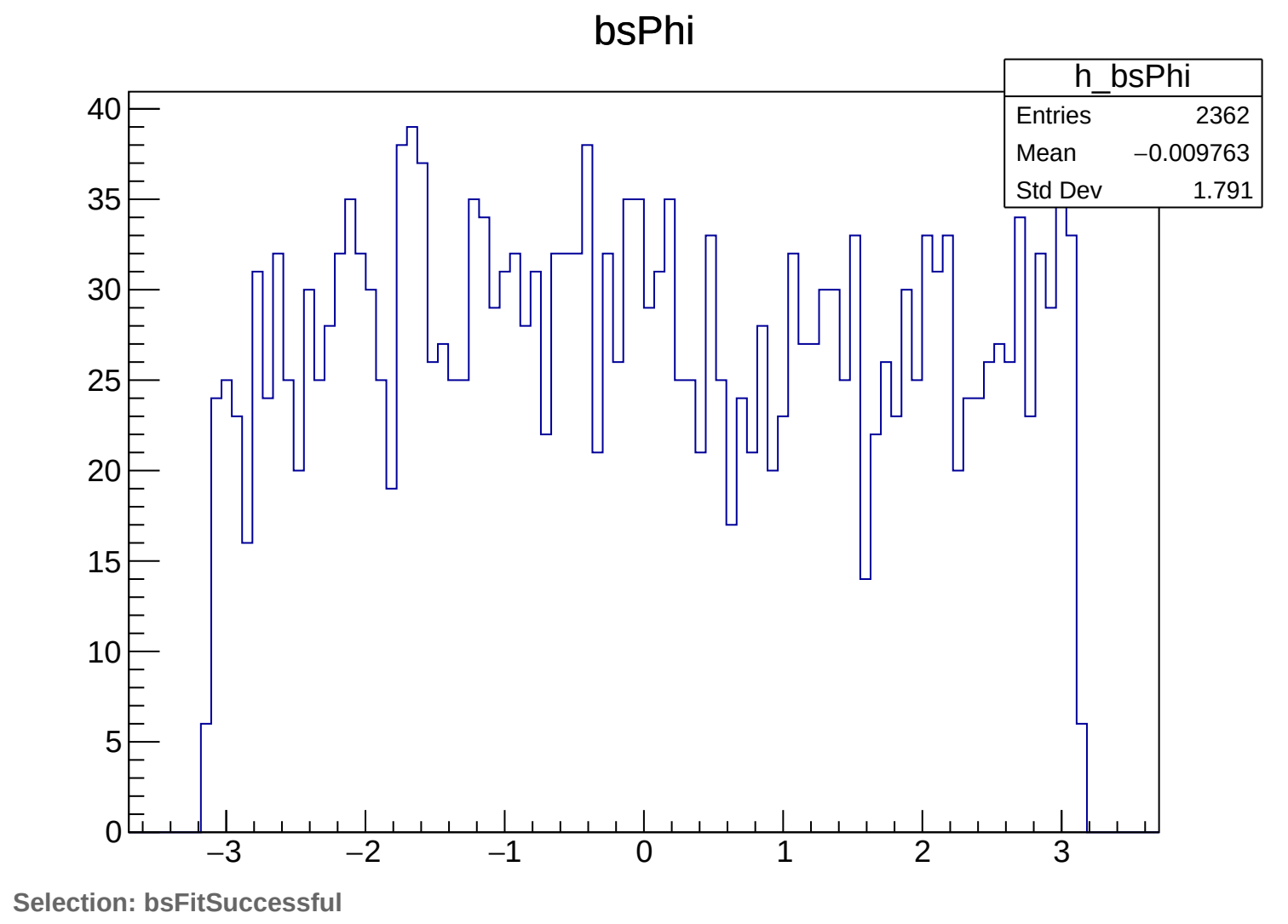

bsCt
h_bsCt
Entries 2362
Mean 0.04788
Std Dev 0.04771
220
200
180
160
140
120
100
80
60
40
20
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4
0.45
Selection: bsFitSuccessful

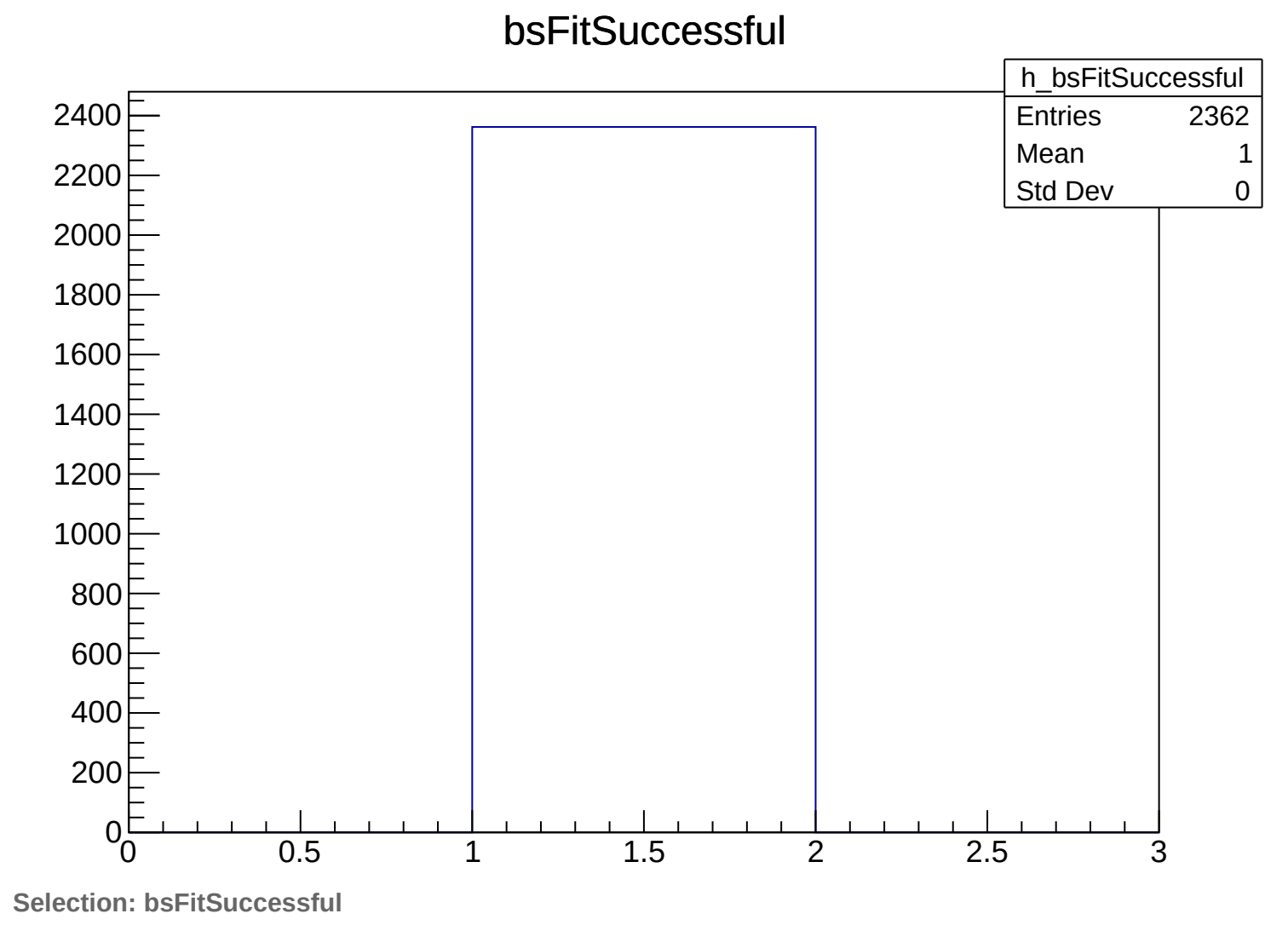

bsMass
h_bsMass
Entries 2362
Mean 5.402
Std Dev 0.1915
300
250
200
150
100
50
0
3
3.5
4
4.5
5
5.5
6
6.5
Selection: bsFitSuccessful


bsEnergy
h_bsEnergy
Entries 2362
Mean 45.29
Std Dev 23.63
220
200
180
160
140
120
100
80
60
40
20
0
0
50
100
150
200
250
Selection: bsFitSuccessful


bsPt
h_bsPt
Entries 2362
Mean 32.25
Std Dev 15.06
250
200
150
100
50
0
0
20
40
60
80
100
120
140
160
180
200
Selection: bsFitSuccessful


bsEta
h_bsEta
Entries 2362
Mean −0.0002069
Std Dev 0.8512
45
40
35
30
25
20
15
10
5
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
Selection: bsFitSuccessful


bsPhi
h_bsPhi
Entries 2362
Mean −0.009763
Std Dev 1.791
40
35
30
25
20
15
10
5
0
−3
−2
−1
0
1
2
3
Selection: bsFitSuccessful


bsFitSuccessful
h_bsFitSuccessful
Entries 2362
Mean 1
Std Dev 0
2400
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
0.5
1
1.5
2
2.5
3
Selection: bsFitSuccessful